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Stamps et al.

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(54) **SYSTEMS, METHODS, AND COMPOSITIONS FOR THE GENERATION OF WATER-SOLUBLE CANNABINOIDS UTILIZING FATTY-ACID BINDING PROTEINS AND FATTY-ACID BINDING-LIKE PROTEINS AS CANNABINOID-CARRIERS**

14/435; C07K 2319/036; C12N 15/62;
C12P 7/22; C12P 7/42; C12P 21/02;
C12P 17/06; C07D 311/80

See application file for complete search history.

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(57) **ABSTRACT**

The inventive technology includes novel systems, methods, and compositions for the generation of water-soluble short-chain fatty acid phenolic compounds, preferably cannabinoids, terpenes, and other volatile compounds produced in *Cannabis* and other cannabinoid producing plants. In particular, the inventive technology includes novel systems and methods to solubilize short-chain fatty acid phenolic compounds, such as cannabinoids, via binding to a water soluble and readily digested carrier protein.

10 Claims, 16 Drawing Sheets

Specification includes a Sequence Listing.

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A61K 31/05 (2006.01)

A61K 31/352 (2006.01)

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(52) **U.S. Cl.**

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CPC **A61K 47/64**; **A61K 31/05**; **A61K 31/352**; **A61K 31/658**; **C07K 14/415**; **C07K**

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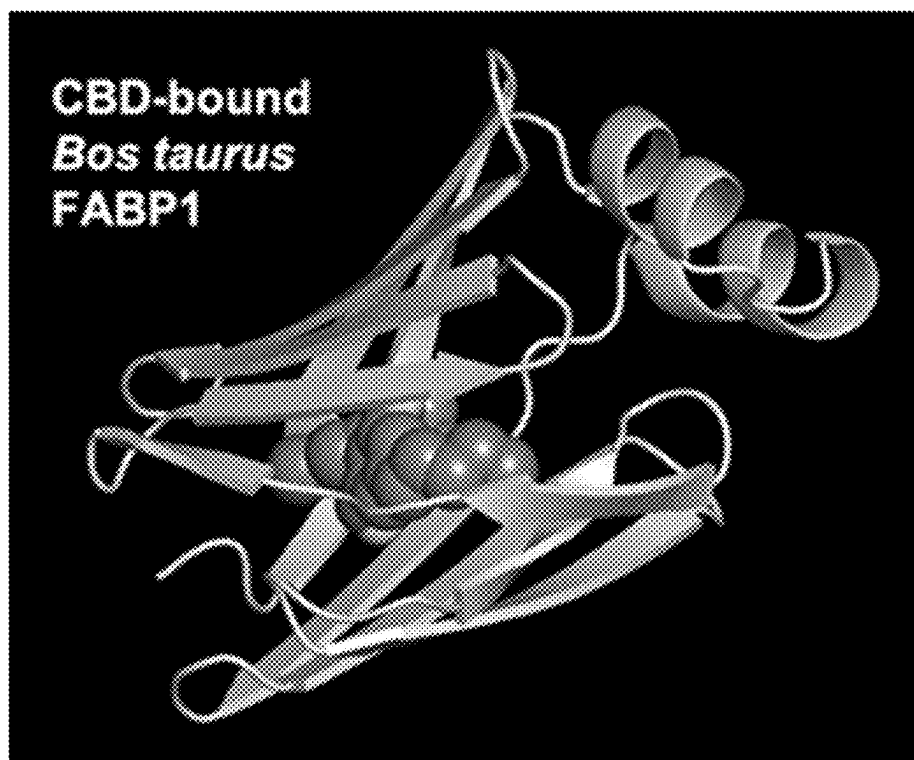


FIGURE 1A

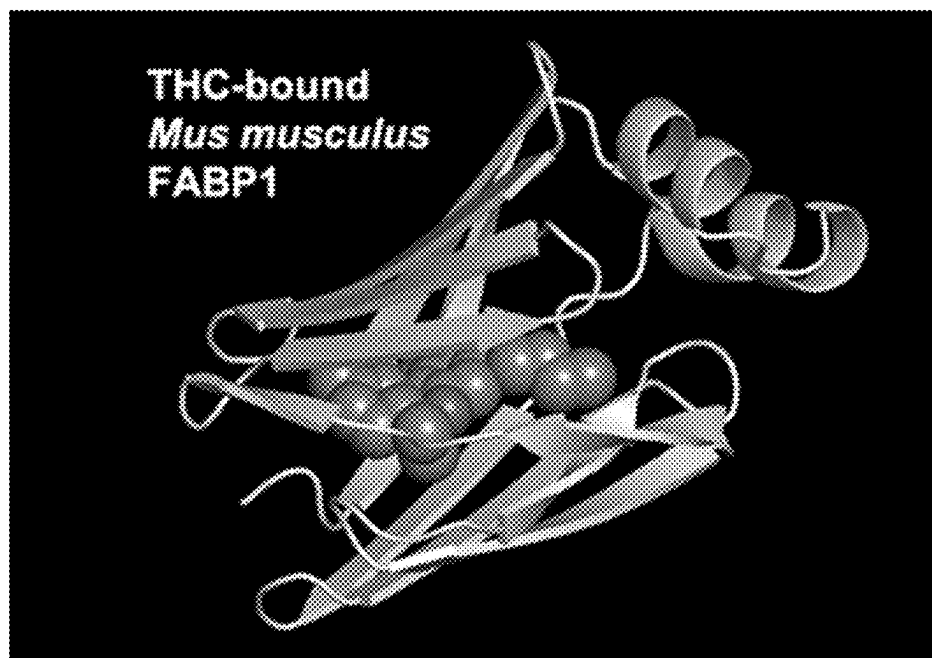


FIGURE 1B

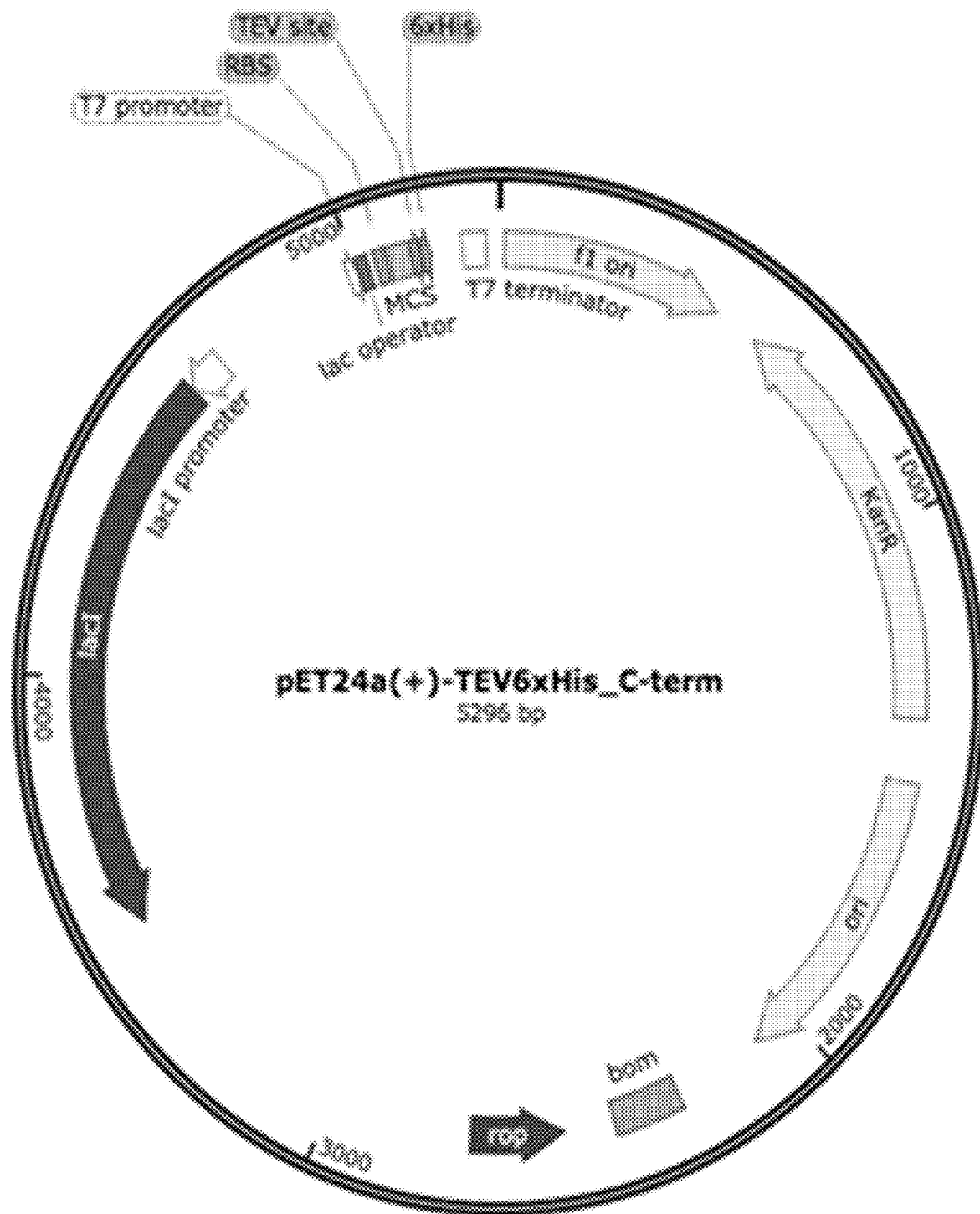


FIGURE 2

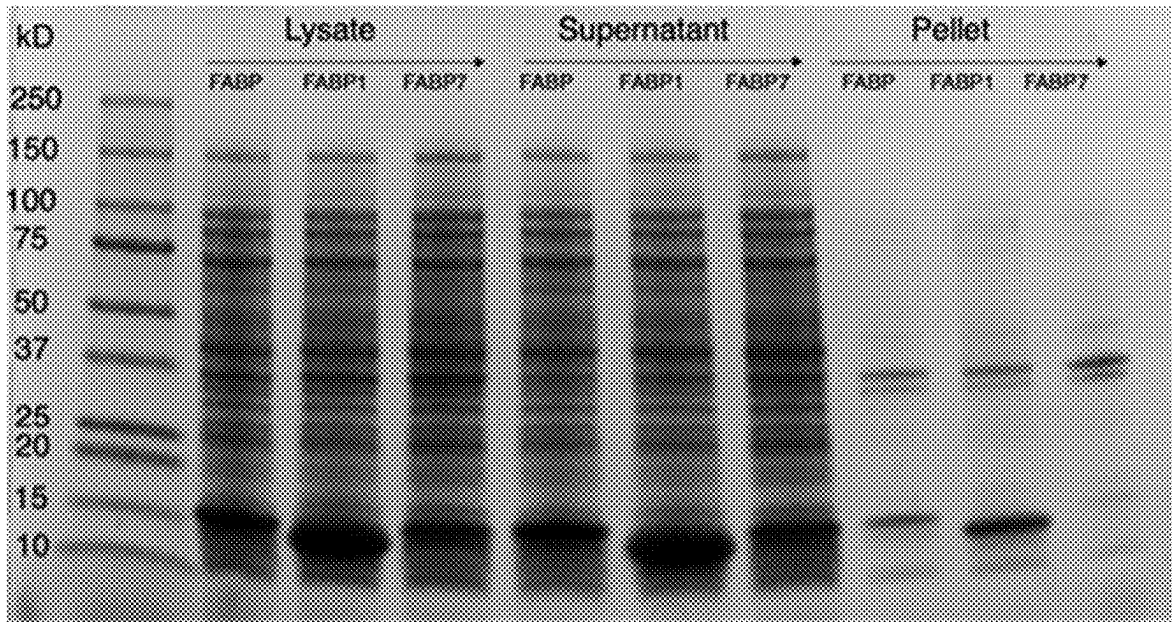


FIGURE 3A

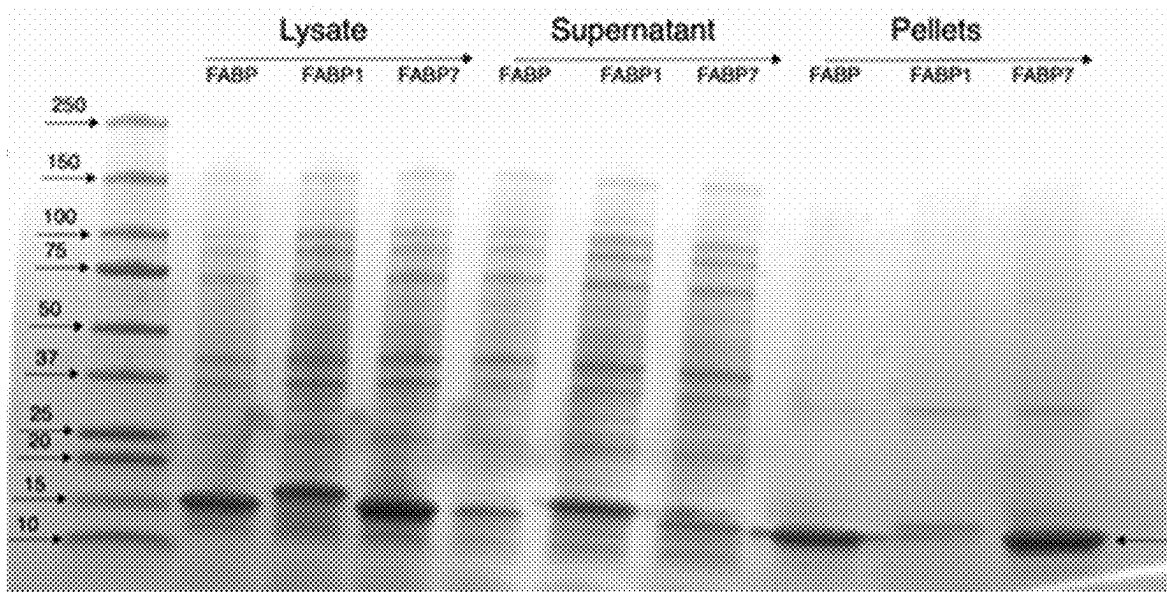


FIGURE 3B

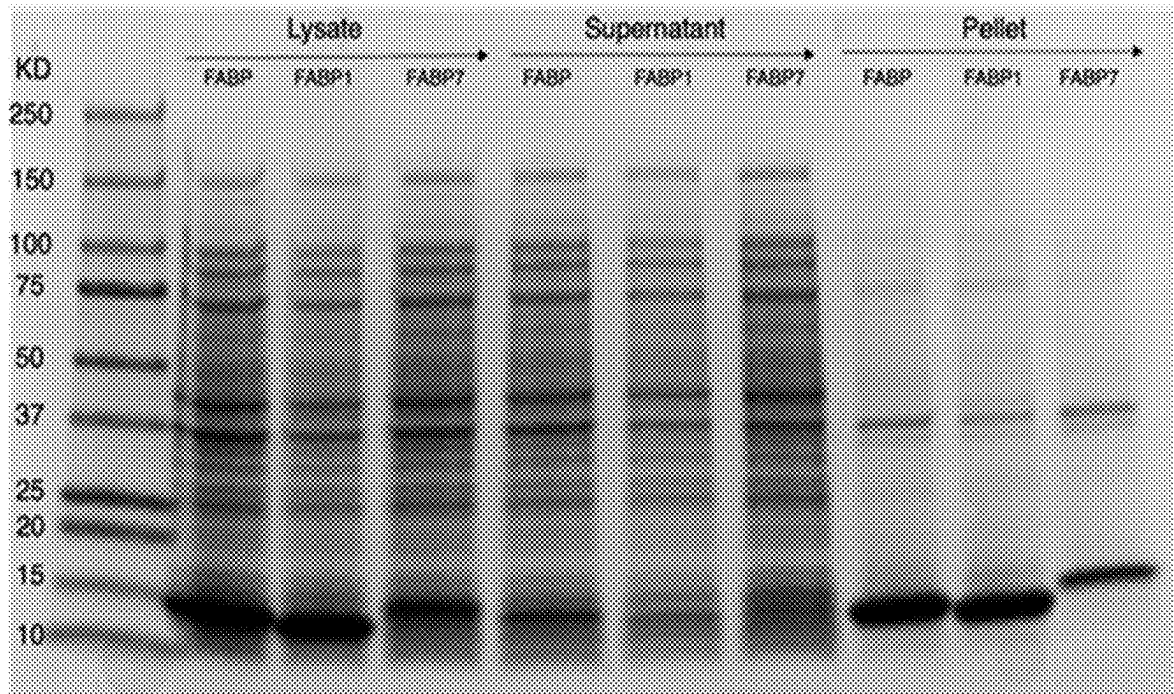


FIGURE 3C

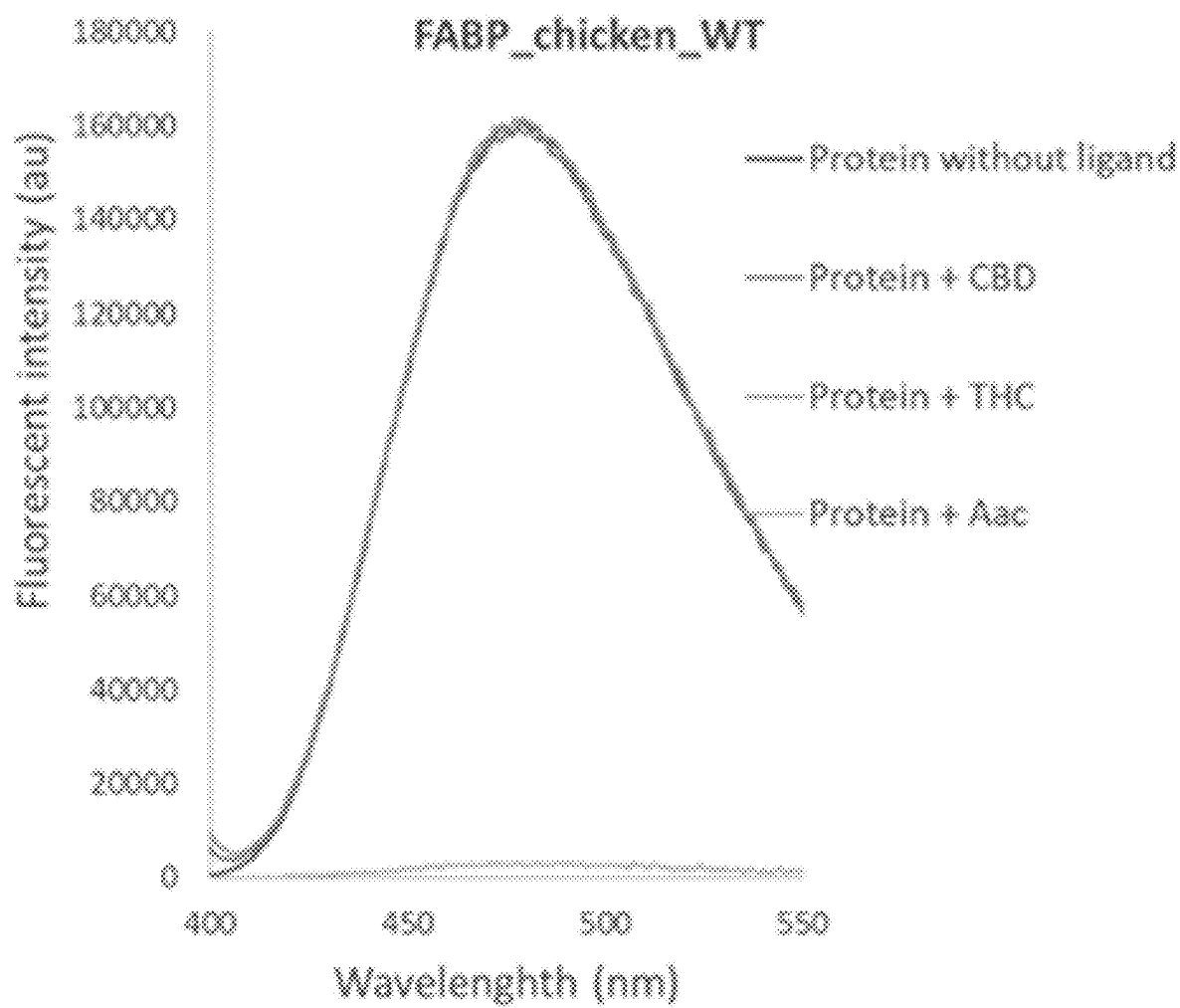


FIGURE 4A

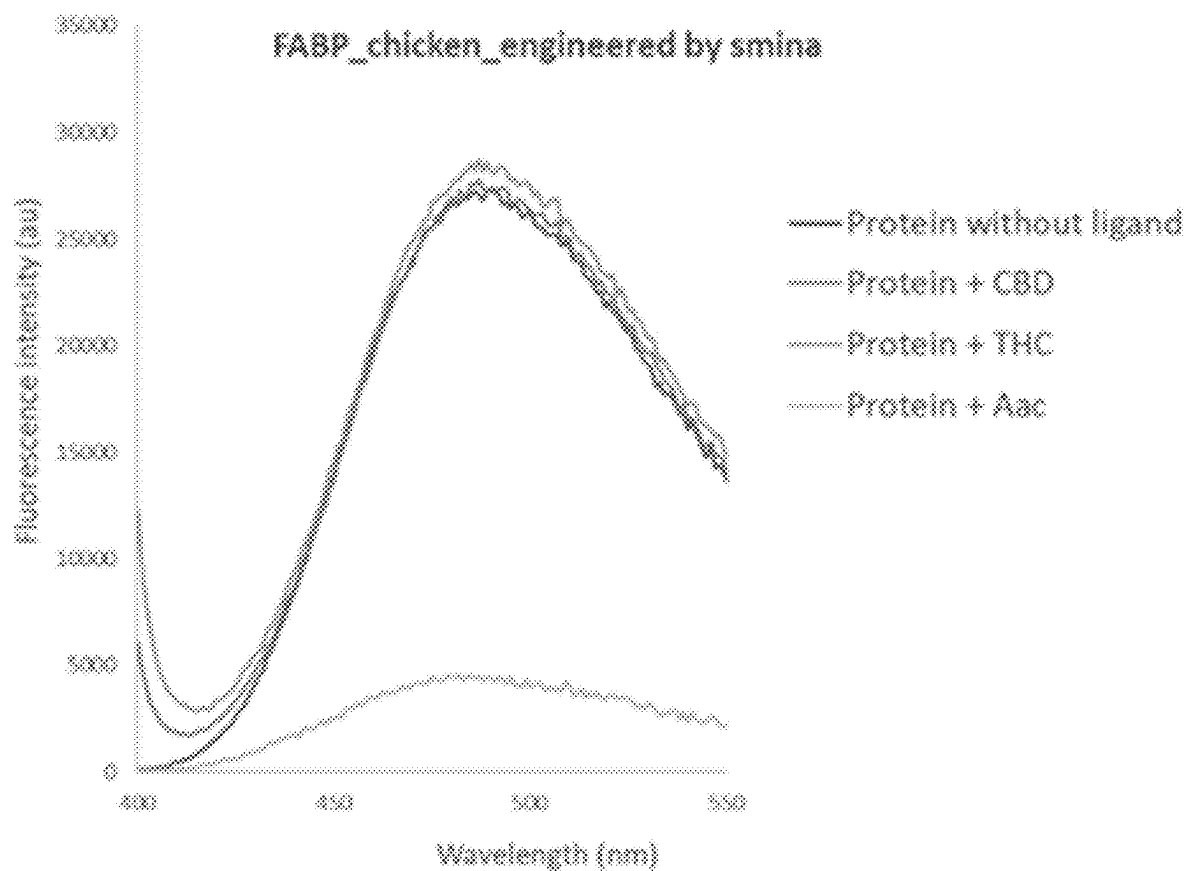


FIGURE 4B

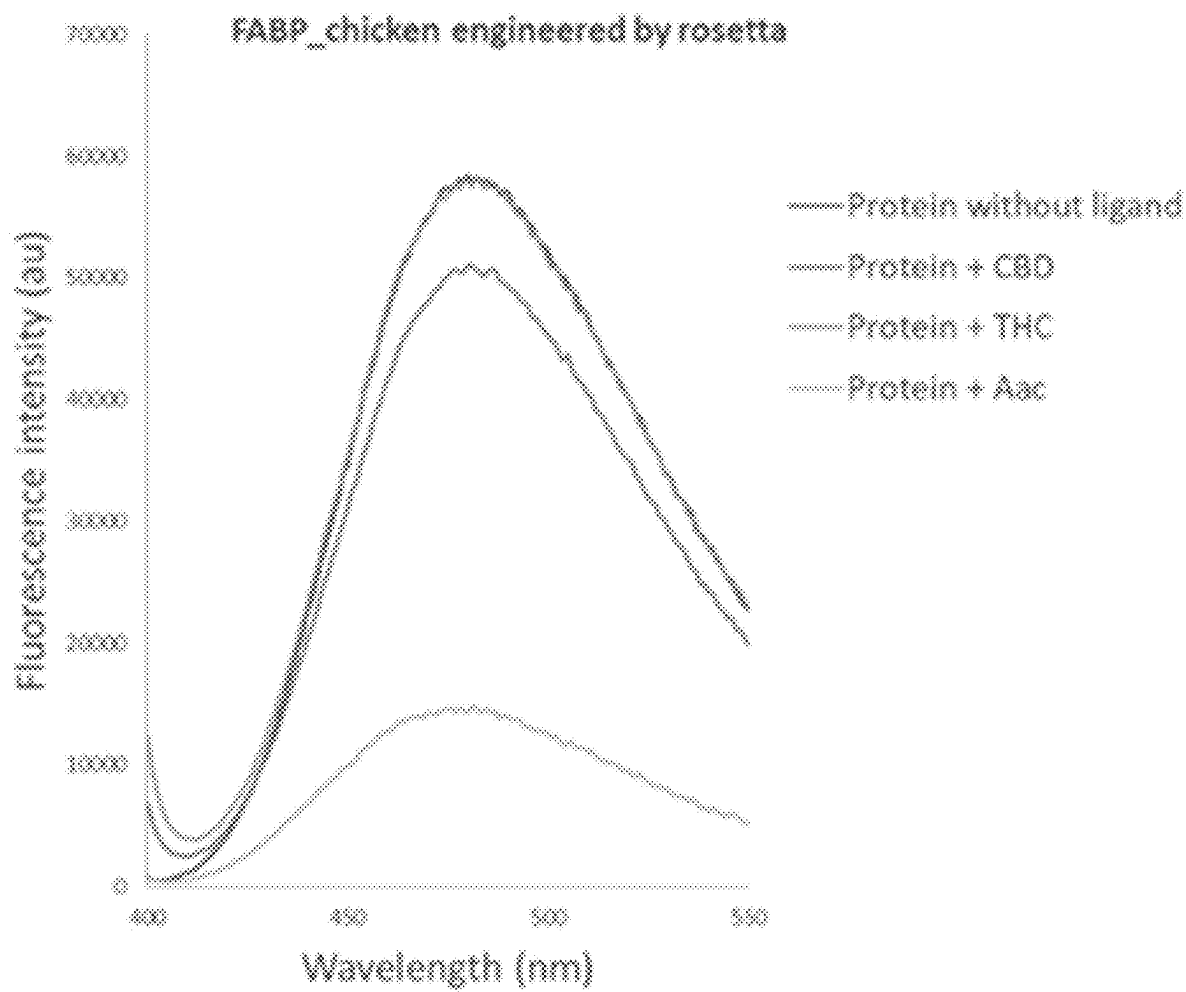


FIGURE 4C

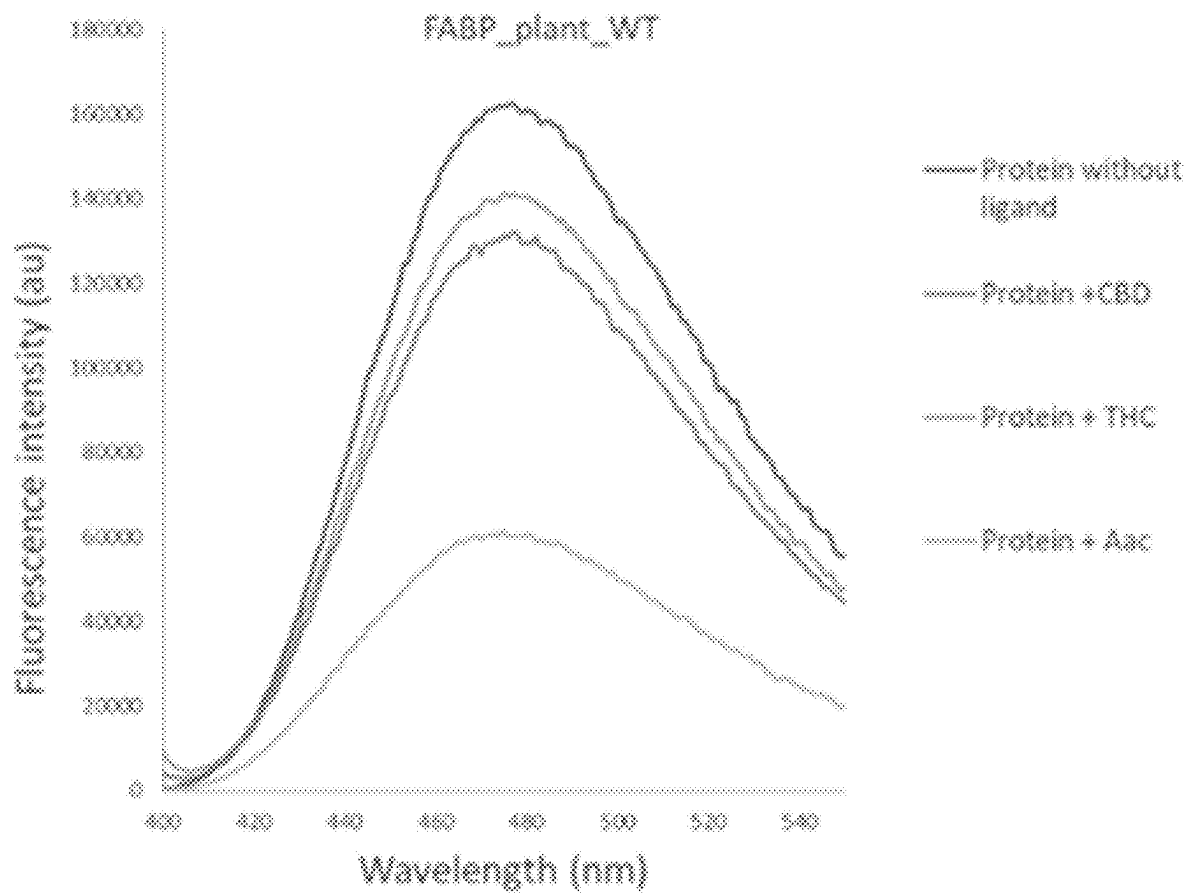


FIGURE 5A

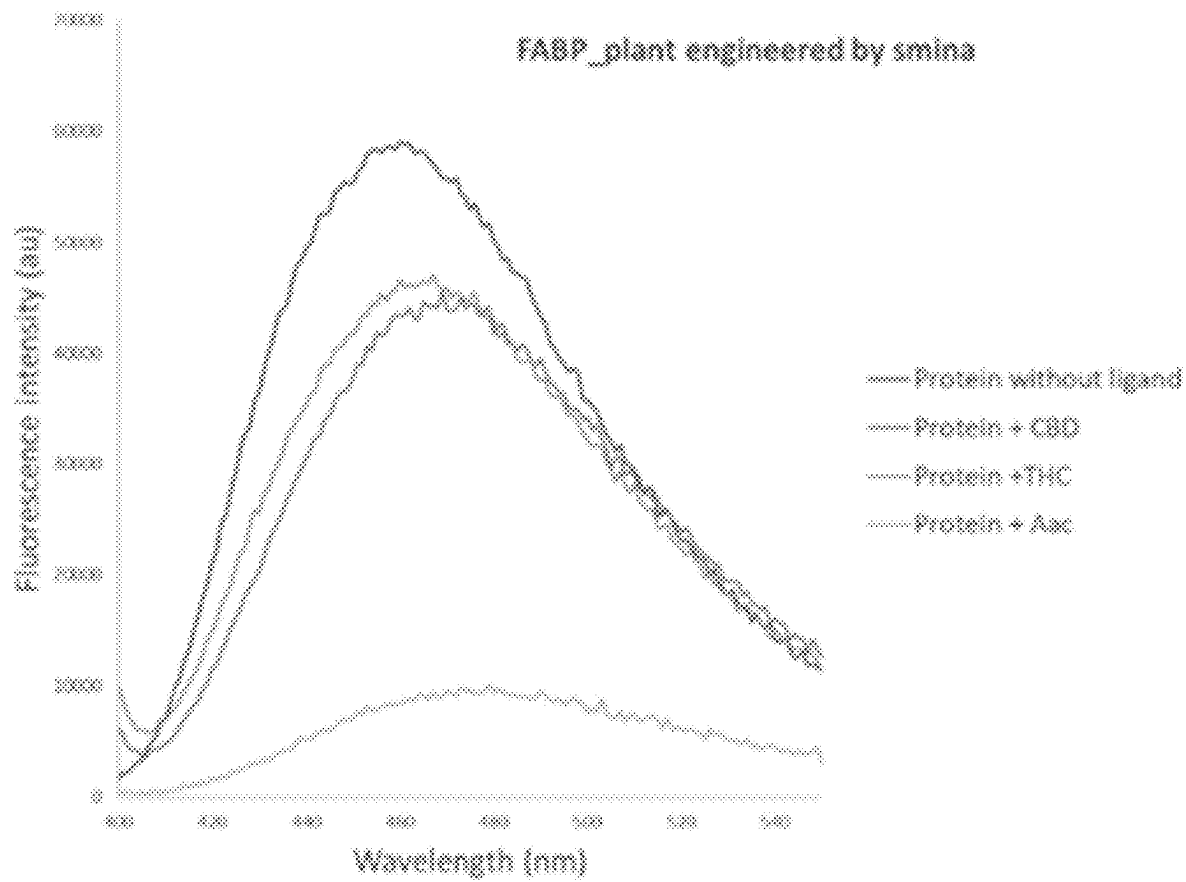


FIGURE 5B

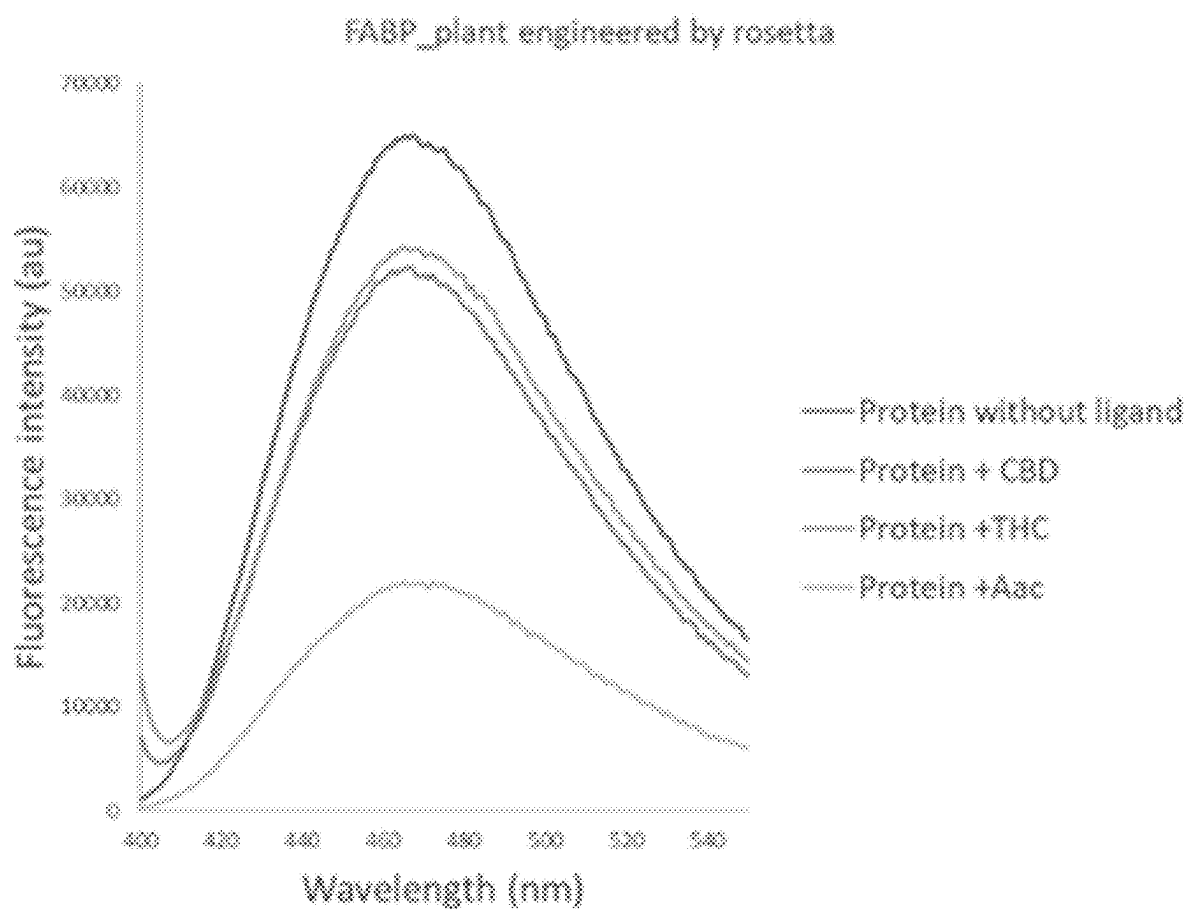


FIGURE 5C

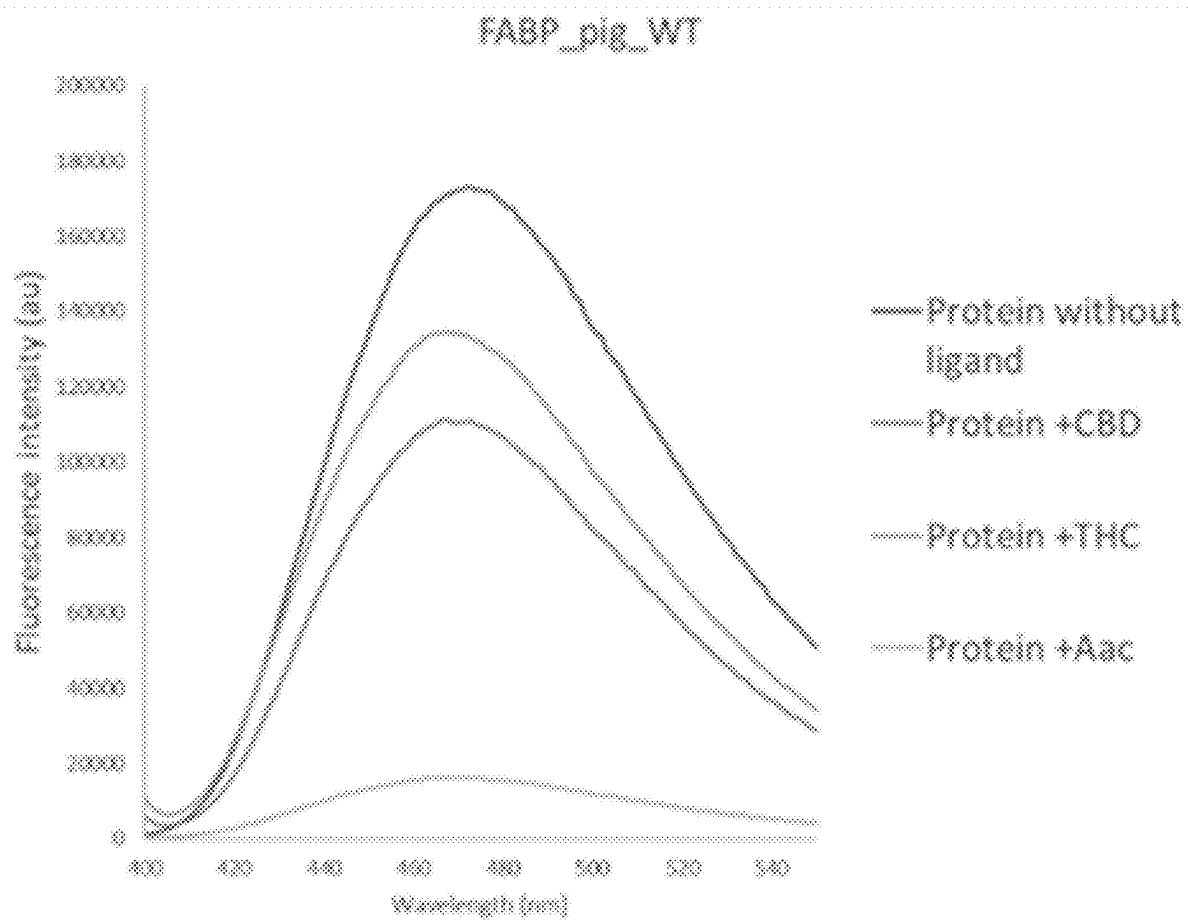


FIGURE 6A

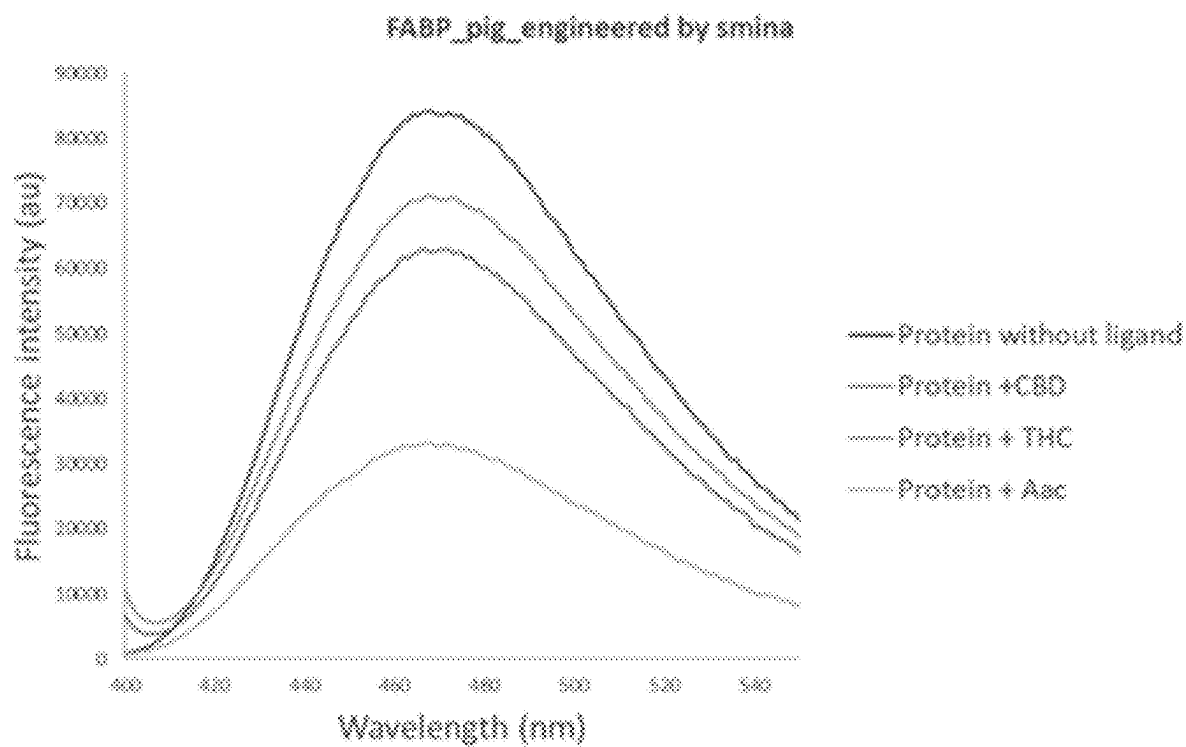


FIGURE 6B

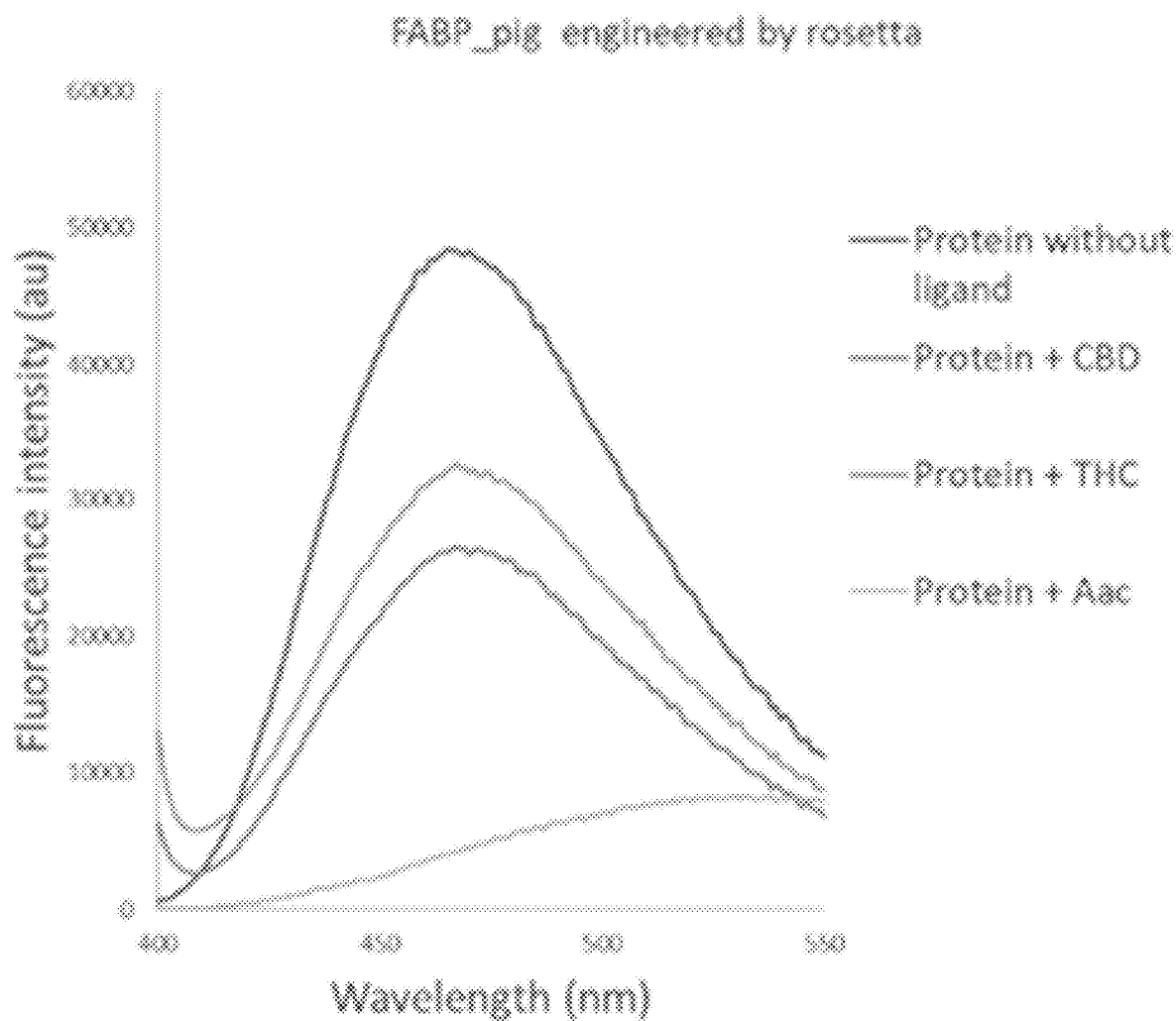


FIGURE 6C

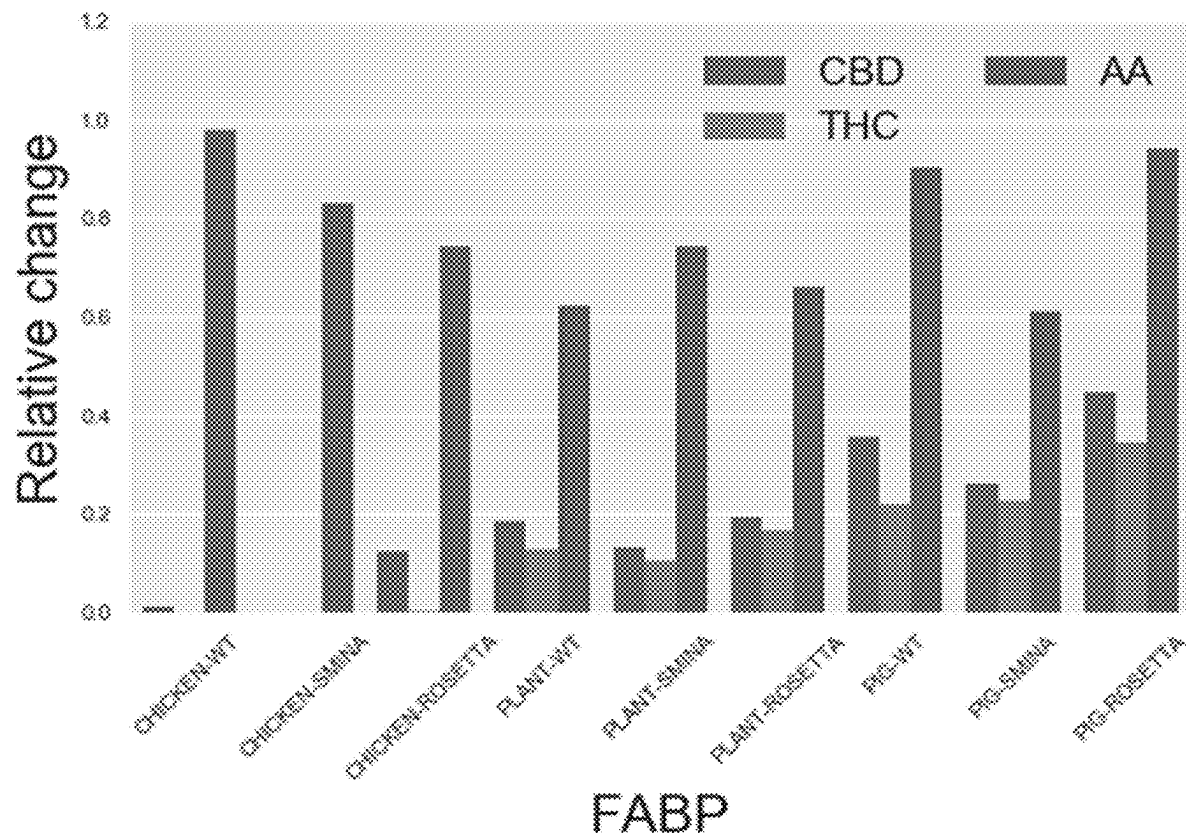


FIGURE 7

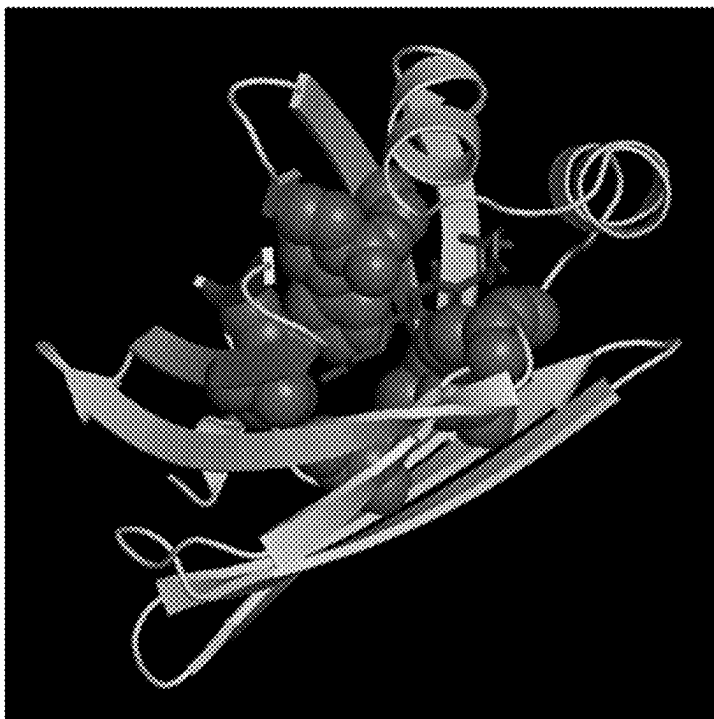


FIGURE 8A

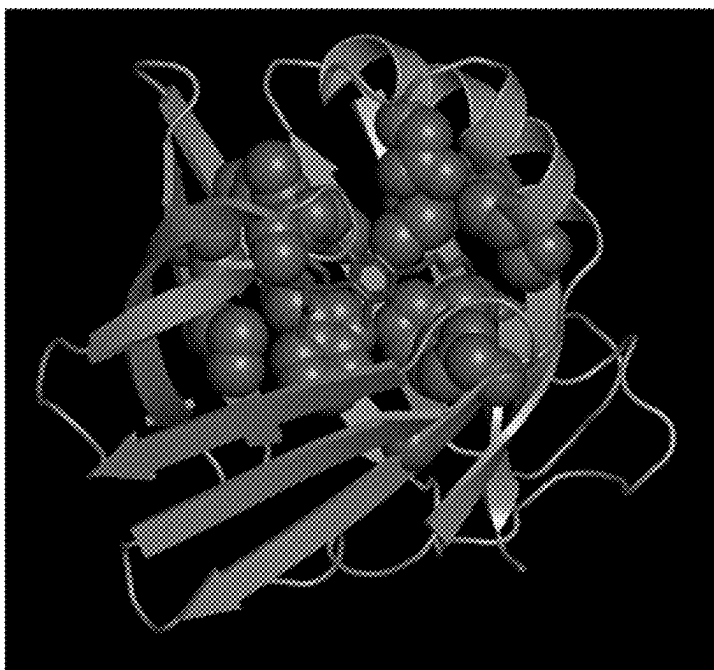


FIGURE 8B

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SYSTEMS, METHODS, AND COMPOSITIONS FOR THE GENERATION OF WATER-SOLUBLE CANNABINOIDS UTILIZING FATTY-ACID BINDING PROTEINS AND FATTY-ACID BINDING-LIKE PROTEINS AS CANNABINOID-CARRIERS

CROSS REFERENCE TO RELATED APPLICATION

This application is a national stage application under 35 U.S.C. 371 of PCT Application No. PCT/US20/16642 having an international filing date of Feb. 4, 2020, which designated the United States, which PCT application claimed the benefit of U.S. Application Ser. No. 62/800,692, filed Feb. 4, 2019, both of which are incorporated by reference in their entirety.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Aug. 4, 2021, is named 90425-00162-Sequence-Listing-AF.txt, and is 213 Kbytes in size.

This International PCT Application claims the benefit of and priority to U.S. Provisional Application No. 62/800,692 filed Feb. 4, 2019. The entire specification and figures of the above-referenced application is hereby incorporated, in its entirety by reference.

TECHNICAL FIELD

The inventive technology includes novel systems, methods, and compositions for the generation of water-soluble short-chain fatty acid phenolic compounds, preferably cannabinoids, terpenes, and other volatile compounds produced in *Cannabis* and other cannabinoid producing plants. In particular, the inventive technology includes novel systems and methods to solubilize short-chain fatty acid phenolic compounds, such as cannabinoids, via binding to a water soluble and readily digested carrier protein.

BACKGROUND OF THE INVENTION

Cannabinoids are a class of specialized compounds synthesized by *Cannabis*. They are formed by condensation of terpene and phenol precursors. They include these more abundant forms: Δ^9 -tetrahydrocannabinol (THC), cannabidiol (CBD), cannabichromene (CBC), and cannabigerol (CBG). Another cannabinoid, cannabinol (CBN), is formed from THC as a degradation product and can be detected in some plant strains. Typically, THC, CBD, CBC, and CBG occur together in different ratios in the various plant strains. These cannabinoids are generally lipophilic, nitrogen-free, mostly phenolic compounds and are derived biogenetically from a monoterpene and phenol, the acid cannabinoids from a monoterpene and phenol carboxylic acid, and have a C21 base. Cannabinoids also find their corresponding carboxylic acids in plant products. In general, the carboxylic acids have the function of a biosynthetic precursor. For example, these compounds arise in vivo from the THC carboxylic acids by decarboxylation the tetrahydrocannabinols Δ^9 - and Δ^8 -THC and CBD from the associated cannabidiol.

Importantly, cannabinoids are hydrophobic small molecules and, as a result, are highly insoluble. Due to this

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insolubility, cannabinoids such as THC and CBD may need to be efficiently solubilized to facilitate transport, storage, and adsorption through certain tissues and organs. As described in, U.S. Pat. No. 8,410,064 by Pandya et al., cannabinoids may be subject to cytochrome P450 oxidation and subsequent UDP-glucuronosyltransferase (UGT)-dependent glucuronidation in the body after consumption. The resulting glucuronide of the oxidized cannabinoid is the main metabolite found in urine, and thus, this solubilization process plays a critical role in the metabolic clearance of cannabinoids. In another embodiments outlined in PCT/US18/24409 and PCT/US18/41710 (both of which are incorporated herein in their entirety by reference), by Sayre et al., cannabinoids may be glycosylated in vivo to form water-soluble glycoside and acetylated cannabinoid glycoside compounds.

As outlined below, cannabinoids may be solubilized by binding to certain carrier proteins. For example, cannabinoids, and other short-chain fatty acid phenolic compounds, may be transported in biological fluids (such as blood) and tissues (including the intracellular milieu) by these so-called carrier proteins. Generally, the binding to these carrier proteins molecules effectively increases the water-solubility of fatty acids and other lipophilic molecules, thereby facilitating their transport through aqueous environments as well as their transfer across cellular membranes. Human and homologous non-human carrier proteins may offer an opportunity for use in the solubilization of cannabinoids among other compounds. One area where water-soluble cannabinoids has seen renewed interest is in the fields of cannabinoid-infused consumer products. However, the ability to effectively solubilize cannabinoids has limited their applicability. To overcome these limitations, many manufacturers of cannabinoid-infused products have adopted the use of traditional pharmaceutical delivery methods of using nanoemulsions of cannabinoids. This nanoemulsion process essentially coats the cannabinoid in a hydrophilic compound, such as oil or other similar compositions. However, the use of nanoemulsions is limited both technically, and from a safety perspective.

First, a large number of surfactants and cosurfactants are required for nanoemulsion stabilization. Moreover, the stability of nanoemulsions is inherently unstable, and may be disturbed by slight fluctuations in temperature and pH, and is further subject to the "oswald ripening effect" or ORE. ORE describes the process whereby molecules on the surface of particles are more energetically unstable than those within. Therefore, the unstable surface molecules often go into solution shrinking the particle over time and increasing the number of free molecules in solution. When the solution is supersaturated with the molecules of the shrinking particles, those free molecules will redeposit on the larger particles. Thus, small particles decrease in size until they disappear and large particles grow even larger. This shrinking and growing of particles will result in a larger mean diameter of a particle size distribution (PSD). Over time, this causes emulsion instability and eventually phase separation.

Second, nanoemulsions may not be safe for human consumption. For example, nanoemulsions were first developed as a method to deliver small quantities of pharmaceutical compounds having poor solubility. However, the ability to "hide" a compound, such as a cannabinoid, in a nanoemulsion may allow the cannabinoid to be delivered to parts of the body where it was previously prevented from entering, as well as accumulating in tissues and organs where cannabinoids and nanoparticles would not typically be found. Additionally, such nanoemulsions, as well as other water-com-

patible strategies, do not address one of the major shortcomings of cannabinoid-infused commercial consumables, namely the strong unpleasant smell and taste. Moreover, such water-compatible strategies deliver inconsistent and delayed cannabinoid uptake in the body which may result in user's ingesting a larger than recommended amount of cannabinoid-infused product, as well as delayed and inconsistent medical and/or psychotropic experiences.

As a result, there is a need for more effective strategies to both solubilize cannabinoids, and other associated compounds, such as terpenes and the like, in a way that is both cost-effective, as well as safe to consumers. In one aspect of the invention, members of the Fatty Acid-Binding Protein (FABP) family may be used in novel systems, methods, and compositions to solubilize cannabinoids and other associated compounds. FABPs generally act as intracellular lipid chaperones and help coordinate lipid responses in cells and are also strongly linked to metabolic and inflammatory pathways. FABPs are abundantly expressed 14-15 kDa proteins that reversibly bind hydrophobic ligands, such as saturated and unsaturated long-chain fatty acids, eicosanoids and other lipids, with high affinity. FABPs are found across species, from *Drosophila melanogaster* and *Caenorhabditis elegans* to mice and humans, demonstrating strong evolutionary conservation. However, little is known about their exact biological functions and mechanisms of action. Studies in cultured cells have suggested potential action of FABPs in fatty-acid import, storage and export as well as cholesterol and phospholipid metabolism. FABPs have also been proposed to sequester and/or distribute ligands to regulate signaling processes and enzyme activities. As will be discussed in more detail below, the current inventive technology overcomes the limitations of traditional cannabinoid emulsion systems while meeting the objectives of a truly effective and scalable cannabinoid production, solubilization, and isolation system.

SUMMARY OF THE INVENTION

Generally, the inventive technology relates to systems, methods and compositions to solubilize short-chain fatty acid phenolic compounds, such as cannabinoids, terpenes and other volatile compounds found in cannabinoid-producing plants such as *Cannabis*. In one aspect, human and homologous non-human FABP proteins, sometimes referred to herein as a carrier, or carrier-proteins, may be used to bind to, and solubilization of one or more cannabinoid compounds. In addition aspects, chimeric FABP proteins may be used to bind to, and solubilization of one or more cannabinoid compounds. In another aspect, rationally engineered FABP proteins having one or more directed mutations may exhibit enhanced binding and solubilization of one or more cannabinoid compounds.

Another aspect of the current invention may include novel methods and compositions for increasing the water solubility of one or more cannabinoid compounds via binding to a select FABP. In this embodiment, FABPs may be utilized to solubilize, transport, and store cannabinoid compounds in vitro, ex vivo, and in vivo systems. In specific preferred aspects, non-human homologs of FABPs, such as plant FABPs, or engineered FABP may be utilized to solubilize, transport, and store, for example, THC, CBD, and other cannabinoids, terpenoids, and volatile compounds produced in *Cannabis* and other cannabinoid producing plants, or even synthetically generated cannabinoids.

Another aspect of the current invention includes novel methods and compositions for increasing the water solubil-

ity of one or more cannabinoid compounds via binding to a select chimeric or genetically modified, sometimes referred to as an engineered, FABP. In this aspect, a novel chimeric FABP construct may be rationally designed from homologs of human FABP to allow for enhanced binding of cannabinoid molecules to a single protein chain. In one specific aspect, a novel chimeric FABP construct may be rationally designed from non-human homologs of FABP to allow for enhanced binding of THC, CBD, or other cannabinoid molecules to a single protein chain. In another aspect, one or more FABPs may be subjected to directed evolutionary process, preferably via in silico random mutagenesis, to identify one or more mutations that may increase the binding affinity, and solubility, of one or more target cannabinoids.

Another aspect of the current invention may include the decrease of aggregation propensity and potential antigenicity for the FABP-carrier proteins that may be specific for short-chain fatty acid phenolic compounds, such as cannabinoids. In this embodiment, mutations on the surface of a carrier protein may be rationally designed to reduce aggregate formation in solution and to reduce any elicited antigenic reactions.

Another aspect of the current invention may include the decrease of aggregation propensity and potential antigenicity for a novel wild-type, or engineered FABP-carrier protein, as discussed below, that may be specific for short-chain fatty acid phenolic compounds, such as cannabinoids. In this embodiment, mutations on the surface of a carrier protein may be rationally designed to reduce aggregate formation in solution and to reduce any elicited antigenic reactions.

Another aspect of the current invention may include systems, methods, and compositions for the solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in cell cultures that express one or more FABP-carrier, or engineered FABP proteins. Exemplary cell cultures may include bacterial, yeast, plant, algae and fungi cell cultures. In another aspect, FABP-carrier, or engineered FABP proteins, may be coupled with secretion signals to allow such proteins to be more easily exported from the cell culture into the surrounding supernatant or media. In this aspect of the invention, a FABP-carrier protein, the terms generally encompassing FABP carrier proteins, or engineered FABP proteins that bind to one or more target compounds, and preferably, cannabinoids, may be exported out of a cell through the action of the secretion signal that may direct posttranslational protein translocation into the endoplasmic reticulum (ER), or in alternative embodiments, a secretion signal that may direct cotranslational translocation across the ER membrane where it may assume its three-dimensional form and bind one or more cannabinoid or other compounds as described herein. In one preferred embodiment, a FABP-carrier protein may be generated in a cell culture, preferably a bacterial, yeast, plant or fungi cell culture, and then be exported out of the cell through natural cellular action, or through the action of the secretion signal where it may assume its three dimensional form and bind one or more cannabinoid or other compounds that may be present, preferably by addition of said compound, such as: a quantity of an isolated cannabinoid; a quantity of a plurality of cannabinoids; or *Cannabis* extract, to the culture's supernatant.

In another aspect of the invention, a FABP-carrier protein may be exported out of a cell through the action of the secretion signal after it has assumed a transitory and or final three dimensional form and may further be bound to one or more cannabinoid or other compounds as described herein. In one preferred embodiment, a FABP-carrier protein may

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be generated in a cell culture, preferably a bacterial, yeast, plant or fungi cell culture, and more preferably a plant suspension culture of a cannabinoid-producing plant such as *Cannabis*, where it may assume a transitory or final three dimensional form and bind one or more cannabinoids or other compounds that may be present or produced in the cell.

Another aspect of the current invention may include systems, methods and compositions for the solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in whole plants and plant cell cultures. In certain embodiments, such plants or cell cultures may be genetically modified to direct cannabinoid synthesis to the cytosol, as opposed to a trichome structure. One or more FABP-carrier proteins may be coupled with a secretion signal, preferable in a plant cell culture, to allow such proteins to be exported from the cell into the surrounding media. Expression of exportable and non-exportable FABP-carrier proteins may be co-expressed with one or more catalase and/or one or more myb transcription factors which may enhance cannabinoid production in a *Cannabis* plant or cell culture.

Another aspect of the current invention may include systems, methods and compositions for the coupled glycosylation and solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in whole cannabinoid-producing plants or cell cultures, preferably a *Cannabis* plant or cell culture. In this embodiment, such *Cannabis* plants or cell cultures may be genetically modified to direct cannabinoid synthesis to the cytosol, as opposed to a trichome structure as generally outlines by the current inventors, Sayre et al. in U.S. Pat. No. 10,378,020, the disclosure of such processes being fully incorporated herein by reference. Such *Cannabis* plant, or *Cannabis* cell culture may be further genetically modified to express one or more heterologous glycosyltransferases having glycosylation activity towards at least one cannabinoid (for example SEQ ID NOs. 32-47, and SEQ ID NOs. 61-62). In additional embodiments, a plant or cell may be further genetically modified to express one or more heterologous glycosyltransferases, wherein in said polynucleotides encoding such glycosyltransferases may be codon-optimized for expression in an exogenous system, such as in yeast (for example SEQ ID NOs. 49-60). In additional embodiments, a heterologous or exogenous, being generally the same term, cytochrome P450 and/or a P450 oxidoreductase may be expressed. In this configuration a heterologous cytochrome P450 (for example SEQ ID NOs. 22-23, and SEQ ID NOs. 26-27) may hydroxylate a cannabinoid to form a hydroxylated cannabinoid and/or oxidize a hydroxylated cannabinoid to form a cannabinoid carboxylic acid. Further, in this embodiment, a heterologous P450 oxidoreductase (for example SEQ ID NOs. 24-25, and SEQ ID NOs. 28-29) may facilitate electron transfer from a nicotinamide adenine dinucleotide phosphate (NADPH) to said cytochrome P450.

As noted above, a heterologous glycosyltransferase may glycosylate a cannabinoid compound and thereby produce a water-soluble cannabinoid glycoside. This glycosylated cannabinoid may bind to a heterologous FABP-carrier also expressed in the *Cannabis* plant or cell that may be coupled with a secretion signal, to allow the carrier proteins to be exported from the cell into the surrounding media. Expression of exportable and non-exportable FABP-carriers may be co-expressed with one or more catalase and/or one or more myb transcription factors. The glycosylated cannabinoids bound to the FABP-carrier, being further coupled with a tag in some embodiments, may be isolated, while in still further embodiments, the FABP-carrier protein may be

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disrupted by a protease, or other protein disrupting detergent and the like, such that the glycosylated cannabinoid may be released from the FABP-carrier and may be further isolated or reconstituted to their original forms through the action of a glycosidase that may remove the sugar moiety.

Another aspect of the current invention may include systems, methods, and compositions for the coupled glycosylation and solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in non-cannabinoid-producing plants and cell cultures, preferably a tobacco cell culture. In this embodiment, a tobacco cell culture may endogenously express one or more glycosyltransferases having glycosylation activity towards at least one cannabinoid. The tobacco cell culture may optionally be genetically modified to express a heterologous cytochrome P450, and a P450 oxidoreductase. In this configuration a heterologous cytochrome P450 may hydroxylate a cannabinoid added to a tobacco cell culture for example, to form a hydroxylated cannabinoid and/or oxidizes a hydroxylated cannabinoid to form a cannabinoid carboxylic acid. Further, in this embodiment, a heterologous P450 oxidoreductase may facilitate electron transfer from a nicotinamide adenine dinucleotide phosphate (NADPH) to said cytochrome P450. As noted above, the endogenously expressed heterologous glycosyltransferases (for example, NtGT1, 2, 3, 4 or 5 as identified below) may glycosylate one or more cannabinoids introduced to the tobacco cell culture converting it into a water-soluble cannabinoid-glycoside. This glycosylated cannabinoid may bind to a heterologous FABP-carrier co-expressed or added to the tobacco cell culture. In this aspect, an expression of an exportable FABP-carrier may be co-expressed with one or more catalase and/or one or more myb transcription factors. The glycosylated cannabinoids bound to the FABP-carrier, being further coupled with a tag in some embodiments, may be isolated, while in still further embodiments, the carrier protein may be disrupted by a protease or other protein disrupting detergent and the like such that the glycosylated cannabinoids may be released from the carrier protein and may be further isolated or reconstituted to their original forms through the action of a glycosidase.

Another aspect of the current invention may include systems, methods and compositions for the coupled glycosylation and solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in a cell cultures, preferably a yeast cell culture. In these embodiments, yeast cultures may be genetically modified to biosynthesize one or more cannabinoids. The yeast cell culture may be further genetically modified to express one or more heterologous glycosyltransferases having glycosylation activity towards at least one cannabinoid, as well as in some embodiments, a heterologous cytochrome P450 and/or a P450 oxidoreductase.

As noted above, heterologous glycosyltransferases may glycosylate the cannabinoid making it water-soluble. This glycosylated cannabinoid may bind to a heterologous FABP-carrier protein also expressed in the yeast culture which may further be coupled with a secretion signal, to allow the carrier proteins to be exported from the yeast cell into the surrounding media. Expression of exportable and non-exportable FABP-carrier may be co-expressed with a catalase. The glycosylated cannabinoids bound to the FABP-carrier being further coupled with a tag in some embodiments, may be isolated, while in still further embodiments, the carrier protein may be disrupted by a protease or other protein disrupting detergent and the like such that the glycosylated cannabinoids may be released from the carrier protein and

may be further isolated or reconstituted to their original forms through the action of a glycosidase.

Another aspect of the current invention may include systems, methods and compositions for the coupled glycosylation and solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in a cell cultures, preferably yeast, bacteria, fungi or algal cell culture. In these embodiments, a yeast cultures may be genetically modified to express one or more heterologous glycosyltransferases having glycosylation activity towards at least one cannabinoid, as well as in some embodiments, a heterologous cytochrome P450 and/or a P450 oxidoreductase.

As noted above, in one preferred embodiment, a quantity of cannabinoids may be added to the cell culture, and preferably a yeast cell culture, where heterologous glycosyltransferases may glycosylate the cannabinoid making it water-soluble. This glycosylated cannabinoid may bind to a heterologous FABP-carrier co-expressed in the yeast culture which may further be coupled with a secretion signal, to allow the carrier proteins to be exported from the yeast cell into the surrounding media. The glycosylated cannabinoids bound to the FABP-carrier, being further coupled with a tag in some embodiments, may be isolated, while in still further embodiments, the carrier protein may be disrupted by a protease or other protein disrupting detergent and the like such that the glycosylated cannabinoids may be released from the carrier protein and may be further isolated or reconstituted to their original forms through the action of a glycosidase.

Another aspect of the current invention may include one or more heterologous glycosyltransferases coupled with the expression of a FABP-carrier optionally having secretion signal, and in some embodiments a tag, which may be expressed in a plant, yeast or bacterial cell culture. Another aspect of the current invention may include one or more heterologous glycosyltransferases coupled with the addition of a FABP-carrier to a plant, yeast, or bacterial cell culture.

Another aspect of the current invention may include one or more endogenously expressed glycosyltransferases coupled with the expression of a FABP-carrier, and preferably an engineered FABP-carrier having secretion signal, and in some embodiments a tag, that may be expressed in a plant, yeast or bacterial cell culture. Another aspect of the current invention may include one or more endogenously expressed glycosyltransferases coupled with the addition of a FABP-carrier to a plant cell culture.

Another aspect of the current invention may include the increase of CBD water solubility for transport via binding to a FABP-carrier. In this embodiment, plant or other non-human homologs of FABPs may be utilized to solubilize, transport, and/or store CBD and closely-related cannabinoids. Another aspect of the current invention may include the increase of CBD water solubility for transport via binding to a FABP-carrier. In this aspect, a novel engineered FABP-carrier construct may be rationally designed from homologs of FABPs to allow for enhanced binding of CBD molecules to a single protein chain.

Another aspect of the current invention may include the increase of THC water solubility for transport via binding to a FABP-carrier. In this embodiment, plant or other non-human homologs of FABPs may be utilized to solubilize, transport, and/or store THC and closely-related cannabinoids. Another aspect of the current invention may include the increase of THC water solubility for transport via binding to a FABP-carrier. In this aspect, a novel engineered FABP-carrier construct may be rationally designed from

homologs of FABPs to allow for enhanced binding of THC molecules to a single protein chain.

Another aspect of the inventive technology may include polynucleotides encoding one or more engineered FABP carrier proteins having enhanced cannabinoid affinity motifs. Another aspect of the inventive technology may include polynucleotides encoding one or more engineered FABP-carrier proteins being heterologously expressed in a genetically modified microorganism, such as yeast, bacteria, fungi, algae and the like.

Another aspect of the inventive technology may include genetically modified bacteria that express at least one polynucleotide encoding one or more heterologous FABP-carrier, and preferably one or more engineered FABP-carrier proteins having enhanced cannabinoid affinity motifs. Another aspect of the inventive technology may include novel engineered FABP-carrier amino acid sequences having enhanced cannabinoid affinity motifs.

Another aspect of the inventive technology provides for a method of enhancing the solubility and stability of cannabinoids, terpenoids and/or other short-chain fatty acid phenolic compounds utilizing FABP-carrier proteins. In a preferred embodiment, a nucleotide sequence encoding a FABP-carrier protein may be genetically engineered to express a rationally designed FABP-carrier protein having cannabinoid affinity or binding sites having enhanced affinity for cannabinoids such that the engineered FABP-carrier carrier protein may bind cannabinoids with a higher affinity thereby increasing the solubility and stability of the cannabinoid in a solution or other form.

Another aspect of the invention includes compositions of novel engineered FABP-carrier polynucleotides and proteins and their method or manufacture. Another aspect of the invention includes compositions of novel engineered FABP-carrier polynucleotides and proteins and their method or manufacture. Another aspect of the invention involves the identification of FABP-carrier proteins that may have endogenous cannabinoid or other affinity sites. Another aspect of the invention involves the rational design of engineered FABP-carrier proteins have enhanced cannabinoid affinity.

Another aspect of the invention includes compositions of novel consumer products that incorporate one or more solubilized cannabinoids or terpenoids bound to FABP-carrier proteins and/or engineered FABP-carrier proteins.

Additional aspects of the invention may include one or more of the following embodiments:

1. A method of solubilizing a cannabinoid comprising the steps of:

generating a Fatty-Acid Binding Protein (FABP)-carrier protein having affinity towards at least one cannabinoid; and

introducing said FABP-carrier protein to said at least one cannabinoid, wherein said FABP-carrier protein binds said at least one cannabinoid to form a water-soluble protein-cannabinoid composition.

2. The method of embodiment 1, wherein the FABP-carrier protein comprises a FABP-carrier protein having an amino acid sequence selected from the group of consisting of: SEQ ID NOs. 1-2, 21, and 74-86, or a homolog having affinity towards at least one cannabinoid thereof.

3. The method of embodiment 2, wherein said step of generating a FABP-carrier protein comprises the step of generating a FABP-carrier protein in a protein production system selected from the group consisting of:

a bacterial cell culture;
a yeast cell culture;
a plant cell culture;

a fungi cell culture;
 an algae cell culture;
 a bioreactor production system;
 a plant.

4. The method of embodiment 3, wherein the FABP-carrier protein is coupled with a secretion signal.

5. The method of embodiment 4, wherein said secretion signal comprises a secretion signal selected from the group consisting of: SEQ ID NO. 3, SEQ ID NO. 65, SEQ ID NOs. 69-73.

6. The method of embodiments 3 and 5, wherein the FABP-carrier protein is introduced to said at least one cannabinoid in said protein production system.

7. The method of embodiment 2, wherein the at least one cannabinoid comprises a cannabinoid selected from the group consisting of: cannabidiol (CBD), cannabidiolic acid (CBDA), Δ^9 -tetrahydrocannabinol (THC), tetrahydrocannabinolic acid (THCA), and (cannabigerolic acid) CBGA).

8. The method of embodiment 1, and further comprising the step of genetically modifying the FABP-carrier protein form to an engineered FABP-carrier protein having enhanced affinity for at least one cannabinoid, such genetic modification comprising one or more of the following:

replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket with side chains pointing towards orientated toward the binding cavity;

replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket having a hydrophilic side chain with amino acid residues having a hydrophobic side chain; and

replacing one or more small hydrophobic amino acid residues of the FABP-carrier protein cannabinoid binding pocket with larger hydrophobic amino acid residues.

9. The method of embodiment 8, wherein said engineered FABP-carrier protein comprises an engineered FABP-carrier protein having an amino acid sequence selected from the group of consisting of: SEQ ID NOs. 79-86.

10. A method of solubilizing a cannabinoid comprising the steps of:

genetically modifying a Fatty-Acid Binding Protein (FABP)-carrier protein having affinity towards at least one cannabinoid to produce an engineered FABP-carrier protein having enhanced affinity towards said at least one cannabinoid;

introducing said engineered FABP-carrier protein to at least one cannabinoid, wherein said engineered FABP-carrier protein binds said at least one cannabinoid forming a water-soluble protein-cannabinoid composition.

11. The method of embodiment 10, wherein the engineered FABP-carrier protein comprises an engineered FABP-carrier protein having an amino acid sequence selected from the group of consisting of: SEQ ID NOs. SEQ ID NOs. 79-86.

12. The method of embodiments 10 and 11, wherein said step of genetically modifying the FABP-carrier protein comprises one or more of the following:

replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket with side chains pointing towards orientated toward the binding cavity;

replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket having a hydrophilic side chain with amino acid residues having a hydrophobic side chain; and

replacing one or more small hydrophobic amino acid residues of the FABP-carrier protein cannabinoid binding pocket with larger hydrophobic amino acid residues.

13. The method of embodiment 11, wherein the engineered FABP-carrier protein is expressed in a protein production system selected from the group consisting of:

a bacterial cell culture;
 a yeast cell culture;
 a plant cell culture;
 a fungi cell culture;
 an algae cell culture;
 a bioreactor production system; and
 a plant.

14. The method of embodiment 13, wherein the engineered FABP-carrier protein is coupled with a secretion signal.

15. The method of embodiment 14, wherein said secretion signal is selected from the group consisting of: SEQ ID NO. 3, SEQ ID NO. 65, SEQ ID NOs. 69-73.

16. The method of embodiments 13-15, wherein the engineered FABP-carrier protein is introduced to said at least one cannabinoid in said protein production system.

17. The method of embodiments 1 and 10, wherein said water-soluble protein-cannabinoid composition undergoes lyophilisation.

18. The method of embodiment 10, and wherein said at least one cannabinoid comprises at least one cannabinoid selected from the group consisting of: cannabidiol (CBD), cannabidiolic acid (CBDA), Δ^9 -tetrahydrocannabinol (THC), tetrahydrocannabinolic acid (THCA), and (cannabigerolic acid) CBGA).

19. An isolated polynucleotide that encodes one or more amino acid sequences selected from the group of consisting of: SEQ ID NOs. 1-2, 21, and 74-86, or a homolog having affinity towards at least one cannabinoid thereof.

20. The polynucleotide of embodiment 19, wherein said polynucleotide is operably linked to a promoter forming an expression vector.

21. The polynucleotide of embodiment 19, wherein said polynucleotide is codon optimized for expression in a microorganism, or plant cell, and is further operably linked to a promoter forming an expression vector.

22. A genetically modified organism expressing at least one of the expression vectors of embodiments 20 and 21

23. The genetically modified organism of embodiment 22, wherein said genetically modified organism is selected from the group consisting of:

a genetically modified bacterial cell
 a genetically modified yeast cell,
 a genetically modified plant cell,
 a genetically modified fungi cell,
 a genetically modified algae cell, and
 a genetically modified plant.

24. A method of solubilizing a cannabinoid comprising the steps of:

establishing a cell culture of genetically modified yeast, plant, or bacteria cells that express a nucleotide sequence encoding a heterologous Fatty-Acid Binding Protein (FABP)-carrier protein operably linked to a promoter wherein said heterologous FABP-carrier protein exhibits activity towards one or more cannabinoids;

introducing one or more cannabinoids to the genetically modified yeast or bacteria cell culture; and

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wherein said FABP-carrier protein binds said one or more cannabinoids to form a water-soluble protein-cannabinoid composition.

25. The method of embodiment 24, wherein the step of introducing comprises the step of introducing one or more cannabinoids to a genetically modified yeast, plant, or bacteria cell culture in a fermenter or suspension cell culture.

26. The method of embodiment 24, wherein the step of introducing comprises the step of biosynthesizing one or more cannabinoids in a genetically modified yeast or bacteria cell culture wherein said heterologous FABP-carrier protein binds said one or more biosynthesized cannabinoids to form a water-soluble protein-cannabinoid composition.

27. The method of embodiment 24, wherein said heterologous FABP-carrier protein comprises a heterologous FABP-carrier protein having an amino acid sequence selected from the group of consisting of: SEQ ID NOs. 1-2, 21, and 74-86, or a homolog having affinity towards at least one cannabinoid thereof.

28. The method of embodiments 24 and 27, wherein said heterologous FABP-carrier protein is coupled with a tag.

29. The method of embodiments 24 and 27, wherein said heterologous FABP-carrier protein is coupled with a secretion signal.

30. The method of embodiment 29, wherein said secretion signal comprises a secretion signal selected from the group consisting of: SEQ ID NO. 3, SEQ ID NO. 65, SEQ ID NOs. 69-73.

31. The method of embodiment 24, wherein the at least one cannabinoid comprises a cannabinoid selected from the group consisting of: cannabidiol (CBD), cannabidiolic acid (CBDA), Δ^9 -tetrahydrocannabinol (THC), tetrahydrocannabinolic acid (THCA), and (cannabigerolic acid) CBGA).

32. The method of embodiment 24, and further comprising the step of genetically modifying the FABP-carrier protein form an engineered FABP-carrier protein having enhanced affinity for at least one cannabinoid, such genetic modification comprising one or more of the following:

replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket with side chains pointing towards orientated toward the binding cavity;

replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket having a hydrophilic side chain with amino acid residues having a hydrophobic side chain; and

replacing one or more small hydrophobic amino acid residues of the FABP-carrier protein cannabinoid binding pocket with larger hydrophobic amino acid residues.

33. The method of embodiment 32, wherein said engineered FABP-carrier protein comprises an engineered FABP-carrier protein having an amino acid sequence selected from the group of consisting of: SEQ ID NOs. 79-86.

34. The FABP-carrier protein of embodiments 1, 2, 10-11, 19, 24, and 27, wherein said FABP-carrier protein is further genetically modified to decrease aggregation propensity and potential antigenicity.

35. The FABP-carrier protein of embodiments 34, wherein said FABP-carrier protein comprises a FABP-carrier protein having one or more of the following mutations: V58E, V42R.

36. The water-soluble protein-cannabinoid composition of any of the embodiments above wherein said water-soluble protein-cannabinoid composition is introduced to a consumer product meant for human-consumption, or a pharma-

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ceutical composition for administration of a therapeutically effective dose to a subject in need thereof; or a prodrug for administration of a therapeutically effective dose to a subject in need thereof.

37. An exportable FABP-carrier protein having a first domain having affinity for at least one cannabinoid, and a second domain comprising a eukaryotic or prokaryotic secretion signal.

38. The exportable FABP-carrier protein of embodiment 37, wherein said first domain is selected from the group of amino acid sequences identified as SEQ ID NO. 1-2, 21, and 74-86.

39. The exportable FABP-carrier protein of embodiment 37, wherein said second domain is selected from the group of amino acid sequences identified as: SEQ ID NO. 3, SEQ ID NO. 65, SEQ ID NOs. 69-73.

40. The exportable polypeptide of embodiments 37-39, wherein said exportable FABP-carrier protein binds at least one cannabinoid.

41. The exportable polypeptide of embodiments 37-39, wherein said exportable FABP-carrier protein binds at least one cannabinoid introduced, or biosynthesized in a cell culture.

42. The exportable polypeptide of embodiment 41, wherein said cell culture comprises a cell culture selected from the group consisting of: a bacterial cell culture, a yeast cell culture, a plant cell culture.

43. A genetically modified *Cannabis* plant expressing a nucleotide sequence operably linked to a promoter encoding at least one FABP-carrier protein.

44. The *Cannabis* plant of embodiment 43 and wherein said FABP-carrier protein comprises a FABP-carrier protein selected from the group consisting of: an amino acid sequence according to SEQ ID NOs. 1-2, 21, and 74-86.

45. The *Cannabis* plant of embodiments 43 and 44, and further comprising the step of expressing a nucleotide sequence operably linked to a promoter encoding one or more cannabinoid synthases having its trichome targeting sequence disrupted or removed.

46. The *Cannabis* plant of embodiment 45, wherein one or more cannabinoid synthase genes has been disrupted or knocked out.

47. The *Cannabis* plant of embodiment 46, wherein said one or more cannabinoid synthases having its trichome targeting sequence disrupted or removed is selected from the group consisting of: the nucleotide sequence identified as: SEQ ID NO. 13, SEQ ID NO. 14, and SEQ ID NO. 15.

48. The *Cannabis* plant of embodiment 44, and further comprising the step of expressing at least one myb transcription factor.

49. The *Cannabis* plant of embodiment 48, wherein said at least one myb transcription factor is selected from the group consisting of: SEQ ID NOs. 16-20.

50. The *Cannabis* plant of embodiment 44, and further comprising the step of expressing at least one catalase.

51. The *Cannabis* plant of embodiment 50, wherein said at least one catalase is selected from the group consisting of: SEQ ID NOs. 6-10.

52. The *Cannabis* plant of embodiment 44, and further comprising the step of expressing at least one heterologous glycosyltransferase,

53. The *Cannabis* plant of embodiment 52, wherein said at least one at least one heterologous glycosyltransferase is selected from the group consisting of: 32-47, and SEQ ID

NOs. 61-62. Additional aspects of the invention may be evident from the specification and figures below.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1. (A) an exemplary cannabinoid (CBD) molecule (purple spheres) is bound to the predicted micromolar affinity site in *Bos taurus* FABP1 (white cartoons). (B) an exemplary cannabinoid (THC) molecule (purple spheres) is bound to the predicted micromolar affinity site in *Mus musculus* FABP1 (white cartoons).

FIG. 2. Vector map of modified pET24a(+).

FIG. 3 Small-scale of protein expression from 10 ml of culture for WT and engineered FABPs. (A) FABP (from the plant *Rhodamnia argentea*), FABP1 (from pig) and FABP7 (from chicken) Wild type (no modification, natural protein). (B) Engineered FABP from the same organism described above with mutations in binding sites using smina (sm) method of binding prediction of affinity binding. (C) Engineered FABP using rosetta (ro) method for prediction of affinity binding. All wild type FABPs expressed as soluble protein, the protein is found in supernatant (soluble).

FIG. 4. FABP binding to cannabinoids by ANS displacement by ligands. (A) Wild type FABP from chicken indicating no binding to either CBD or THC. (B) Chicken FABP engineered by smima showing no binding to THC or CBD. (C) Chicken FABP engineered by rosetta showing binding to CBD but not THC.

FIG. 5. Binding of the plant wild type (A) and engineered (B and C) FABP to cannabinoids by ANS displacement. Both wild-type and engineered proteins bind to CBD and THC.

FIG. 6. Binding of pig wild type (A) and engineered (B and C) FABP to CBD and THC by ANS displacement.

FIG. 7. Summary of ANS-displacement by ligand, showing the relative change in fluorescence. 2 uM of protein in 50 mM Tris-HCl buffer was labelled with 20 uM ANS. 33 uM of CBD, THC and Arachidonic acid (AAc) was used as ligand diluted in 10% of MeOH.

FIG. 8. Structural models of engineered variants of (A) plant FABP (images encompass modified amino acid residues for SEQ ID NOs. 85-86) and (B) horse FABP1 (images encompass modified amino acid residues for both SEQ ID NOs. 83-84) with predicted enhanced binding affinity to CBD relative to the WT proteins. Protein backbones are shown in green or blue cartoons, while bound CBD is shown with magenta sticks. The mutations in each protein that were introduced by the in silico directed evolution procedure are shown as red spheres. Note that all these mutations, which are predicted to help improve each protein's binding affinity to CBD, occur within the binding cavity.

DETAILED DESCRIPTION OF THE INVENTION

In certain embodiments, the invention may include the use of a FABP as a carrier protein, generally referred to as a FABP-carrier, to solubilize cannabinoids, terpenes/terpenoids, and other short-chain fatty acid phenolic compounds.

In one embodiment, the invention may include systems, methods and compositions to solubilize cannabinoids, terpenes/terpenoids, and other short-chain fatty acid phenolic compounds utilizing FABPs an, preferably non-human homologs of FABPs to solubilize cannabinoids and other compounds generally identified herein. In this embodiment, the use of FABPs compositions to solubilize cannabinoids may facilitate the transport, and/or storage of cannabinoids

in in vitro, and in vivo systems, as well as their use in commercial products where enhanced solubility may enhance the product's characteristics or price.

In another embodiment, the current invention may include the rational design of a novel chimeric FABP construct, preferably based on non-human homologs of FABPs, to increase cannabinoid water solubility via binding. In a preferred embodiment, a human homolog of hepatic FABP1 (SEQ ID NO. 21), for example as identified in SEQ ID NO. 1-2, or a homolog thereof, may be used to solubilize cannabinoids and other compounds in both in vitro and in vivo systems generally identified herein. Additional embodiments may include the generation of genetically modified or engineered FABPs that may be used to solubilize cannabinoids.

In a preferred embodiment, such site-directed mutations may be rationally designed such that one or more mutations may be made near a cannabinoid, or other binding site. Such rationally designed mutations may modulate the compounds binding affinity with the FABP-carrier protein. In a preferred embodiment, rationally designed mutations may increase its strength of binding with a cannabinoid, terpene, or other short-chain fatty acid phenolic compound. In some further embodiments, rationally designed mutations may enhance binding affinity for the FABP-carrier protein that is compound specific. In this embodiment, mutations at and/or near the binding site may be rationally designed to increase its strength of binding with THC, CBD or other cannabinoids as identified herein.

In one example, the directed evolution algorithm may identify amino acid substitution mutations that aid in enhancing the binding affinity to one or more target cannabinoids, such as CBD based quantitatively on the Metropolis criterion. As generally shown in FIG. 8, which shows 11 three-dimensional models of the FABP-carriers according to SEQ ID NOs. 84-85, and 85-86, the specifically identified mutations sites followed a pattern. In these examples, the set of residues at, or near the cannabinoid binding pocket and with side chains pointing towards this cavity were targeted for mutations. Thus, when looking at any two variants (that started from the same WT parent), there is an expected higher chance of seeing a mutation occur at a particular position. For instance, both chicken FABP7 variants (SEQ ID NOs. 79 & 80) have mutations at positions 34 and 54, and both pig FABP1 variants (SEQ ID NOs. 81 & 82) have mutations at positions 100 and 102. Such mutations may induce conformational or other changes that enhance the affinity of the cannabinoid binding pocket on the respective FABP-carrier proteins.

Such mutations may induce conformational or other changes that enhance the affinity of the cannabinoid binding pocket on the respective FABP-carrier proteins for a specific ligand cannabinoid, such as CBD. Since CBD is highly nonpolar or hydrophobic, as generally shown in SEQ ID NOs. 79-86, the specific mutations may: i) replace a residue with a hydrophilic side chain to another residue with a hydrophobic side chain; or ii) replace a small hydrophobic residue with a bulkier/larger hydrophobic residue. These mutations may enhance the overall hydrophobicity of the cannabinoid binding pocket, thereby making it more attractive to the ligand, for example CBD, for binding. In one example, the FABP-carrier identified a chicken FABP7 (SEQ ID NO. 74) for instance, the wild-type Gly at position 34 is mutated to either Ala or Phe in the variants (case B), and the wild-type Thr at position 54 is replaced with either Met or Ile in the variants (case A). In another example, the FABP-carrier identified a pig FABP1 (SEQ ID NO. 75), the

wild-type Ser at position 100 is mutated to Ala in both variants and the wild-type Thr at position 102 is replaced with either Phe or Ile in the variants.

In another example, as noted in the SEQUENCING LISTING below, amino acid residues that have been modified from their wild type versions are underlined (see for example the amino acid sequences according to SEQ ID NO. 79-86, compared to wild type amino acid sequences according to SEQ ID NO. 74-78). For example, in one embodiment, the amino acid sequence for a wild type FABP-carrier having affinity for one or more cannabinoids from the plant *Rhodamnia argentea*, being identified as SEQ ID NO. 77, may be engineered to form an engineered FABP-carrier protein through rationally designed mutations derived from the in silico directed evolution of the proteins binding site such that the rationally designed proteins may exhibit enhanced binding affinity for the FABP-carrier protein. For example, in one embodiment, specific rationally designed mutations identified in the FABP-carrier according to SEQ ID NO 85 may include: A34M, G35M, P39A, T41V, F53I, T55V, L77F, Q95I, R107A, V117F, W119E, and R128A. In another example, specific rationally designed mutations identified in the FABP-carrier according to SEQ ID NO. 86 may include: T41I, F53I, T55M, I64V, N74A, R76A, 84GA, Q95I, Q96N, K104T, and A128A.

In certain embodiment, the invention may include an engineered FABP-carrier protein having one or more mutations that enhance the overall hydrophobicity of the cannabinoid binding pocket, thereby making it more attractive to the cannabinoid ligand. In certain embodiment, the invention may include an engineered FABP-carrier protein having at least one mutation that: (i) that replace a residue with a hydrophilic side chain to another residue with a hydrophobic side chain thereby enhancing the overall hydrophobicity of the cannabinoid binding pocket, thereby making it more attractive to the cannabinoid ligand; and/or that replace a small hydrophobic residue with a bulkier/larger hydrophobic residue thereby enhancing the overall hydrophobicity of the cannabinoid binding pocket, thereby making it more attractive to a cannabinoid ligand. As noted above, in certain embodiments the invention may include an engineered FABP-carrier protein having at least one of the substitution mutations identified in SEQ ID NO. 79-86. In alternative embodiments, invention may include an engineered FABP-carrier protein according to SEQ ID NO. 79-86.

In one embodiment, the current invention may include the increase of THC water solubility for transport via binding to a non-human homolog of an FABP. In a preferred embodiment, the FABP1 (hepatic FABP) homolog from *Mus musculus* may be used due to its sequence homology with *Homo sapiens* FABP1 (high sequence identity of around 84%) and the presence of a site with predicted micromolar affinity for binding the cannabinoid THC. The protein sequence for *Mus musculus* FABP1 is identified as SEQ ID NO. 1. The left panel of FIG. 1 above demonstrates a structural model of a THC molecule bound to *Mus musculus* FABP1.

In another embodiment, the current invention may include the rational design of mutations to the binding site of an FABP-carrier protein to enhance its binding affinity for THC or other related cannabinoids. In one preferred embodiment, these mutations may be designed into wild type FABP-carriers identified by amino acid SEQ ID NO. 1-2, 21, and 74-78, using a combination of in vitro, in vivo studies as well as bioinformatics approaches such as computational docking, binding affinity estimation, and molecular dynamics simulations. Such bioinformatics applications may be further employed to identify additional potential FABP

carrier proteins, as well as direct specific point-mutations to modulate or enhance cannabinoid binding affinity.

In another embodiment, the current invention may include the increase of CBD water solubility for transport via binding to a non-human homolog of an FABP. In a preferred embodiment, the FABP1 (hepatic FABP) homolog from *Bos taurus* may be used due to its sequence homology with *Homo sapiens* FABP1 (high sequence identity of around 81%) and the presence of a site with predicted micromolar affinity for binding CBD. The protein sequence for *Bos taurus* FABP1 is identified as amino acid sequence SEQ ID NO. 2. The right panel of FIG. 1 above demonstrates a structural model of a CBD molecule bound to *Bos taurus* FABP1.

In another embodiment, the current invention may include the increase of CBD water solubility for transport via binding to a non-human homolog of an FABP selected from the amino acid sequences according to SEQ ID NO. 1-2, 21, 74-86. In one preferred embodiment, the aforementioned sequences may be coupled with a tag, and/or secretion signal.

In another embodiment, the current invention may include the rational design of mutations at and/or near the binding site of an FABP-based carrier protein to enhance its binding affinity for CBD. In a preferred embodiment, these mutations will be designed into the FABP1 homolog identified as amino acid sequence SEQ ID NO. 1-2, 21, and 74-78 using a combination of in vitro, in vivo studies as well as bioinformatics approaches such as computational docking, binding affinity estimation, and molecular dynamics simulations. Such bioinformatics applications may be further employed to identify additional potential FABP carrier proteins, as well as direct specific point-mutations to modulate or enhance cannabinoid binding affinity.

In another embodiment, the current invention may include the rational design of mutations on the surface of a FABP-carrier protein to decrease its aggregation propensity and potential antigenicity. In a preferred embodiment, one or more of the mutations V58E, or V42R, may be introduced into one or more of the FABP-carriers identified herein. In a preferred embodiment, one or more of the mutations V58E, or V42R, may be introduced into one or more of the FABP-carriers according to SEQ ID NOs. 21, 75-76, 78, and 81-84. Each mutation may reduce the tendency of this protein to form aggregates in solution, without destabilizing its conformational stability. Preliminary analyses of linear epitopes for this FABP1 homolog suggest a high false positive prediction rate, based on a comparison with corresponding predictions for *Homo sapiens* FABP1 (SEQ ID NO. 21).

The above FABP-carrier proteins are provided as exemplary embodiments only. Nor are they limiting as to the number of punitive cannabinoid, or other short-fatty-acid phenolic compound affinity sites that may be engineered in a FABP-carrier. Consideration of which may include the desired type of short-fatty-acid phenolic compound to be bound by the FABP-carrier, as well as steric considerations resulting from the addition of such modified affinity motifs presented in the three-dimensional folded protein. Naturally, certain modifications may be made to a FABP-carrier that may alter the affinity strength of one or more existing cannabinoid affinity sites.

In another embodiment, the current invention may include the rational design of mutations on the surface of a cannabinoid-specific FABP-carrier protein that may decrease its aggregation propensity and potential antigenicity. In a preferred embodiment, the mutation V58K may be introduced

into the FABP-carrier which may further include SEQ ID NOs. SEQ ID NOs. 21, 75-76, 78, and 81-84, or a homolog thereof. Such a mutation may reduce the tendency of a FABP-carrier to form aggregates in solution, without destabilizing its conformational stability. Analyses of the linear epitopes for this modified FABP-carrier suggest a high false positive prediction rate, based on a comparison with corresponding predictions for *Homo sapiens* FABP1 (SEQ ID NO. 21).

In another embodiment, the current invention may include the increase of cannabinoid water solubility for transport via binding to an artificial or non-human homolog of FABP. In this embodiment of the current invention may include the increase of CBD water solubility for transport via binding to a non-human homolog of FABP.

In another embodiment, a FABP-carrier protein may be identified an engineered FABP-carrier protein having at least one modified cannabinoid-affinity sites or motifs, such as a THC, or CBD-affinity site or motif, when the protein is in its folded three-dimensional configuration. In one embodiment, a cannabinoid affinity from an engineered FABP-carrier may be modified to have increased or decreased cannabinoid affinity. In one preferred embodiment, a FABP-carrier protein, preferably a non-human homolog such as identified in SEQ ID NO. 74-86, or a homolog thereof may be identified and established as an engineered FABP-carrier protein. This engineered FABP-carrier protein may be genetically modified to construct a FABP-chimera or fusion protein having one or more modified cannabinoid-affinity sites or motifs having, preferable enhanced cannabinoid affinity. For example, as shown in FIG. 1, a FABP-carrier may have a micromolar affinity for a cannabinoid, while an engineered FABP-carrier may be engineered to have a nanomolar or greater affinity for cannabinoids.

As one of ordinary skill in the art would recognize, a ligand, such as a cannabinoid, or other short-chain fatty acid phenolic compound, with nanomolar (nM) dissociation constant may bind more tightly to a particular protein than a ligand with micromolar (μ M) dissociation constant. As a result, in certain embodiments of the inventive technology, customized FABP-carrier proteins, and in particular engineered FABP-carrier proteins may be generated that have a customized dissociation constant. This customized dissociation constant may be engineered according to the specifications of a particular application. For example, in one application a FABP-carrier may be engineered to be a fusion, or chimeric protein having one or more cannabinoid affinity sites having nanomolar (nM) or greater dissociation constant. Such FABP-carrier may be useful for long-term storage of cannabinoids in solution, or for applications inclusion in various commercial and other consumer products where the FABP-carrier may be exposed to artificial, or natural environmental conditions or other chemical processes that might degrade the protein structure and prematurely released the cannabinoid. Alternatively, in one application a FABP-carrier may be engineered to have one or more a cannabinoid affinity sites having micromolar (μ M) dissociation constant. Such FABP-carrier may allow for one or more cannabinoid compounds to be more easily release from the FABP-carrier. In one preferred embodiment, a FABP-carrier may include one or more a cannabinoid affinity sites having a macro- or micromolar (μ M) dissociation that may allow for greater release, as compared for example to a nanomolar (nM) dissociation, and bioavailability of the cannabinoid upon consumption.

As noted above, cannabinoid producing strains of *Cannabis*, as well as other plants may be utilized with the

inventive technology. In certain preferred embodiments, *Cannabis* plant material may be harvested and undergo cannabinoid extraction through one or more of the methods generally known in the art. These extracted cannabinoids, terpenoids and other short chain fatty acid phenolic compounds, may be introduced to a quantity of FABP-carrier proteins to be solubilized as described herein.

In one embodiment, cells of a microorganism, such as yeast or bacteria, may be transformed with artificially created expression vectors encoding one or more FABP-carrier proteins. In this preferred embodiment, the nucleotide sequences encoding a FABP-carrier protein may be codon optimized for exogenous expression. Additional embodiments may include genetic control elements such as promoters and/or enhancers as well as post-transcriptional regulatory elements that may also be expressed in the genetically engineered microorganisms, such as yeast or bacteria, such that the presence, quantity and activity of any FABP-carrier proteins present in a culture may be controlled. In a preferred embodiment, the microorganism, such as yeast or bacteria strain, may be further modified to generate high-levels of FABP-carrier proteins. In a preferred embodiment, the bacterial strain may include a high expression strain of bacteria, such as *E. coli* strain BL21(DE3) for optimal protein expression. In another preferred embodiment, the yeast strain may include genetically modified yeast cells selected from the group consisting of: genetically modified *Pichia pastoris* cells, genetically modified *Saccharomyces cerevisiae* cells, and/or genetically modified *Kluyveromyces marxianus* cells.

As noted above, in one embodiment the inventive technology may include individual expression of one or more FABP-carrier proteins each having a selected molecular tag. In a preferred embodiment, an FABP-carrier-protein, for example engineered from the amino acid sequences SEQ ID NO. 1-2, 21, and 74-86, or other homologous sequences, may each be configured to contain a poly-His or His-6 tag, which may be used later for protein purification. In this embodiment, the expressed FABP-carrier-protein may be detected and purified because the string of histidine residues bind to several types of immobilized metal ions, including nickel, cobalt and copper, under appropriate buffer conditions.

In one embodiment of the inventive technology, a cell culture, such as a plant, yeast or bacterial culture, may be genetically modified to express a tagged heterologous FABP-carrier and may further be allowed to grow to a desired level of cell or optical density, or in other instances until a desired level of FABP-carrier proteins have accumulated in the cultured cells and/or media, for example through the addition of a secretion signal that directs the FABP-carrier to be exported from the cell. All, or a portion of the accumulated FABP-carrier proteins may then be harvested from the culture and/or media, which in a preferred embodiment may be an industrial-scale fermenter or other apparatus suitable for the large-scale culturing of microorganisms. Optionally, the harvested cells may be lysed such that the accumulated FABP-carrier proteins may be released to the surrounding lysate. Additional steps may include treating this lysate. Examples of such treatment may include filtering, centrifugation or screening to remove extraneous cellular material as well as chemical treatments to improve later FABP-carrier yields.

The FABP-carrier may be further isolated and purified. In one preferred embodiment, the cell lysate or media may be processed utilizing affinity chromatography or other purification methods. In this preferred embodiment, an affinity

column having a ligand configured to bind with one or more of the tags coupled with the FABP-carrier protein, for example, a poly-His or His-6 tag, among others, may be immobilized or coupled to a solid support. The lysate may then be passed over the column such that the tagged FABP-carrier, having specific binding affinity to the ligand becomes bound and immobilized. In some embodiments, non-binding and non-specific binding proteins that may have been present in the lysate may be removed. Finally, the FABP-carrier may be eluted or displaced from the affinity column by, for example, a corresponding protein, tag or other compound that may displace or disrupt the tag-ligand bond. The eluted FABP-carrier proteins may be collected and further purified or processed. Notably, in other embodiments, FABP-carrier proteins may be commercially obtained or synthesized and used consistent with the embodiments described herein.

All FABP and FABP-carrier amino sequences described herein include homologs of said sequences which may have between 75-99% homology, while retaining specific activity towards one or more short-chain fatty-acid phenolic compound, and preferably one or more cannabinoid compounds.

One embodiment of the inventive technology includes the generation of novel genetically modified short-chain fatty-acid phenolic compound-carrier proteins that may have enhanced affinity for short-chain fatty-acid phenolic compounds, and preferably a terpenoid or cannabinoid from *Cannabis*. In one preferred embodiment, the inventive technology includes the generation of novel genetically modified cannabinoid-carrier FABPs engineered from one or more wild type FABP-carriers, for example SEQ ID NO. 1-2, and 74-78, or homologs thereof that may have enhanced affinity for cannabinoids.

Another embodiment of the inventive technology provides for systems and methods of high-capacity cannabinoid solubilization. In this preferred embodiment, a polynucleotide configured to express one or more FABP-carrier proteins, for example the amino acid sequences SEQ ID NO. 1-2, 74-86, or a homolog thereof, which may optionally be coupled with a tag for purification or isolation purposes, and may be operably linked to a promoter forming an expression vector. This expression vector may be used to transform a microorganism which may express one or more tagged FABP-carrier proteins, which may be further isolated, preferably through affinity purification. The isolated tagged FABP-carrier may be placed into a bio-reactor or other suitable environment where they may be introduced to one or more cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds. The tagged FABP-carrier proteins may solubilize the cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds through affinity binding to one or more affinity site. The solubilized cannabinoids may be isolated and used for commercial, pharmaceutical and other applications as generally described herein.

Another embodiment of the invention provides for methods of masking the typical unpleasant small and taste of cannabinoid-infused commercial products and beverages. For example, in this embodiment a FABP-carrier, for example as identified in the amino acid sequences 1-2, 21, and 74-86, may bind to one or more cannabinoids and allow it to be solubilized. In this solubilized state, the carrier protein allows for the masking of the cannabinoids natural small and taste. Moreover, in additional embodiments, a FABP-carrier may bind to, and solubilize one or more terpenes, the compounds in *Cannabis* primarily responsible for its distinctive smell. In this manner, the invention may

generate cannabinoid-infused commercial products, such as consumables and beverages that eliminate, mask or ameliorate the undesired smell and taste of the cannabinoid and terpene compounds.

Another embodiment of the invention provides for methods of generating solubilized cannabinoids, terpenes and other short-chain fatty-acid phenolic compounds that may have a more rapid metabolic uptake or bioavailability upon ingestion. In this embodiment, a FABP-carrier, such as an engineered FABP-carrier, may bind to one or more cannabinoids and/or terpenes and allow it to be solubilized such that upon ingestion, it may be more readily taken up by the body, for example, through the association with the carrier protein. This embodiment may allow for not only a more rapid uptake of the target compound, but allow for consistent consumer experiences, as well as facilitating safe and effective consumer-controlled dosing of cannabinoids and other compounds. This embodiment may further allow for lower amounts of cannabinoid and terpenes to be used in infused consumables and beverages as a result of this improved bioavailability. For example, absent this enhance bioavailability of the solubilized cannabinoids and terpenes, a large portion of the compounds may not be efficiently taken up by the body and may be eventually eliminated through natural chemical degradation or other strategies to metabolically clear the compounds from the body.

Another embodiment of the invention provides for methods of generating precise doses and/or formulations and/or ratios of cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds. In a preferred embodiment, a polynucleotide may be generated that is configured to express one or more FABP-carrier proteins configured to have binding affinity motifs that selectively bind an individual or class of cannabinoid, terpenoids, and/or other short-chain fatty-acid phenolic compounds. Again, this selective FABP-carrier-protein may be coupled with a tag for purification or isolation purposes and may be operably linked to a promoter forming an expression vector. This expression vector may be used to transform a microorganism, such as bacteria, yeast, fungi, or algae, which may express the tagged selective FABP-carrier protein which may be further isolated, preferably through affinity purification. The isolated selective FABP-carrier-protein may be placed into a bio-reactor or other suitable environment where they may be introduced to one or more cannabinoid, terpenoids, and/or other short-chain fatty-acid phenolic compounds. The FABP-carrier may selectively solubilize a quantity of cannabinoid, terpenoids, and/or other short-chain fatty-acid phenolic compounds, consistent with its endogenous and/or engineered affinity sites. The solubilized cannabinoid, terpenoids, and/or other short-chain fatty-acid phenolic compounds may be used for commercial, pharmaceutical, and other applications as generally described herein.

Another aspect of the invention provides for methods of generating precise mixed doses, ratios, and/or formulations of cannabinoids, terpenoids, and/or other short-chain fatty acid phenolic compounds. In a preferred embodiment, a first polynucleotide may be generated that is configured to express a FABP-carrier configured to have a selective binding affinity motif(s) that selectively bind an individual or class of cannabinoid, terpenoid, and/or other short-chain fatty-acid phenolic compound. An additional polynucleotide may be generated that is configured to express a FABP-carrier configured to have a cannabinoid binding affinity motif(s) that selectively bind a different individual or class of cannabinoid, terpenoid, and/or other short-chain fatty-

acid phenolic compound. Both selective FABP-carrier proteins may be coupled with a tag for purification or isolation purposes and may be incorporated into one or more expression vectors being operably linked to a promoter. Such expression vector(s) may be used to transform a microorganism, such as bacteria, yeast, fungi, or algae, which may express the tagged selective FABP-carrier proteins which may be further isolated, preferably through affinity purification. The isolated selective FABP-carrier proteins may be placed into a bio-reactor or other suitable environment where they may be introduced to one or more cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds. The first FABP-carrier protein may selectively solubilize a quantity of individual or class of cannabinoid, terpenoid, and/or other short-chain fatty-acid phenolic compound consistent with the number and type of its endogenous and/or engineered affinity sites. The additional FABP-carrier protein may selectively solubilize a quantity of a separate individual or class of cannabinoid, terpenoid, and/or other short-chain fatty-acid phenolic compound consistent with the number and type of its endogenous and/or engineered affinity sites. The solubilized cannabinoid, terpenoids, and/or other short-chain fatty-acid phenolic compounds may be used for commercial, pharmaceutical, and other applications as generally described herein.

Another aspect of the invention may include in vitro systems and methods to solubilize cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds. In a preferred embodiment, FABP-carrier proteins, for example the proteins identified according to SEQ ID NO. 1-2, 21, and 74-86, may be artificially synthesized in vitro and then placed into a bio-reactor, or other suitable environment, where they may be introduced to one or more cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds. The FABP-carrier proteins carrier proteins may solubilize the cannabinoids, terpenoids, and/or other short-chain fatty acid phenolic compounds as generally described herein.

Another embodiment of the inventive technology provides for direct systems and methods of high-capacity cannabinoid solubilization. In this preferred embodiment, a polynucleotide configured to express one or more FABP-carriers, for example SEQ ID NO. 1-2, 21, and 74-86, may be coupled with a tag for purification or isolation purposes. This polynucleotide may be operably linked to a promoter forming an expression vector. This expression vector may be used to transform a microorganism which may be grown in an industrial scale fermenter or other like apparatus known in the art for high-level protein production. While in culture, the genetically modified microorganism may express one or more tagged FABP-carrier proteins. Short-chain fatty-acid phenolic compounds, such as cannabinoids, terpenes, and other volatiles may be extracted from cannabinoid-producing plants or artificially biosynthesized and added to the cell culture. The isolated FABP-carrier proteins produced in culture may be introduced to one or more cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds in the culture. In one embodiment, one or more FABP-carrier proteins produced in culture may be coupled with a secretion signal, for example as identified in SEQ ID NO. 3 or 65, so as to be able to be more easily exported to the cultures media or supernatant, preferably in a yeast culture. The FABP-carrier proteins may bind to, and solubilize one or more cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds in the culture's supernatant or media. The tagged FABP-carrier proteins,

and their bound compounds, may be isolated utilizing affinity chromatography or other purification methods.

Another embodiment of the inventive technology provides for direct systems and methods of high-capacity cannabinoid solubilization. In this preferred embodiment, a polynucleotide configured to express one or more FABP-carrier proteins may be coupled with a tag for purification or isolation purposes. Such polynucleotide may be operably linked to a promoter forming an expression vector. This expression vector may be used to transform a bacterium which may be grown in an industrial scale fermenter or other like apparatus known in the art for high-level protein production. While in culture, the genetically modified bacteria may express one or more tagged FABP-carriers, for example one or more of the amino acid sequences according to SEQ ID NO. 1-2, 74-86, that may also be coupled with a secretion signal. Short-chain fatty-acid phenolic compounds, such as cannabinoids, terpenes, and other volatiles, may be extracted from cannabinoid-producing plants or artificially biosynthesized and added to the cell culture. The isolated FABP-carrier proteins produced in culture may be introduced to one or more cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds in the culture. The FABP-carrier proteins may bind to and solubilize one or more cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds. The tagged FABP-carrier proteins carrier proteins, and their bound compounds, may be isolated utilizing affinity chromatography or other purification methods.

Another embodiment of the inventive technology provides for systems and methods of high-capacity cannabinoid solubilization coupled with cannabinoid biosynthesis in microorganisms genetically engineered to produce cannabinoids. Implementing cannabinoid biosynthesis strategies proposed by: Carvalho Â, et al.; US Pat. App. No. US20180371507, by Paulos et al.; and WO2017139496, by Hussain et al.; (all of which are incorporated herein by reference) for the generation of cannabinoids in microorganisms such as yeast, fungi, and bacteria, in one embodiment the inventive technology may include systems and methods for solubilization of cannabinoids produced in non-cannabinoid producing microorganisms or artificial chemically-synthesized cannabinoids.

In one embodiment, one or more metabolic pathways for cannabinoid biosynthesis may be reconstructed in microorganisms, such as bacteria, fungi, algae, or yeast. Such pathways may be reconstructed through the expression of a plurality of heterologous genes necessary for the biosynthesis of precursor and cannabinoid compounds. In one preferred embodiment, a microorganism, such as bacteria, yeast, or fungi, may be genetically engineered to produce one or more cannabinoids, terpenes, or other short-chain fatty acid phenolic compounds. The microorganism may be further genetically modified to express a polynucleotide encoding a cannabinoid-carrier FABPs or a homolog thereof, such as those identified in SEQ ID NO. 1-2, 74-86 or more preferably an engineered FABP-carrier protein, such as those identified in SEQ ID NO. 79-86, or other homolog thereof. In one preferred embodiment, a FABP-carrier protein may bind to and solubilize one or more exogenously biosynthesized cannabinoids. This FABP-carrier protein may be tagged to facilitate isolation and purification as generally described herein and may further be coupled with a secretion signal as discussed below.

In another aspect of the invention, a FABP-carrier protein may be exported out of a cell through the natural action of the cell, gradient movement, or diffusion, as well as the

action of the secretion signal, where it may bind to one or more cannabinoid or other compounds located externally to a cell. In one preferred embodiment, a FABP-carrier protein may be generated in a cell culture, preferably a bacterial, yeast, plant or fungi cell culture, and more preferably a plant suspension culture of a cannabinoid-producing plant such as *Cannabis*, where it may be exported out of the cell and bind one or more cannabinoid or other compounds that may be present in the external cellular environment. In additional embodiments, a FABP-carrier protein may bind one or more cannabinoid or other compounds that may be present in the intracellular environment.

In another aspect of the invention, a FABP-carrier protein having a secretion signal may be expressed in a genetically modified yeast culture and exported out of a cell through the action of the secretion signal, or other natural action of the yeast cell. In one preferred embodiment, a heterologous polynucleotide may express one or more exportable FABP-carrier proteins having a secretion signal. In one embodiment, a secretion signal may direct posttranslational protein translocation into the endoplasmic reticulum (ER). In additional embodiments, a secretion signal may direct cotranslational translocation of the carrier protein across the ER membrane. Notably, protein translocation is the process by which peptides are transported across a membrane bilayer. Translocation of proteins across the membrane of the membrane of the ER is known to occur in one of two ways: cotranslationally, in which translocation is concurrent with peptide synthesis by the ribosome, or posttranslationally, in which the protein is first synthesized in the cytosol and later is transported into the ER.

In eukaryotic organisms such as yeast, proteins that are targeted for translocation across the ER membrane have a distinctive amino-terminal signal sequence, such as the amino acid sequence identified in SEQ ID NO. 65, which is recognized by the signal recognition particle (SRP). The SRP in eukaryotes is a large ribonucleoprotein which, when bound to the ribosome and the signal sequence of the nascent peptide, is able to arrest protein translation by blocking tRNA entry. The ribosome is targeted to the ER membrane through a series of interactions, starting with the binding of the SRP by the SRP receptor. The signal sequence of the nascent peptide chain is then transferred to the protein channel, Sec61. The binding of SRP to its receptor causes the SRP to dissociate from the ribosome, and the SRP and SRP receptor also dissociate from each other following GTP hydrolysis. As the SRP and SRP receptor dissociate from the ribosome, the ribosome is able to bind directly Sec61.

The Sec61 translocation channel (known as SecY in prokaryotes) is a highly conserved heterotrimeric complex composed of α -, β - and γ -subunits. The pore of the channel, formed by the α -subunit, is blocked by a short helical segment which may become unstructured during the beginning of protein translocation, allowing the peptide to pass through the channel. The signal sequence of the nascent peptide intercalates into the walls of the channel, through a side opening known as the lateral gate. During translocation, the signal sequence is cleaved by a signal peptide peptidase, freeing the amino terminus of the growing peptide.

During cotranslational translocation in eukaryotes, the ribosome provides the motive power that pushes the growing peptide into the ER lumen. During posttranslational translocation, additional proteins are necessary to ensure that the peptide moves unidirectionally into the ER membrane. In eukaryotes, posttranslational translocation requires the Sec62/Sec63 complex and the chaperone protein BiP. BiP is a member of the Hsp70 family of ATPases, a group

which is characterized as having an N-terminal nucleotide-binding domain (NBD), and a C-terminal substrate-binding domain (SBD) which binds to peptides. The nucleotide binding state of the NBD determines whether the SBD can bind to a substrate peptide, in this case a FABP-carrier or engineered FABP-carrier. While the NBD is bound to ATP, the SBD is in an open state, allowing for peptide release, while in the ADP state, the SBD is closed and peptide-bound. The primary role of the membrane protein complex Sec62/Sec63 is to activate the ATPase activity of BiP via a J-domain located on the lumen-facing portion of Sec63. The SBD of BiP binds non-specifically to the peptide as it enters the ER lumen, and keeps the peptide from sliding backwards in a ratchet-type mechanism.

Again, in one preferred embodiment, a FABP-carrier may be modified to include at least one secretion signal that may facilitate vesicle transport of the protein out of the cell, preferably a yeast cell. In one embodiment, a FABP-carrier may be modified to include a secretion signal which directs posttranslational protein translocation into the ER. In one preferred embodiment, a secretion signal which directs posttranslational protein translocation into the ER may be identified in amino acid SEQ ID NO. 3 (see below) which encodes an N-terminal secretion signal from α -factor mating pheromone in *S. cerevisiae*. The secretion signal is made up of a 19 amino acid 'presequence' which directs posttranslational protein translocation into the ER, and a 66-amino acid 'pro region' mediating receptor-dependent packaging into ER-derived COPII transport vesicles.

SEQ ID NO. 3:

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MRFPSIFTAVLFAASSALAAPVNTTTEDETAQIPA
EAVIGYSDLEGDFDVAVLPPFSNSTNNGLLFINTTI
ASIAAKEEGVSLEKR
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In another embodiment, a FABP-carrier may be modified to include a secretion signal which directs cotranslational translocation across the ER membrane. In one preferred embodiment, an enhanced secretion signal which directs cotranslational translocation across the ER membrane may be identified in amino acid sequence of SEQ ID NO. 65, where the 19 amino acid 'presequence' is replaced with the enhance 'presequence' with the Ost1 (OST=oligosaccharyltransferase) signal sequence identified by amino acid SEQ ID NO. 66.

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MRQVWFSWIVGLFLCFNVSSA
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In this preferred embodiment, an enhanced secretion signal may be identified as SEQ ID NO. 65:

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MRQVWFSWIVGLFLCFNVSSAAPVNTTTEDETA
QIPAEAVIGYSDLEGDFDVAVLPPFSNSTNNGLLF
INTTASIAAKEEGVSLEKR
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Again, in a preferred embodiment, one or more of the FABP-carrier proteins identified herein may be modified and expressed, preferably in a yeast cell, to include a secretion signal which directs posttranslational protein translocation into the ER, such signal preferably being SEQ ID NO. 32 which encode an N-terminal secretion signal from α -factor mating pheromone in *S. cerevisiae*. Such exportable FABP-carrier carrier proteins may include exemplary amino acid

sequence identified as SEQ ID NO. 4-5, and amino acid sequences 1-2, 21, and 74-86 coupled with the N-terminal secretion signal according to SEQ ID NO. 3, may bind to, and solubilize one or more cannabinoids located in the cell, or more preferably they may solubilize one or more cannabinoids outside in the cell, such as cannabinoids added to a cell culture supernatant. The exportable FABP-carrier proteins, having solubilized one or more target cannabinoids or other compounds identified herein may be further isolated as generally described herein.

In another embodiment, a FABP-carrier protein may be modified and expressed, preferably in a yeast cell, to include an enhanced secretion signal which directs cotranslational translocation across the ER membrane, such signal preferably being, SEQ ID NO. 65 which include the Ost1 signal sequence identified as amino acid sequence SEQ ID NO. 66 coupled with the 66-amino acid 'pro region' of the N-terminal secretion signal from α -factor mating pheromone in *S. cerevisiae*. Such enhanced exportable FABP-carrier proteins may include exemplary amino acid sequence identified as SEQ ID NO. 67-68, and amino acid sequences 1-2, 21, and 74-86 coupled with the enhanced N-terminal secretion signal according to SEQ ID NO. 65, may bind to, and solubilize one or more cannabinoids located in the cell, or more preferably one or more cannabinoids located outside in the cell, such as cannabinoids added to a cell culture supernatant. The exportable FABP-carrier, having solubilized one or more target cannabinoids or other compound identified herein, may be further isolated as generally described herein.

Additional embodiments also feature a method for producing FABP-carrier polypeptides. The method includes culturing a recombinant bacterium cell in a culture medium under conditions that allow the FABP-carrier polypeptides to be secreted into the culture medium, the recombinant bacterium cell comprising at least one exogenous nucleic acid, the exogenous nucleic acid comprising first and second nucleic acid sequences, wherein the first nucleic acid sequence encodes a signal peptide and the second nucleic acid sequence encodes a FABP-carrier carrier polypeptides, wherein the first and second nucleic acid sequences are operably linked to produce a fusion polypeptide comprising the signal peptide and the FABP-carrier polypeptide, and wherein upon secretion of the fusion polypeptide from the cell into the culture medium, the signal peptide may be removed from the -containing polypeptide. The method further can include isolating the FABP-carrier polypeptides from the culture medium.

In another aspect of the invention, a FABP-carrier protein may be exported out of a bacterial cell through the action of a secretion signal where the FABP-carrier protein may be secreted in an unfolded conformation and bind to one or more cannabinoid or other compounds located externally to a cell. In one preferred embodiment, a FABP-carrier protein may be generated in a cell culture, preferably a bacterial cell culture, where it may be exported out of the cell and bind one or more cannabinoid or other compounds that may be present in the external cellular environment. In this embodiment, a FABP-carrier protein may be coupled with a secretion signal that may direct the carrier protein to be secreted from a bacterium through a SEC-mediated secretion pathway.

Notably, in bacteria, translated peptides may be actively translocated post-translationally through a SecY channel by a protein called SecA. SecA is composed of a nucleotide-binding domain, a polypeptide crosslinking domain, and helical wing and scaffold domains. During translocation, a region of the helical scaffold domain forms a two-finger

helix which inserts into the cytoplasmic side of the SecY channel, thereby pushing the translocating carrier peptide through. A tyrosine found on the tip of the two-finger helix plays a critical role in translocation, and is thought to make direct contact with the translocating peptide. The polypeptide crosslinking domain (PPXD) forms a clamp which may open as the translocating peptide is being pushed into the SecY channel by the two-finger helix, and close as the two-finger helix resets to its "up" position. The conformational changes of SecA are powered by its nuclease activity, with one ATP being hydrolyzed during each cycle. This SEC system secretes proteins having a consensus signal peptide that is similar to, but distinct from, that of the Tat system as described below. The Sec signal sequence lacks an N-terminal consecutive-arginine sequence and has a relatively hydrophobic central region and a relatively short signal sequence compared with that of Tat. Exemplary Sec signal sequences may be identified as SEQ ID NO. 69.

Again, in one preferred embodiment, a FABP-carrier may be modified to include at least one Sec-mediated secretion signal that may facilitate translocation of transport of the unfolded carrier protein out of a bacterial cell via a Sec-secretion pathway. In one embodiment, a FABP-carrier may be modified to include a secretion signal which directs posttranslational protein translocation. In one preferred embodiment, a secretion signal which directs posttranslational protein translocation may be identified in amino acid SEQ ID NO. 69 which encodes an exemplary Sec-signal sequence from *E. coli* L-asparaginase II.

Again, in a preferred embodiment, one or more of the FABP-carrier carrier proteins may be selected from SEQ ID NO. 1-2, 21, and 74-86, and may be modified and expressed, preferably in a bacterial cell, to include a secretion signal which directs posttranslational protein translocation of the unfolded protein, such signal preferably being SEQ ID NO. 70, or homologous or similar Tat-secretion signal sequence, which may encode an exemplary Sec-secretion signal sequence. Such exportable FABP-carrier proteins may be translocated from a bacterial cell to the external environment where they may come into contact with, bind to, and solubilize one or more cannabinoids located outside in the cell, such as cannabinoids added to a cell culture supernatant.

In another aspect of the invention, a FABP-carrier protein may be exported out of a bacterial cell through the action of a secretion signal where the FABP-carrier protein may assume its folded three-dimensional configuration prior to secretion. In this embodiment, a FABP-carrier protein may bind to one or more cannabinoid or other compounds located internally or externally to the cell. In one preferred embodiment, a FABP-carrier protein may be generated in a cell culture, preferably a bacterial cell culture, where it may be exported out of the cell and into the external cellular environment. In this embodiment, a FABP-carrier protein may be coupled with a secretion signal that may direct the carrier protein to be secreted from a bacterium through a TAT-mediated secretion pathway.

Unlike the Sec system, the Tat system is involved in the transport of pre-folded protein substrates. Proteins are targeted to the Tat pathway by possession of N-terminal tripartite signal peptides. The signal peptides include a conserved twin-arginine motif in the N-region of Tat signal peptide. The motif has been defined as R-R-x- Φ - Φ , where Φ represents a hydrophobic amino acid. In *E. coli* the Tat pathway comprises the three-membrane protein TatA, TatB and TatC. A fourth protein TatE forms a minor component of the Tat machinery and has a similar function to TatA.

Because of the ability to secrete pre-folded protein substrates, the Tat pathway may be especially suited for secreting a high level of heterologous FABP-carrier proteins. Estimates of Tat substrates in organisms other than *Bacillus subtilis* and *E. coli* have been based predominantly in in silico analysis of genome sequences using programs trained to recognize specific features of tat targeting sequences. An exemplary Tat signal sequences may be identified as SEQ ID NO. 70.

Again, in a preferred embodiment, a FABP-carrier may be modified to include at least one Tat-mediated secretion signal that may facilitate translocation of transport of the folded carrier protein out of a bacterial cell. In one embodiment, a FABP-carrier may be modified to include a secretion signal which directs posttranslational protein translocation via a Tet-secretion pathway. In one preferred embodiment, a secretion signal which directs posttranslational protein translocation may be identified in amino acid SEQ ID NO. 70 or homologous or similar Sec-secretion signal sequence which encodes an exemplary tat signal peptide for *E. coli* strain k12 periplasmic nitrate reductase.

Again, in a preferred embodiment, one or more of the FABP-carrier proteins may be selected from SEQ ID NO. 1-2, 21, and 74-86 and may be modified and expressed, preferably in a bacterial cell, to include a secretion signal which directs posttranslational protein translocation of the folded protein via a Tet-secretion pathway, such signal preferably being SEQ ID NO. 70 or homologous or similar Sec-secretion signal sequence. Such exportable FABP-carrier proteins may be translocated from a bacterial cell already having one or more bound cannabinoids, or other compounds. In alternative embodiments, an exportable FABP-carrier protein may be translocated from a bacterial cell where it may come into contact with, bind to, and solubilize one or more cannabinoids located outside in the cell, such as cannabinoids added to a cell culture supernatant. The exportable FABP-carrier proteins, having solubilized one or more target cannabinoids or other compounds identified herein may be further isolated.

In another embodiment, the invention includes a recombinant plant, or plant cell producing FABP-carrier proteins. The plant or plant cell can include at least one heterologous nucleic acid encoding a FABP-carrier, wherein the plant or plant cell is from a species of *Cannabis*. The plant or plant cell can include at least one heterologous nucleic acid encoding a FABP-carrier, wherein the plant or plant cell is from a species of *Nicotiana*. The plant or plant cell can include at least one heterologous nucleic acid encoding a FABP-carrier, wherein the plant or plant cell is from a species other than *Nicotiana*. The heterologous nucleic acid further can include a regulatory control element such as a promoter (e.g., a tissue-specific promoter such as leaves, roots, stems, or seeds).

A polypeptide can be expressed in monocot plants and/or dicot plants. Techniques for introducing nucleic acids into plants are known in the art, and include, without limitation, *Agrobacterium*-mediated transformation, viral vector-mediated transformation, electroporation, and particle gun transformation (also referred to as biolistic transformation). See, for example, U.S. Pat. Nos. 5,538,880; 5,204,253; 6,329,571; and 6,013,863; Richards et al., Plant Cell. Rep. 20:48-54 (2001); Somleva et al., Crop Sci. 42:2080-2087 (2002); Sinagawa-Garcia et al., Plant Mol Biol (2009) 70:487-498; and Lutz et al., Plant Physiol., 2007, Vol. 145, pp. 1201-1210. In some instances, intergenic transformation of plastids can be used as a method of introducing a polynucleotide into a plant cell. In some instances, the

method of introduction of a polynucleotide into a plant comprises chloroplast transformation. In some instances, the leaves and/or stems can be the target tissue of the introduced polynucleotide. If a cell or cultured tissue is used as the recipient tissue for transformation, plants can be regenerated from transformed cultures if desired, by techniques known to those skilled in the art.

Other suitable methods for introduce polynucleotides include electroporation of protoplasts, polyethylene glycol-mediated delivery of naked DNA into plant protoplasts, direct gene transformation through imbibition (e.g., introducing a polynucleotide to a dehydrated plant), transformation into protoplasts (which can comprise transferring a polynucleotide through osmotic or electric shocks), chemical transformation (which can comprise the use of a polybrene-spermidine composition), microinjection, pollen-tube pathway transformation (which can comprise delivery of a polynucleotide to the plant ovule), transformation via liposomes, shoot apex method of transformation (which can comprise introduction of a polynucleotide into the shoot and regeneration of the shoot), sonication-assisted *Agrobacterium* transformation (SAAT) method of transformation, infiltration (which can comprise a floral dip, or injection by syringe into a particular part of the plant (e.g., leaf)), silicon-carbide mediated transformation (SCMT) (which can comprise the addition of silicon carbide fibers to plant tissue and the polynucleotide of interest), electroporation, and electrophoresis. Such expression may be from transient or stable transformations.

Additional embodiments also feature a method for producing FABP-carrier polypeptides in plants and preferably a plant cell in culture. The method includes culturing a recombinant plant cell in a culture medium under conditions that allow the FABP-carrier polypeptides to be secreted into the culture medium, the recombinant plant cell comprising at least one heterologous nucleic acid, the heterologous nucleic acid encoding a FABP-carrier polypeptide, wherein the nucleic acid sequence is operably linked to a promoter and optionally may further produce a fusion polypeptide comprising the signal peptide and the FABP-carrier polypeptide, and wherein upon secretion of the polypeptide from the plant cell into the culture medium, the signal peptide may be removed from the FABP-carrier polypeptide. The method further can include isolating the FABP-carrier polypeptides from the culture medium.

In another aspect of the invention, a FABP-carrier protein may be exported out of a plant cell through the action of a secretion signal where the FABP-carrier protein may be secreted via a plant protein secretion pathway. In a preferred embodiment, FABP-carrier may be coupled with an N-terminal signal peptide which may direct their translocation to the extracellular region via the Endoplasmic Reticulum-Golgi apparatus and the subsequent endomembrane system. In one preferred embodiment, a FABP-carrier protein may be generated in a plant, and preferably a plant cell culture, where it may be exported out of the cell and bind one or more cannabinoid or other compounds that may be present in the external cellular environment. In this embodiment, a FABP-carrier protein may be coupled with a secretion signal that may direct the carrier protein to be secreted from a plant cell via the Endoplasmic Reticulum-Golgi apparatus and the subsequent endomembrane system.

Again, in one preferred embodiment, a FABP-carrier may be modified to include at least one plant secretion signal that may facilitate translocation of transport of the protein out of a plant cell. In one embodiment, a FABP-carrier may be modified to include a secretion signal which directs. In one

preferred embodiment, a secretion signal which directs protein translocation from a plant cell may be identified in amino acid SEQ ID NO. 71, which encodes an exemplary secretion signal from an extracellular *Arabidopsis* protease Ara12 (At5g67360). Additional examples include the amino acid SEQ ID NO. 72, which encodes an exemplary secretion signal from a barley (*Hordeum vulgare*) alpha amylase. Still further examples include the amino acid SEQ ID NO. 73, which encodes an exemplary secretion signal from a rice a-Amylase.

Again, in a preferred embodiment, one or more of the FABP-carrier carrier proteins may be selected from SEQ ID NO. 1-2, 21, and 74-86, and may be further modified and expressed, preferably in a plant cell, to include a secretion signal which directs protein translocation out of the plant cell, such signal preferably being SEQ ID NO. 71, 72-74. Such exportable FABP-carrier proteins may be translocated from a plant cell already having one or more bound cannabinoids, or other compounds. In alternative embodiments, an exportable FABP-carrier protein may be translocated from a plant cell where it may come into contact with, bind to, and solubilize one or more cannabinoids located outside in the cell, such as cannabinoids added to a cell culture supernatant. The exportable FABP-carrier protein, having solubilized one or more target cannabinoids or other compounds identified herein, may be further isolated.

In another embodiment, one or more of the FABP-carrier proteins may be secreted from a plant cell in culture using the Hydroxyproline-Glycosylation (Hyp-Glyco) technology. In this embodiment, one or more of the FABP-carrier carrier proteins may be selected from SEQ ID NO. 1-2, 21, and 74-86 and may be modified and expressed, preferably in a plant cell and further fused with Hyp-rich repetitive peptide (HypRP) tag that directs extensive Hyp-O-glycosylation in plant cells resulting in arabinogalactan polysaccharides populating this repetitive peptide fusion facilitating the secretion of the expressed protein from cultured plant cells.

In certain embodiments, a catalase enzyme may be co-expressed with cannabinoid biosynthesis genes and FABP-carrier proteins, as well as optionally FABP-transporters or other genes that may reduce cannabinoid biosynthesis toxicity and/or facilitate transport of the solubilized cannabinoids through or out of the cell. In one embodiment a heterologous catalase is selected from the group consisting of: the amino acid sequence SEQ ID NO. 6, the amino acid sequence SEQ ID NO. 7, the amino acid sequence SEQ ID NO. 8, the amino acid sequence SEQ ID NO. 9, the amino acid sequence SEQ ID NO. 10 and a sequence having at least 80% homology to amino acid sequence SEQ ID NO. 6, SEQ ID NO. 7, SEQ ID NO. 8, SEQ ID NO. 9 and SEQ ID NO. 10.

Another embodiment of the inventive technology provides for systems and methods of high-capacity cannabinoid solubilization coupled with cannabinoid biosynthesis in cannabinoid producing plants or plants engineered to produce cannabinoids. In this preferred embodiment, cannabinoid biosynthesis may be redirected from the plant's trichome to be localized in the plant cell's cytosol. In certain embodiments, a cytosolic cannabinoid production system may be established as directed in PCT/US18/24409 and PCT/US18/41710, both by the present inventors Sayre et al. (These applications are both incorporated by reference with respect to their disclosure related to cytosolic cannabinoid production and/or modification in whole, and plant cell systems).

In one embodiment, a cytosolic cannabinoid production and solubilization system may include the in vivo creation of one or more recombinant proteins that may allow cannabi-

noid biosynthesis to be localized to the cytosol where one or more heterologous FABP-carrier proteins may also be expressed and present in the cytosol. This inventive feature allows not only higher levels of cannabinoid production and accumulation, but efficient production of cannabinoids in suspension cell cultures. Even more importantly, this inventive feature allows cannabinoid production and accumulation without a trichome structure in whole plants, allowing cells that would not traditionally produce cannabinoids, such as cells in *Cannabis* leaves and stalks, to become cannabinoid-producing cells

More specifically, in this preferred embodiment, one or more cannabinoid synthases may be modified to remove all or part of an N-terminal extracellular trichome targeting. An exemplary N-terminal trichome targeting sequence for THCA synthase is identified as SEQ ID NO. 11, while an N-terminal trichome targeting sequence for CBDA synthase is identified as SEQ ID NO. 12. Co-expression with this cytosolic-targeted synthase (for example SEQ ID NOS. 13-15) with a heterologous FABP-carrier protein, and preferably an engineered FABP-carrier protein, may allow the localization of cannabinoid synthesis, accumulation and solubilization to the cytosol. The FABP cannabinoid carrier proteins may be later isolated with their bound cannabinoid molecules through a water-based extraction process due to their solubility, as opposed to traditional chemical or supercritical CO₂ extractions methods. In this way, a cell culture of a cannabinoid producing plant may form a continuous production platform for solubilized cannabinoids.

As noted below, in certain embodiments cannabinoid biosynthesis may be coupled with cannabinoid glycosylation in a cell cytosol. For example, in one preferred embodiment a cytosol-target glycosyltransferase (for example SEQ ID NOS. 63-64) may be expressed in a cell, preferably a cannabinoid producing cell, and even more preferably a *Cannabis* cell.

Such cytosolic target enzymes may be co-expressed with heterologous catalase and cannabinoid transporters or other genes that may reduce cannabinoid biosynthesis toxicity and/or facilitate transport through or out of the cell. In one embodiment a heterologous catalase is selected from the group consisting of: the amino acid sequence SEQ ID NO. 6, the amino acid sequence SEQ ID NO. 7, the amino acid sequence SEQ ID NO. 8, the amino acid sequence SEQ ID NO. 9, the amino acid sequence SEQ ID NO. 10 and a sequence having at least 80% homology to amino acid sequence SEQ ID NO. 6, SEQ ID NO. 7, SEQ ID NO. 8, SEQ ID NO. 9 and SEQ ID NO. 10.

Such cytosolic target enzymes may also be co-expressed with one or more myb transcription factors that may enhance metabolite flux through the cannabinoid biosynthetic pathway which may increase cannabinoid production. In one embodiment a myb transcription factor may be endogenous to *Cannabis*, or an ortholog thereof. Examples of endogenous myb transcription factor may include SEQ ID NO. 16 and 17, or orthologs thereof. In one embodiment a myb transcription factor may be heterologous to *Cannabis*. A heterologous myb transcription factor may be selected from the group consisting of a nucleotide sequence that expresses: amino acid sequence SEQ ID NO. 18, amino acid sequence SEQ ID NO. 19, amino acid sequence SEQ ID NO. 20.

In an alternative embodiment, isolated heterologous FABP-carrier proteins may be added to a cell culture of a cannabinoid-producing plant, preferably a *Cannabis* suspension cell culture, having a cytosolic cannabinoid production system. In this preferred embodiment, one or more cannabi-

noid may be produced in the cytosol and transported into the surrounding culture media through passive or active transport mechanisms. Once the cannabinoids have been transported to the surrounding culture media, a quantity of FABP-carrier proteins may be added to the media and bind to and solubilize one or more cannabinoids. This media may then be removed and replenished, such that the solubilized cannabinoids bound to FABP-carrier proteins may be further isolated from the media as generally described herein. In one embodiment, the FABP-carrier proteins may be later isolated with their bound cannabinoid molecules through a water-based extraction process due to their solubility, as opposed to traditional chemical or super-critical CO₂ extractions methods. In this way, a cell culture of a cannabinoid producing plant may form a continuous production platform for solubilized cannabinoids.

Another embodiment of the invention may include the generation of an expression vector comprising this polynucleotide, namely a cannabinoid synthase lacking an N-terminal extracellular trichome targeting sequence and a heterologous FABP-carrier gene, for example a nucleotide sequence that encodes one or more of the amino acid sequences identified in SEQ ID NOs. 1-2, 21, and 74-86, as well as optionally one or more secretion signals as discussed herein, operably linked to a promoter. This expression vector may be used to create a genetically altered plant or parts thereof and its progeny comprising this polynucleotide operably linked to a promoter, wherein said plant or parts thereof and its progeny produce said proteins. For example, seeds and pollen contain this expression vector, a genetically altered plant cell comprising this expression vector such that said plant cell produces said chimeric protein. Another embodiment comprises a tissue culture comprising a plurality of the genetically altered plant cells having this expression vector.

One preferred embodiment of the invention may include a genetically altered cannabinoid-producing plant or cell expressing a cytosolic-targeted cannabinoid synthase protein having a cannabinoid synthase N-terminal extracellular targeting sequence (SEQ ID NOs. 11-12) inactivated or removed. In one embodiment, a cytosolic targeted THCA synthase (ctTHCAs) may be identified as SEQ ID NO. 13, while in another embodiment, cytosolic targeted CBDA synthase (cytCBDAs) are identified as SEQ ID NO. 14-15, respectively. Such cytosolic-targeted cannabinoid synthase proteins may be operably linked to a promoter. Another embodiment provides a method for constructing a genetically altered plant or part thereof having solubilization of cannabinoids in the plant's cytosol compared to a non-genetically altered plant or part thereof, the method comprising the steps of: introducing a polynucleotide encoding a cannabinoid synthase into a plant or part thereof to provide a genetically altered plant or part thereof, wherein the cannabinoid synthase N-terminal extracellular targeting sequence has been disrupted or removed and further expressing a polynucleotide encoding a cannabinoid-carrier FABPs, such as those identified in SEQ ID NO. 1-2, 21, and 74-86, or a homolog thereof.

Notably, in a preferred embodiment, one or more endogenous cannabinoid synthase genes may be disrupted and/or knocked out and replaced with cytosolic-targeted cannabinoid synthase proteins as described herein. The disrupted endogenous cannabinoid synthase gene(s) may be the same or different than the expressed cytosolic-targeted cannabinoid synthase protein. Methods of disrupting or knocking-out a gene are known in the art and could be accomplished by one of ordinary skill without undue experimentation.

In another embodiment, one or more endogenous cannabinoid synthase genes may be disrupted and/or knocked out in a *Cannabis* plant or suspension cell culture wherein one or more cannabinoid synthase genes has been disrupted and/or knocked out is selected from the group consisting of: a CBG synthase gene; a THCA synthase, a CBDA synthase, and a CBCA synthase. In this embodiment, the *Cannabis* plant or suspension cell culture may express a polynucleotide encoding one or more cannabinoid synthases having its trichome targeting sequence disrupted and/or removed which may be selected from the group consisting of: a CBG synthase gene having its trichome targeting sequence disrupted and/or removed; a THCA synthase having its trichome targeting sequence disrupted and/or removed; a CBDA synthase having its trichome targeting sequence disrupted and/or removed; and a CBCA synthase having its trichome targeting sequence disrupted and/or removed.

The current invention may further include systems, methods and compositions for the solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in cell cultures. Exemplary cell cultures may include bacterial, yeast, plant, algae and fungi cell cultures. FABP-carrier may be coupled with secretion signals to allow such proteins to be exported from the cell culture into the surrounding media. In this embodiment, a FABP-carrier protein may be engineered to include a secretion signal that may allow it to be exported from a cell. In one preferred embodiment, one or more of sequences identified as SEQ ID NO. 1-19, and 74-86, may be coupled with a secretion signal. In one preferred embodiment, one or more of sequences identified as SEQ ID NO. 1-2, 21, and 74-86, may be coupled with the N-terminal secretion signal identified in SEQ ID NO. 3 or SEQ ID NO. 65. One exemplary exportable FABP-carrier protein may include SEQ ID NO. 4 or SEQ ID NO. 5 or a FABP-carrier protein engineered from SEQ ID NO. 4 or SEQ ID NO. 5 or may be coupled with the secretion signal identified as amino acid sequence SEQ ID NO. 66, to form an enhanced exportable a FABP-carrier protein being identified herein as SEQ ID NO. 67 and SEQ ID NO. 68. Naturally, such examples are meant to be illustrative of the type and number of exportable FABP-carrier and FABP-carrier proteins within the scope of the current invention.

Another aspect of the current invention may include systems, methods and compositions for the solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in whole plants and plant cell cultures. In certain embodiments, such plants or cell cultures may be genetically modified to direct cannabinoid synthesis to the cytosol, as opposed to a trichome structure. Further, FABP-carrier proteins may be coupled with a secretion signal, preferable in a plant cell culture, to allow such proteins to be exported from the cell into the surrounding media. Expression of exportable and non-exportable FABP-carrier proteins may be co-expressed with one or more catalase and/or myb transcription factors

Another embodiment of the inventive technology may include the generation of a powder containing one or more solubilized cannabinoids. In one preferred embodiment, cannabinoids, terpenes, and other short-chain fatty acid phenolic compounds may be solubilized by association with a FABP-carrier protein. FABP-carrier proteins, having solubilized a quantity of cannabinoids, may undergo lyophilisation, to form a FABP-carrier protein powder containing the solubilized cannabinoids. In a preferred embodiment, a FABP-carrier protein may solubilize a quantity of cannabinoids through one of the methods generally described herein

and then may further undergo lyophilisation, to form a FABP-carrier protein powder containing the solubilized cannabinoids. This powder may have enhanced properties, such as enhanced cannabinoid affinity to provide greater retention and shelf-life to the cannabinoids in the powdered composition. Additionally, this cannabinoid infused powder may be reintroduced to a liquid such that the cannabinoids are re-dissolved in the liquid. This powder may be used, for example, by consumers that wish to add a quantity of one or more cannabinoids to a beverage or other consumable product. It may also be used for pharmaceutical preparations and for proper cannabinoid dosing. This type of soluble cannabinoid-infused powder may be used as a food additive, or even coupled with flavoring agents to be used as a beverage additive. The presence of the FABP-carrier proteins, as well as the enhanced cannabinoid affinity and binding capacity, may allow less powder to be used to achieve an equivalent dose, whether in a pharmaceutical or consumer beverage/consumable product.

Other embodiments may allow for the creation of high-concentration solutions of solubilized cannabinoids bound to FABP-carrier proteins. Such solutions may allow a user to generate liquid-based food and beverage additives of varying concentrations. Such solutions may further allow a user to generate liquid-based food and beverage additives of varying types of cannabinoids or combinations of cannabinoids and/or terpenes and the like. Due to the enhanced characteristics of certain FABP-carriers, in particular the enhanced ability to bind and solubilize greater numbers of individual cannabinoid molecules per protein chain, such solutions may achieve higher than normal concentrations of solubilized cannabinoids. Also, due to the enhanced affinity characteristics of certain FABP-carriers, liquid solutions having solubilized cannabinoids may achieve a longer-shelf life.

In another embodiment, the inventive technology may include novel systems, methods and compositions to decrease potential antigenicity for the FABP-carrier proteins. In one preferred embodiment, the recognition sequences of one or more FABP-carrier, or preferably FABP-carrier proteins that correspond to the formation of one or more post-translational glycosylation sites or motifs may be disrupted. In this embodiment, site-directed mutagenesis of recognition sequences that allow for post-translational glycosylation for the sequences identified as SEQ ID NO. 1-2, 21, and 74-86, or a homolog thereof may be accomplished. The removal of such glycosylation sites in a FABP-carrier may result in decreased antigenicity.

The inventive technology may further include novel engineered FABP-carrier proteins that may exhibit enhance cannabinoid affinity and/or novel affinity motifs allowing for a greater number of individual cannabinoid compounds to be bound and solubilized at one time. In one preferred embodiment, the invention may include a pharmaceutical composition as active ingredient an effective amount or dose of one or more FABP-carrier proteins coupled with one or more cannabinoids, terpenes or other short-chain fatty acid phenolic compounds. In some instances, the active ingredient may be provided together with pharmaceutically tolerable adjuvants and/or excipients in the pharmaceutical composition. Such pharmaceutical composition may optionally be in combination with one or more further active ingredients. In one embodiment, one of the aforementioned FABP-carrier proteins coupled with one or more cannabinoids, terpenes or other short-chain fatty acid phenolic compounds may act as a prodrug. The term "prodrug" is taken to mean FABP-carrier proteins according to the invention which

have been modified by, for example, the additional of novel cannabinoid affinity sites, or enhanced cannabinoid affinity sites, and wherein the FABP-carrier proteins are sterically modified in the organism to released the bound cannabinoid or other compound to form the therapeutically effective dose according to the invention. The terms "effective amount" or "effective dose" or "dose" are interchangeably used herein and denote an amount of the pharmaceutical compound having a prophylactically or therapeutically relevant effect on a disease or pathological conditions, i.e. which causes in a tissue, system, animal or human a biological or medical response which is sought or desired, for example, by a researcher or physician. Pharmaceutical formulations can be administered in the form of dosage units which comprise a predetermined amount of active ingredient per dosage unit. The concentration of the prophylactically or therapeutically active ingredient in the formulation may vary from about 0.1 to 100 wt %. Preferably, the compound of formula (I) or the pharmaceutically acceptable salts thereof are administered in doses of approximately 0.5 to 1000 mg, more preferably between 1 and 700 mg, and most preferably 5 and 100 mg per dose unit. Generally, such a dose range is appropriate for total daily incorporation. In other terms, the daily dose is preferably between approximately 0.02 and 100 mg/kg of body weight. The specific dose for each patient depends, however, on a wide variety of factors as already described in the present specification (e.g. depending on the condition treated, the method of administration and the age, weight and condition of the patient). Preferred dosage unit formulations are those which comprise a daily dose or part-dose, as indicated above, or a corresponding fraction thereof of an active ingredient. Furthermore, pharmaceutical formulations of this type can be prepared using a process which is generally known in the pharmaceutical art.

As noted above, the present invention allows the scaled production of water-soluble or solubilized cannabinoids (the terms being generally used to denote a cannabinoid or other compound, such as a terpene of short-chain fatty acid phenolic compound that is water-soluble or may be dissolved in water). Because of this solubility, the invention allows for the addition of such solubilized cannabinoid to a variety of compositions without requiring oils and/or emulsions that are generally required to maintain the generally hydrophobic cannabinoid compounds in suspension. As a result, the present invention may allow for the production of a variety of compositions for the food and beverage industry, as well as pharmaceutical applications that do not required oils or emulsion suspensions and the like.

In one embodiment, the invention may include aqueous compositions containing one or more solubilized cannabinoids that may be introduced to a food or beverage. In a preferred embodiment, the invention may include an aqueous solution containing one or more solubilized cannabinoids. In this embodiment, one or more cannabinoids, terpenes, or other short-chain fatty acid phenolic compounds may be solubilized through binding to a FABP-carrier protein. Here, the solubilized cannabinoids may be generated in vivo as generally described herein, or in vitro. In additional embodiments, the solubilized cannabinoid may be an isolated non-psychoactive, such as CBD and the like. Such selection of one or more cannabinoids may be due to specific affinity specificities in a FABP-carrier protein for one cannabinoid over another. Moreover, in this embodiment, the aqueous solution may contain one or more of the following: saline, purified water, propylene glycol, deionized water, and/or an alcohol such as ethanol, as well as a pH buffer that may allow the aqueous solution to be maintained at a pH

below 7.4. Additional embodiments may include the addition of an acid or base, such as formic acid, or ammonium hydroxide.

In another embodiment, the invention may include a consumable food additive having at least one FABP-carrier solubilized cannabinoid. In this embodiment, one or more cannabinoids, terpenes or other short-chain fatty acid phenolic compounds may be solubilized through binding to a FABP-carrier protein. Here, the solubilized cannabinoids may be generated in vivo as generally described herein, or in vitro. This consumable food additive may further include one or more food additive polysaccharides, such as dextrin and/or maltodextrin, as well as an emulsifier. Example emulsifiers may include, but not be limited to: gum arabic, modified starch, pectin, xanthan gum, gum ghatti, gum tragacanth, fenugreek gum, mesquite gum, mono-glycerides and di-glycerides of long chain fatty acids, sucrose monoesters, sorbitan esters, polyethoxylated glycerols, stearic acid, palmitic acid, mono-glycerides, di-glycerides, propylene glycol esters, lecithin, lactylated mono- and di-glycerides, propylene glycol monoesters, polyglycerol esters, diacetylated tartaric acid esters of mono- and di-glycerides, citric acid esters of monoglycerides, stearyl-2-lactylates, polysorbates, succinylated monoglycerides, acetylated monoglycerides, ethoxylated monoglycerides, quillaia, whey protein isolate, casein, soy protein, vegetable protein, pullulan, sodium alginate, guar gum, locust bean gum, tragacanth gum, tamarind gum, carrageenan, furcellaran, Gellan gum, psyllium, curdlan, konjac mannan, agar, and cellulose derivatives, or combinations thereof.

The consumable food additive of the invention may be a homogenous composition and may further comprise a flavoring agent. Exemplary flavoring agents may include: sucrose (sugar), glucose, fructose, sorbitol, mannitol, corn syrup, high fructose corn syrup, saccharin, aspartame, sucralose, acesulfame potassium (acesulfame-K), neotame. The consumable food additive of the invention may also contain one or more coloring agents. Exemplary coloring agents may include: FD&C Blue Nos. 1 and 2, FD&C Green No. 3, FD&C Red Nos. 3 and 40, FD&C Yellow Nos. 5 and 6, Orange B, Citrus Red No. 2, annatto extract, beta-carotene, grape skin extract, cochineal extract or carmine, paprika oleoresin, caramel color, fruit and vegetable juices, saffron, Monosodium glutamate (MSG), hydrolyzed soy protein, autolyzed yeast extract, disodium guanylate or inosinate. In one embodiment, this powdered lyophilized FABP-carrier protein, having solubilized a quantity of cannabinoids, may be a food additive. In certain preferred embodiments, one or more flavoring agents may be added to a quantity of powdered or lyophilized FABP-carrier proteins having solubilized a quantity of cannabinoids.

The consumable food additive of the invention may also contain one or more surfactants, such as glycerol monostearate and polysorbate 80. The consumable food additive of the invention may also contain one or more preservatives. Exemplary preservatives may include ascorbic acid, citric acid, sodium benzoate, calcium propionate, sodium erythorbate, sodium nitrite, calcium sorbate, potassium sorbate, BHA, BHT, EDTA, or tocopherols. The consumable food additive of the invention may also contain one or more nutrient supplements, such as: thiamine hydrochloride, riboflavin, niacin, niacinamide, folate or folic acid, beta carotene, potassium iodide, iron or ferrous sulfate, alpha tocopherols, ascorbic acid, Vitamin D, amino acids, multi-vitamin, fish oil, co-enzyme Q-10, and calcium.

In one embodiment, the invention may include a consumable fluid containing at least one FABP-carrier solubilized

cannabinoid, terpenoid, or other short chain fatty acid phenolic compound. In one preferred embodiment, this consumable fluid may be added to a drink or beverage to infuse it with the solubilized cannabinoid generated through binding to a FABP-carrier protein, in an in vitro, ex vivo, or in vivo system as generally herein described, or through an in vitro process. The consumable fluid may include a food additive polysaccharide such as maltodextrin and/or dextrin, which may further be in an aqueous form and/or solution. For example, in one embodiment, an aqueous maltodextrin solution may include a quantity of sorbic acid and an acidifying agent to provide a food grade aqueous solution of maltodextrin having a pH of 2-4 and a sorbic acid content of 0.02-0.1% by weight.

In certain embodiments, the consumable fluid may include water, as well as an alcoholic beverage; a non-alcoholic beverage, a noncarbonated beverage, a carbonated beverage, a cola, a root beer, a fruit-flavored beverage, a citrus-flavored beverage, a fruit juice, a fruit-containing beverage, a vegetable juice, a vegetable containing beverage, a tea, a coffee, a dairy beverage, a protein containing beverage, a shake, a sports drink, an energy drink, and a flavored water. The consumable fluid may further include at least one additional ingredient, including but not limited to: xanthan gum, cellulose gum, whey protein hydrolysate, ascorbic acid, citric acid, malic acid, sodium benzoate, sodium citrate, sugar, phosphoric acid, and water. In certain embodiments, the consumable fluid of the invention may be generated by addition of a quantity of solubilized cannabinoid in powder or liquid form as generally described herein to an existing consumable fluid, such as a branded beverage or drink.

In one embodiment, the invention may include a consumable gel having at least one FABP-carrier solubilized cannabinoid and gelatin in an aqueous solution. In a preferred embodiment, the consumable gel may include a one or more cannabinoids, terpenes or other short-chain fatty acid phenolic compounds solubilized through binding to a FABP-carrier protein. Here, the solubilized cannabinoids may be generated in vivo as generally described herein, or in vitro.

Additional embodiments may include a liquid composition having at least one cannabinoid solubilized by a FABP-carrier protein, in a first quantity of water; and at least one of: xanthan gum, cellulose gum, whey protein hydrolysate, ascorbic acid, citric acid, malic acid, sodium benzoate, sodium citrate, sugar, phosphoric acid, and/or a sugar alcohol. In one preferred embodiment, the composition may further include a quantity of ethanol. Here, the amount of solubilized cannabinoids may include: less than 10 mass % water; more than 95 mass % water; about 0.1 mg to about 1000 mg of the solubilized cannabinoid; about 0.1 mg to about 500 mg of the solubilized cannabinoid; about 0.1 mg to about 200 mg of the solubilized cannabinoid; about 0.1 mg to about 100 mg of the solubilized cannabinoid; about 0.1 mg to about 100 mg of the solubilized cannabinoid; about 0.1 mg to about 10 mg of the solubilized cannabinoid; about 0.5 mg to about 5 mg of the solubilized cannabinoid; about 1 mg/kg to 5 mg/kg (body weight) in a human of the solubilized cannabinoid.

In alternative embodiments, the composition may include at least one cannabinoid solubilized by a FABP-carrier protein, in the range of 50 mg/L to 300 mg/L; at least one FABP-carrier solubilized cannabinoid in the range of 50 mg/L to 100 mg/L; at least one FABP-carrier solubilized cannabinoid in the range of 50 mg/L to 500 mg/L; at least one FABP-carrier solubilized cannabinoid over 500 mg/L; at least one FABP-carrier solubilized cannabinoid under 50

mg/L. Additional embodiments may include one or more of the following additional components: a flavoring agent; a coloring agent; and/or caffeine.

In one embodiment, the invention may include a liquid composition having at least one cannabinoid solubilized by a FABP-carrier protein, being solubilized in said first quantity of water and a first quantity of ethanol in a liquid state. In a preferred embodiment, a first quantity of ethanol in a liquid state may be between 1% to 20% weight by volume of the liquid composition. In this embodiment, a solubilized cannabinoid may include a cannabinoid solubilized by a FABP-carrier protein, a terpenoid/terpene solubilized by a FABP-carrier protein, or a mixture of both. Such solubilized cannabinoids may be generated in an in vivo, ex vivo, and/or in vitro system as herein identified. In a preferred embodiment, the ethanol or ethyl alcohol component may be up to about ninety-nine point nine-five percent (99.95%) by weight and the solubilized cannabinoid about zero point zero five percent (0.05%) by weight.

Examples of the preferred embodiment may include liquid ethyl alcohol compositions having at least one cannabinoid solubilized by a FABP-carrier protein, wherein said ethyl alcohol has a proof greater than 100, and/or less than 100. Additional examples of a liquid composition containing ethyl alcohol and at least one cannabinoid solubilized by a FABP-carrier protein, may include, beer, wine and/or distilled spirits.

Additional embodiments of the invention may include a chewing gum composition having a first quantity of at least one cannabinoid solubilized by a FABP-carrier protein. In a preferred embodiment, a chewing gum composition may further include a gum base comprising a buffering agent selected from the group consisting of acetates, glycinates, phosphates, carbonates, glycerophosphates, citrates, borates, and mixtures thereof. Additional components may include at least one sweetening agent and at least one flavoring agent. As noted above, in a preferred embodiment, at least one cannabinoid solubilized by a FABP-carrier protein, may be generated in vitro, or in vivo respectively.

In one embodiment, the chewing gum composition described above may include:

0.01 to 1% by weight of at least one FABP-carrier solubilized cannabinoid;

25 to 85% by weight of a gum base;

10 to 35% by weight of at least one sweetening agent; and 1 to 10% by weight of a flavoring agent.

Here, such flavoring agents may include: menthol flavor, eucalyptus, mint flavor and/or L-menthol. Sweetening agents may include one or more of the following: xylitol, sorbitol, isomalt, aspartame, sucralose, acesulfame potassium, and saccharin. Additional preferred embodiment may include a chewing gum having a pharmaceutically acceptable excipient selected from the group consisting of: fillers, disintegrants, binders, lubricants, and antioxidants. The chewing gum composition may further be non-disintegrating and also include one or more coloring and/or flavoring agents.

The invention may further include a composition for a cannabinoid infused solution comprising essentially of: water and/or purified water, at least one cannabinoid solubilized by a FABP-carrier protein and at least one flavoring agent. A FABP-carrier solubilized cannabinoid infused solution of the invention may further include a sweetener selected from the group consisting of: glucose, sucrose, invert sugar, corn syrup, stevia extract powder, stevioside, steviol, aspartame, saccharin, saccharin salts, sucralose, potassium acetosulfam, sorbitol, xylitol, mannitol, erythri-

tol, lactitol, alitame, miraculin, monellin, and thaumatin or a combination of the same. Additional components of the solubilized cannabinoid infused solution may include, but not be limited to: sodium chloride, sodium chloride solution, glycerin, a coloring agent, and a demulcent. As to this last potential component, in certain embodiments, a demulcent may include: pectin, glycerin, honey, methylcellulose, and/or propylene glycol. As noted above, in a preferred embodiment, a FABP-carrier solubilized cannabinoid may include at least one FABP-carrier solubilized cannabinoid wherein such solubilized cannabinoids may be generated in vivo, ex vivo, and/or in vitro respectively.

The invention may further include a composition for a FABP-carrier solubilized cannabinoid infused anesthetic solution having water, or purified water, at least one FABP-carrier solubilized cannabinoid, and at least one oral anesthetic. In a preferred embodiment, an anesthetic may include benzocaine, and/or phenol in a quantity of between 0.1% to 15% volume by weight.

Additional embodiments may include a FABP-carrier solubilized cannabinoid infused anesthetic solution having a sweetener which may be selected from the group consisting of: glucose, sucrose, invert sugar, corn syrup, stevia extract powder, stevioside, steviol, aspartame, saccharin, saccharin salts, sucralose, potassium acetosulfam, sorbitol, xylitol, mannitol, erythritol, lactitol, alitame, miraculin, monellin, and thaumatin or a combination of the same. Additional components of a FABP-carrier solubilized cannabinoid infused solution may include, but not be limited to: sodium chloride, sodium chloride solution, glycerin, a coloring agent, and a demulcent. In a preferred embodiment, a demulcent may be selected from the group consisting of: pectin, glycerin, honey, methylcellulose, and propylene glycol. As noted above, in a preferred embodiment, a FABP-carrier solubilized cannabinoid may include at least one cannabinoid solubilized by a FABP-carrier protein. In this embodiment, such solubilized cannabinoids may have been generated in vivo, ex vivo, and/or in vitro respectively.

The invention may further include a composition for a hard lozenge for rapid delivery of solubilized cannabinoids through the oral mucosa. In this embodiment, such a hard lozenge composition may include: a crystalized sugar base and at least one FABP-carrier solubilized cannabinoid, wherein the hard lozenge has moisture content between 0.1 to 2%. In this embodiment, the solubilized cannabinoid may be added to the sugar base when it is in a liquefied form and prior to the evaporation of the majority of water content. Such a hard lozenge may further be referred to as a candy.

In a preferred embodiment, a crystalized sugar base may be formed from one or more of the following: sucrose, invert sugar, corn syrup, and isomalt or a combination of the same. Additional components may include at least one acidulant. Examples of acidulants may include, but not be limited to: citric acid, tartaric acid, fumaric acid, and malic acid. Additional components may include at least one pH adjuster. Examples of pH adjusters may include, but not be limited to: calcium carbonate, sodium bicarbonate, and magnesium trisilicate.

In another preferred embodiment, the composition may include at least one anesthetic. Example of such anesthetics may include benzocaine, and phenol. In this embodiment, first quantity of anesthetic may be between 1 mg to 15 mg per lozenge. Additional embodiments may include a quantity of menthol. In this embodiment, such a quantity of menthol may be between 1 mg to 20 mg. The hard lozenge composition may also include a demulcent, for example: pectin, glycerin, honey, methylcellulose, propylene glycol,

and glycerin. In this embodiment, a demulcent may be in a quantity between 1 mg to 10 mg. As noted above, in a preferred embodiment, a FABP-carrier solubilized cannabinoid may include at least one cannabinoid solubilized by a FABP-carrier protein. In this embodiment, such solubilized cannabinoid may have been generated in vivo and/or in vitro respectively.

The invention may include a chewable lozenge for rapid delivery of solubilized cannabinoids through the oral mucosa. In a preferred embodiment, the compositions may include: a glycerinated gelatin base, at least one sweetener, and at least one FABP-carrier solubilized cannabinoid dissolved in a first quantity of water. In this embodiment, a sweetener may include a sweetener selected from the group consisting of: glucose, sucrose, invert sugar, corn syrup, stevia extract powder, stevioside, steviol, aspartame, saccharin, saccharin salts, sucralose, potassium acetosulfam, sorbitol, xylitol, mannitol, erythritol, lactitol, alitame, miraculin, monellin, and thaumatin or a combination of the same.

Additional components may include at least one acidulant. Examples of acidulants may include, but not be limited to: citric acid, tartaric acid, fumaric acid, and malic acid. Additional components may include at least one pH adjuster. Examples of pH adjusters may include, but not be limited to: calcium carbonate, sodium bicarbonate, and magnesium trisilicate.

In another preferred embodiment, the composition may include at least one anesthetic. Example of such anesthetics may include benzocaine and phenol. In this embodiment, first quantity of anesthetic may be between 1 mg to 15 mg per lozenge. Additional embodiments may include a quantity of menthol. In this embodiment, such a quantity of menthol may be between 1 mg to 20 mg. The chewable lozenge composition may also include a demulcent, for example: pectin, glycerin, honey, methylcellulose, propylene glycol, and glycerin. In this embodiment, a demulcent may be in a quantity between 1 mg to 10 mg. As noted above, in a preferred embodiment, a FABP-carrier solubilized cannabinoid may include at least one cannabinoid solubilized by a FABP-carrier protein. In this embodiment, such solubilized cannabinoid may be generated in vivo, ex vivo, or in vitro respectively.

The invention may include a soft lozenge for rapid delivery of solubilized cannabinoids through the oral mucosa. In a preferred embodiment, the compositions may include: a polyethylene glycol base, at least one sweetener, and at least one FABP-carrier solubilized cannabinoid, referring generally to a cannabinoid solubilized by a FABP-carrier, dissolved in a first quantity of water. In this embodiment, a sweetener may include sweetener selected from the group consisting of: glucose, sucrose, invert sugar, corn syrup, stevia extract powder, stevioside, steviol, aspartame, saccharin, saccharin salts, sucralose, potassium acetosulfam, sorbitol, xylitol, mannitol, erythritol, lactitol, alitame, miraculin, monellin, and thaumatin or a combination of the same. Additional components may include at least one acidulant. Examples of acidulants may include, but not be limited to: citric acid, tartaric acid, fumaric acid, and malic acid. Additional components may include at least one pH adjuster. Examples of pH adjusters may include, but not be limited to: calcium carbonate, sodium bicarbonate, and magnesium trisilicate.

In another preferred embodiment, the composition may include at least one anesthetic. Example of such anesthetics may include benzocaine and phenol. In this embodiment, first quantity of anesthetic may be between 1 mg to 15 mg per lozenge. Additional embodiments may include a quan-

tity of menthol. In this embodiment, such a quantity of menthol may be between 1 mg to 20 mg. The soft lozenge composition may also include a demulcent, for example: pectin, glycerin, honey, methylcellulose, propylene glycol, and glycerin. In this embodiment, a demulcent may be in a quantity between 1 mg to 10 mg. As noted above, in a preferred embodiment, a FABP-carrier solubilized cannabinoid may include at least one cannabinoid solubilized by a FABP-carrier protein. In this embodiment, such solubilized cannabinoid may be generated in vivo, ex vivo, or in vitro respectively.

In another embodiment, the invention may include a tablet or capsule consisting essentially of a solubilized cannabinoid and a pharmaceutically acceptable excipient. Examples may include solid, semi-solid, and aqueous excipients such as: maltodextrin, whey protein isolate, xanthan gum, guar gum, diglycerides, monoglycerides, carboxymethyl cellulose, glycerin, gelatin, polyethylene glycol and water-based excipients. In this embodiment, the cannabinoid solubilized by a FABP-carrier protein, may have an improved shelf-life, composition stability, and bioavailability upon injection.

In a preferred embodiment, a solubilized cannabinoid may include at least one cannabinoid solubilized by a FABP-carrier protein. In this embodiment, such solubilized cannabinoids may be generated in vivo, ex vivo, or in vitro respectively. Examples of such in vivo systems being generally described herein, including in plant, as well as cell culture systems including *Cannabis* cell culture, tobacco cell culture, bacterial cell cultures, fungal cell cultures, and yeast cell culture systems. In one embodiment, a tablet or capsule may include an amount of solubilized cannabinoid of 5 milligrams or less. Alternative embodiments may include an amount of solubilized cannabinoid between 5 milligrams and 200 milligrams. Still other embodiments may include a tablet or capsule having an amount of solubilized cannabinoid that is more than 200 milligrams. Still other embodiments may include a tablet or capsule having an amount of solubilized cannabinoid that is more than 500 milligrams.

The invention may further include a method of manufacturing and packaging a solubilized cannabinoid dosage, consisting of the following steps: 1) preparing a fill solution with a desired concentration of one or more FABP-carrier solubilized cannabinoid in a liquid carrier wherein said cannabinoid is dissolved in said liquid carrier; 2) encapsulating said fill solution in capsules; 3) packaging said capsules in a closed packaging system; and 4) removing atmospheric air from the capsules. In one embodiment, the step of removing atmospheric air consists of purging the packaging system with an inert gas, such as, for example, nitrogen gas, such that said packaging system provides a room temperature stable product. In one preferred embodiment, the packaging system may include a plaster package, which may be constructed of material that minimizes exposure to moisture and air.

In one embodiment, a preferred liquid carrier may include a water-based carrier, such as for example an aqueous sodium chloride solution. In a preferred embodiment, a solubilized cannabinoid may include at least one cannabinoid solubilized by a FABP-carrier protein. In this embodiment, such solubilized cannabinoids may be generated in vivo, ex vivo, or in vitro respectively. In one embodiment, a desired solubilized cannabinoid concentration may be about 1-10% w/w, while in other embodiments it may be about 1.5-6.5% w/w. Alternative embodiments may include an amount of solubilized cannabinoid between 5 milligrams and 200 milligrams. Still, other embodiments may include a

tablet or capsule having amount of solubilized cannabinoid that is more than 200 milligrams. Other embodiments may include a tablet or capsule having an amount of solubilized cannabinoid that is more than 500 milligrams.

The invention may include an oral pharmaceutical solution, such as a sub-lingual spray having solubilized cannabinoids and a liquid carrier. One embodiment may include a solubilized cannabinoid, 30-33% w/w water, about 50% w/w alcohol, 0.01% w/w butylated hydroxyanisole (BHA) or 0.1% w/w ethylenediaminetetraacetic acid (EDTA) and 5-21% w/w co-solvent, having a combined total of 100%, wherein said co-solvent is selected from the group consisting of propylene glycol, polyethylene glycol, and combinations thereof, and wherein said solubilized cannabinoid is at least one cannabinoid solubilized by a FABP-carrier protein. In an alternative embodiment, such an oral pharmaceutical solution may consist essentially of 0.1 to 5% w/w of said solubilized cannabinoid, about 50% w/w alcohol, 5.5% w/w propylene glycol, 12% w/w polyethylene glycol and 30-33% w/w water. In a preferred composition, the alcohol component may be ethanol.

The invention may include an oral pharmaceutical solution, such as a sublingual spray, consisting essentially of about 0.1% to 1% w/w solubilized cannabinoids, about 50% w/w alcohol, 5.5% w/w propylene glycol, 12% w/w polyethylene glycol, 30-33% w/w water, 0.01% w/w butylated hydroxyanisole, having a combined total of 100%, and wherein said solubilized cannabinoid is at least one cannabinoid solubilized by a FABP-carrier protein, that may be further generated in vivo, ex vivo, or in vitro respectively. In an alternative embodiment, such an oral pharmaceutical solution may consist essentially of 0.54% w/w solubilized cannabinoid, 31.9% w/w water, 12% w/w polyethylene glycol 400, 5.5% w/w propylene glycol, 0.01% w/w butylated hydroxyanisole, 0.05% w/w sucralose, and 50% w/w alcohol, wherein the alcohol components may be ethanol.

The invention may include a solution for nasal and/or sublingual administration of a solubilized cannabinoid including: 1) an excipient of propylene glycol, ethanol anhydrous, or a mixture of both; and 2) a solubilized cannabinoid which may include at least one cannabinoid solubilized by a FABP-carrier protein, that may be further generated in vivo, ex vivo, or in vitro respectively. In a preferred embodiment, the composition may further include a topical decongestant, which may include phenylephrine hydrochloride, Oxymetazoline hydrochloride, and Xylometazoline in certain preferred embodiments. The composition may further include an antihistamine, and/or a steroid. Preferably, the steroid component is a corticosteroid selected from the group consisting of: neclomethasone dipropionate, budesonide, ciclesonide, flunisolide, fluticasone furoate, fluticasone propionate, mometasone, and triamcinolone acetonide. In alternative embodiments, the solution for nasal and/or sublingual administration of a solubilized cannabinoid may further comprise at least one of the following: benzalkonium chloride solution, benzyl alcohol, boric acid, purified water, sodium borate, polysorbate 80, phenylethyl alcohol, microcrystalline cellulose, carboxymethylcellulose sodium, dextrose, dipasic, sodium phosphate, edetate disodium, monobasic sodium phosphate, and propylene glycol.

The invention may further include an aqueous solution for nasal and/or sublingual administration of a solubilized cannabinoid comprising: a water and/or saline solution; and a solubilized cannabinoid which may include at least one cannabinoid solubilized by a FABP-carrier protein that may be further generated in vivo, ex vivo, or in vitro, respectively. In a preferred embodiment, the composition may

further include a topical decongestant, which may include phenylephrine hydrochloride, Oxymetazoline hydrochloride, and Xylometazoline in certain preferred embodiments. The composition may further include an antihistamine and/or a steroid. Preferably, the steroid component is a corticosteroid selected from the group consisting of: neclomethasone dipropionate, budesonide, ciclesonide, flunisolide, fluticasone furoate, fluticasone propionate, mometasone, and triamcinolone acetonide. In alternative embodiments, the aqueous solution may further comprise at least one of the following: benzalkonium chloride solution, benzyl alcohol, boric acid, purified water, sodium borate, polysorbate 80, phenylethyl alcohol, microcrystalline cellulose, carboxymethylcellulose sodium, dextrose, dipasic, sodium phosphate, edetate disodium, monobasic sodium phosphate, or propylene glycol.

The invention may include a topical formulation for the transdermal delivery of solubilized cannabinoids. In a preferred embodiment, a topical formulation for the transdermal delivery of solubilized cannabinoids which may include at least one cannabinoid solubilized by a FABP-carrier protein, and a pharmaceutically acceptable excipient. The solubilized cannabinoids may be generated in vivo, ex vivo, or in vitro respectively. Preferably a pharmaceutically acceptable excipient may include one or more: gels, ointments, cataplasms, poultices, pastes, creams, lotions, plasters and jellies or even polyethylene glycol. Additional embodiments may further include one or more of the following components: a quantity of capsaicin; a quantity of benzocaine; a quantity of lidocaine; a quantity of camphor; a quantity of benzoin resin; a quantity of methylsalicylate; a quantity of triethanolamine salicylate; a quantity of hydrocortisone; or a quantity of salicylic acid.

The invention may include a gel for transdermal administration of a solubilized cannabinoid which may include at least one cannabinoid solubilized by a FABP-carrier protein, and which may be generated in vivo, ex vivo, or in vitro. In this embodiment, the mixture preferably contains from 15% to about 90% ethanol, about 10% to about 60% buffered aqueous solution or water, about 0.1 to about 25% propylene glycol, from about 0.1 to about 20% of a gelling agent, from about 0.1 to about 20% of a base, from about 0.1 to about 20% of an absorption enhancer and from about 1% to about 25% polyethylene glycol, and a solubilized cannabinoid as generally described herein.

In another embodiment, the invention may further include a transdermal composition having a pharmaceutically effective amount of a solubilized cannabinoid for delivery of the cannabinoid to the bloodstream of a user. This transdermal composition may include a pharmaceutically acceptable excipient and at least one FABP-carrier solubilized cannabinoid, which may include at least one cannabinoid solubilized by a FABP-carrier protein, and which may be generated in vivo, ex vivo, or in vitro, wherein the solubilized cannabinoid is capable of diffusing from the composition into the bloodstream of the user. In a preferred embodiment, a pharmaceutically acceptable excipient to create a transdermal dosage form selected from the group consisting of: gels, ointments, cataplasms, poultices, pastes, creams, lotions, plasters and jellies. The transdermal composition may further include one or more surfactants. In one preferred embodiment, the surfactant may include a surfactant-lecithin organogel, which may further be present in an amount of between about 95% and about 98% w/w. In an alternative embodiment, a surfactant-lecithin organogel comprises lecithin and PPG-2 myristyl ether propionate

and/or high molecular weight polyacrylic acid polymers. The transdermal composition may further include a quantity of isopropyl myristate.

The invention may further include transdermal composition having one or more permeation enhancers to facilitate transfer of the solubilized cannabinoid across a dermal layer. In a preferred embodiment, a permeation enhancer may include one or more of the following: propylene glycol monolaurate, diethylene glycol monoethyl ether, an oleoyl macroglyceride, a caprylocaproyl macroglyceride, and an oleyl alcohol.

The invention may also include a liquid cannabinoid liniment composition consisting of water, isopropyl alcohol solution, and a solubilized cannabinoid, which may include at least one cannabinoid solubilized by a FABP-carrier protein, and which may be generated in vivo, ex vivo, or in vitro. This liquid cannabinoid liniment composition may further include approximately 97.5% to about 99.5% by weight of 70% isopropyl alcohol solution and from about 0.5% to about 2.5% by weight of a solubilized cannabinoid mixture.

Based on the improved solubility and other physical properties, as well as cost advantages, improved cannabinoid affinity and capacity, extended shelf-life, and scalability of the inventions in vivo, ex vivo, or in vitro solubilized cannabinoid production platform, the invention may include one or more commercial infusions. For example, commercially available products, such as a lip balm, soap, shampoos, lotions, creams, and cosmetics may be infused with one or more solubilized cannabinoids.

The invention may further include a novel composition that may be used to supplement a cigarette or other tobacco-based product. In this embodiment, the composition may include at least one FABP-carrier solubilized cannabinoid in a powder as already described, or dissolved in an aqueous solution. This aqueous solution may be introduced to a tobacco product, such as a cigarette and/or a tobacco leaf such that the aqueous solution may evaporate generating a cigarette and/or a tobacco leaf that contains the aforementioned solubilized cannabinoid(s), which may further have been generated in vivo as generally described herein.

In one embodiment, the invention may include one or more methods of treating a medical condition in a mammal. In this embodiment, the novel method may include of administering a therapeutically effective amount of a solubilized cannabinoid, such as an in vivo, ex vivo, or in vitro cannabinoid solubilized by a FABP-carrier protein, wherein the medical condition is selected from the group consisting of: obesity, post-traumatic stress syndrome, anorexia, nausea, emesis, pain, wasting syndrome, HIV-wasting, chemotherapy induced nausea and vomiting, alcohol use disorders, anti-tumor, amyotrophic lateral sclerosis, glioblastoma multiforme, glioma, increased intraocular pressure, glaucoma, *Cannabis* use disorders, Tourette's syndrome, dystonia, multiple sclerosis, inflammatory bowel disorders, arthritis, dermatitis, Rheumatoid arthritis, systemic lupus erythematosus, anti-inflammatory, anti-convulsant, anti-psychotic, anti-oxidant, neuroprotective, anti-cancer, immunomodulatory effects, peripheral neuropathic pain, neuropathic pain associated with post-herpetic neuralgia, diabetic neuropathy, shingles, burns, actinic keratosis, oral cavity sores and ulcers, post-episiotomy pain, psoriasis, pruritis, contact dermatitis, eczema, bullous dermatitis herpetiformis, exfoliative dermatitis, mycosis fungoides, pemphigus, severe erythema multiforme (e.g., Stevens-Johnson syndrome), seborrheic dermatitis, ankylosing spondylitis, psoriatic arthritis, Reiter's syndrome, gout, chondrocalcinosis, joint

pain secondary to dysmenorrhea, fibromyalgia, musculoskeletal pain, neuropathic-postoperative complications, polymyositis, acute nonspecific tenosynovitis, bursitis, epicondylitis, post-traumatic osteoarthritis, synovitis, and juvenile rheumatoid arthritis. In a preferred embodiment, the pharmaceutical composition may be administered by a route selected from the group consisting of: transdermal, topical, oral, buccal, sublingual, intra-venous, intra-muscular, vaginal, rectal, ocular, nasal and follicular. The amount of solubilized cannabinoids may be a therapeutically effective amount, which may be determined by the patient's age, weight, medical condition cannabinoid-delivered, route of delivery, and the like. In one embodiment, a therapeutically effective amount may be 50 mg or less of a solubilized cannabinoid. In another embodiment, a therapeutically effective amount may be 50 mg or more of a solubilized cannabinoid.

It should be noted that for any of the above composition, unless otherwise stated, an effective amount of solubilized cannabinoids may include amounts between: 0.01 mg to 0.1 mg; 0.01 mg to 0.5 mg; 0.01 mg to 1 mg; 0.01 mg to 5 mg; 0.01 mg to 10 mg; 0.01 mg to 25 mg; 0.01 mg to 50 mg; 0.01 mg to 75 mg; 0.01 mg to 100 mg; 0.01 mg to 125 mg; 0.01 mg to 150 mg; 0.01 mg to 175 mg; 0.01 mg to 200 mg; 0.01 mg to 225 mg; 0.01 mg to 250 mg; 0.01 mg to 275 mg; 0.01 mg to 300 mg; 0.01 mg to 325 mg; 0.01 mg to 350 mg; 0.01 mg to 375 mg; 0.01 mg to 400 mg; 0.01 mg to 425 mg; 0.01 mg to 450 mg; 0.01 mg to 475 mg; 0.01 mg to 500 mg; 0.01 mg to 525 mg; 0.01 mg to 550 mg; 0.01 mg to 575 mg; 0.01 mg to 600 mg; 0.01 mg to 625 mg; 0.01 mg to 650 mg; 0.01 mg to 675 mg; 0.01 mg to 700 mg; 0.01 mg to 725 mg; 0.01 mg to 750 mg; 0.01 mg to 775 mg; 0.01 mg to 800 mg; 0.01 mg to 825 mg; 0.01 mg to 850 mg; 0.01 mg to 875 mg; 0.01 mg to 900 mg; 0.01 mg to 925 mg; 0.01 mg to 950 mg; 0.01 mg to 975 mg; 0.01 mg to 1000 mg; 0.01 mg to 2000 mg; 0.01 mg to 3000 mg; 0.01 mg to 4000 mg; 0.01 mg to 5000 mg; 0.01 mg to 0.1 mg/kg; 0.01 mg to 0.5 mg/kg; 0.01 mg to 1 mg/kg; 0.01 mg to 5 mg/kg; 0.01 mg to 10 mg/kg; 0.01 mg to 25 mg/kg; 0.01 mg to 50 mg/kg; 0.01 mg to 75 mg/kg; and 0.01 mg to 100 mg/kg.

The FABP solubilized cannabinoid compounds of the present invention are useful for a variety of therapeutic applications. For example, the compounds are useful for treating or alleviating symptoms of diseases and disorders involving CB1 and CB2 receptors, including appetite loss, nausea and vomiting, pain, multiple sclerosis and epilepsy. For example, they may be used to treat pain (i.e. as analgesics) in a variety of applications including but not limited to pain management. In additional embodiments, such solubilized cannabinoids may be used as an appetite suppressant. Additional embodiments may include administering the FABP solubilized cannabinoid compounds of the invention.

By "treating," the present inventors mean that the compound is administered in order to alleviate symptoms of the disease or disorder being treated. Those of skill in the art will recognize that the symptoms of the disease or disorder that is treated may be completely eliminated or may simply be lessened. Further, the compounds may be administered in combination with other drugs or treatment modalities, such as with chemotherapy or other cancer-fighting drugs.

Implementation may generally involve identifying patients suffering from the indicated disorders and administering the compounds of the present invention in an acceptable form by an appropriate route. The exact dosage to be administered may vary depending on the age, gender, weight, and overall health status of the individual patient, as well as the precise etiology of the disease. However, in

general, for administration in mammals (e.g. humans), dosages in the range of from about 0.01 to about 300 mg of compound per kg of body weight per 24 hr., and more preferably about 0.01 to about 100 mg of compound per kg of body weight per 24 hr., may be effective.

Administration may be oral or parenteral, including intravenously, intramuscularly, subcutaneously, intradermal injection, intraperitoneal injection, etc., or by other routes (e.g. transdermal, sublingual, oral, rectal and buccal delivery, inhalation of an aerosol, etc.). In a preferred embodiment of the invention, the solubilized cannabinoid are provided orally or intravenously.

The compounds, for example a cannabinoid solubilized by a FABP-carrier, may be administered in the pure form or in a pharmaceutically acceptable formulation including suitable elixirs, binders, and the like (generally referred to as a "secondary carrier") or as pharmaceutically acceptable salts (e.g. alkali metal salts such as sodium, potassium, calcium or lithium salts, ammonium, etc.) or other complexes. It should be understood that the pharmaceutically acceptable formulations include liquid and solid materials conventionally utilized to prepare both injectable dosage forms and solid dosage forms such as tablets and capsules and aerosolized dosage forms. In addition, the compounds may be formulated with aqueous or oil based vehicles. Water may be used as the carrier for the preparation of compositions (e.g. injectable compositions), which may also include conventional buffers and agents to render the composition isotonic. Other potential additives and other materials (preferably those which are generally regarded as safe [GRAS]) include: colorants; flavorings; surfactants (TWEEN, oleic acid, etc.); solvents, stabilizers, elixirs, and binders or encapsulants (lactose, liposomes, etc). Solid diluents and excipients include lactose, starch, conventional disintegrating agents, coatings and the like. Preservatives such as methyl paraben or benzalkonium chloride may also be used. Depending on the formulation, it is expected that the active composition will consist of about 1% to about 99% of the composition and the secondary carrier will constitute about 1% to about 99% of the composition. The pharmaceutical compositions of the present invention may include any suitable pharmaceutically acceptable additives or adjuncts to the extent that they do not hinder or interfere with the therapeutic effect of the active compound.

The administration of the solubilized compounds of the present invention may be intermittent, bolus dose, or at a gradual or continuous, constant, or controlled rate to a patient. In addition, the time of day and the number of times per day that the pharmaceutical formulation is administered may vary and are best determined by a skilled practitioner such as a physician. Further, the effective dose can vary depending upon factors such as the mode of delivery, gender, age, and other conditions of the patient, as well as the extent or progression of the disease. The compounds may be provided alone, in a mixture containing two or more of the compounds, or in combination with other medications or treatment modalities.

Notably, as used herein, "FABP," or "FABP-carrier" or "FABP-carrier protein" explicitly includes any FABPs that have affinity for a cannabinoid, terpene or other short-chain fatty acid phenolic compounds. Additionally, "FABP-carrier" may be generically used to explicitly describe FABP-like proteins, regardless of family or classification, that exhibit an affinity for a cannabinoid, terpene or other short-chain fatty acid phenolic compounds. The term "FABP-carrier" explicitly encompasses FABP-like proteins, FABP homologs, and FABP orthologs having affinity for a can-

nabinoid, terpene or other short-chain fatty acid phenolic compounds. As used herein a "FABP," or "FABP-carrier" or "FABP-carrier protein" specifically encompasses wild-type, as well as "engineered" or "engineered FABP-carrier" or "engineered FABP-carrier proteins." As used herein, a "engineered" or "engineered FABP-carrier" or "engineered FABP-carrier protein" may be a FABP-carrier protein that may bind or, to have some affinity for a target compound, such as a cannabinoid, that may be further modified to have enhanced affinity, or additional affinity sites for the target compound, such as a mutation that may modulate a characteristic of that protein, such as cannabinoid binding characteristic, or antigenicity characteristic, or for example may be modified to form a chimera or fusions protein through the addition of a secretion signal. In one embodiment, as "engineered" may be established through bioinformatics approaches such as computational docking, binding affinity estimation, and molecular dynamics simulations. Additional embodiments may include the site-directed mutations of an "engineered," "engineered FABP-carrier," or "engineered carrier protein" may also be determined by through bioinformatics approaches such as computational docking, binding affinity estimation, and molecular dynamics simulations as well as in high-throughput binding assays and the like. According to some embodiments of the present invention, a suitable computational platform for executing the method presented herein, in one embodiment may include the generation of an engineered FABP-carrier, may include the Rosetta™ software suite platform, publically available from the "Rosetta@home" at the Baker laboratory, University of Washington, U.S.A. Briefly, Rosetta™ is a molecular modeling software package for understanding protein structures, protein design, protein docking, protein-DNA and protein-protein interactions. The Rosetta software contains multiple functional modules, including RosettaAbinitio, RosettaDesign, RosettaDock, RosettaAntibody, RosettaFragments, RosettaNMR, RosettaDNA, RosettaRNA, RosettaLigand, RosettaSymmetry, and more. According to some embodiments of the present invention, a suitable computational platform for executing the method presented herein, is the SMINA software suite platform, which generally encompasses a version of AutoDock Vina specially optimized to support high-throughput scoring and user-specified custom scoring functions. Additional embodiment may further include software suites for AutoDock Vina, and idock.

As used herein, the term "enhanced" as used to described the binding of a cannabinoid to a FABP-carrier, or engineered FABP-carrier means having a binding affinity greater than wild-type binding affinity.

As used herein, a "cannabinoid" is a chemical compound (such as cannabinol, THC or cannabidiol) that is found in the plant species *Cannabis* among others like: *Echinacea*; *Acmella Oleracea*; *Helichrysum umbraculigerum*; *Radula marginata* (Liverwort) and *Theobroma cacao*, and metabolites and synthetic analogues thereof that may or may not have psychoactive properties. Cannabinoids therefore include (without limitation) compounds (such as THC) that have high affinity for the cannabinoid receptor (for example $K_i < 250$ nM), and compounds that do not have significant affinity for the cannabinoid receptor (such as cannabidiol, CBD). Cannabinoids also include compounds that have a characteristic dibenzopyran ring structure (of the type seen in THC) and cannabinoids which do not possess a pyran ring (such as cannabidiol). Hence a partial list of cannabinoids includes THC, CBD, dimethyl heptylpentyl cannabidiol (DMHP-CBD), 6,12-dihydro-6-hydroxy-cannabidiol (described in U.S. Pat. No. 5,227,537, incorporated by refer-

ence); (3S,4R)-7-hydroxy- Δ 6-tetrahydrocannabinol homologs and derivatives described in U.S. Pat. No. 4,876, 276, incorporated by reference; (+)-4-[4-DMH-2,6-diacetoxy-phenyl]-2-carboxy-6,6-dimethylbicyclo[3.1.1]hept-2-en, and other 4-phenylpinene derivatives disclosed in U.S. Pat. No. 5,434,295, which is incorporated by reference; and cannabidiol (-)(CBD) analogs such as (-)CBD-monomethylether, (-)CBD dimethyl ether; (-)CBD diacetate; (-)3'-acetyl-CBD monoacetate; and $\pm$ AF11, all of which are disclosed in Consroe et al., J. Clin. Pharmacol. 21:428S-436S, 1981, which is also incorporated by reference. Many other cannabinoids are similarly disclosed in Agurell et al., Pharmacol. Rev. 38:31-43, 1986, which is also incorporated by reference.

As claimed herein, the term "cannabinoid" may also be generically applied to describe all cannabinoids, short-chain fatty acid phenolic compounds, endocannabinoids, phytocannabinoids, as well as terpenes that have affinity for one or more FABP-carrier proteins, or their homologs as generally described herein. Moreover, as used herein, the term "solubilized cannabinoid" describes a "cannabinoid," that binds to or interacts with one or more FABP-carrier proteins, or their homologs as generally described herein.

Examples of cannabinoids are tetrahydrocannabinol, cannabidiol, cannabigerol, cannabichromene, cannabicyclol, cannabivarin, cannabielsoin, cannabicitran, cannabigerolic acid, cannabigerolic acid monomethylether, cannabigerol monomethylether, cannabigerovarinic acid, cannabigerovarin, cannabichromenic acid, cannabichromevarinic acid, cannabichromevarin, cannabidolic acid, cannabidiol monomethylether, cannabidiol-C4, cannabidivarinic acid, cannabidiolcol, delta-9-tetrahydrocannabinolic acid A, delta-9-tetrahydrocannabinolic acid B, delta-9-tetrahydrocannabinolic acid-C4, delta-9-tetrahydrocannabivarinic acid, delta-9-tetrahydrocannabivarin, delta-9-tetrahydrocannabiorcolic acid, delta-9-tetrahydrocannabiorcol, delta-7-cis-iso-tetrahydrocannabivarin, delta-8-tetrahydrocannabinolic acid, delta-8-tetrahydrocannabinol, cannabicyclolic acid, cannabicyclovarin, cannabielsoic acid A, cannabielsoic acid B, cannabiniolic acid, cannabinol methylether, cannabinol-C4, cannabinol-C2, cannabiorcol, 10-ethoxy-9-hydroxy-delta-6a-tetrahydrocannabinol, 8,9-dihydroxy-delta-6a-tetrahydrocannabinol, cannabitolvarin, ethoxy-cannabitolvarin, dehydrocannabifuran, cannabifuran, cannabichromanon, cannabicitran, 10-oxo-delta-6a-tetrahydrocannabinol, delta-9-cis-tetrahydrocannabinol, 3,4,5,6-tetrahydro-7-hydroxy-alpha-alpha-2-trimethyl-9-n-propyl-2,6-methano-2H-1-benzoxocin-5-methanol-cannabiripsol, trihydroxy-delta-9-tetrahydrocannabinol, and cannabinol. Examples of cannabinoids within the context of this disclosure include tetrahydrocannabinol and cannabidiol.

The term "endocannabinoid" refers to compounds including arachidonoyl ethanolamide (anandamide, AEA), 2-arachidonoyl ethanolamide (2-AG), 1-arachidonoyl ethanolamide (1-AG), and docosahexaenoyl ethanolamide (DHEA, synaptamide), oleoyl ethanolamide (OEA), eicapentaenoyl ethanolamide, prostaglandin ethanolamide, docosahexaenoyl ethanolamide, linolenoyl ethanolamide, 5(Z),8(Z),11(Z)-eicosatrienoic acid ethanolamide (mead acid ethanolamide), heptadecanoyl ethanolamide, stearoyl ethanolamide, docosaenoyl ethanolamide, nervonoyl ethanolamide, tricosanoyl ethanolamide, lignoceroyl ethanolamide, myristoyl ethanolamide, pentadecanoyl ethanolamide, palmitoleoyl ethanolamide, docosahexaenoic acid (DHA). Particularly preferred endocannabinoids are AEA, 2-AG, 1-AG, and DHEA.

Terpenoids a.k.a. isoprenoids, are a large and diverse class of naturally occurring organic chemicals similar to terpenes, derived from five-carbon isoprene units assembled and modified in a number of varying configurations. Most are multi-cyclic structures that differ from one another not only in functional groups but also in their basic carbon skeletons. Terpenoids are essential for plant metabolism, influencing general development, herbivory defense, pollination and stress response. These compounds have been extensively used as flavoring and scenting agents in cosmetics, detergents, food and pharmaceutical products. They also display multiple biological activities in humans, such as anti-inflammatory, anti-microbial, antifungal and antiviral.

Cannabis terpene profiles define the aroma of each plant and share the same precursor (geranyl pyrophosphate) and the same synthesis location (glandular trichomes) as phytocannabinoids. The terpenoids most commonly found in *Cannabis* extracts include: limonene, myrcene, alpha-pinene, linalool, beta-caryophyllene, caryophyllene oxide, nerolidol, and phytol. Terpenoids are mainly synthesized in two metabolic pathways: mevalonic acid pathway (a.k.a. HMG-CoA reductase pathway, which takes place in the cytosol) and MEP/DOXP pathway (a.k.a. The 2-C-methyl-D-erythritol 4-phosphate/1-deoxy-D-xylulose 5-phosphate pathway, non-mevalonate pathway, or mevalonic acid-independent pathway, which takes place in plastids). Geranyl pyrophosphate (GPP), which is used by *Cannabis* plants to produce cannabinoids, is formed by condensation of dimethylallyl pyrophosphate (DMAPP) and isopentenyl pyrophosphate (IPP) via the catalysis of GPP synthase. Alternatively, DMAPP and IPP are ligated by FPP synthase to produce farnesyl pyrophosphate (FPP), which can be used to produce sesquiterpenoids. Geranyl pyrophosphate (GPP) can also be converted into monoterpenoids by limonene synthase.

Some examples of terpenes, and their classification, are as follows. Hemiterpenes: Examples of hemiterpenes, which do not necessarily have an odor, are 2-methyl-1,3-butadiene, hemialboside, and hymenoside. Monoterpenes: pinene, a-pinene, β -pinene, cis-pinane, trans-pinane, cis-pinanol, trans-pinanol (Erman and Kane (2008) Chem. Biodivers. 5:910-919), limonene; linalool; myrcene; eucalyptol; a-phellandrene; β -phellandrene; a-ocimene; β -ocimene, cis-ocimene, ocimene, Δ -3-carene; fenchol; sabinene, borneol, isoborneol, camphene, camphor, phellandrene, a-phellandrene, a-terpinene, geraniol, linalool, nerol, menthol, myrcene, terpinolene, a-terpinolene, β -terpinolene, γ -terpinolene, Δ -terpinolene, α -terpineol, and trans-2-pinanol. Sesquiterpenes: caryophyllene, caryophyllene oxide, humulene, a-humulene, a-bisabolene; β -bisabolene; santalol; selinene; nerolidol, bisabolol; a-cedrene, β -cedrene, β -eudesmol, eudesm-7(II)-en-4-ol, selina-3,7(II)-diene, guaial, valencene, a-guaiene, β -guaiene, Δ -guaiene, guaiene, farnesene, a-farnesene, β -farnesene, elemene, a-elemene, β -elemene, γ -elemene, Δ -elemene, germacrene, germacrene A, germacrene B, germacrene C, germacrene D, and germacrene E. Diterpenes: oridonin, phytol, and isophytol. Triterpenes: ursolic acid, oleanolic acid. Terpenoids, also known as isoprenoids, are a large and diverse class of naturally occurring organic chemicals similar to terpenes, derived from five-carbon isoprene units assembled and modified in a number of ways. Most are multicyclic structures that differ from one another not only in functional groups but also in their basic carbon skeletons. Plant terpenoids are used extensively for their aromatic qualities.

The term "prodrug" refers to a precursor of a biologically active pharmaceutical agent (drug). Prodrugs must undergo

a chemical or a metabolic conversion to become a biologically active pharmaceutical agent. A prodrug can be converted ex vivo to the biologically active pharmaceutical agent by chemical transformative processes. In vivo, a prodrug is converted to the biologically active pharmaceutical agent by the action of a metabolic process, an enzymatic process, or a degradative process that removes the prodrug moiety to form the biologically active pharmaceutical agent.

As used herein, the term “transformation” refers to the transfer of one or more nucleic acid molecule(s) into a cell. A plant is “transformed” or “genetically modified” by a nucleic acid molecule transduced into the plant when the nucleic acid molecule becomes stably replicated by the plant. As used herein, the term “transformation” or “genetically modified” encompasses all techniques by which a nucleic acid molecule can be introduced into an organism or cell. A protein has “homology” or is “homologous” to a second protein if the amino acid sequence encoded by a gene has a similar amino acid sequence to that of the second gene. Alternatively, a protein has homology to a second protein if the two proteins have “similar” amino acid sequences. (Thus, the term “homologous proteins” is defined to mean that the two proteins have similar amino acid sequences). More specifically, in certain embodiments, the term “homologous” with regard to a contiguous nucleic acid sequence, refers to contiguous nucleotide sequences that hybridize under appropriate conditions to the reference nucleic acid sequence. For example, homologous sequences may have from about 75%-100, or more generally 80% to 100% sequence identity, such as about 81%; about 82%; about 83%; about 84%; about 85%; about 86%; about 87%; about 88%; about 89%; about 90%; about 91%; about 92%; about 93%; about 94%; about 95%; about 96%; about 97%; about 98%; about 98.5%; about 99%; about 99.5%; and about 100%. The property of substantial homology is closely related to specific hybridization. For example, a nucleic acid molecule is specifically hybridizable when there is a sufficient degree of complementarity to avoid non-specific binding of the nucleic acid to non-target sequences under conditions where specific binding is desired, for example, under stringent hybridization conditions. For example, in one embodiment, SEQ ID NO. 1 is 84% homologous to *Homo sapiens* FABP1 identified as SEQ ID NO. 21), and would fall within the range of a homolog. In another embodiment, expression optimization, for example for a mammalian FABP, to be expressed in yeast may be considered homologous and having a variable sequence identity due to the variable codon positions. Additional embodiments may also include homology to include redundant nucleotide codons.

The term “homolog”, used with respect to an original enzyme or gene of a first family or species, refers to distinct enzymes or genes of a second family or species which are determined by functional, structural or genomic analyses to be an enzyme or gene of the second family or species which corresponds to the original enzyme or gene of the first family or species. Most often, homologs will have functional, structural or genomic similarities. Techniques are known by which homologs of an enzyme or gene can readily be cloned using genetic probes and PCR. Identity of cloned sequences as homolog can be confirmed using functional assays and/or by genomic mapping of the genes.

The term “operably linked,” when used in reference to a regulatory sequence and a coding sequence, means that the regulatory sequence affects the expression of the linked coding sequence. “Regulatory sequences,” or “control elements,” refer to nucleotide sequences that influence the

timing and level/amount of transcription, RNA processing or stability, or translation of the associated coding sequence. Regulatory sequences may include promoters; translation leader sequences; introns; enhancers; stem-loop structures; repressor binding sequences; termination sequences; polyadenylation recognition sequences; etc. Particular regulatory sequences may be located upstream and/or downstream of a coding sequence operably linked thereto. Also, particular regulatory sequences operably linked to a coding sequence may be located on the associated complementary strand of a double-stranded nucleic acid molecule.

As used herein, the term “promoter” refers to a region of DNA that may be upstream from the start of transcription, and that may be involved in recognition and binding of RNA polymerase and other proteins to initiate transcription. A promoter may be operably linked to a coding sequence for expression in a cell, or a promoter may be operably linked to a nucleotide sequence encoding a signal sequence which may be operably linked to a coding sequence for expression in a cell. An “inducible” promoter may be a promoter which may be under environmental control. Tissue-specific, tissue-preferred, cell type specific, and inducible promoters constitute the class of “non-constitutive” promoters. A “constitutive” promoter is a promoter which may be active under most environmental conditions or in most cell or tissue types.

The term “vector” refers to some means by which DNA, RNA, a protein, or polypeptide can be introduced into a host. The polynucleotides, protein, and polypeptide which are to be introduced into a host can be therapeutic or prophylactic in nature; can encode or be an antigen; or can be regulatory in nature, etc. There are various types of vectors including virus, plasmid, bacteriophages, cosmids, and bacteria.

As is known in the art, different organisms preferentially utilize different codons for generating polypeptides. Such “codon usage” preferences may be used in the design of nucleic acid molecules encoding the proteins of the invention in order to optimize expression in a particular host cell system.

An “expression vector” is nucleic acid capable of replicating in a selected host cell or organism. An expression vector can replicate as an autonomous structure, or alternatively can integrate, in whole or in part, into the host cell chromosomes or the nucleic acids of an organelle, or it is used as a shuttle for delivering foreign DNA to cells, and thus replicate along with the host cell genome. Thus, an expression vector are polynucleotides capable of replicating in a selected host cell, organelle, or organism, e.g., a plasmid, virus, artificial chromosome, nucleic acid fragment, and for which certain genes on the expression vector (including genes of interest) are transcribed and translated into a polypeptide or protein within the cell, organelle or organism; or any suitable construct known in the art, which comprises an “expression cassette.” In contrast, as described in the examples herein, a “cassette” is a polynucleotide containing a section of an expression vector of this invention. The use of a cassette assists in the assembly of the expression vectors. An expression vector is a replicon, such as plasmid, phage, virus, chimeric virus, or cosmid, and which contains the desired polynucleotide sequence operably linked to the expression control sequence(s).

A polynucleotide sequence is operably linked to an expression control sequence(s) (e.g., a promoter and, optionally, an enhancer) when the expression control sequence controls and regulates the transcription and/or translation of that polynucleotide sequence.

Unless otherwise indicated, a particular nucleic acid sequence also implicitly encompasses conservatively modified variants thereof (e.g., degenerate codon substitutions), the complementary (or complement) sequence, and the reverse complement sequence, as well as the sequence explicitly indicated. Specifically, degenerate codon substitutions may be achieved by generating sequences in which the third position of one or more selected (or all) codons is substituted with mixed-base and/or deoxyinosine residues (see e.g., Batzer et al., *Nucleic Acid Res.* 19:5081 (1991); Ohtsuka et al., *J. Biol. Chem.* 260:2605-2608 (1985); and Rossolini et al., *Mol. Cell. Probes* 8:91-98 (1994)). Because of the degeneracy of nucleic acid codons, one can use various different polynucleotides to encode identical polypeptides. The Table below, contains information about which nucleic acid codons encode which amino acids.

Amino Acid Nucleic Acid Codons

Amino Acid	Nucleic Acid Codons
Ala/A	GCT, GCC, GCA, GCG
Arg/R	CGT, CGC, CGA, CGG, AGA, AGG
Asn/N	AAT, AAC
Asp/D	GAT, GAC
Cys/C	TGT, TGC
Gln/Q	CAA, CAG
Glu/E	GAA, GAG
Gly/G	GGT, GGC, GGA, GGG
His/H	CAT, CAC
Ile/I	ATT, ATC, ATA
Leu/L	TTA, TTG, CTT, CTC, CTA, CTG
Lys/K	AAA, AAG
Met/M	ATG
Phe/F	TTT, TTC
Pro/P	CCT, CCC, CCA, CCG
Ser/S	TCT, TCC, TCA, TCG, AGT, AGC
Thr/T	ACT, ACC, ACA, ACG
Trp/W	TGG
Tyr/Y	TAT, TAC
Val/V	GTT, GTC, GTA, GTG

Moreover, because the proteins are described herein, one can chemically synthesize a polynucleotide which encodes these polypeptides described herein. Oligonucleotides and polynucleotides that are not commercially available can be chemically synthesized e.g., according to the solid phase phosphoramidite triester method first described by Beaucage and Caruthers, *Tetrahedron Letts.* 22:1859-1862 (1981), or using an automated synthesizer, as described in Van Devanter et al., *Nucleic Acids Res.* 12:6159-6168 (1984). Other methods for synthesizing oligonucleotides and polynucleotides are known in the art. Purification of oligonucleotides is by either native acrylamide gel electrophoresis or by anion-exchange HPLC as described in Pearson & Reanier, *J. Chrom.* 255:137-149 (1983).

The term "plant" or "plant system" includes whole plants, plant organs, progeny of whole plants or plant organs, embryos, somatic embryos, embryo-like structures, protocorms, protocorm-like bodies (PLBs), and culture and/or suspensions of plant cells. Plant organs comprise, e.g., shoot vegetative organs/structures (e.g., leaves, stems and tubers), roots, flowers and floral organs/structures (e.g., bracts, sepals, petals, stamens, carpels, anthers and ovules), seed (including embryo, endosperm, and seed coat) and fruit (the mature ovary), plant tissue (e.g., vascular tissue, ground tissue, and the like) and cells (e.g., guard cells, egg cells, trichomes and the like). The invention may also include Cannabaceae and other *Cannabis* strains, such as *C. sativa* generally.

The term "expression," as used herein, or "expression of a coding sequence" (for example, a gene or a transgene) refer to the process by which the coded information of a nucleic acid transcriptional unit (including, e.g., genomic DNA or cDNA) is converted into an operational, non-operational, or structural part of a cell, often including the synthesis of a protein. Gene expression can be influenced by external signals; for example, exposure of a cell, tissue, or organism to an agent that increases or decreases gene expression. Expression of a gene can also be regulated anywhere in the pathway from DNA to RNA to protein. Regulation of gene expression occurs, for example, through controls acting on transcription, translation, RNA transport and processing, degradation of intermediary molecules such as mRNA, or through activation, inactivation, compartmentalization, or degradation of specific protein molecules after they have been made, or by combinations thereof. Gene expression can be measured at the RNA level or the protein level by any method known in the art, including, without limitation, Northern blot, RT-PCR, Western blot, or in vitro, in situ, or in vivo protein activity assay(s).

The term "nucleic acid" or "nucleic acid molecules" include single- and double-stranded forms of DNA; single-stranded forms of RNA; and double-stranded forms of RNA (dsRNA). The term "nucleotide sequence" or "nucleic acid sequence" refers to both the sense and antisense strands of a nucleic acid as either individual single strands or in the duplex. The term "ribonucleic acid" (RNA) is inclusive of iRNA (inhibitory RNA), dsRNA (double stranded RNA), siRNA (small interfering RNA), mRNA (messenger RNA), miRNA (micro-RNA), hpRNA (hairpin RNA), tRNA (transfer RNA), whether charged or discharged with a corresponding acetylated amino acid), and cRNA (complementary RNA). The term "deoxyribonucleic acid" (DNA) is inclusive of cDNA, genomic DNA, and DNA-RNA hybrids. The terms "nucleic acid segment" and "nucleotide sequence segment," or more generally "segment," will be understood by those in the art as a functional term that includes both genomic sequences, ribosomal RNA sequences, transfer RNA sequences, messenger RNA sequences, operon sequences, and smaller engineered nucleotide sequences that encoded or may be adapted to encode, peptides, polypeptides, or proteins.

The term "gene" or "sequence" refers to a coding region operably joined to appropriate regulatory sequences capable of regulating the expression of the gene product (e.g., a polypeptide or a functional RNA) in some manner. A gene includes untranslated regulatory regions of DNA (e.g., promoters, enhancers, repressors, etc.) preceding (up-stream) and following (down-stream) the coding region (open reading frame, ORF) as well as, where applicable, intervening sequences (i.e., introns) between individual coding regions (i.e., exons). The term "structural gene" as used herein is intended to mean a DNA sequence that is transcribed into mRNA which is then translated into a sequence of amino acids characteristic of a specific polypeptide. It should be noted that any reference to a SEQ ID or sequence specifically encompasses that sequence, as well as all corresponding sequences that correspond to that first sequence. For example, for any amino acid sequence identified, the specific specifically includes all compatible nucleotide (DNA and RNA) sequences that give rise to that amino acid sequence or protein, and vice versa.

A nucleic acid molecule may include either or both naturally occurring and modified nucleotides linked together by naturally occurring and/or non-naturally occurring nucleotide linkages. Nucleic acid molecules may be modi-

fied chemically or biochemically, or may contain non-natural or derivative nucleotide bases, as will be readily appreciated by those of skill in the art. Such modifications include, for example, labels, methylation, substitution of one or more of the naturally occurring nucleotides with an analog, internucleotide modifications (e.g., uncharged linkages: for example, methyl phosphonates, phosphotriesters, phosphoramidates, carbamates, etc.; charged linkages: for example, phosphorothioates, phosphorodithioates, etc.; pendent moieties: for example, peptides; intercalators: for example, acridine, psoralen, etc.; chelators; alkylators; and modified linkages: for example, alpha anomeric nucleic acids, etc.). The term “nucleic acid molecule” also includes any topological conformation, including single-stranded, double-stranded, partially duplexed, triplexed, hair-pinned, circular, and padlocked conformations.

As used herein with respect to DNA, the term “coding sequence,” “structural nucleotide sequence,” or “structural nucleic acid molecule” refers to a nucleotide sequence that is ultimately translated into a polypeptide, via transcription and mRNA, when placed under the control of appropriate regulatory sequences. With respect to RNA, the term “coding sequence” refers to a nucleotide sequence that is translated into a peptide, polypeptide, or protein. The boundaries of a coding sequence are determined by a translation start codon at the 5'-terminus and a translation stop codon at the 3'-terminus. Coding sequences include, but are not limited to: genomic DNA; cDNA; EST; and recombinant nucleotide sequences.

The term “sequence identity” or “identity,” as used herein in the context of two nucleic acid or polypeptide sequences, refers to the residues in the two sequences that are the same when aligned for maximum correspondence over a specified comparison window.

The term “recombinant” when used with reference, e.g., to a cell, or nucleic acid, protein, or vector, indicates that the cell, organism, nucleic acid, protein, or vector has been modified by the introduction of a heterologous nucleic acid or protein, or the alteration of a native nucleic acid or protein, or that the cell is derived from a cell so modified. Thus, for example, recombinant cells may express genes that are not found within the native (nonrecombinant or wild-type) form of the cell or express native genes that are otherwise abnormally expressed—over-expressed, under expressed, or not expressed at all.

The terms “approximately” and “about” refer to a quantity, level, value, or amount that varies by as much as 30%, or in another embodiment by as much as 20%, and in a third embodiment by as much as 10% to a reference quantity, level, value or amount. As used herein, the singular form “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

As used herein, “heterologous” or “exogenous” in reference to a nucleic acid is a nucleic acid that originates from a foreign species, or is synthetically designed, or, if from the same species, is substantially modified from its native form in composition and/or genomic locus by deliberate human intervention. A heterologous protein may originate from a foreign species or, if from the same species, is substantially modified from its original form by deliberate human intervention. By “host cell” is meant a cell which contains an introduced nucleic acid construct and supports the replication and/or expression of the construct.

The invention now being generally described will be more readily understood by reference to the following examples, which are included merely for the purposes of illustration of certain aspects of the embodiments of the present invention.

The examples are not intended to limit the invention, as one of skill in the art would recognize from the above teachings and the following examples that other techniques and methods can satisfy the claims and can be employed without departing from the scope of the claimed invention. Indeed, while this invention has been particularly shown and described with references to preferred embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the invention encompassed by the appended claims.

EXAMPLES

Example 1: Genetic Modification of FABP and Predictive Binding Affinity Determination Utilizing in Silico Directed Evolution

In one embodiment the present invention demonstrated an increase water solubility, transport, and storage stability of cannabinoids, terpenoids, and volatiles through the use of genetically modified, or engineered fatty acid-binding proteins (FABPs). The present inventors selected FABP isoforms 1, 3, 5, and 7 as an initial set of exemplary targets as these have been experimentally shown to bind the cannabinoids THC and CBD. The present inventors generated crystal structures for the human versions of these four FABP isoforms, while corresponding homology models were built for various mammalian versions (cow, pig, sheep, horse, rabbit, and chicken) as well as for a plant-based FABP identified from a rainforest tree named *Rhodamnia argentea*. Computational docking of CBD was then combined with in silico directed evolution (via random mutagenesis) to identify FABP variants with enhanced binding affinity to CBD as an exemplary cannabinoids. FABP variants were screened using two separate scoring schemes. Variants selected for experimental testing comprised those with the best predicted binding affinities based on wither score. As generally shown in Table 1, exemplary top predicted CBD-binding variants were identified for chicken FABP7 (SEQ ID NOs. 74, and 79-80), pig FABP1 (SEQ ID NOs. 75, and 81-82), horse FABP1 (SEQ ID NOs. 76, and 83-84) and plant FABP (SEQ ID NOs. 77, and 85-86). The top scoring variants using either scoring scheme were selected, leading to 8 engineered FABP sequences provided for experimental testing.

Example 2: FABP Binding to Cannabinoids by ANS Displacement

As noted above, FABPs with higher predicted binding affinity to cannabinoids were selected for overexpression, purification and binding assays. In addition, select FABPs were modified to enhance binding affinity to cannabinoids. Binding of cannabinoids to FABPs was measured by ANS displacement. Arachidonic acid was used as a positive control. The decrease in fluorescence compared to protein without ligand provides a measure of binding of the ligand, for example one or more cannabinoid such as CBD or THC to the target protein. As shown generally in FIGS. 4-6, representative spectral images for binding experiments conducted by the present inventors with the selected FABPs are demonstrated. As specifically noted in FIG. 7, the present inventors demonstrated the quantified binding results as changes in fluorescence from protein without ligand.

As shown in FIG. 4A, no, or negligible binding to cannabinoids, which in this exemplary embodiment included CBD and THC, was observed with the wild type

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target FABP identified from chicken (SEQ ID NO. 74). Notably, the present inventor demonstrated that the genetically modified target FABP identified from chicken (SEQ ID NO. 79) demonstrated cannabinoid binding, and particularly binding to CBD (FIG. 4C). As further shown in FIG. 7, an exemplary rosetta-engineered FABP from pig (SEQ ID NO. 81) also demonstrated improved binding to both CBD and THC relative to the wild type based on relative change in fluorescence.

Example 3: Materials and Methods

Cloning, transformation and protein expression in *E. coli*: and FABPs were cloned in a bacteria expression system using a modified pET 24a(+) vector (from GenScript, FIG. 2) and transformed in BL21 (DE3) competent cells. This vector is under the control of the strong T7 promoter, and has 6xHis tag at the C-terminal of the protein sequence for purification. One colony was inoculated in 10 ml of LB and grown overnight for small scale protein expression. Next day, the culture was diluted 1:100 in LB medium and grown until OD reached 0.5. Protein expression was induced with 400 μ M of isopropyl- β -D-thio-galactoside (IPTG) for 3 hours at 30 C and with shaking at 250 rpm. After 3 hours of growth, the cells were harvested and washed with 50 mM Tris-HCl and cell pellets were stored at -80° C. for further protein purification.

Protein purification: Cell pellets of 500 ml cell culture were thawed and resuspended in 15 ml of cell lysis containing 50 mM of Tris-HCl and protease inhibitors. Cells were lysed using Ultrasonic-Homogenizer, Biologics Inc Model 3000. After sonication lysed cells were spun down at 14,000 rpm for 10 min. For soluble proteins, supernatants were used for affinity chromatography purification using TALON resin. For insoluble proteins, pellets were dissolved in the detergent-based buffer SoluLyse (Genlantis, San Diego, CA) with multiple washing steps to extract protein from inclusion bodies according to SoluLyse manufacturers. Proteins from inclusion bodies were unfolded in 9M Urea and 5 mM DTT and refolded by dilution with 50 mM Tris-HCl and 150 mM NaCl pH 8.

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Ligand binding assays-ANS binding studies: Binding assays of cannabinoids to proteins were assessed by 8-anilino-1-naphthalenesulfonic acid (ANS, Thermofisher scientific, Waltham, MA) displacement. ANS is a fluorescent probe commonly used to measure conformational changes due to ligand binding. ANS binds mostly to hydrophobic sites in the protein (Yu and Strobel, 1996; Huang et al., 2016). 2 μ M of protein was labelled with 20 μ M of ANS. 100 μ M stocks of cannabidiol (CBD), delta 9 tetrahydrocannabinol (THC) and Arachidonic acid were prepared in 10% of MeOH. Protein-ANS complex were excited at 390 nm and emission scan were recorded from 400 to 550 nm. All the experiments were done at 20 C on a FluroMax Spectrofluorometer.

TABLE 1

Predicted CBD binding affinities for the WT and engineered FABPs.			
Species and isoform	Variant	CBD binding affinity using Rosetta (REU)	CBD binding affinity using Smina (kcal/mol)
Chicken FABP7	mdl_r5_0225_ro	-31.444	-8.074
	mdl_r3_1403_sm	-29.481	-10.116
	WT	-20.864	-7.606
Pig FABP1	mdl_r5_1652_ro	-33.233	-7.516
	mdl_r4_0372_sm	-31.764	-10.065
	WT	-24.231	-7.265
Horse FABP1	mdl_r5_0954_ro	-33.131	-9.451
	mdl_r5_1179_sm	-31.288	-10.995
	WT	-27.447	-8.759
Plant FABP	mdl_r3_0887_ro_sm	-35.135	-10.536
	mdl_r4_1489_ro	-31.257	-9.026
	WT	-23.775	-9.022

SEQUENCE LISTINGS

Amino Acid

Hepatic Fatty acid-binding protein 1

Mus musculus

SEQ ID NO. 1

MNFSGKYQLQSQENFEPFMKAIGLPEDLIQKGKDIKGVSEIVHEGKKIKLTITYGPKVVRNEFT

LGEECELETMTGEKVKAVVKLEGDNKMMVTTFKGIKSVTELNQDITNTMTLGDIVYKRVSKRI

Amino Acid

Hepatic Fatty acid-binding protein 1

Bos taurus

SEQ ID NO. 2

MNFSGKYQVQTQENYEAFAFMKAVGMPDDIIQKGKDIKGVSEIVQNGKHPKFIITAGSKVIQNEFT

LGEECEMEFMTGEKIKAVVQQEGDNKLVTTTFKGIKSVTEFNQDITVTMTKGDVVFVKRVSKRI

N-terminal secretion signal

S. cerevisiae

SEQ ID NO. 3

MRFPSIFTAVLFAASSALAAPVNTTTEDETAQIPAEAVIGYSDLEGGDFDVAVLPSFNSTNNGLL

FINTTASIAAKEEGVSLEKR

-continued

Amino Acid
Exportable Hepatic Fatty acid-binding protein 1
Mus musculus

SEQ ID NO. 4

MRFPSIFTAVLFAASSALAAPVNTTTEDETAQIPAEAVIGYSDLEGDFDVAVL PFSNSTNNGLL
FINTTIASIAAKEEGVSLEKRMNFGSKYQLQSQENFEPFMKAIGLPEDLIQKGKDIKGVSEIVH
EGKKIKLTITYGPKVVRNEFTLGEECELETMTGEKVKAVVKLEGDNKMVTTFKGIKSVTELNGD
TITNTMTLGDIVYKRVSKRI

Amino Acid
Exportable Hepatic Fatty acid-binding protein 1
Bos taurus

SEQ ID NO. 5

MRFPSIFTAVLFAASSALAAPVNTTTEDETAQIPAEAVIGYSDLEGDFDVAVL PFSNSTNNGLL
FINTTIASIAAKEEGVSLEKRMNFGSKYQVQTQENYEA FMKAVGMPDDIIQKGKDIKGVSEIVQ
NGKHFKPIITAGSKVIQNEFTLGEECEMEFMTGEKIKAVVQQEGDNKLVTTFKGIKSVTEFNGD
TVTSTMTKGDVVFKRVSKRI

Amino Acid
Catalase
Arabidopsis thaliana

SEQ ID NO. 6

MDPYKYRPASSYNSPFFTNSGAPVWNNSSMTVGPRGLIILEDYHLVEKLANFDRERIPERVV
HARGASAKGFFEVTHTDISNLTCADFLRAPGVQTPVIVRFSTVIHARGSPETLRDPRGFAVKFYT
REGNFDLVGNFPVFVIRDMKFPDIVHALKPNPKSHIQENWRILDFSSHHPESLNMFTFLFDD
IGIPQDYRHMDGSGVNTYMLINKAGKAHYVKFHWKPTCGVKSLEEDAIRLGGTNHSHATQDLY
DSIAAGNYPEWKLFIQIIDPADEKDFDPLDVTKTWPEDILPLQPVGRMVLNKNIDNFFAENE
QLAFCPAIIIVPGIHYSDKLLQTRVFSYADTQRHRLGPNYLQLPVNAPKCAHHNNHHEGFMNFM
HRDEEVNYFPSRYDQVRHAEKYPTPPAVCSGKRERCIIEKENNFKEPGERYRTFTPERQERFIQ
RWIDALSDPRITHEIRSIWISYWSQADKSLGQKLASRLNVRPSI

Amino Acid
Catalase HPII (KatE)
Escherichia coli

SEQ ID NO. 7

MSQHNEKNPHQHQSPLHDSSEAKPGMDSLAPEDGSHRPAAEPTPPGAQPTAPGSLKAPDTRNEK
LNSLEDVRKGSENYALTNTNQGVRIADDQNSLRAGSRGPTLLEDFILREKITHFDHERIPERIVH
ARGSAAHGYFPQPKSLSDITKADFLSDPNKITPVFVRFSTVQGGAGSADTVRDIRGFATKIFYTE
EGIFDLVGNNTPIFFIQDAHKFPDFVHAVKPEPHWAIPQGQSAHDTFWDYVSLQPETLHNVMWA
MSDRGIPRSYRTMEGFGIHTFRLINAEKATFVRFWKPLAGKASLVWDEAQKLTGRDPDFHRR
ELWEAIEAGDFPEYELGFLIPEEDEFKFDPLDPTKLIPEELVPVQVRGKMVLNRNPDNFFA
ENEQAAPHGPHIVPGLDFTNDPLLGRLFSYTDQISRLGGPNFHEIPINRPTCPYHNFQRDGM
HRMGIDTNPANYEPNSINDNWPRETPPGPKRGGFESYQERVEGNKVRERSPSFGYYSHPRLFW
LSQTPFEQRHIVDGFSELSKVVRPYIRERVVDQLAHIDLTLAQAVAKNLGIELTDDQLNITPP
PDVNLGKKDPSLSLYAIPDGDVKGRVVAILLNDEVRSADLLAILKALKAKGVHAKLLYSRMGEV
TADDGTVLPIAATFAGAPSLTVDAIVPCGNIADIADNGDANYYLMEAYKHLKPIALAGDARKF
KATIKIADQGEEGIVEADSADGSFMDLLTLMAHRVWSRIPKIDKIPA

Amino Acid
Catalase 1
Arabidopsis thaliana

SEQ ID NO. 8

MDPYRVRPSSAHSPPFFTNSGAPVWNNSSSLTVGTRGPILLEDYHLLKLANFDRERIPERVV
HARGASAKGFFEVTHTDITQLTSADFLRGPGVQTPVIVRFSTVIHERGSPETLRDPRGFAVKFYT
REGNFDLVGNFPVFVVRDMKFPDMVHALKPNPKSHIQENWRILDFSSHHPESLHMFSLFDD

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LGIPQDYRHMEGAGVNTYMLINKAGKAHYVKFHWKPTCGIKCLSDEEAIKRVGGANHSHATKDLV
DSIAAGNYPQWNLFVQVMDPAHEDKFDPLDVTKIWPEDILPLQPVGRVLNKNIDNFFNENE
QIAFCPALVVPGIHYSDDKLLQTRIFSYADSRHRLGPNYLQLPVNAPKCAHHNNHHDGFMNFM
HRDEEVNYPFSRLDPVRHAEKYPTTPIVCSGNREKCFIKENNFQKPGERYRSWSDRQERFVK
RFVEALSEPRVTHEIRSIWISYWSQADKSLGQKLATRLNVRPNF

Amino Acid
Catalase 2
Arabidopsis thaliana

SEQ ID NO. 9

MDPYKYRPASSYNSPFFTNSGAPVWNNNSMTVGPRGPILLEHYHLVEKLANFDRERIPERVV
HARGASAKGFFEVTHTDISNLTADFLRAPGVQTPVIVRFSTVIHERGSPETLRDPRGFAVKFYT
REGNFDLVGNFPVFFIIRDGMKFPDMVHALKPNPKSHIQENWRILDDFFSHHPESLNMFTFLFDD
IGIPQDYRHMDSGSGVNTYMLINKAGKAHYVKFHWKPTCGVKSLEEDAIRVGGTNHSHATQDLV
DSIAAGNYPEWKLFIQIIDPADEKFDPLDVTKTWPEDILPLQPVGRMVLNKNIDNFFAENE
QLAFCPAIIVPGIHYSDDKLLQTRVFSYADTQRHRLGPNYLQLPVNAPKCAHHNNHHEGFMNFM
HRDEEVNYPFSRYDQVRHAEKYPTTPAVCSGKRERCIIEKENNFKEPGERYRTFTPERQERFIQ
RWIDALSDPRITHEIRSIWISYWSQADKSLGQKLASRLNVRPSI

Amino Acid
Catalase 3
Arabidopsis thaliana

SEQ ID NO. 10

MDPYKYRPSSAYNAPFYTTNGGAPVSNNISLTIGERGVLLEDYHLIEKVANFTRERIPERVV
HARGISAKGFFEVTHTDISNLTADFLRAPGVQTPVIVRFSTVVHERASPETMRDIRGFAVKFYT
REGNFDLVGNNTPVFFIIRDDGIQFPDVVHALKPNPKTNIQEYWRILDYMSHLPESLLTWCWMFDD
VGIPQDYRHMEGFGVHTYTLIAKSGKVLVVKFHWKPTCGIKNLTDEEAKVVGANHSHATKDLH
DAIASGNYPEWKLFIQITMDPADEKFDPLDVTKIWPEDILPLQPVGRVLNRTIDNFFNETE
QLAFNPGLVVPGIYSDDKLLQCRIFAYGDTQRHRLGPNYLQLPVNAPKCAHHNNHHEGFMNFM
HRDEEINYPSKFPDVRCAEKVPTPTNSYTGIRTKCVIKKENNFQAGDRYRSWAPDRQDRFVK
RWVEILSEPRLTHERGIWISYWSQADKSLGQKLASRLNVRPSI

Amino Acid
THCA Synthase Trichome targeting domain
Cannabis

SEQ ID NO. 11

MNCSAFSFWFVCKIIFFFLSFHIQISIA

Amino Acid
CBDA Synthase Trichome targeting domain
Cannabis

SEQ ID NO. 12

MKCSTFSFWFVCKIIFFFFSFNIQTSIA

Amino Acid
Cytosolic targeted THCA Synthase (ctTHCAs)
Cannabis

SEQ ID NO. 13

NPRENFKCFKSHIPNNVANPKLVYTQHDQLYMSILNSTIQNLRFISDTPKPLVIVTPSNNSH
IQATILCSKKVGLQIRTRSGGHAEGMSYISQVPFVVVDLRNMHSIKIDVHSQTAWVEAGATLG
EVYYWINEKNENLSFPGGYCPTVGVGGHFSGGGYGALMRNYGLAADNII DAHLVNVDGKVLDRK
SMGEDLFWAIRGGGGENFGIIAAWKIKLVDVPSKSTIFSVKKNMEIHGLVKLFNKWQNIAYKYD
KDLVLMTHFITKNITDNHGKNKTTVHGYFSSIFHGGVDSLVDLMNKSFPPELGIKKTDCKEFSWI

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DTTIFYSGVNFNTANFKKEILLDRSAGKKTAFS IKLDYVKKPIPETAMVKILEKLYEEDVGAG
 MYVLYPYGGIMEEISESAIPPHRAGIMYELWYTASWEKQEDNEKHINWVRSVYNFTTPYVSQN
 PRLAYLNYRDLDLGKTNHASPNNYTQARIWGEKYFGKNFNRLVKVKTQVDPNNFFRNEQSIPPL
 PPHHH

DNA
 Cytosolic CBDA synthase (cytCBDAs)
Cannabis sativa

SEQ ID NO. 14

ATGAATCCTCGAGAAAACCTCCTTAAATGCTTCTCGCAATATATTTCCCAATAATGCAACAAATC
 TAAAACTCGTATACACTCAAAACAACCCATTGTATATGTCTGTCTCTAAATTCGACAATACACAA
 TCTTAGATTACCTCTGACACAACCCAAAACCACTTGTATCGTCACTCCTTCACATGTCTCT
 CATATCCAAGGCATATTCTATGCTCCAAGAAAGTTGGCTTGCAGATTGGAAGTGAAGTGGTG
 GTCATGATTCTGAGGGCATGTCTTACATATCTCAAGTCCCATTTGTTATAGTAGACTTGAGAAA
 CATGCGTTCAATCAAAATAGATGTTTCATAGCCAACTGCATGGGTTGAAGCCGGAGCTACCCCTT
 GGAGAAGTTTATTATTGGGTAAATGAGAAAAATGAGAATCTTAGTTTGGCGGCTGGGTATTGCC
 CTACTGTTTGGCGAGGTGGACACTTTGGTGGAGGAGGCTATGGACCATTGATGAGAACTATGG
 CCTCGCGGCTGATAATATCATTGATGCACACTTAGTCAACGTTTCATGGAAAAGTCTAGATCGA
 AAATCTATGGGGGAAGATCTCTTTTGGGCTTTACGTGGTGGTGGAGCAGAAAGCTTCGGAATCA
 TTGTAGCATGGAATAATAGACTGGTTGCTGTCTCCAAAGTCTACTATGTTTAGTGTAAAAAGAT
 CATGGAGATACATGAGCTTGTCAAGTTAGTTAACAAATGGCAAAATATTGCTTACAAGTATGAC
 AAAGATTTATTACTCATGACTCACTTCATAACTAGGAACATTACAGATAATCAAGGGAAGAATA
 AGACAGCAATACACACTTACTTCTCTTCAGTTTTCCTTGGTGGAGTGGATAGTCTAGTCGACTT
 GATGAACAAGAGTTTCTCTGAGTTGGGTATTAAAAAACGGATTGCAGACAATTGAGCTGGATT
 GATACTATCATCTTCTATAGTGGTGTGTAATTACGACACTGATAATTTTAAACAAGGAAATTT
 TGCTTGATAGATCCGCTGGGCAGAACGGTGCTTTCAAGATTAAGTTAGACTACGTTAAGAAACC
 AATTCAGAAATCTGTATTGTCCAAATTTTGGAATAATATATGAAGAAGATATAGGAGCTGGG
 ATGTATGCGTTGTACCCCTTACGGTGGTATAATGGATGAGATTTCAGAATCAGCAATTCATTCC
 CTCATCGAGCTGGAATCTTGTATGAGTTATGGTACATATGTAGTTGGGAGAAGCAAGAAGATAA
 CGAAAAGCATCTAACTGGATTAGAAATATTTATAACTTCATGACTCCTTATGTGTCCAAAAT
 CCAAGATTGGCATACTCAATTATAGAGACCTTGATATAGGAATAAATGATCCCAAGAAATCCAA
 ATAATTACACACAAGCACGTATTGGGGTGAGAAGTATTTTGGTAAAAATTTTGACAGGCTAGT
 AAAAGTGAAAACCTGGTTGATCCCAATAACTTTTTTAGAAACGAACAAAGCATCCACCTCTA
 CCACGGCATCGTCATTAA

Amino Acid
 Cytosolic CBDA synthase (cytCBDAs)
Cannabis sativa

SEQ ID NO. 15

MNPRENFLKCFSQYIPNNATNLKLVYTQNNPLYMSVLNSTIHNLRFTSDTTPKPLVIVTPSHVS
 HIQGTILCSKKVGLQIRTRSGGHDSEGMSYISQVPFVIVDLRNMRSIKIDVHSQTAWVEAGATL
 GEVYYVWNEKNENLSLAAGYCPTVCAGGHFGGGGYGPLMRNYGLAADNI IDAHLNVNVHGKVLDR
 KSMGEDLFWALRGGAESFGIIVAWKIRLVAVPKSTMFSVKKIMEIHELVLNVNWKQNIAYKYD
 KDLLLMTHFITRNI TDNQGNKTAIHTYFSSVFLGGVDSLVDLMNKSFPPELGIKKTDRCRLSWI

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DTIIFYSGVVNYDTDNFNKEILLDRSAGQNGAFKIKLDYVKKPIPESVFVQILEKLYEEDIGAG
 MYALYPYGGIMDEISESAIPFPHRAGILYELWYICSWEKQEDNEKHLNWIRNIYNFMTPYVSKN
 PRLAYLNYRDLDIGINDPKNPNNYTQARIWGEKYFGKNFDRLVKVKTLVDPNNFFRNEQSIPPL
 PRHRH

DNA
 MYB12-like
Cannabis

SEQ ID NO. 16

ATGAAGAAGAACAATCAACTAGTAATAATAAGAACAACAACAGTAATAATATCATCAAAACG
 ACATCGTATCATCATCATCAACAACAACATCATCAACAACACAGCAACATCATCATT
 TCATAATGAGAAAGTTACTGTGACTGATCATATTATTAATCTTGATGATAAGCAGAAACGA
 CAATTATGTCGTTGTCGTTTAGAAAAAGAAGAAGAAGAAGGAAGTGGTGGTGTGGTGAGA
 CAGTAGTAATGATGCTAGGTCAGTATCTCTGCTGCTGCTACTGCTGCTGCAGCTGGGGCTC
 ATCAAGTTGTGATGAAGACATGTGGGTGGTCATGATCAACTGTTGTTGTTGTTGTTGTTGAG
 AAAAAACGACAGAAATTTATCAGTGGTGAACTTTAATAATAATAATAATAATAAGGAAA
 ATGGTGACGAAGTTTCAGGACCGTACGATTATCATCATATAAGAAGAGGAAGAAGAAGA
 AGAAGATGAAGCATCTGCATCAGTAGCAGCTGTTGATGAAGGGATGTTGTTGCTTTGATGAC
 ATAATAGATAGCCACTTGCTAAATCCAATGAGGTTTGGACTTTAAGAGAAGATAGCCATAATG
 AAGGTGGGCAGCTGATCAGATTGACAAGACTTGTGAATAACTACTATTACTACTAATGA
 TGATTATAACAATAACTTGATGATGTTGAGCTGCAATAAACGGAGATTATGTTATTAGTGAT
 GATCATGATGATCAGTACTGGATAGACGACGTCGTTGGAGTTGACTTTTGGAGTTGGGAGAGTT
 CGACTACTACTGTTATTACCCAAGAACAAGAACAAGAACAAGATCAAGTTCAAGAACAGAGAA
 TATGTGGGATAATGAGAAAGAGAACTGTTGTCTTGCTATGGGATAATAGTGATAACAGCAGC
 AGTTGGGAGTTACAAGATAAAGCAATAATAATAATAATAATGTTCTTAACAAATGTCAAG
 AGATTACCTCTGATAAAGAAATGCTATGGTTGCATGGCTTCTCTCTGA

Amino Acid
 MYB12
Cannabis

SEQ ID NO. 17

MKKNKSTSNKNNNNNNIKNDIVSSSSSTTTSTTTATSSPHNEKVTVSTDHIINLDDKQKR
 QLCRCRLEKEEEEEEGSGCGETVVMMLGVSVPAAATAAAGSSSCDEMDLGGHDQLLLLCCSE
 KKTEISSVVNFNNNNNNKENGDEVSGPYDYHHHKEEEEEEEDEASASVAAVDEGMMLCFDD
 IIDSHLLNPNEVLTLREDSHNEGGAADQIDKTCNNNTTITNDYNNLMMLSCNNNGDYVISD
 DHDDQYWIDDVVGVDVFWSWESSTTVITQEQEQEQDQVQEQKNMWDNEKEKLLSLLWDNSDNSS
 SWELQDKSNNNNNNVNPNCQEITSDKENAMVAWLLS

Amino Acid
 MYB8 - orthologue for CAN738
Humulus lupulus

SEQ ID NO. 18

MGRAPCCEKVGLKKGRWTSEEDILTKYIQSNGEGCWRSLPKNAGLLRCGKSCRLRWINYLRAD
 LKRGNISSEEDIIKLHSLGNRWSLIASHLPGRTDNEIKNYWNSHLSRKIHTFRRCNNTTTH
 HHHLPNLVTVTKVNLPIPKRKGRTSRLAMKKNKSSTSNQSSVIKNDVGSSSTTTTSVHQRT
 TTTTPTMDQQRQLSRCLREKEDQDGASTGTVVMMLGQAAAVGSSCDEMDLGGHDQLSFLCCS
 EEKTENSMTNLKENGDEHVS GPYDYDHRYEKETSVDEGMMLCFNDIIDSNNLNPNEVLTLSEE
 SLNLGGALMDTTTSTTTNNNNYLSYNNNGDCVISDDHDQYWLDDVVGVDVFWSWESSTTVTQEQ
 EQEQEQEQEQEQEQEQEHHQDQKNTWDNEKEKMLALLWDSDNSWELQDNNNYHKCQEI
 TSDKENAMVAWLLS

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Amino Acid
atMYB12 - orthologue for CAN739
Arabidopsis thaliana

SEQ ID NO. 19

MGRAPCCEKVGIKRGRWTAEDQILSNYIQSNGEGSWRSLPKNAGLKRCGKSCRLRWINYLRSD
LKRGNITPEEEELVVKLHSTLGNRWSLIAGHLPGRTDNEIKNYWNSHLRKLHNFIRKPSISQD
VSAVIMTNASSAPPPQAKRRLGRTSRSAMKPKIHRKTRKTKTSAPPEPNADVAGADKEALM
VESSGAEAEALGRPCDYDDCNKNLMSINGDNGVLTFDDDIIDLLLEDSDPGHLYTNTTCGGDG
ELHNIRDSSEGARGFSDTNQGNLDCLLQSCPSVESFLNYDHQVNDASTDEFIDWDCVWQEGSDN
NLWHEKENPDSMVSWLLDGDDEATIGNSNCFGEPLDHDDESALVAWLLS

Amino Acid
MYB112 - orthologue for CAN833
Arabidopsis thaliana

SEQ ID NO. 20

MNISRTEFANCKTLINHKEEVEEVEKKMEIEIRRGPTVEEDMKLVSYISLHGEGRWNSLSRSA
GLNRTGKSCRLRWLNLYRPDIRRGDISLQEQFIILELHSRWGNRWSKIAQHLPGRTDNEIKNYW
RTRVQKHAKLLKCDVNSKQFKDTIKHLWMPRIERIAATQSVQFTSNHYPENSSVATATSSTS
SSEAVRSSFYGGDQVEFGTLDHMTNGGYWFGGDTFETLCSFDELNKWLIQ

Amino Acid
Hepatic Fatty acid-binding protein 1
Homo Sapiens

SEQ ID NO. 21

MSFSGKYQLQSQENFEAFMKAIGLPEELIQKGKDIKGVSEIVQNGKHFKFTITAGSKVIQNEFT
VGEECELEMTGKVKTVVQLEGDNKLVTFKNIKSVTELNGDIIINTMTLGDIVFKRISKRI

DNA
Cytochrome P450 (CYP3A4)
Mus musculus

SEQ ID NO. 22

ATGAACCTGTTTCTGCTTTGTCTTTGGATACTTTGGTTTTGTTGGCTATTATTTTGGTTTTGT
TGTACAGATACGGTACTAGAACTCATGGTTTGTAAAGAAGCAAGGTATTCAGGTCCTAAAGCC
ATTGCCATTTTGGGTACTGTTTGAACACTACACTGGTATTTGGAAGTTTGATATGGAATGT
TACGAAAAGTACGGTAAGACTTGGGTTTGTGTTGATGGTCAAACTCCATTGTTGGTTATTACTG
ATCCAGAACTATTAAGAACGTTTTGGTTAAGGATTGTTGTCTGTTTTTACTAACAGAAGAGA
ATTTGGTCCAGTTGGTATTATGTCTAAGGCTATTTCTATTTCTAAGGATGAAGAATGGAAGAGA
TACAGAGCTTTGTTGTCTCCAACCTTTACTTCTGGTAGATTGAAGGAATGTTTCCAGTTATTG
AACAATACGGTGATATTTTGGTTAAGTACTTGAGACAAGAAGCTGAAAAGGGTATGCCAGTTGC
TATGAAGGATGTTTTGGGTGCTTACTCTATGGATGTTATTACTTCTACTTCTTTTGGTGTAAAC
GTTGATTCTTTGAACAACCCAGAAGATCCATTTGTTGAAGAAGCTAAGAAGTTTTTGAGAGTTG
ATTTTTTTGATCCATTGTTGTTTTCTGTTGTTTTGTTCCATTGTTGACTCCAGTTTACGAAAT
GTTGAACATTTGTATGTTTCCAAACGATTCTATTGAATTTTTTAAGAAGTTTGTGATAGAAATG
CAAGAATCTAGATTGGATTCTAACCAAAAGCATAGAGTTGATTTTTTGCAATTGATGATGAAC
CTCATAACAACCTAAGGATAAGGATTCTCATAAGGCTTTTCTAACATGGAAATTACTGTTCA
ATCTATTATTTTTATTCTGCTGGTTACGAACTACTTCTTCTACTTTGTCTTTTACTTTGTAC
TGTTTGGCTACTCATCCAGATATTCAAAAGAAGTTGCAAGCTGAAATTGATAAGGCTTTGCCAA
ACAAGGCTACTCCAACCTGTGATACTGTTATGGAAATGGAATACTTGATATGGTTTTGAACGA
AACTTTGAGATTGTACCAATTTGTTACTAGATTGGAAAGAGTTTGTAAAGAAGGATGTTGAATTG
AACGGTGTTCATATCCAAGGGTTCTATGGTTATGATTCATCTACGCTTTGCATCATGATC
CACAAATTTGGCCAGATCCAGAAGAATTTCAACCAGAAAGATTTTCTAAGGAAAACAAGGGTTC

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TATTGATCCATACGTTTACTTGCCATTTGGTATTGGTCCAAGAACTGTATTGGTATGAGATTT
 GCTTTGATGAACATGAAGTTGGCTGTTACTAAGGTTTGCAGAACTTTCTTTTCAACCATGTC
 AAGAACTCAAATTCATTGAAGTTGTCTAGACAAGGTATTTGCAACCAGAAAAGCCAATTGT
 TTTGAAGTTGTTCCAAGAGATGCTGTTATTACTGGTGCTTAA

Amino Acid
 Cytochrome P450 (CYP3A4)
Mus musculus

SEQ ID NO. 23

MNLFSAISLDTLVLLAIILVLLYRYGTRTHGLFKKQIGIPGPKLPFLGTVLNYTGIWKFDMEC
 YEKYKGTWGLFDGQTPLLIVITDPETIKNVLVKDCLSVFTNRREFGPGVIMSKAISISKDEWKR
 YRALLSPTFTSGRLKEMFPVIEQYGDILVKYLRQEAEGMPVAMKDVLGAYSMDVITSTSPGVN
 VDSLNNPEDPFVEEAKFLRVDFDPLLFVVLFPLLTPVYEMLNICMFPNDSIEFFKKFVDRM
 QESRLDSNQKHRVDFLQLMMNSHNNSKDKDSHKAFSNMEITVQSIIFISAGYETTSSTLSPTLY
 CLATHPDIQKKLQAEIDKALPNKATPTCDTVMEMEYLDMLNETLRLYPITRRLERVCKKDVEL
 NGVYIPKGSMMIPSYALHHDPOHWPDPPEEFQPERFSKENKGSIDPYVYLPFGIGPRNCIGMRF
 ALMMMKLAVTKVLQNFSPQCQETQIPLKLSRQGIQPEKPIVLKVVPRDAVITGA

DNA
 P450 oxidoreductase gene (CYP oxidoreductase)
Mus musculus

SEQ ID NO. 24

ATGGGTGATTCTCATGAAGATACTTCTGCTACTGTTCCAGAAGCTGTTGCTGAAGAAGTTTCTT
 TGTCTTCTACTACTGATATTGTTTTGTTTTCTTTGATTGTTGGTGTGTTGACTTACTGGTTTAT
 TTTTAAGAAGAAGAAGGAAGAAATTCAGAATTTCTAAGATTCAACTACTGCTCCACCAGTT
 AAGGAATCTTCTTTGTTGAAAGATGAAGAAGACTGGTAGAAACATTATTGTTTTTACGGTT
 CTCAAACCTGCTACTGCTGAAGAATTGCTAACAGATTGTCTAAGGATGCTCATAGATACGGTAT
 GAGAGGTATGCTGCTGATCCAGAAGAATACGATTTGGCTGATTTGTCTCTTTGCCAGAAATT
 GATAAGTCTTTGGTGTGTTTTTGTATGGCTACTTACGGTGAAGGTGATCCAACTGATAACGCTC
 AAGATTTTACGATTGGTTGCAAGAACTGATGTTGATTGACTGGTGTAAAGTTGCTGTTTT
 TGGTTTGGGTAACAAGACTTACGAACATTTTAACGCTATGGGTAAGTACGTTGATCAAAGATTG
 GAACAATTGGGTGCTCAAAGAATTTTGAATTGGGTTGGGTGATGATGATGGTAACCTGGAAG
 AAGATTTTATTACTTGGAGAGAACAATTTGGCCAGCTGTTGTGAATTTTGGTGTGAAGC
 TACTGGTGAAGAATCTTCTATTAGACAATACGAATTGGTTGTTTATGAAGATATGGATACTGCT
 AAGGTTTACACTGGTGAATGGGTAGATTGAAGTCTTACGAAAACCAAAGCCACCATTGATG
 CTAAGAACCATTTTTGGCTGCTGTTACTACTAACAGAAAGTTGAACCAAGGTACTGAAAGACA
 TTTGATGCATTTGGAATTGGATATTTCTGATTCTAAGATTAGATACGAATCTGGTGATCATGTT
 GCTGTTTACCCAGCTAACGATTCTACTTTGGTTAACCAAATTGGTGAAATTTGGGTGCTGATT
 TGGATGTTATTATGCTCTTGAACAACCTGGATGAAGAATCTAACAAGAAGCATCCATTTCCATG
 TCCAACTACTTACAGAAGTCTTTGACTTACTACTTGGATATTACTAACCACCAAGAAGTAAC
 GTTTGTACGAATTGGCTCAATACGCTTCTGAACCATCTGAACAAGAATTTGCATAAGATGG
 CTTCTTCTCTGGTGAAGTAAGGAATTGTACTTGTCTTGGGTTGTTGAAGCTAGAAGACATAT
 TTTGGCTATTTTGCAAGATTACCCATCTTTGAGACCACCAATTGATCATTTGTGTGAATTGTTG
 CCAAGATTGCAAGCTAGATACTACTTATTGCTTCTTCTTCTAAGGTTTCATCCAACTCTGTTC
 ATATTTGTGCTGTTGCTGTTGAATACGAAGCTAAGTCTGGTAGAGTTAACAAGGTGTTGCTAC
 TTCTTGGTTGAGAACTAAGGAACCGCTGGTGAACCGGTAGAAGAGCTTTGGTTCCAATGTTT
 GTTAGAAAGTCTCAATTTAGATTGCCATTTAAGCCAACCTACTCCAGTTATTATGGTTGGTCCAG

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GTACTGGTGTGCTCCATTATGGGTTTATTCAAGAAAGAGCTTGGTTGAGAGAACAAGGTAA
GGAAGTTGGTGAAGCTTTGTTGTACTACGGTTGTAGAAGATCTGATGAAGATTACTTGTACAGA
GAAGAATTGGCTAGATTTTCATAAGGATGGTGCTTTGACTCAATTGAACGTTGCTTTTTCTAGAG
AACAAGCTCATAAGGTTTACGTTCAACATTTGTTGAAGAGAGATAAGGAACATTTGTGGAAGTT
GATTCATGAAGTGGTGCTCATATTTACGTTTGTGGTGATGCTAGAAACATGGCTAAGGATGTT
CAAAACACTTTTTACGATATTGTTGCTGAATTGGTCCAATGGAACATACTCAAGCTGTTGATT
ACGTTAAGAAGTTGATGACTAAGGGTAGATACTCTTGGATGTTTGGTCTTAA

Amino Acid
P450 oxidoreductase (CYP oxidoreductase)
Mus musculus

SEQ ID NO. 25

MGDSHEDTSATVPEAAVEVSLFSTTDIVLFLVGLTYWFIKFKKKEIPEFSKIQTAPPV
KESFVEKMKKTGRNIIIVYGSQTGAEEFANRLSKDAHRYGMRGMSADPEEYDLADLSSSLPEI
DKSLVVFCMATYGECDPTDNAQDFYDWLQETDVLDTGVKFAVFGNGKTYEHFNAMKYVDQRL
EQLGAQRIFELGLGDDDNLEEDFITWREQFWPAVCEFFGVEATGEESSIRQYELVVHEDMDTA
KVYTGEMGRLKSYENQKPPFDKPNFLAAVTTNRKLNQGTERRHLMHLELDISDSKIRYESGDHV
AVYPANDSTLVNQIGEILGADLDVIMSLNLDDESNKKHPFCPTTYRTALTYLDITNPPRTN
VLYELAQYASEPSEQEHLHKMASSGEGKELYLSWVVEARRHILAILQDYPSLRPPIDHLCCELL
PRLQARYYSIASSSKVHPNSVHICAVAVEYEAKSGRVNKGVATSWLRTKEPAGENGRRALVPMF
VRKSQFRLPFKPTTPVIMVGPGTGVAPFMGFIQERAWLREQKEVGETLLYYGCRRSDEDYLYR
EELARFHKDALTQLNVAFSREQAHKVYVQHLLKRDKEHLWLKLIHEGGAHIYVCGDARNMAKDV
QNTFYDIVAEFGPMEHTQAVDVVKKLMTKGYSLDVWS

DNA
Cytochrome P450 (CYP3A4)
Human

SEQ ID NO. 26

ATGGCTTTGATTCTGATTGGCTATGGAACTAGATTGTTGTTGGCTGTTTCATTGGTTTTGT
TGTATTTGTATGGAACCTATTACATGGATTGTTTAAAAAATTGGGAATTCCTGGACCTACTCC
TTTGCTTTTTTTGGGAAATATTTTGTATATCATAAAGGATTTTGCATGTTTGATATGGAATGC
CATAAAAAATATGGAAGGTTTGGGGATTTTATGATGGACAACAACCTGTTTTGGCTATTACTG
ATCCTGATATGATTAAGTGTGTTTAAAGATGCTATTAGTTTTTACTAATAGAAGACC
TTTTGGACCTGTTGGATTATGAAATCAGCTATTTCAATTGCTGAAGATGAAGATGGAAGAA
TTGAGATCATTGTTGTACCTACTTTTACTTCAGGAAAATTGAAAGAAATGGTTCCTATTATTG
CTCAATATGGAGATGTTTTGGTTAGAAATTTGAGAAGAGAAGCTGAACTGGAAAACCTGTTAC
TTTGAAGATGTTTTTGGAGCTTATTCAATGGATGTATTACTTCACTTCATTGGAGTTAAT
ATTGATTCATTGAATAATCCTCAAGATCCTTTTGTGAAAATACTAAAAAATTGTTGAGATTG
ATTTTTTGGATCCTTTTTTTTGTCAATTACTGTTTTTCTTTTTTGATTCTATTTTGAAGT
TTTGAATATTTGCGTTTTTCTAGAGAAGTTACTAATTTTTTGAGAAAATCAGTTAAAAGAATG
AAAGAATCAAGATTGGAAGATACTCAAAAACATAGAGTTGATTTTTTGCAATTGATGATTGATT
CACAAAATTCAAAAGAACTGAATCACATAAAGCTTTGTCAGATTTGGAATTGGTTGCTCAATC
AATTATTTTTATTTTGTGATGCGAACTACTTCATCAGTTTTGTCAATTATTATGTATGAA
TTGGCTACTCATCCTGATGTTCAACAAAAATTGCAAGAAGAAATTGATGCTGTTTTGCCTAATA
AAGCTCCTCCTACTTATGATACTGTTTTGCAATGGAATATTTGGATATGGTTGTTAATGAAAC
TTTGAGATTGTTTCCTATTGCTATGAGATTGGAAAGAGTTTGCAAAAAGATGTTGAAATTAAT

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GGAATGTTTATTCTAAAGGAGTTGTTGTTATGATTCCTTCATATGCTTTGCATAGAGATCCTA
AATATTGGACTGAACCTGAAAAATTTTGCCTGAAAGATTTTCAAAAAAATAAGATAATAT
TGATCCTTATATTACTCCTTTTGGATCAGGACCTAGAAATTGCATTGGAATGAGATTTGCT
TTGATGAATATGAAATTGGCTTTGATTAGAGTTTGC AAAATTTTCATTTAAACCTTGCAAAG
AAACTCAAATTCCTTTGAAATTGTCATTGGGAGGATTGTTGCAACCTGAAAAACCTGTGTTTT
GAAAGTTGAATCAAGAGATGGAACGTGTTTCAGGAGCT

Amino Acid
Cytochrome P450 (CYP3A4)
Human

SEQ ID NO. 27

MALIPDLAMETRLLAVSLVLLYLYGTHSHGLFKKLGI PGPTPLPFLGNILSYHKGF CMFDMEC
HKYGVWGFYDQQPVLAITDPDMIKTVLVKECYSVFTNRRPFGVGMKSAISIAEDEEWKR
LRSLLSPTFTSGKLKEMVPIIAQYGDVLRNLRREAETGKPVTLKDVPGAYSMDVITSTSPGVN
IDSLNNPQDPFVENTKKLLRFDLDPFPLSITVFPFLIPILEVNICVFPREVTNFLRKS VKRM
KESRLEDTKQHRVDFLQLMIDSQNSKETESHKALSDLELVAQSIIFIFAGCETTSSVLSFIMYE
LATHPDVQKQLQEEIDAVLPNKAPPTYDVLQMEYLD MVVNETLRLFP IAMRLERVCKKDVEIN
GMFIPKGVVVMIPSYALHRDPKYWTEPEKFLPERFSKKNKDIDPYIYTPFGSGPRNCIGMRFA
LMNMKLALIRVLQNFSPKPKETQIPLKLSLGGLLQPEKPVVLKVESRDGT VSGA

DNA
P450 oxidoreductase gene (oxred)
Human

SEQ ID NO. 28

ATGATTAATATGGGAGATTCACATGTTGATACTTCATCAACTGTTTCAGAAGCTGTGCTGAAG
AAGTTTCATTGTTTCAATGACTGATATGATTTTGTTCATTGATTGTTGGATTGTTGACTTA
TTGGTTTTGTTTAGGAAAAAAGAGAATCCTGAATTACTAAAAATCAAACCTTGACT
TCATCAGTTAGAGAATCATCATTTGTTGAAAAATGAAAAAAGTGAAGAAATATTATTGTTT
TTTATGGATCACAACTGGAAGTCTGAAGAATTGCTAATAGATTGTCAAAGATGCTCATAG
ATATGGAATGAGAGGAATGTCAGCTGATCCTGAAGAATATGATTGGCTGATTGTCATCATTG
CCTGAAATTGATAATGCTTTGTTGTTTTTGCATGGCTACTTATGGAGAAGGAGATCCTACTG
ATAATGCTCAAGATTTTATGATTGTTGCAAGAACTGATGTTGATTGTGAGAGTTAAATT
TGCTGTTTTGATTGGGAAATAAACTTATGAACATTTAATGCTATGGGAAATATGTTGAT
AAAAGATTGGAACAATTGGGAGCTCAAAGAATTTTGAATTGGGATTGGGAGATGATGATGGAA
ATTTGGAAGAAGATTTTATTACTTGAGAGACAATTTGTTGGCTGTTTGC GAACATTTTGG
AGTTGAAGCTACTGGAGAAGATCATCAATTAGACAATATGAATTGGTTGTTTACTACTGATATT
GATGCTGCTAAAGTTTATATGGGAGAAATGGGAAGATTGAAATCATATGAAATCAAAACCTC
CTTTTGATGCTAAAAATCCTTTTTTGGCTGCTGTACTACTAATAGAAAATGAATCAAGGAAC
TGAAAGACATTTGATGCATTTGGAATTGGATATTTAGATTCAAAAAATGATATGAATCAGGA
GATCATGTTGCTGTTTATCCTGCTAATGATTCAGCTTTGGTTAATCAATTGGGAAAAATTTTGG
GAGCTGATTTGGATGTTGTTATGTCATTGAATAATTGGATGAAGAATCAAATAAAAAACATCC
TTTTCTTGCCCTACTTCATATAGAACTGCTTTGACTTATTATTTGGATATTACTAATCCTCCT
AGAACTAATGTTTGTATGAATTGGCTCAATATGCTTCAGAACCTTCAGAACAGAATTGTTGA
GAAAAATGGCTTCATCATCAGGAGAAGGAAAAGAATTGATTTGTGTCATGGGTTGTTGAAGCTAG
AAGACATATTTGGCTATTTTGCAAGATTGCCCTTCATTGAGACCTCCTATTGATCATTTGTGC
GAATTGTTGCCTAGATTGCAAGCTAGATATTATTCAATTGCTTCATCATCAAAAGTTTCATCCTA
ATTCAGTTTCATATTTGCGCTGTTGTTGTTGAATGAAACTAAAGCTGGAAGAATTAATAAAGG

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AGTTGCTACTAATTGGTTGAGAGCTAAAGAACCTGTTGGAGAAAAATGGAGGAAGAGCTTTGGTT
CCTATGTTTGTGTAGAAAATCACAATTTAGATTGCCTTTTAAAGCTACTACTCCTGTTATTATGG
TTGGACCTGGAAGCTGGAGTTGCTCCTTTTATTGGATTTATTCAAGAAAGAGCTTGGTTGAGACA
ACAAGGAAAAGAAGTTGGAGAACTTTGTTGTATTATGGATGCAGAAGATCAGATGAAGATTAT
TTGTATAGAGAAGAATTGGCTCAATTCATAGAGATGGAGCTTTGACTCAATTGAATGTTGCTT
TTTCAAGAGAACAAATCACATAAAGTTTATGTTCAACATTTGTTGAAACAAGATAGAGAACATTT
GTGGAAATTGATTGAAGGAGGAGCTCATATTTATGTTTGCAGAGATGCTAGAAATATGGCTAGA
GATGTTCAAAATACTTTTTATGATATTGTTGCTGAATTGGGAGCTATGGAACATGCTCAAGCTG
TTGATTATATTAATAAATTGATGACTAAAGGAAGATATTCATTGGATGTTTGGTCA

Amino Acid
P450 oxidoreductase
Human

SEQ ID NO. 29

MINMGDSHVDTSSTVSEAVAEVSLFSMTDMILFSLIVGLLTYWFLPRKKKEEVPEFTKIQTLT
SSVRESSFVEKMKKTGRNIIIVFYGSQTGTAEFANRLSKDAHRYGMRGMSADPEEYDLADLSSL
PEIDNALVVFCMATYGECDPTDNAQDFYDWLQETDVLDSGVKFAVFGLGNKTYEHFNAMGKYVD
KRLEQLGAQRIFELGLGDDDNLEEDFITWREQFWLAVCEHFGVEATGEESIRQYELVVHTDI
DAAKVMGEMGRKLSYENQKPPFDKPNPFLAAVTTNRKLNQTERHLMHLELDISDSKIRYESG
DHSVAVPANDSALVNQLGKILGADLDVMSLNNLDEESNKKHPPPCPTSYRTALTYLDITNPP
RTNVLYELAQYASEPSEQELLRKMASSGEGKELYLSWVVEARRHILAILQDCPSLRPPIDHLC
ELLPLRLQARYYSIASSSKVHPNSVHICAVVVEYETKAGRINKGVATNWLRAKEPVGENGGRALV
PMFVRKSQFRLPFKATTPVIMVGPGTGVAPFIGFIQERAWLRQQKEVGETLLYGCRRSDEDY
LYREELAQFHRDALTQLNVAFSREQSHKVYVQHLLKQDREHLWKLIBGGAHIYVCGDARNMAR
DVQNTFYDIVAELGAMEHAQAVDYIKKLMTKGRYSLDVWS

DNA
cannabidiolic acid (CBDA) synthase
Cannabis sativa

SEQ ID NO. 43

ATGAATCCTCGAGAAAACCTTCTTAAATGCTTCTCGCAATATATCCCAATAATGCAACAAATC
TAAAACTCGTATACACTCAAAACAACCCATTGTATATGTCTGTCTCTAAATTCGACAATACACAA
TCTTAGATTCACTCTGACACAACCCAAAACCACTTGTATCGTCACTCCTTCACATGTCTCT
CATATCCAAGGCATATTCTATGCTCCAAGAAAGTTGCTTGCAGATTGCAACTCGAAGTGGTG
GTCATGATTCTGAGGGCATGTCTACATATCTCAAGTCCCATTGTTATAGTAGACTTGAGAAA
CATGCGTTCAATCAAATAGATGTTCATAGCCAACTGCATGGGTGAAGCCGGAGCTACCCCTT
GGAGAAGTTTATTATTGGGTAAATGAGAAAAATGAGAATCTTAGTTTGGCGGCTGGGTATTGCC
CTACTGTTTGGCGAGGTGGACACTTTGGTGGAGGAGGCTATGGACCATTGATGAGAACTATGG
CCTCGCGGCTGATAATATCATTGATGCACACTTAGTCAACGTTTATGGAAGTGTAGATCGA
AAATCTATGGGGGAAGATCTCTTTTGGGCTTACGTGGTGGTGGAGCAGAAAGCTTCGAATCA
TTGTAGCATGGAAAATTAGACTGGTTGCTGTCCAAAGTCTACTATGTTTAGTGTAAAAAGAT
CATGGAGATACATGAGCTTGTCAAGTTAGTTAACAAATGGCAAAATATTGCTTACAAGTATGAC
AAAGATTTATTACTCATGACTCACTTCATAACTAGGAACATTACAGATAATCAAGGGAAGAATA
AGACAGCAATACACACTTACTTCTTTCAGTTTTCCTTGGTGGAGTGGATAGTCTAGTCGACTT
GATGAACAAGAGTTTCTCTGAGTTGGGTATTAATAAACGGATTGCAGACAATTGAGCTGGATT
GATACTATCATCTTCTATAGTGGTGTGTAATACGACACTGATAATTTTAACAAGGAAATTT

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TGCTTGATAGATCCGCTGGGCAGAACGGTGCTTTCAAGATTAAGTTAGACTACGTAAAGAAACC
 AATTCAGAATCTGTATTTGTCCAATTTTGAAAAATTATATGAAGAAGATATAGGAGCTGGG
 ATGTATGCGTGTACCCCTACGGTGGTATAATGGATGAGATTTCAGAATCAGCAATTCATTCC
 CTCATCGAGCTGGAATCTTGTATGAGTTATGGTACATATGTAGTTGGGAGAAGCAAGAAGATAA
 CGAAAAGCATCTAAACTGGATTAGAAATATTTATAACTTCATGACTCCTTATGTGTCCAAAAAT
 TCAAGATTGGCATATCTCAATTATAGAGACCTTGATATAGGAATAAATGATCCCAAGAAATCCAA
 ATAATTACACACAAGCACGTATTTGGGGTGAGAAGTATTTTGGTAAAAATTTTGACAGGCTAGT
 AAAAGTGAAACCCCTGGTTGATCCCAATAACTTTTTTAGAAACGAACAAAGCATCCACCTCAA
 CCACGGCATCGTCATTAA

Amino Acid
 Cannabidiolic acid (CBDA) synthase
Cannabis sativa

SEQ ID NO. 31

MNPRENFLKCFSQYIPNNATNLKLVYTQNNPLYMSVLNSTIHNLRFTSDTTPKPLVIVTPSHVS
 HIQGTILCSKKVGLQIRTRSGGHDSEGMSYISQVPFVIVDLRNMRSIKIDVHSQTAWVEAGATL
 GEVYYVWNEKNENLSLAAGYCTVCAGGHFGGGGYGPLMRNYGLAADNI IDAHLVNVHGKVLDR
 KSMGEDLFWALRGGGAESFGIIVAWKIRLVAVPKSTMFSVKKIMEIHELVLVKNQNIAYKYD
 KDLLLMTHFITRNTDNQGNKTAIHTYFSSVFLGGVDSLVDLMNKSFPPELGIKKTDRCQLSWI
 DTIIIFYSGVVNYDTDNFNKEILLDRSAGQNGAFKIKLDYVKKPIPESVFVQILEKLYEEDIGAG
 MYALYPYGGIMDEISESAIPFPHRAGILYELWYICSWEKQEDNEKHLNWIRNIYNFMTPYVSKN
 SRLAYLNYRDLDIGINDPKNPNNYTQARIWGEKYFGKNFDRLVKVKTLVDPNNFPRNEQSIPPQ
 PRHRH

DNA
 UDP glycosyltransferase 76G1
Stevia rebaudiana

SEQ ID NO. 32

ATGGAAAATAAACTGAACTACTGTTAGAAGAAGAAGAATTATTTGTTTCCTGTTCTCTT
 TTCAAGGACATATTAATCCTATTTTGCATTTGGCTAATGTTTGTATTCAAAGGATTTTCAAT
 TACTATTTTTCATACTAATTTTAATAAACCTAAACTTCAAATTATCCTCATTTTACTTTTAGA
 TTTATTTTGGATAATGATCCTCAAGATGAAAGAATTTCAAATTTGCCTACTCATGGACCTTTGG
 CTGGAATGAGAATTCCTATTATTAATGAACATGGAGCTGATGAATTGAGAAGAGAATTGGAATT
 GTTGATGTTGGCTTCAGAAGAAGATGAAGAAGTTTCATGCTTGATTACTGATGCTTTGTGGTAT
 TTTGCTCAATCAGTTGCTGATTCATTGAATTTGAGAAGATTGGTTTTGATGACTTCATCATTGT
 TTAATTTTCATGCTCATGTTTCATTGCCTCAATTTGATGAATTGGGATATTGGATCCTGATGA
 TAAACTAGATTGGAAGAACAGCTTCAGGATTTCCATGTTGAAAGTTAAAGATATTAAATCA
 GCTTATTCAAATTGGCAAATTTTGAAAGAAATTTTGGGAAAATGATTAAACAACTAGAGCTT
 CATCAGGAGTTATTTGGAATTCATTTAAAGAATTGGAAGAATCAGAATTGGAACTGTTATTAG
 AGAAATTCCTGCTCCTTCATTTTGATTCCCTTGCCCTAAACATTTGACTGCTTCATCATCATCA
 TTGTTGGATCATGATAGAAGCTGTTTTCAATGGTTGGATCAACAACCTCCTTCATCAGTTTGT
 ATGTTTCATTGGATCAACTTCAGAAGTTGATGAAAAAGATTTTTTGAAATTGCTAGAGGATT
 GGTTGATTCAAAACAATCATTTTGTGGGTTGTTAGACCTGGATTGTTAAAGGATCAACTTGG
 GTTGAACCTTTGCCTGATGGATTTTGGGAGAAAGAGGAAGAATTGTTAAATGGGTTCTCTCAAC
 AAGAAGTTTGTGCTCATGGAGCTATTGGAGCTTTTGGACTCATTAGGATGGAATTCAACTTT
 GGAATCAGTTTGCAGAGGAGTTCTATGATTTTTTCAGATTTTGGATTGGATCAACCTTTGAAT
 GCTAGATATATGTCAGATGTTTTGAAAGTTGGAGTTTATTTGGAAAATGGATGGGAAAGAGGAG

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AAATTGCTAATGCTATTAGAAGAGTTATGGTTGATGAAGAAGGAGAATATATTAGACAAAATGC
TAGAGTTTTGAAACAAAAAGCTGATGTTTCATTGATGAAAGGAGGATCATCATATGAATCATTG
GAATCATTGGTTTCATATATTTTCATCATTG

Amino Acid
UPD glycosyltransferase 76G1
Stevia rebaudiana

SEQ ID NO. 33

MENKTETTVRRRRRIILFPVPFQGHINPILQLANVLYSKGFSITIFHTNFNKPKTSNYPHFTFR
FILDNDPQDERISNLPTHGPLAGMRIPINEHGADELRRLELLMLASEEDEEVSLITDALWY
FAQSVADSLNLRRLVMTSSLFNFHAHVSLPQFDELGYLDPDDKTRLEEQASGFPMKVKDIKS
AYSNWQILKEILGKMIKQTRASSGVIWNSFKELESELETVIREIPAPSFLIPLPKHLTASSSS
LLDHDRTVFQWLDQQPPSSVLYVSFGSTSEVDEKDFLEIARGLVDSKQSFLWVVRPGFVKGSTW
VEPLPDGFLGERGRIVKWVPQQEVLAHGAIGAFWTHSGWNSTLESVCEGVPMIFSDFGLDQPLN
ARYMSDVLKVGVYLENGWERGEIANAIRVMVDEEGEYIRQNARVLKQKADVSLMKGGSSYESL
ESLVSYSISSL

Amino Acid
Glycosyltransferase (NtGT5a)
Nicotiana tabacum

SEQ ID NO. 34

MGSIGAECLKPHAVCIPYPAQGHINPMLKLAKILHHKGFHITFVNTEFNHRRLLKSRGPDCLKG
LSSFRFETIPDGLPPCEADATQDIPSLCESTTNTCLAPFRDLLAKLNDTNTSNVPPVSCIVSDG
VMSFTLAAQELGVPEVLFWTTSACGFLGYMHYCKVIEKGYAPLKDASDLTNGYLETTLDFIG
MKDVRRLDLPFLRTTNDEFMIKFVLQETERARKASAIILNTFETLEAEVLESRLNLLPPVYP
IGPLHFLVKHVDENLKGRLSSLWKEEPECIQWLDTKEPNSVVYVNFSGITVMTPNQLIEPAWG
LANSQQTFLWIIRPDIVSGDASILPPEFVEETKNRGLASWCSQEVLVSHPAIVGFLTHSGWNS
TLESISSGVPMICWPFPAEQQTNCWFVSVTKWDVGMEIDSDVKRDEVESLVREL MVGKGKMKKK
KAMEWKELAEASAKEHSGSSSVNIEKLVNDILLSSKH

DNA
Glycosyltransferase (NtGT5a)
Nicotiana tabacum

SEQ ID NO. 35

ATGGGTTCATTGGTGCTGAATTAACAAAGCCACATGCAGTTTGCATACCATATCCCGCCAAG
GCCATATTAACCCCATGTTAAAGCTAGCCAAAATCCTTCATCACAAGGCTTTCACATCACTTT
TGTCAACTACTGAATTTAACCCGACGTCTCCTTAAATCTCGTGCCCTGATTCTCTCAAGGGT
CTTTCTTCTTTCCGTTTTGAGACCATTCTGATGGACTTCCGCCATGTGAGGCAGATGCCACAC
AAGATATACCTTCTTGTGTGAATCTACAACCAATACTGCTTGGCTCCTTTTAGGGATCTTCT
TGCGAAACTCAATGATACTAACACATCTAACGTGCCACCCGTTTCGTGCATCGTCTCGGATGGT
GTCATGAGCTTCACCTTAGCCGCTGCACAAGAATTGGGAGTCCCTGAAGTTCTGTTTTGGACCA
CTAGTGCTTGTGGTTTCTTAGGTTACATGCATTACTGCAAGGTTATTGAAAAGGATATGCTCC
ACTTAAAGATGCGAGTGACTTGACAAATGGATACCTAGAGACAACATTGGATTTTATACCAGGC
ATGAAAGACGTACGTTTAAGGGATCTTCCAAGTTCTTGAGAACTACAAATCCAGATGAATTCA
TGATCAAATTTGCTCTCCAAGAAACAGAGAGAGCAAGAAAGGCTTCTGCAATTATCCTCAACAC
ATTTGAAACACTAGAGGCTGAAGTTCTTGATCGCTCCGAAATCTTCTCTCCAGTCTACCCC
ATAGGGCCCTTGCAATTTCTAGTGAAACATGTTGATGATGAGAATTTGAAGGGACTTAGATCCA
GCCTTGGAAAGAGGAACAGAGTGTATACAATGGCTTGATACCAAAGAACCAAATCTGTTGT
TTATGTTAACTTTGGAAGCATTACTGTTATGACTCCTAATCAGCTTATTGAGTTTGCTTGGGGA

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CTTGCAAAACAGCCAGCAAAACATTCTTATGGATCATAAGACCTGATATTGTTTCAGGTGATGCAT
CGATTCTTCCACCCGAATTCTGTGAAGAAACGAAGAACAGAGGTATGCTTGCTAGTTGGTGTTC
ACAAGAAGAAGTACTTAGTCACCCCTGCAATAGTAGGATTCTTGACTCACAGTGGATGGAATTCG
ACACTCGAAAGTATAAGCAGTGGGGTGCCTATGATTGCTGGCCATTTTCGCTGAACAGCAAA
CAAATTGTTGGTTTTCCGTCATAAATGGGATGTTGGAATGGAGATTGACAGTGTGAAGAG
AGATGAAGTGGAAAGCCTTGTAAGGGAATTGATGGTTGGGGGAAAAGGCAAAAAGATGAAGAAA
AAGGCAATGGAATGGAAGGAATTGGCTGAAGCATCTGCTAAAGAACATTACAGGGTCATCTTATG
TGAACATTGAAAAGTTGGTCAATGATATTCTTCTTTCATCCAAACATTAA

Amino Acid
Glycosyltransferase (NtGT5b)
Nicotiana tabacum

SEQ ID NO. 36

MGSIGAEFTKPHAVCIPYPAQGHINPMLKLAKILHHKGFHITFVNTEPNHRLLKSRGPDCLKG
LSSFRFETIPDGLPPCDADATQDIPSLCESTTNTCLGPFRDLLAKLNDTNTSNVPPVSCIISDG
VMSFTLAAQELGVPEVLFWTTSSACGFLGYMHYKVKIEKGYAPLKDASDLTNGYLETTLDIFPC
MKDVRRLDLPSPRLTNPDEFMIKPVLPQETERARKASAIILNTYETLEAEVLESRLNLLPPVYP
IGPLHFLVKHVDENLKLRLSSSLWKEEPECIQWLDTKEPNSVVYVNFSGITVMTPNQLIEFAWG
LANSQQSFLWIIRPDIVSGDASILPPEFVEETKKRGLASWCSQEEVLSHPAIGGFLTHSGWNS
TLESISSGVPMICWPFPAEQQTNCWFSVTKWDVGMEIDCDVKRDEVESLVRELMVGGKGMKKK
KAMEWKELAEASAKEHSGSSYVNIKVVNDILLSSKH

DNA
Glycosyltransferase (NtGT5b)
Nicotiana tabacum

SEQ ID NO. 37

ATGGGTTCCATTGGTGTGAATTTACAAGCCACATGCAGTTTGATACCATATCCCGCCCAAG
GCCATATTAACCCCATGTTAAAGCTAGCCAAAATCCTTCATCACAAAGGCTTTCACATCACTTT
TGTCAACTACTGAATTTAACCACAGACGTCTGCTTAAATCTCGTGGCCCTGATTCTCTCAAGGT
CTTTCTTTTCCGTTTTGAGACAATTCTGATGGACTTCCGCCATGTGATGCAGATGCCACAC
AAGATATACCTTCTTTGTGTGAATCTACAACCAATACTTGCTTGGGTCTTTTAGGGATCTTCT
TGCAGAACTCAATGATACTAACACATCTAACGTGCCACCCGTTTCGTGCATCATCTCAGATGGT
GTCATGAGCTTACCTTAGCCGCTGCACAAGAATTGGGAGTCCCTGAAGTTCTGTTTTGACCA
CTAGTGCTTGTGGTTTCTTAGGTTACATGCATTATTACAAGGTTATTGAAAAGGATACGCTCC
ACTTAAAGATGCGAGTGACTTGACAAATGGATACCTAGAGACAACATTGGATTTTATACCATGC
ATGAAAGACGTACGTTTAAAGGATCTTCCAAGTTTCTTGAGAACTACAAATCCAGATGAATTCA
TGATCAAATTTGTCTCTCAAGAAACAGAGAGCAAGAAAGGCTTCTGCAATTATCCTCAACAC
ATATGAAACACTAGAGGCTGAAGTTCTTGAATCGCTCCGAAATCTTCTCCTCCAGTCTACCCC
ATTGGGCCCTTGCAATTTCTAGTGAAACATGTTGATGATGAGAATTTGAAGGGACTTAGATCCA
GCCTTTGGAAAGAGGAACCAGAGTGTATACAATGGCTTGATACCAAAGAACCAAATCTGTTGT
TTATGTTAACTTTGGAAGCATTACTGTTATGACTCCTAATCAACTATTGAATTTGCTTGGGGA
CTTGCAAAACAGCCAACAATCATTCTTATGGATCATAAGACCTGATATTGTTTCAGGTGATGCAT
CGATTCTTCCCCCGAATTCTGTGAAGAAACGAAGAAGAGAGGTATGCTTGCTAGTTGGTGTTC
ACAAGAAGAAGTACTTAGTCACCCCTGCAATAGGAGGATTCTTGACTCACAGTGGATGGAATTCG
ACACTCGAAAGTATAAGCAGTGGGGTGCCTATGATTGCTGGCCATTTTCGCTGAACAGCAAA
CAAATTGTTGGTTTTCCGTCATAAATGGGATGTTGGAATGGAGATTGACTGTGATGTGAAGAG
GGATGAAGTGGAAAGCCTTGTAAGGGAATTGATGGTTGGGGGAAAAGGCAAAAAGATGAAGAAA

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AAGGCAATGGAATGGAAGGAATTGGCTGAAGCATCTGCTAAAGAACATTCAGGGTCATCTTATG
TGAACATTGAGAAGGTGGTCAATGATATTCTTCTTCGTCCAAACATTAA

Amino Acid
UDP-glycosyltransferase 73C3 (NtGT4)
Nicotiana tabacum

SEQ ID NO. 38

MATQVHKLHFILFPLMAPGHMIPMIDIAKLLANRGVITTIITTPVNANRFSSTITRAIKSGLRI
QILTLKFPSVEVGLPEGCENIDMLPSLDLASKFFAAISMLKQOVENLLEGINPSPSCVISDMGF
PWTQTIAQNFNIPRIVFHGTCFSLCSYKILSSNILENITSDSEYFVVPDLPDRVELTKAQVS
GSTKNNTTSVSSSVLKVEVTEQIRLAEESYGVIVNSFEELEQVYEKEYRKARGKKVWCVGPVSLC
NKEIEDLVTRGNKTAIDNQDCLKWLDNFETESVVYASLGSLSRLTLLQMVELGLGLEESNRPFV
WVLGGGDKLNDLEKWILENGFEQRIKERVLRGWAPQVLILSHPAIGGVLTCHGWNSTLEGIS
AGLPMVTWPLFAEQFCNEKLVVQVLKIGVSLGVKVPVKWGDEENVGVLVKKDDVKKALDKLMDE
GEEGQVRRTKAKELGELAKKAFGEGSSVNLTSIEDIEQQNHKEK

DNA
UDP-glycosyltransferase 73C3 (NtGT4)
Nicotiana tabacum

SEQ ID NO. 39

ATGGCAACTCAAGTGCACAACTTCATTCATACTATTCCCTTTAATGGCTCCAGGCCACATGA
TTCTTATGATAGACATAGCTAACTTCTAGCAAATCGCGGTGTCTATTACCACTATCATCAACCAC
TCCAGTAAACGCCAATCGTTTCAGTTCAACAATTACTCGTGCCATAAAATCCGGTCTAAGAATC
CAAATTCTTACTACTCAAATTTCCAAGTGTAGAAGTAGGATTACCAGAAGGTTGCGAAAATATTG
ACATGCTTCTCTCTTGACTTGGCTTCAAAGTTTTTGTCTGCAATTAGTATGCTGAAACAACA
AGTTGAAAATCTCTTAGAAGGAATAAATCCAAGTCCAAGTTGTGTTATTTAGATATGGGATTT
CCTTGGACTIONTCAAATTGCACAAAATTTAATATCCCAAGAATTGTTTTTCATGGTACTTGT
GTTTCTCACTTTTATGTTCTTATAAAATCTTCTCTCAACATTCTTGAAAATATAACCTCAGA
TTCAGAGTATTTGTGTTCTTGATTTACCCGATAGAGTTGAACTAACGAAAGCTCAGGTTTCA
GGATCGACGAAAAATACTACTTCTGTTAGTCTTCTGTATTGAAAGAAGTTACTGAGCAAATCA
GATTAGCCGAGGAATCATCATATGGTGTAATTGTTAATAGTTTTGAGGAGTTGGAGCAAGTGTA
TGAGAAAGAATATAGGAAAGCTAGAGGGAAAAAGTTTGGTGTGTGGTCTGTTTCTTTGTGT
AATAAGGAAATTGAAGATTGGTTACAAGGGTAATAAACTGCAATTGATAATCAAGATTGCT
TGAAATGGTTAGATAATTTTGAACAGAACTGTGGTTTATGCAAGTCTTGAAGTTTATCTCG
TTTGACATTATTGCAATGGTGGAACCTGGTCTTGGTTTAGAAGAGTCAAATAGGCCTTTTGTA
TGGGTATTAGGAGGAGGTGATAAATAAATGATTAGAGAAATGGATTCTTGAGAATGGATTG
AGCAAAGAATTAAAGAAAGAGGAGTTTTGATTAGAGGATGGGCTCCTCAAGTGCTTATACCTTC
ACACCCTGCAATTGGTGGAGTATTGACTCATTGCGGATGGAATTCTACATTGGAAGGTATTTCA
GCAGGATTACCAATGGTAACATGGCCACTATTGCTGAGCAATTTGCAATGAGAAGTTAGTAG
TCCAAGTGCTAAAAATTGGAGTGAGCCTAGGTGTGAAGTGCCGTCAAATGGGAGATGAGGA
AAATGTTGGAGTTTTTGGTAAAAAGGATGATGTTAAGAAAGCATTAGACAACTAATGGATGAA

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GGAGAAGAAGGACAAGTAAGAAGAACAAAGCAAAGAGTTAGGAGAATTGGCTAAAAAGGCAT
TTGGAGAAGGTGGTTCTTCTTATGTTAACTTAACATCTCTGATTGAAGACATCATTGAGCAACA
AAATCACAAGGAAAAATAG

Amino Acid
Glycosyltransferase (NtGT1b)
Nicotiana tabacum

SEQ ID NO. 40

MKTAELVFIPAPGMGHLVPTVEVAKQLVDRHEQLSITVLIMTIPLNETNIPSYTKSLSSDYSSRI
TLLPLSQPETSVMSSFNAINFFEYISSYKGRVKDAVSETSFSSSNSVKLAGFVIDMFCTAMID
VANEFGIPSYVFTSSAAMLGLQLHFQSLSECSKPVHNYVEPESEVLITYMNPVPVKCLPGI
ILVNDESSTMFVNHARRPRETKGIMVNTFTELESHALKALSDDEKIPPIYPVGPILNLENGNED
HNQEYDAIMKWLDEKPNSSVFLCFGSKGSFEEDQVKEIANALESSGYHFLWSLRRPPPKDKLQ
FPSEFENPEEVLPEGFFQRTKGRGKVI GWAPQLAILSHPSVGGFVSHCGWNSTLESVRSGVPIA
TWPLYAEQQSNAPQLVKDLGMAVEIKMDYREDFNTRNPPLVKAEEIEDGIRKLMDSENKIRAKV
TEMKDKSRAALLEGGSSYVALGHFVETVMKN

DNA
Glycosyltransferase (NtGT1b)
Nicotiana tabacum

SEQ ID NO. 41

ATGAAGACAGCAGAGTTAGTATTCTTCCTGCTCCTGGGATGGGTACCTTGTACCAACTGTGG
AGGTGGCAAAGCAACTAGTCGACAGACACGAGCAGCTTTCGATCACAGTTCTAATCATGACAAT
TCCTTTGGAACAAATATTCCATCATATACTAAATCACTGTCCTCAGACTACAGTTCTCGTATA
ACGCTGCTTCCACTCTCTCAACCTGAGACCTCTGTTACTATGAGCAGTTTAAATGCCATCAATT
TTTTTGAGTACATCTCCAGCTACAAGGGTCGTGTCAAAGATGCTGTTAGTGAAACCTCCTTAG
TTCGTCAAATTCTGTGAAACTTGCAGGATTTGTAATAGACATGTTCTGCACTGCGATGATTGAT
GTAGCGAACGAGTTTGGAATCCCAAGTTATGTGTTCTACACTTCTAGTGACGCTATGCTTGGAC
TACAACCTGCATTTTCAAAGTCTTAGCATTGAATGCAGTCCGAAAGTTCATAACTACGTTGAACC
TGAATCAGAAGTCTGATCTCAACTTACATGAATCCGGTTCAGTCAAATGTTTGCCCGGAATT
ATACTAGTAAATGATGAAAGTAGCACCATGTTTGTCAATCATGCACGAAGATTGAGGAGACGA
AAGGAATTATGGTGAAACACGTTCACTGAGCTTGAATCACACGCTTGAAGCCCTTTCCGATGA
TGAAAAAATCCCACCAATCTACCCAGTTGGACCTATACTTAACCTTGAAAATGGGAATGAAGAT
CACAAATCAAGAAATATGATGCGATTATGAAGTGGCTTGACGAGAAGCCTAATTCATCAGTGGTGT
TCTTATGCTTTGGAAGCAAGGGTCTTTCGAAGAAGATCAGGTGAAGGAAATAGCAAATGCTCT
AGAGAGCAGTGGCTACCACTTCTTGTGGTCGCTAAGGCGACCGCCACCAAAGACAAGCTACAA
TTCCCAAGCGAATTGAGAATCCAGAGGAAGTCTTACCAGAGGGATTCTTTCAAAGGACTAAAG
GAAGAGGAAAGGTGATAGGATGGGCACCCAGTTGGCTATTTTGTCTCATCCTTCAGTAGGAGG
ATTCTGTGCGATTGTTGGGTGAATTCAACTCTGGAGAGCGTTGAAGTGGAGTGCCGATAGCA
ACATGGCCATTGTATGCAGAGCAACAGAGCAATGCATTTCAACTGGTGAAGGATTGGGTATGG
CAGTAGAGATTAAGATGGATTACAGGGAAGATTTTAATACGAGAAATCCACCACTGGTTAAAGC

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TGAGGAGATAGAAGATGGAATTAGGAAGCTGATGGATTGAGAGAATAAAATCAGGGCTAAGGTG
ACGGAGATGAAGGACAAAAGTAGAGCAGCACTGCTGGAGGGCGGATCATCATATGTAGCTCTTG
GGCATTTTGTTGAGACTGTCATGAAAACTAG

Amino Acid
Glycosyltransferase (NtGT1a)
Nicotiana tabacum

SEQ ID NO. 42

MKTTELVPFIPAPGMGHLVPTVEVAKQLVDRDEQLSITVLIMTLPLETNIPSYTKSLSSDYSSRI
TLLQLSQPETSVMSSFNAINFFEYISSYKDRVKDAVNETFSSSSSVKLKGFVIDMFCTAMIDV
ANEFGIPSYVFFYTSNAAMLGLQLHFQSLSEYSPKVHNYLDPESEVAISTYINPIPVKCLPGII
LDNDKSGTMFVNHARRFRETKGIMVNTFALESHALKALSDEKIPPIYPVGPILNLGDGNEDH
NQEYDMMIKWLDEQPHSSVVFLLCFGSKGSFEEDQVKEIANALERSGNRFLWSLRPPPKDTLQF
PSEFENPEEVLVPVGGFQRTKGRKVGWAPQLAILSHPAVGGFVSHCGWNTLESVRSVPIAT
WPLYAEQQSNAPQLVKDLGMAVEIKMDYREDFNKTNPLLVKAEIEDGIRKLMDSENKIRAKVM
EMKDKSRAALLEGGSSYVALGHFVETVMKN

DNA
Glycosyltransferase (NtGT1a)
Nicotiana tabacum

SEQ ID NO. 43

ATGAAGACAACAGAGTTAGTATTCTTCCTGCTCCTGGCATGGGTACCTTGTACCCACTGTGG
AGGTGGCAAAGCAACTAGTCGACAGAGACGAACAGCTTTCAATCACAGTTCTCATCATGACGCT
TCCTTTGGAACAAATATTCCATCATATACTAAATCACTGTCCTCAGACTACAGTTCTCGTATA
ACGCTGCTTCAACTTTCTCAACCTGAGACCTCTGTTAGTATGAGCAGTTTAAATGCCATCAATT
TTTTTGAGTACATCTCCAGCTACAAGGATCGTGTCAAAGATGCTGTTAATGAAACCTTTAGTTC
GTCAAGTTCTGTGAAACTCAAAGGATTTGTAATAGACATGTTCTGCACTGCGATGATTGATGTG
GCGAACGAGTTTGGAATCCCAAGTTATGTCTTCTACACTTCTAATGCAGCTATGCTTGGACTCC
AACTCCATTTTCAAAGTCTTAGTATTGAATACAGTCCGAAAGTTCATAATTACCTAGACCCTGA
ATCAGAAGTAGCGATCTCAACTTACATTAATCCGATTCCAGTCAAATGTTTGCCCGGGATTATA
CTAGACAATGATAAAGTGGCACCATGTTTCGTCAATCATGCACGAAGATTCAGGGAGACGAAAG
GAATTATGGTGAACACATTCGCTGAGCTTGAATCACACGCTTTGAAAGCCCTTTCCGATGATGA
GAAAATCCCACCAATCTACCCAGTTGGGCCCTATACTTAACCTTGAGATGGGAATGAAGATCAC
AATCAAGAATATGATATGATTATGAAGTGGCTCGACGAGCAGCCTCATTATCAGTGGTGTTC
TATGCTTTGGAAGCAAGGATCTTTCGAAGAAGATCAAGTGAAGGAAATAGCAAATGCTCTAGA
GAGAAGTGGTAACCGGTCTTGTGGTCGCTAAGACGACCGCCACCAAAAGACACGCTACAATTC
CCAAGCGAATTCGAGAATCCAGAGGAAGTCTTGCCGGTGGGATCTTTCAAAGGACTAAAGGAA
GAGGAAAGGTGATAGATGGGCACCCAGTTGGCTATTTGTCTCATCCTGCAGTAGGAGGATT
CGTGTGCAATTGTGGGTGGAATTCAACTTTGGAGAGTGTTCGTAGTGGAGTACCGATAGCAACA
TGGCCATTGTAATGCAGAGCAACAGAGCAATGCATTTCACTGGTGAAGGATTTGGGGATGGCAG
TGGAGATTAAAGATGGATTACAGGGAAGATTTTAATAAGACAAATCCACCCTGGTTAAAGCTGA

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GGAGATAGAAGATGGAATTAGGAAGCTGATGGATTGAGAGAATAAAATCAGGGCTAAGGTGATG
GAGATGAAGGACAAAAGTAGAGCAGCGTTATTAGAAGGCGGATCATCATATGTAGCTCTCGGGC
ATTTTGTGAGACTGTCATGAAAACTAA

Amino Acid
Glycosyltransferase (NtGT3)
Nicotiana tabacum

SEQ ID NO. 44

MKETKKIELVFIPSPGIGHLVSTVEMAKLLIAREEQLSITVLI IQWPNDKKLDSYIQSVANFSS
RLKFIRLPQDDSIMQLKSNIFTTFIASHKPAVRDAVADILKSENNTLAGIVIDLFTCSMIDV
ANEFELPTYVFTYSGAATLGLHYHIQNLRFENKDI TKYKDEPEEKLSIATYLNPPPAKCLPSV
ALDKEGGSTMFLDLAKRPRETKGIMINTFLELESYALNLSLRDKNLPPIYPVGPVLNINNVEGD
NLGSSDQNTMKWLDDQPASSVFLCFGSGGSFEKHQVKEIAYALESSGCRPLWSLRPPTEDAR
FPSNYENLEEILPEGFLERTKGIGKVI GWAPQLAILSHKSTGGFVSHCGWNSTLESTYFGVPIA
TWPMYAEQQANAFQLVKDLRMGVEIKMDYRKDMKVMGKEVIVKAEIEKAIREIMDSESEIRVK
VKEMKEKSRAAQMEGGSSYTSIGGFIQIIMENSQ

DNA
Glycosyltransferase (NtGT3)
Nicotiana tabacum

SEQ ID NO. 45

ATGAAAGAAACCAAGAAATAGAGTTAGTCTTCATTCTTCACCAGGAATTGGCCATTAGTAT
CCACAGTTGAAATGGCAAAGCTTCTTATAGCTAGAGAAGAGCAGCTATCTATCACAGTCCTCAT
CATCCAATGGCCTAACGACAAGAAGCTCGATTCTTATATCCAATCAGTCGCCAATTTAGCTCG
CGTTTGAAATTCATTCGACTCCCTCAGGATGATTCCATTATGCAGCTACTCAAAGCAACATTT
TCACCACGTTTATTGCCAGTCATAAGCCTGCAGTTAGAGATGCTGTTGCTGATATTCTCAAGTC
AGAATCAATAATACGCTAGCAGGTATTGTTATCGACTTGTTCTGCACCTCAATGATAGACGTG
GCCAATGAGTTCGAGCTACCAACCTATGTTTTCTACACGTCTGGTGCAGCAACCCTTGGTCTTC
ATTATCATATACAGAATCTCAGGGATGAATTTAACAAAGATATTACCAAGTACAAAGACGAACC
TGAAGAAAACTCTCTATAGCAACATATCTCAATCCATTTCCAGCAAAATGTTTGCCTGTCTGTA
GCCTTAGACAAAGAAGGTGGTTCAACAATGTTTCTTGATCTCGCAAAAAGGTTTCGAGAAACCA
AAGGTATTATGATAAACACATTTCTAGAGCTCGAATCCTATGCATTAAACTCGCTCTCACGAGA
CAAGAATCTTCCACCTATATACCTGTGCGACCAAGTATTGAACCTTAACAATGTTGAAGGTGAC
AACTTAGGTTTCATCTGACCAGAATACTATGAAATGGTTAGATGATCAGCCCGCTTCATCTGTAG
TGTTCCCTTTGTTTTGGTAGTGGTGAAGCTTTGAAAAACATCAAGTTAAGGAAATAGCCTATGC
TCTGGAGAGCAGTGGGTGTCGGTTTTTGTGGTCTGTTAAGGCGACCACCAACCGAAGATGCAAGA
TTTCCAAGCAACTATGAAAATCTTGAAGAAATTTTGCCAGAAGGATTCTTGGAAGAACAAAAG
GGATTGGAAGATGATAGGATGGGCACCTCAGTTGGCGATTTTGTACATAAATCGACGGGGGG
ATTTGTGTCGCACTGTGGATGGAATTCGACTTTGGAAAGTACATATTTTGGAGTGCCAATAGCA
ACCTGGCCAATGTACGCGGAGCAACAAGCGAATGCATTTCAATTGGTTAAGGATTTGAGAATGG
GAGTTGAGATTAAGATGGATTATAGGAAGGATATGAAAGTATGGGCAAGAAGTTATAGTGAA
AGCTGAGGAGATTGAGAAAGCAATAAGAGAAATTATGGATTCCGAGAGTGAAATTCGGGTGAAG

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GTGAAAGAGATGAAGGAGAAGAGCAGAGCAGCACAAATGGAAGGTGGCTCTTCTTACACTTCTA

TTGGAGGTTTCATCCAAATTATCATGGAGAATTCTCAATAA

Amino Acid
Glycosyltransferase (NtGT2)
Nicotiana tabacum

SEQ ID NO. 46

MVQPHVLLVTFPAQGHINPCLQFAKRLIRMGIEVTFATSVFAHRRMAKTTTSLSKGLNFAAFS

DGYDDGFKADEHDSQHYMSEIKSRGSKTLKDIILKSDEGRPVTSLVYSLLLPWAAKVAREFHI

PCALLWIQPATVLDIYYYYFNGYEDAIKGSTNDPNWCIQLPRLPLLSQDLPSFLLSSSNEEKY

SPALPTFKQLDLDLVEENPKVLVNTFDALPEKELKAIKYNLIGIGLIPSTFLDGKDPLDSS

FGGDLFQKSNDYIEWLNSKANSSVVYISFGSLLNLSKNQKEEIAKGLIEIKPFLWVIRDQENG

KGDEKEEKLSCMMELEKQKIVPWCSQLVLTHTPSIGCFVSHCGWNSTLESLSGVSVVAFPHW

TDQGTNAKLIEDVWKTGVRLLKNEDGVVESEEEKRCIEMVMDGGEKGEEMRNAQKWKELAREA

VKEGGSSEMNLKAFVQEVGKGC

DNA
Glycosyltransferase (NtGT2)
Nicotiana tabacum

SEQ ID NO. 47

ATGGTGCAACCCCATGTCTCTTGGTGACTTTTCCAGCACAAAGCCATATTAATCCATGTCTCC

AATTTGCCAAGAGGCTAATTAGAATGGGCATTGAGGTAACTTTGGCCACGAGCGTTTTCGCCCA

TCGTCGTATGGCAAAACTACGACTTCCACTCTATCCAAGGGCTTAAATTTGCGGCATTCTCT

GATGGGTACGACGATGGTTTCAAGGCCGATGAGCATGATTCTCAACATTACATGTCGGAGATAA

AAAGTCGCGGTTCTAAAACCTAAAAGATATCATTTGAAGAGCTCAGACGAGGACGTCCTGT

GACATCCCTCGTCTATTCTCTTTTGCTTCCATGGGCTGCAAAGGTAGCGCGTGAATTTACATA

CCGTGCGCGTTACTATGGATTCAACCAGCAACTGTGCTAGACATATATTATTACTTCAATG

GCTATGAGGATGCCATAAAGGTAGCACCAATGATCCAAATTGGTGTATTCAATTGCCTAGGCT

TCCACTACTAAAAAGCCAAGATCTTCTTCTTTTACTTTCTTCTAGTAATGAAGAAAAATAT

AGCTTTGCTCTACCAACATTTAAAGAGCAACTTGACACATTAGATGTTGAAGAAAATCCTAAAG

TACTTGTGAACACATTTGATGCATTAGAGCCAAAGGAACCTCAAAGCTATTGAAAAGTACAATTT

AATTGGGATTGGACCATTGATTCCTTCAACATTTTGGACGGAAAAGACCCCTTGGATTCTTCC

TTTGGTGGTGATCTTTTCAAAAGTCTAATGACTATATTGAATGGTTGAACCTCAAAGGCTAACT

CATCTGTGGTTTATATCTCATTTGGGAGTCTCTTGAATTTGTCAAAAAATCAAAGGAGGAGAT

TGCAAAAGGGTTGATAGAGATTAAAAGCCATTCTTGTGGGTAATAAGAGATCAAGAAAATGGT

AAGGGAGATGAAAAAGAAGAGAAATTAAGTTGTATGATGGAGTTGGAAAAGCAAGGGAAAATAG

TACCATGGTGTTCACAACTTGAAGTCTTAACACATCCATCTATAGGATGTTTCGTGTCACATTG

TGGATGGAATTCGACTCTGGAAAAGTTATCGTCAGGCGTGTGAGTAGTGGCATTTCCTCATTTG

ACGGATCAAGGGACAAATGCTAAACTAATTGAAGATGTTTGAAGACAGGTGTAAGGTTGAAAA

AGAATGAAGATGGTGTGGTTGAGAGTGAAGAGATAAAAAGGTGCATAGAAATGGTAATGGATGG

TGGAGAGAAAAGGAGAAGAAATGAGAAGAAATGCTCAAAAATGGAAGAATTGGCAAGGGAAGCT

GTAAAAAGAAGCGGATCTTCGGAAATGAATCTAAAAGCTTTTGTTCAGAAGATTGGCAAGGTT

GCTGA

Amino Acid
THCA Synthase
Cannabis

SEQ ID NO. 48

MNCSAFSEFWVCKIIFFFLSFHIQISIANPRENFKCFKSHIPNNVANPKLVYTOHDQLYMSIL

NSTIQNLRFISDTPKPLVIVTPSNNSHIQATILCSKKVGLQIRTRSGGHDAGGMSYISQVFPV

SEQ ID NO. 50

MKTTTLVFIPAPGMGHLVPTVEVAKQLVDRDEQLSITVLMITLPLETNIPSYTKSLSSDYSSRI

TLLQLSQPETSVMSSFNAINFFEYISSYKDRVKDAVNETPSSSSSVKLGKGFVIDMFCTAMIDV

ANEFGIPSYVFYTSNAAMLGLQLHFAQSLIEYSPKVHNYLDPESEVAISTYINPIPVKCLPGII

LDNDKSGTMFVNHARRFRETKGIMVNTFAELESALKALSDDKIPPIYPVGPILNLGDGNEHDH

NQEYDMIMKWLD EQPHSSVVFLCPFGSKGSFEEDQVKEIANALERSGNRFLWSLRPPPKDTLQF

PSEFENPEEVLPGVFFORTKGRGKVIWGAQOLAILSHPAVGGFVSHCGWNSTLESVRSGVPIAT

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WPLYAEQQSNAPQLVKDLGMAVEIKMDYREDFNKTNPPLVKAEEIEDGIRKLMDSENKIRAKVM
EMKDKSRAALLEGGSSYVALGHFVETVMKN

DNA

Glycosyltransferase (NtGT2 - codon optimized for yeast expression)
Nicotiana tabacum

SEQ ID NO. 51

ATGGTTCAACCACACGTCTTACTGGTTACTTTTCCAGCACAAAGCCATATCAACCTTGCCTAC
AATTCGCCAAAAGACTAATAAGGATGGGCATCGAAGTAACCTTTGCCACGAGTGTATTCGCACA
TAGGCGTATGGCTAAACTACGACATCAACTTTGTCCAAAGGACTAACTTCGCCGCCTTCAGT
GATGGCTATGACGATGGATTCAAAGCCGACGAACATGACAGTCAACACTACATGAGTGAAATAA
AGTCCCGTGGATCTAAACACTTAAGGATATTATACTTAAATCCTCCGATGAGGGAAGACCCGT
TACCTCTTTAGTTTATTCACTGTTACTGCCCTGGGCTGCAAAAGTCGCCAGAGAGTTTCATATT
CCTTGCCTTTATTGTGGATCCAACAGCTACGGTATTAGACATCTACTATTACTACTTCAATG
GATACGAGGATGCAATAAAGGGATCAACAAACGACCCCAACTGGTGTATTCAACTGCCTAGACT
TCCTCTATTAAAAAGTCAGGACTTACCTAGTTTTTTACTGTCATCCAGTAACGAAGAAAAATAT
TCATTCGCTTTACCCACCTTCAAAGAGCAGCTTGACACTTTGGATGTTGAAGAGAACCCCAAGG
TTTTGGTCAATACTTTTGACGCTTTGGAGCCAAAAGAGCTAAAGGCTATTGAAAAATATAACCT
TATCGGCATAGACCTTTAATCCCCTCTACTTTCTTAGATGGCAAAGACCCCTCTAGATTCAAGT
TTCGGAGGTGATTGTTTCAAAGAGTAACGATTATATCGAGTGGCTAAATAGTAAAGCCAACT
CCAGTGTGGTCTACATTTCTTTCGGAAGTCTTCTGAATTTATCAAAAAACCAAAGGAAGAGAT
CGCAAAGGACTGATAGAGATAAAAAAACCTTTCTTATGGGTGATCAGAGACCAGGAAAACGGT
AAAGCGCATGAGAAGGAGGAAAAACTGTCTGTATGATGGAGCTAGAGAAACAAGGAAAAATCG
TTCCCTGGTGTTACAGTTAGAAGTGTTAACCCTCCATCCATAGGTTGCTTCGTATCACATTG
TGTTTGAATAGTACACTTGAAAGTCTTTCATCAGGCGTCTCTGTGTCGATTCCTCCCACTGG
ACGGACCAGGGCACAAACGCCAAACTGATCGAAGATGTATGGAAGACGGGCGTCAGGCTAAAAA
AAAATGAGGATGGCGTGGTAGAGAGTGAAGAGATAAAGCGTTGCATAGAAATGGTCATGGATGG
CGGTGAAAAGGGAGAGGAAATGAGGCGTAACGCACAAAAGTGAAGGAAGTACCCCGTGAAGCA
GTGAAGAAGGAGGTTCTAGTGAGATGAATTTAAAGCTTTTCGTGCAGGAAGTTGAAAAGGCT
GCTGA

Amino Acid

Glycosyltransferase (NtGT2 - generated from codon optimized sequence
for yeast expression)
Nicotiana tabacum

SEQ ID NO. 52

MVQPHVLLVTFPAQGHINPCLQFAKRLIRMGIEVTFATSVFAHRRMAKTTTSLSKGLNFAAFS
DGYDDGFKADEHDSQHMYSEIKSRGSKTLKDIILKSSDEGRPVTSLVYSLLLPWAAKVAREFHI
PCALLWIQPATVLDIYYYYFNGYEDAIKGSTNDPNWCIQLPRLPLLKSQDLPSFLLSSSNEEKY
SFALPTFKEQLDLDVEENPKVLVNTFDALPEKELKAIKYNLIGIGLIPSTFLDGKPLDSS
FGGDLFQKSNDYIEWLNSKANSSVYISFGSLNLNLSKNQKEEIAKGLIEIKKPLWVIRDQENG
KGDEKEEKLSCMMELEKQKIWPWCSQLEVLTHPSIGCFVSHCGWNSTLESLSGGVSVVAFPHW

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TDQGTNAKLIEDVWVKTVRLKKNEDGVVESEEIKRCIEMVMDGGKEGEMRRNAQKWKELAREA

VKEGGSSSEMNLKAFVQEVGKGC

DNA

Glycosyltransferase (NtGT3 - codon optimized for yeast expression)

Nicotiana tabacum

SEQ ID NO. 53

ATGAAAGAGACTAAAAAATTGAGTTAGTTTTTATCCCCAGTCCTGGTATAGGACACTTAGTCT

CAACTGTGGAGATGGCCAAACTGTTGATAGCCCGTGAAGAGCAACTTTCTATTACTGTCCTGAT

TATACAATGGCCTAATGATAAAAAGCTAGACAGTTATATCCAGTCCGTCGCAAACTTTAGTTCT

AGACTGAAGTTTATACGCTGCCCCAAGATGACTCAATCATGCAACTTTTGAAATCAAACATTT

TCACGACATTCATCGCCTCTCACAAGCCAGCTGTAAGAGACGCCGTTGCTGACATACTAAAGAG

TGAAAGTAATAACACATTGGCAGGCATTGTAATCGATCTTTTCTGCACATCCATGATCGATGTA

GCCAATGAGTTTGAGCTGCCTACTTATGTGTTTTACACTAGTGGCGCAGCCACGTTGGGTCTGC

ACTACCATATTCAAATCTGCGTGATGAGTTTAATAAAGACATTACCAAATATAAGGATGAGCC

AGAAGAAAAATTAAGTATAGCCACGTACCTTAACCCATTCCCTGCTAAGTGTCTACCTCCGTG

GCATTGGATAAGGAAGGAGGATCAACGATGTTCTTAGACTTAGCTAAGAGGTTAGGGAGACCA

AAGGCATAATGATTAACACTTTTCTTGAGCTGGAATCATACGCTCTAAACTCATTGTCTAGAGA

TAAAAACTTGCCCCCTATATACCCTGTAGGCCCTGTTTTGAACCTGAACAACGTTGAGGGTGAT

AACTTGGGCTCTAGTGATCAAAATACCATGAAATGGCTGGACGACCAGCCAGCTTCTTCCGTTG

TGTTCTTATGTTTTGGCTCAGGAGGAAGTTTCGAAAAACACCAAGTCAAAGAAATAGCTTATGC

CTTAGAATCTTCCGGATGCAGGTTCTTGTTGGAGTTTGCCTAGACCCCCACGGAAGATGCTAGG

TTCCCTTCTAATTACGAAAACCTAGAGGAAATTTTACCAGAGGGATTCTGGAAAGAACGAAAG

GCATTGGTAAGGTCATTGGATGGGCCCCACAGTTAGCAATCTTGTCTCACAAGTCCACAGGAGG

ATTCTGTCTCATTGCGGATGGAACCTACCCCTGAAAGTACCTATTTTCGGCGTTCTTATTGCT

ACTTGCCCAATGTATGCTGAACAACAGGCCAACGCTTTTCAACTTGTTAAAGATTGAGGATGG

GTGTTGAGATCAAAATGGATTATAGGAAGGATATGAAGTAAATGGGCAAGGAGGTTATCGTTAA

GGCAGAAGAAATTGAAAAGGCCATAAGGGAATCATGGACTCAGAATCAGAAATCAGGGTCAAG

GTCAAAGAGATGAAGGAGAAAAGTCGTGCAGCCCAATGGAAGGAGGATCATCATATACCTCTA

TCGGCGGCTTCATTCAAATAATCATGGAGAACTCACAGTAA

Amino Acid

Glycosyltransferase (NtGT3 - generated from codon optimized sequence

for yeast expression)

Nicotiana tabacum

SEQ ID NO. 54

MKETKKIELVFIPSPGIGHLVSTVEMAKLLIAREEQLSITVLIQWPNDKKLDSYIQSVANFSS

RLKFIRLPQDDSIMQLKSNIFTTFIASHKPAVRDAVADILKSENNTLAGIVIDLFCTSMIDV

ANEFELPTYVFYTSGAATLGLHYHIQNLRDEFNKDITKYKDEPEEKLSIATYLNPPPAKCLPSV

ALDKEGGSTMFLDLAKRFRRETKGIMINTFLELESYALNSLSRDKNLPPIYPVGPVLNLNVEGD

NLGSSDQNTMKWLDDQPASSVFLCFGSGGSFEKHQVKEIAYALESSGCRFLWSLRPPPTEDAR

FPSNYENLEEILPEGFLERTKGIGKVIQWAPQLAILSHKSTGGFVSHCGWNSTLESTYFGVPIA

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TWPMYAEQQANAFQLVKDLRMGVEIKMDYRKDMKVMGKEVIVKAEIEKAIREIMDSESEIRVK
VKEMKEKSRAAQMEGSSSYTSIGGFIQIIMENSQ

DNA

UDP-glycosyltransferase 73C3 (NtGT4 - codon optimized for yeast
expression)

Nicotiana tabacum

SEQ ID NO. 55

ATGGCTACTCAGGTGCATAAATTCGATTTCTTCTGTTCCCACTGATGGCTCCCGGTCACATGA
TCCCTATGATAGACATCGCAAACTATTGGCTAACCGTGGCGTGATACTACCATAATAACTAC
GCCCCGTTAACGCCAATCGTTTTTCTCTACGATCACTAGGGCCATTAAATCAGGCCTAAGAATC
CAGATTTTAACTTAAATTCCTCATGTTGAGGTAGGCCTGCCTGAAGGATGTGAAAACATCG
ACATGTTGCCATCTTTGGACTTAGCCTCTAAATTTCTTGCTGCTATTTCTATGCTTAAACAACA
AGTGGAGAACTTGTAGAGGGTATTAACCCTAGTCCCTCATGCGTTATTTCTGACATGGGCTTC
CCATGGACGACACAGATCGCTCAAAATTTCAATATTCCTCGTATCGTATTTTCATGGCACGTGTT
GCTTTTCTCTTCTTGTCTTACAAAATCCTGTCATCCAATATCTTAGAGAACATTACTAGTGA
CTCAGAGTATTTTGTCTGCCAGATCTGCCAGACCGTGTGAGCTAACTAAGGCCCAAGTCTCT
GGATCTACAAAGAATACTACATCAGTAAGTAGTTAGTACTGAAGGAGGTTACAGAGCAGATCA
GGCTTGACAGGAATCATCCTACGGTGTGATAGTTAATTCCTTCGAAGAACTGGAACAGGTGTA
TGAAAAAGAGTACAGAAAAGCCAGGGGCAAAAAGGTCTGGTGCCTGGGTCTGTCTCTTTGTGC
AACAAGGAGATTGAAGATCTTGTACTAGAGGAAACAAAACCGCTATAGACAATCAGGATTGTC
TTAAGTGGTTAGACAACCTCGAGACTGAATCCGTCGTCTATGCAAGTTTAGGCTCACTAAGTAG
GCTTACGTTACTGCAAAATGGTTGAGCTGGGATTGGGACTGGAGGAGAGTAATAGGCCATTTGTA
TGGGTTCTGGGAGGAGGAGACAACTAAATGATCTTGAGAAATGGATATTGGAGAATGGCTTTG
AACAGCGTATAAAGGAGAGAGGTGTCTGATACGTGGCTGGGCACCTCAAGTATTGATTTTAAG
TCACCCCGCAATTGGAGGAGTTTAAACGCATTGTGGATGGAACCTACATTAGAGGGCATTCA
GCCGGACTACCCATGGTCACCTGGCCACTATTTGCCGAACAGTTCTGTAACGAAAAATTAGTAG
TGCAGGTTCTTAAATCGGTGTCTCACTTGGAGTGAAGTCCCTGTTAAGTGGGTGACGAAGA
GAACGTAGGTGTCTTAGTGAAAAAGGATGACGTTAAAAAGCACTGGATAAGCTAATGGATGAG
GGTGAGGAGGGCCAGGTTAGGAGGACCAAGCCAAAGAGCTTGGTGAGTTAGCTAAAAAGCCT
TTGGAGAGGGCGGATCATCCTACGTGAACCTAACGTCCCTAATTGAAGATATAATCGAGCAGCA
GACCATAAGGAGAAGTAG

Amino Acid

UDP-glycosyltransferase 73C3 (NtGT4 - generated from codon optimized
sequence for yeast expression)

Nicotiana tabacum

SEQ ID NO. 56

MATQVHKLHFIPLPMPGHMIPMIDIAKLLANRGVITTIITTPVNANRFSSTITRAIKSGLRI
QILTLKFPSEVEGLPEGECENIDMLPSLDLASKFFAAISMLKQQVENLLEGINPSPSCVISDMGF
PWTQTIAQNFNIPRIVFHGTCCFSLKSYKILSSNILENITSDSEYFVVPDLPDRVELTKAQVS
GSTKNNTSVSSSVLKEVTEQIRLAEESSYGVIVNSFEELEQVYEKEYRKARGKKVWCVGPVSLC
NKEIEDLVTRGNKTAIDNQDCLKWLDNFETESVVYASLSLSRLTLQMVGLGLEESNRPFV
WVLGGGDKLNDLEKWILENGFEQRIKRGVLIRGWAPQVLILSHPAIGGVLTCHGWNSTLEGIS

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AGLPMVTWPLFAEQFCNEKLVVQVLKIGVSLGVKVPVKWGEENVGVLVKKDDVKKALDKLMD

GEEGQVRRTKAKELGELAKKAFGEAGSSVYNLTSLIEDIIEQQNHKEK

DNA

Glycosyltransferase (NtGT5 - codon optimized for yeast expression)
Nicotiana tabacum

SEQ ID NO. 57

ATGGGCTCTATCGGTGCAGAACTAACCAAGCCACACGCCGTATGCATTCCCTATCCCGCCAGG

GACACATAAATCCTATGTGAAGTTAGCTAAGATACTGCATCACAAGGGCTTCCATATAACCTT

CGTAAATACGGAATTTAATCACAGGCGTCTGCTGAAGTCCAGAGGTCCTGACTCCCTGAAAGGT

CTTTCAAGTTTCAGGTTGAGACGATACCTGACGGACTGCCCCCATGCGAAGCTGACGCTACAC

AGGACATTCCTTCACTGTGTGAATCCACGACTAATACATGTCTAGCTCCTTTTAGAGACCTACT

TGCTAAGCTAAATGATACGAATACTTCTAACGTCCTCCCGTAAGTTGTATTGTCAAGTACGGA

GTGATGTCAATTTACCTTGCAGCTGCACAGGAAGTGGGTGTCCAGAGGTTTTATTTTGGACTA

CATCTGCTTGTGGATTCTTAGGTTACATGCACTATTGCAAGTCATTGAAAAGGATATGCTCC

ATTAAAAGACGCATCAGACCTGACGAATGGCTATCTTGAGACAACCTTGGACTTCATCCCGGC

ATGAAGGACGTGAGGCTGAGAGACTTACCTTCCTTTCTTAGGACCACCAATCCAGACGAATTA

TGATTAAGTTTGTACTACAGGAACTGAGCGTGTCTGTAAGGCCAGTGCCATAATACTTAATAC

CTTTGAAACCTTAGAGGCAGAGGTATTAGAATCATTAAAGGAACCTTCTACCCCCGTCTATCCA

ATCGGCCCCCTGCATTTCTTGTCAAACACGTAGACGATGAGAACCTAAAAGGTCTACGTTCCCT

CACTTTGGAAGGAGGAACCTGAATGTATTCAATGGTTAGACACCAAGAACCTAACTCTGTCGT

GTACGTGAATTTGCGATCCATTACTGTGATGACTCCCAATCAATTAATAGAGTTGCTTGGGGA

CTGGCAAACCTCTCAACAGACCTTCCTTTGGATCATAAGGCCTGACATCGTAAGTGGTGATGCTT

CCATATTACCTCCCGAGTTTGTGAGGAGACTAAGAACAGAGGCATGCTTGCCCTCTGGTGCTC

TCAGGAGGAGGTACTATCCCATCCCGCAATAGTGGGATTTTTGACGCACCTCTGTTGGAACTCA

ACTTTAGAATCAATTTCTAGTGGCGTCCCCATGATCTGTTGGCCTTTCTTTGCTGAGCAGCAA

CGAAGTGTGGTTTTTCAGTGACGAAGTGGGACGTTGGAATGGAATTGATTAGATGTGAAGAG

AGATGAAGTAGAGAGTTTAGTAAGAGAGTTAATGGTGGGTGTAAGGCAAGAAGATGAAGAAG

AAGGCAATGGAGTGAAGGAATGGCCGAGGCTICAGCAAAAGAACACTCTGGCTCCTCTTACG

TCAATATCGAGAAGTTGGTTAACGATATATTACTATCTAGTAAGCACTAA

Amino Acid

Glycosyltransferase (NtGT5 - generated from codon optimized
sequence for yeast expression)
Nicotiana tabacum

SEQ ID NO. 58

MGSIGAE LTKPHAVCIPYPAQGHINPMLKLAKILHHKGFHITFVNTEFNHRRLLKSRGPD SLKG

LSSFRFETIPDGLPPCEADATQDIPSLCESTTNTCLAPFRDLLAKLNDTNTSNVPPVSCIVSDG

VMSFTLAAQELGVPEVLFWTTACGFLGYMHYCKVIEKGYAPLKDASDLTNGYLETTLD FIPG

MKDVRRLDLPSPFLRTTNPDFMIKFVLQETERARKASAILNTFETLEAEVLESRLNLLPPVYP

IGPLHFLVKHVDENLKGRLSSLWKEEPECIQWLDTKEPNSVVVYNFGSITVMTPNQLIEFAWG

LANSQOTFLWIIRPDIVSGDASILPPEFVEETKNRGLASWCSQEEVL SHPAIVGFLTHSGWNS

-continued

TLESISSGVPMICWPFPAEQQTNCWFVSVTKWDVGMEDSDVKRDEVESLVRELMVGGKGGKMKK

KAMEWKELAEASAKEHSGSSVYVNIKLVNDILLSSKH

DNA

UDP glycosyltransferase 76G1 (UGT76G1 - codon optimized for yeast expression)

Stevia rebaudiana

SEQ ID NO. 59

ATGGAGAACAAAACCGAGACAACCGTTAGGCGTAGACGTAGGATAATATTGTTTCCCGTGCCCT

TTCAAGGCCATATAAACCCAATCCTGCAGCTAGCCAACGTATTGTACTCAAAGGGCTTCAGTAT

AACGATCTTCCACACCACTTTAATAAGCCAAAACGTCTAATTATCCACACTTCACATTTAGA

TTTATACTTGATAACGACCCACAGGATGAAAGAATATCAAACCTGCCCCACGCACGGCCCACTAG

CCGGAATGAGAATACCAATAATCAATGAGCATGGCGCCGACGAGTTGCGTAGAGAGCTGGAATT

GTTGATGCTAGCCAGTGAGGAAGACGAAGAGGTGTCCTGCTTAATAACGGATGCACTTTGGTAT

TTTGCTCAATCTGTGGCCGACTCCCTTAACCTGAGGCGTCTTGTCCTTATGACCTCCAGTCTAT

TCAACTTTTCATGCCCATGTCTCATTGCCCAATTTGATGAGCTTGGCTATTTGGATCCTGATGA

CAAACTAGGCTGGAGGAACAGGCTTCCGGTTTTCCCATGCTAAAGGTTAAGGACATCAAATCC

GCCTACTCAAACCTGGCAGATCCTTAAGGAAATCTTGGCAAAATGATCAAACAGACGAGGGCAT

CCAGTGGCGTCATCTGGAACCTCTTAAGGAACCTGAAGAATCAGAACTTGAAACAGTAATCAG

AGAAATACCTGCCCAAGTTTCTTGATCCCTCTACCTAAGCACCTTACGGCTTCTAGTTCTTCT

TTGTTGGACCACGATCGTACTGTCTTTCAATGGTTAGATCAGCAACCCCCCTCATCAGTGCTAT

ATGTGTCATTTCGTTAGTACATCAGAAGTGACGAAAAGGATTTCTTGAGATAGCCCGTGGATT

GGTGGACTCTAAACAGTCTTTTTTATGGGTTGTGAGACCTGGATTGTAAAGGGATCCACGTGG

GTCGAACCCCTTGCCCGATGGTTTCTGGGTGAAAGAGGAAGGATAGTGAAGTGGTCCCTCAGC

AAGAGGTACTGGCCCATGGTGTCTATAGGTGCTTTCTGGACCCACTCCGGCTGGAATAGTACACT

AGAATCCGTTTGCAGGGGTGTCCCTATGATTTTTTCTGATTTTGGTTTAGATCAACCCCTGAAT

GCTAGGTACATGTGACAGCTCCTTAAAGTCGGCGTCTACCTAGAAAATGGCTGGGAGAGGGGTG

AGATAGCAAAACGCTATCAGACGTGTTATGGTAGACGAAGAGGGAGGTACATAAGGCAAAACGC

CAGGGTCTGAAACAAAAGCCGATGTGTCCTTGATGAAGGCGGCTCTTCATACGAAAGTCTA

GAAAGTCTTGTTTCTTATATTTCTCTCACTATAA

Amino Acid

UDP glycosyltransferase 76G1 (UGT76G1 - generated from codon optimized sequence for yeast expression)

Stevia rebaudiana

SEQ ID NO. 60

MENKTETTVRRRRRIILFPVPFQGHINPILQLANLVYSKGFSTIFHTNFKPKTSNYPHPTFR

FILDNDPQDERISNLPTHGPLAGMRIPINIIEHGADELRRLELLMLASEEDEEVSLITDALWY

FAQSVADSLNLRRLVMTSSLFNFHAHVSLPQFDELGYLDPDDKTRLEEQASGFPMKVKDIKS

AYSNWQILKEILGKMIKQTRASSGVIWNSFKELEESELETVIREIPAPSFILPLPKHLTASSSS

LLDHDRTVPQWLDQQPPSSVLVVSFGSTSEVDEKDFLEIARGLVDSKQSFLWVVRPGFVKGSTW

VEPLPDGFLGERGRIVKWVPQQEVLAHGAIGAFWTHSGWNSTLESVCEGVPMIFSDFGLDQPLN

ARYMSDVLKVGYLENGWERGEIANAIRVMVDEEGEYIRQNARVLKQKADVSLMKGSSYESL

ESLVSYSISL

DNA

glycosyltransferase (UGT73A10)

Lycium barbarum

SEQ ID NO. 61

ATGGGTCAATGCATTTTTTTTTGTTTCCAATGATGGCTCAAGGTCATATGATTCCAACCTTGG

ATATGGCTAAGTTGATGTCTCTAGAGGTGTTAAGGCTACTATTATTACTACTCCATTGAACGA

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ATCTGTTTTTCTAAGGCTATTCAAAGAAACAAGCAATTGGGTATTGAAATTGAAATTGAAATT
AGATTGATTAAGTTTCCAGCTTTGGAAAACGATTTGCCAGAAGATTGTGAAAGATTGGATTGGA
TTCCAACTGAAGCTCATTGCCAAACTTTTTAAGGCTGCTGCTATGATGCAAGAACCATTGGA
ACAATTGATTCAAGAATGTAGACCAGATTGTTTGGTTTCTGATATGTTTTGCCATGGACTACT
GATACTGCTGCTAAGTTTAACATTCCAAGAATTGTTTTTCATGGTACTAACTACTTTGCTTTGT
GTGTTGGTGATTCTATGAGAAGAAACAAGCCATTTAAGAACGTTTCTTCTGATTCTGAACTTT
TGTTGTTCCAACTTGCCACATGAAATTAAGTTGACTAGAACTCAAGTTTCTCCATTGAAACAA
TCTGATGAAGAATCTGTTATGTCTAGAGTTTTGAAGGAAGTTAGAGAATCTGATTGAACTCTT
ACGGTGTTATTTTTAACTCTTTTTACGAATTGGAACCAGATTACGTTGAACATTACACTAAGGT
TATGGGTAGAAAGTCTTGGGCTATTGGTCCATTGTCTTGTGTAACAGAGATGTTGAAGATAAG
GCTGAAAGAGGTAAGAAGTCTTCTATTGATAAGCATGAATGTTTGGAAATGGTTGGATTCTAAGA
AGCCATCTTCTATTGTTTACGTTTGTGTTGCTTCTGTTGCTAACTTTACTGTTACTCAAATGAG
AGAATTGGCTTTGGGTTTGGGAAGCTTCTGGTTTGGATTTTATTTGGGCTGTTAGAGCTGATAAC
GAAGATTGGTTGCCAGAAGGTTTTGAAGAAAGAACTAAGGAAAAGGGTTTGATTATTAGAGGTT
GGGCTCCACAAGTTTTGATTTTGGATCATGAATCTGTTGGTGCTTTTGTACTCATTGTGGTTG
GAACTCTACTTTGGAAGGTATTTCTGCTGGTGTTCCAATGGTTACTTGCCAGTTTTTGCTGAA
CAATTTTTTAACGAAAAGTTGGTTACTCAAGTTATGAGAAGTGGTGCTGGTGTTGGTTCTGTTT
AATGGAAGAGATCTGCTTCTGAAGGTGTTGAAAAGGAAGCTATTGCTAAGGCTATTAAGAGAT
TATGTTTTCTGAAGAAGCTGAAGGTTTTAGAAACAGAGCTAGAGCTTACAAGGAAATGGCTAGA
CAAGCTATTGAAGAAGGTGGTTCTTCTTACACTGGTTTGACTACTTTGTTGGAAGATATTTCTT
CTTACGAATCTTTGTCTTCTGATTAA

Amino Acid
Glycosyltransferase (UGT73A10)
Lycium barbarum

SEQ ID NO. 62

MGQLHFFLPFPMMAQGHMPTLDMAKLIASRGVKATIITPLNESVFSKAIQRNKQLGIEIEIEI
RLIKFPALENDLPEDCERLDLIPTEAHLNFFKAAAMMQEPLEQLIQECRPDCLVSDMFLPWTT
DTAAKFNIPIRVFHGTNYFALCVGDSMRNKPFPKNVSSDSETFVVPNLPHEIKLRTQVSPFEQ
SDEESVMSRVLKEVRESDLKSYGVI FNSFYELEPDYVEHYTKVMGRKSWAIGPLSLCNRDVEDK
AERGKSSIDKHECLEWLD SKKPSSIYVVCFGSVANFTVTQMRELALGLEASGLDFIWA VRADN
EDWLPEGFEERTKEKGLIIRGWAPQVLILDHESVGAFVTHCGWNSTLEGISAGVPMVTWPVFAE
QFFNEKLVTQVMRTGAGVGSVQWKRSA SEGVEKEAIAKAIKRMVSEEAEGFRNRARAYKEMAR
QAIEEGSSSYTGLTTLLEDISSYESLSSD

DNA
Cytosolic-targeted UDP glycosyltransferase 76G1 (cytUTG)
Stevia rebaudiana

SEQ ID NO. 63

ATGGAAAATAAAACCGAAACACCGTCCGCCGTCGTCGCCGTATCATTCTGTTCCCGGTCCCGT
TCCAGGGCCACATCAACCCGATTCTGCAACTGGCGAACGTGCTGTATTGAAAAGGTTTCAGCAT
CACCATCTTCCATACGAACCTCAACAAGCCGAAGACCAGCAATTACCCGCACCTTTACGTTCCGT
TTTATTCTGGATAACGACCCGAGGATGAACGCATCTCTAATCTGCCGACCCACGCCCCGCTGG
CGGGTATGCGTATTCCGATTATCAACGAACACGGCGCAGATGAACTGCGTCGCGAACTGGAACT
GCTGATGCTGGCCAGCGAAGAAGATGAAGAAGTTTCTTGCCTGATCACCGACGCACCTGTGGTAT
TTTGCCAGTCTGTTGCAGATAGTCTGAACCTGCGTCGCTGGTCCTGATGACCAGCAGCCTGT

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TCAATTTTCATGCCACGTTAGTCTGCCGAGTTCGATGAACTGGGTTATCTGGACCCGGATGA
CAAAACCCGCCTGGAAGAACAGGCGAGCGGCTTCCGATGCTGAAAGTCAAGGATATTAAGTCA
GCGTACTCGAACTGGCAGATTCTGAAAGAAATCCTGGGTAAAATGATTAAGCAAACCAAAGCAA
GTTCCGCGCTCATCTGGAATAGTTTCAAAGAACTGGAAGAATCCGAACTGGAAACGGTGATTCTG
TGAAATCCCGGCTCCGAGTTTTCTGATTCCGCTGCCGAAGCATCTGACCGCGAGCAGCAGCAGC
CTGCTGGATCACGACCCGACCGTGTTTCAGTGGCTGGATCAGCAACCGCCGAGTTCCGTGCTGT
ATGTTAGCTTCGTTAGTACCTCGGAAGTGATGAAAAGGACTTCTGGAAATCGCTCGTGGCCT
GGTTGATAGCAAAACATCTTCTGTGGGTGGTTGCCCCGGTTTTGTGAAGGGCTCTACGTGG
GTTGAACCGCTGCCGAGCGGCTTCTGGGTGAACGTGGCCGCATTGTCAAATGGGTGCCGCAGC
AAGAAGTGCTGGCGCATGGCGGATTGGCGCGTTTTGGACCCACTCCGTTGGAACCAACGCT
GGAATCGGTTTGTGAAGGTGTCCCGATGATTTTCTCAGATTTTGGCCTGGACCAGCCGCTGAAT
GCACGTTATATGTCGGATGTTCTGAAAGTCGGTGTGTACCTGAAAAACGGTTGGGAACGCGGCG
AAATTGCGAATGCCATCCGTCGCGTTATGGTCGATGAAGAAGGCGAATACATTCGTCAGAATGC
TCGCGTCTGAAACAAAAGGCGGACGTGAGCCTGATGAAAGGCGGTTTCATCGTATGAAAGTCTG
GAATCCCTGGTTTCATACATCAGCTCTCTGTAA

Amino Acid
Cytosolic-targeted UDP glycosyltransferase 76G1 (cytUTG)
Stevia rebaudiana

SEQ ID NO. 64

MENKTETTVRRRRRIILFPVPQGHINPILQLANVLYSKGFSITIFHTNFKPKTSNYPHPTFR
FILDNDPQDERISNLPTHGPLAGMRIPIINEHGADELRLRELELLMLASEEDEEVSLITDALWY
FAQSVADSLNLRRLVMTSSLFNFHAHVSLPQFDELGYLDPDDKTRLEEQASGFPMKVKDIKS
AYSNWQILKEILGKMIKQTKASSGVIWNSFKELEESELETVIREIPAPSFLIPLPKHLTASSSS
LLDHDRTVFQWLDQQPPSSVLVVSFGSTSEVDEKDFLEIARGLVDSKQSFLWVVRPGFVKGSTW
VEPLPDGFLGERGRIVKWVPQQEVLHAIGAIGAFWTHSGWNSTLESVCEGVPMIFSDFGLDQPLN
ARYMSDVLKVGYLENGWERGEIANAIRVMVDEEGEYIRQARVLKQKADVSLMKGSSYESL
ESLVSYSISL

Enhanced N-terminal chimera secretion signal with Ost1 signal sequence
S. cerevisiae

SEQ ID NO. 65

MRQVWFSWIVGLFLCFFNVSSAAPVNTTTEDETAQIPAEAVIGYSDLEGDFDVAVLFPNSTNNG
LLFINTTIIASIAAKEEGVSLEKR

Enhanced Ost1 secretion signal presequence
S. cerevisiae

SEQ ID NO. 66

MRQVWFSWIVGLFLCFFNVSSA

Amino Acid
Enhanced Exportable Hepatic Fatty acid-binding protein 1
Mus musculus

SEQ ID NO. 67

MRQVWFSWIVGLFLCFFNVSSAAPVNTTTEDETAQIPAEAVIGYSDLEGDFDVAVLFPNSTNN
GLLFINTTIIASIAAKEEGVSLEKRMNFSGKYQLQSQENFEPFMKAIGLPEDLIQKGDKIKGVSE
IVHEGKKIKLTITYGPKVVRNEFTLGEECELETMTGEKVKAIVKLEGDNKMVTFKGIKSVTEL
NGDTITNTMTLGDIVYKRVSKRI

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Amino Acid
Enhanced Exportable Hepatic Fatty acid-binding protein 1
Bos taurus
SEQ ID NO. 68
MRQVWFSWIVGLFLCFNVSAAAPVNTTTEDETAQIPAEAVIGYSDLEGDFDVAVLPPFSNSTNN
GLLFINTTIIASIAAKEEGVSLEKRMNFSQKYQVQTQENYEAFMKAVGMPDDIIQKGDIKGVSE
IVQNGKHKFKFIITAGSKVIQNEFTLGEECEMEFMTGEKIKAVVQQEGDNKLVTTFKGIKSVTEF
NGDTVTSTMTKGDVVFVKRVSKRI

Amino Acid
Sec signal peptide for *E coli* L-asparaginase II
E. Coli
SEQ ID NO. 69
MEFFKKTALAALVMGFSGAALA

Amino Acid
Tat signal peptide for *E coli* strain k12 periplasmic nitrate reductase
E. Coli
SEQ ID NO. 70
MKLSRRSFMKANAVAAAAAGLSVPGVARAVVGQQ

Amino Acid
secretion signal from an extracellular protease Ara12 (At5g67360)
Arabidopsis thaliana
SEQ ID NO. 71
MSSSFLSSTAFFLLLLCLGFCHVSSS

Amino Acid
secretion signal from a alpha amylase
barley (*Hordeum vulgare*)
SEQ ID NO. 72
MGKKSHICCFSLLLLLFAGLASG

Amino Acid
secretion signal from a a-Amylase
rice
SEQ ID NO. 73
MKNTSSLCLLLLVLCSLTCNSGQAAQV

Amino Acid
Chicken FABP7 WT
Gallus gallus
SEQ ID NO. 74
MVEAFCATWKLADSHNFDEYMKALGVGFAMRQVGNVTKPTVIISSEGDKVVIRTQSTFKNTEIS
FKLGEEFDETTDDRCKSVVTLGDGKLVHVQKWDGKETNFVREIKDGRMVMTLTFGDVVAVRH
YEKA

Amino Acid
Pig FABP1 WT
Sus scrofa
SEQ ID NO. 75
MNFSGKYQVSQENFEAFMKAVGLPDELIQKGDIKGTSEIVQNGKHKLTITTTGSKVVQNEFT
LGEECEMETLTGEKVKTVVQLEGDNKLVTTFKGIKSVTELNGDIITSTMTLGDIVFKRISKRI

Amino Acid
Horse FABP1 WT
Equus caballus
SEQ ID NO. 76
MNFSGKYQLQSQENFEPFMKAVGMSDDLQKGDLDKETSEIVHNGNHFKITITTTGPKVVHHEFT
LGEEFELESFTGEKVKAITHMEGDKLVSTIKGIKSVTELKGDIIITNQLSVWHTPVVCATSALVP
ACRPGPVLDLGTMTLGDIVFKRVSKRI

Amino Acid
Plant FABP WT
Rhodamnia argentea
SEQ ID NO. 77
MTEFAGVYEFVKDDGKFEEILKAMDVNFLIRKVAGKMNPNTAIDVLDGKMAFKTITPLKTVEI
HFELGKEYENKRLDGTVVKGIVTRDGNKLIQEQMSEPKFKVIRELDGPEKLIVTMCKDIVCVR
EYKRIQT

-continued

Amino Acid
 Mouse FABP1 WT
Mus musculus

SEQ ID NO. 78

MNFSGKYQLQSQENFEPFMKAIGLPEDLIQKGKDIKGVSEIVHEGKKIKLTITYGPKVVRNEFT
 LGEECELETMTGEKVKAVVKLEGDNKMVTTFKGIKSVTELNGDTITNTMTLGDIVYKRVSKRI

Amino Acid
 Modified Chicken FABP7 mdl_r5_0225_ro
Gallus gallus

SEQ ID NO. 79

MVEAPCATWKLADSHNMDEALKAWGVGFARQVAVNVTKPTVIISSEGDKVVIRMQSTFKNTEIS
 FKLGEFEDET~~T~~ADDRNCKSVVTLGDGKLVHVQKWDGKETNVVREIKDGRMVMTVTFGDVVAVRH
 YEKA

Amino Acid
 Modified Chicken FABP7 mdl_r3_1403_sm
Gallus gallus

SEQ ID NO. 80

MVEAPCATWKLADSHNQDEGLKALGVGFAMRQLFNV~~D~~KPTVIISSEGDKVVIRIEGT~~V~~KNTEIS
 FKLGEFEDET~~T~~ADDRNCKSVVTLGDGKLVHVQKWDGKETNVVREIKDGRMVATFTFGD~~V~~VAVAH
 YEKA

Amino Acid
 Modified Pig FABP1 mdl_r5_1652_ro
Sus scrofa

SEQ ID NO. 81

MNFSGKYQVQSQENFEAFMKAVGQ~~P~~DELIQKGKDIKGTSEIVQNGKHFKVTIT~~T~~GSKVVQAEFT
 LGEECEAET~~D~~TGEKVKVVQLEGDNKA~~V~~VTAKGIKAVFELNGDII~~T~~ATATLGDIVFKRISKRI

Amino Acid
 Modified Pig FABP1 mdl_r4_0372_sm
Sus scrofa

SEQ ID NO. 82

MNFSGKYQVQSQENFEAAKAAGVPDEEIQKGKDIKGTLEIVQNGKHFKLT~~T~~TGSKVVQNEFT
 LGEECEMETL~~G~~GEKAKTVVQLEGDNKA~~V~~VTFKGIKAVIELNGDII~~T~~LT~~V~~TTLGDIVFKRISKRI

Amino Acid
 Modified Horse FABP1 mdl_r5_0954_ro
Equus caballus

SEQ ID NO. 83

MNFSGKYQLQSQENAEPPFKAVGMSDDLIQ~~L~~AKDLKVTGEIVHNGNHFKITHTSGAAV~~V~~HHEFT
 LGEEFELES~~L~~TGEKVKAITHMEGDKLVSTIKGIKSVTELKGDII~~T~~NQLSVWHTPVVCATSALVP
 ACRPGPVLDLGTMTLGDIVMKFVSKRI

Amino Acid
 Modified Horse FABP1 mdl_r5_1179_sm
Equus caballus

SEQ ID NO. 84

MNFSGKYQLQSS~~EN~~VEPPFNKAVGMSDDLIQ~~K~~AKDLKFTGEIVHNGNHFKITIT~~T~~GPKVVHHEFT
 LGEEFELEHFTGEKVKAITHMEGDKLVMTVKGIKSV~~V~~WVKGDII~~T~~NQLSVWHTPVVCATSALVP
 ACRPGPVLDLGTGV~~L~~GDIVMKHVSKRI

Amino Acid
 Modified Plant FABP mdl_r3_0887_ro_sm
Rhodamnia argentea

SEQ ID NO. 85

MTEFAGVVEFVKDDGKFEEILKAMDVNFLIRKVG~~M~~KMNANVAIDVLDDGKMAIKVITPLKTVEI
 HPFELGKEYENKRF~~D~~GTGVVKGIVTRDGNKLIIEQMSEPKFKVIAELDGPEKLI~~F~~TEMCKDIVCVA
 EYKRIQT

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Amino Acid
Modified Plant FBP mdl_r4_1489_ro
Rhodamnia argentea

SEQ ID NO. 86

MTEFAGVYEFVKDDGKFEILKAMDVNFLIRKVAGKMNPNI~~A~~IDVLDDGKMAI~~K~~MITPLKTVEV

HFELGKEYE~~A~~~~K~~~~A~~LDGTVV~~K~~~~A~~IVTRDGNKLI~~I~~~~E~~~~N~~MSEPKF~~T~~VIRELDGPEKLIVTWMCKDIVCVA

EYKRIQT

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- [13] Chen M H, Huang L F, Li H M, Chen Y R, Yu S M. Signal peptide-dependent targeting of a rice alpha-amylase and cargo proteins to plastids and extracellular compartments of plant cells. *Plant Physiol.* 2004; 135(3): 1367-77

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 86

<210> SEQ ID NO 1

<211> LENGTH: 127

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 1

Met Asn Phe Ser Gly Lys Tyr Gln Leu Gln Ser Gln Glu Asn Phe Glu
1 5 10 15

Pro Phe Met Lys Ala Ile Gly Leu Pro Glu Asp Leu Ile Gln Lys Gly
20 25 30

Lys Asp Ile Lys Gly Val Ser Glu Ile Val His Glu Gly Lys Lys Ile
35 40 45

Lys Leu Thr Ile Thr Tyr Gly Pro Lys Val Val Arg Asn Glu Phe Thr
50 55 60

Leu Gly Glu Glu Cys Glu Leu Glu Thr Met Thr Gly Glu Lys Val Lys
65 70 75 80

Ala Val Val Lys Leu Glu Gly Asp Asn Lys Met Val Thr Thr Phe Lys
85 90 95

Gly Ile Lys Ser Val Thr Glu Leu Asn Gly Asp Thr Ile Thr Asn Thr
100 105 110

Met Thr Leu Gly Asp Ile Val Tyr Lys Arg Val Ser Lys Arg Ile

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115                               120                               125

<210> SEQ ID NO 2
<211> LENGTH: 127
<212> TYPE: PRT
<213> ORGANISM: Bos taurus

<400> SEQUENCE: 2

Met Asn Phe Ser Gly Lys Tyr Gln Val Gln Thr Gln Glu Asn Tyr Glu
1          5          10          15

Ala Phe Met Lys Ala Val Gly Met Pro Asp Asp Ile Ile Gln Lys Gly
          20          25          30

Lys Asp Ile Lys Gly Val Ser Glu Ile Val Gln Asn Gly Lys His Phe
          35          40          45

Lys Phe Ile Ile Thr Ala Gly Ser Lys Val Ile Gln Asn Glu Phe Thr
          50          55          60

Leu Gly Glu Glu Cys Glu Met Glu Phe Met Thr Gly Glu Lys Ile Lys
65          70          75          80

Ala Val Val Gln Gln Glu Gly Asp Asn Lys Leu Val Thr Thr Phe Lys
          85          90          95

Gly Ile Lys Ser Val Thr Glu Phe Asn Gly Asp Thr Val Thr Ser Thr
          100          105          110

Met Thr Lys Gly Asp Val Val Phe Lys Arg Val Ser Lys Arg Ile
          115          120          125

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<210> SEQ ID NO 3
<211> LENGTH: 85
<212> TYPE: PRT
<213> ORGANISM: S. cerevisiae

<400> SEQUENCE: 3

Met Arg Phe Pro Ser Ile Phe Thr Ala Val Leu Phe Ala Ala Ser Ser
1          5              10              15

Ala Leu Ala Ala Pro Val Asn Thr Thr Thr Glu Asp Glu Thr Ala Gln
          20              25              30

Ile Pro Ala Glu Ala Val Ile Gly Tyr Ser Asp Leu Glu Gly Asp Phe
          35              40              45

Asp Val Ala Val Leu Pro Phe Ser Asn Ser Thr Asn Asn Gly Leu Leu
          50              55              60

Phe Ile Asn Thr Thr Ile Ala Ser Ile Ala Ala Lys Glu Glu Gly Val
65          70              75              80

Ser Leu Glu Lys Arg
          85

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<210> SEQ ID NO 4
<211> LENGTH: 212
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 4

Met Arg Phe Pro Ser Ile Phe Thr Ala Val Leu Phe Ala Ala Ser Ser
1          5          10          15
Ala Leu Ala Ala Pro Val Asn Thr Thr Glu Asp Glu Thr Ala Gln
          20          25          30
Ile Pro Ala Glu Ala Val Ile Gly Tyr Ser Asp Leu Glu Gly Asp Phe
          35          40          45
Asp Val Ala Val Leu Pro Phe Ser Asn Ser Thr Asn Asn Gly Leu Leu
          50          55          60

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Phe Ile Asn Thr Thr Ile Ala Ser Ile Ala Ala Lys Glu Glu Gly Val
65                               70                               75                               80

Ser Leu Glu Lys Arg Met Asn Phe Ser Gly Lys Tyr Gln Leu Gln Ser
                               85                               90                               95

Gln Glu Asn Phe Glu Pro Phe Met Lys Ala Ile Gly Leu Pro Glu Asp
                               100                              105                              110

Leu Ile Gln Lys Gly Lys Asp Ile Lys Gly Val Ser Glu Ile Val His
                               115                              120                              125

Glu Gly Lys Lys Ile Lys Leu Thr Ile Thr Tyr Gly Pro Lys Val Val
130                               135                               140

Arg Asn Glu Phe Thr Leu Gly Glu Glu Cys Glu Leu Glu Thr Met Thr
145                               150                              155                              160

Gly Glu Lys Val Lys Ala Val Val Lys Leu Glu Gly Asp Asn Lys Met
                               165                              170                              175

Val Thr Thr Phe Lys Gly Ile Lys Ser Val Thr Glu Leu Asn Gly Asp
                               180                              185                              190

Thr Ile Thr Asn Thr Met Thr Leu Gly Asp Ile Val Tyr Lys Arg Val
195                               200                              205

Ser Lys Arg Ile
210

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<210> SEQ ID NO 5
<211> LENGTH: 212
<212> TYPE: PRT
<213> ORGANISM: Bos taurus

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<400> SEQUENCE: 5

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Met Arg Phe Pro Ser Ile Phe Thr Ala Val Leu Phe Ala Ala Ser Ser
1                               5                               10                               15

Ala Leu Ala Ala Pro Val Asn Thr Thr Thr Glu Asp Glu Thr Ala Gln
20                               25                               30

Ile Pro Ala Glu Ala Val Ile Gly Tyr Ser Asp Leu Glu Gly Asp Phe
35                               40                               45

Asp Val Ala Val Leu Pro Phe Ser Asn Ser Thr Asn Asn Gly Leu Leu
50                               55                               60

Phe Ile Asn Thr Thr Ile Ala Ser Ile Ala Ala Lys Glu Glu Gly Val
65                               70                               75                               80

Ser Leu Glu Lys Arg Met Asn Phe Ser Gly Lys Tyr Gln Val Gln Thr
85                               90                               95

Gln Glu Asn Tyr Glu Ala Phe Met Lys Ala Val Gly Met Pro Asp Asp
100                              105                              110

Ile Ile Gln Lys Gly Lys Asp Ile Lys Gly Val Ser Glu Ile Val Gln
115                              120                              125

Asn Gly Lys His Phe Lys Phe Ile Ile Thr Ala Gly Ser Lys Val Ile
130                              135                              140

Gln Asn Glu Phe Thr Leu Gly Glu Glu Cys Glu Met Glu Phe Met Thr
145                              150                              155                              160

Gly Glu Lys Ile Lys Ala Val Val Gln Gln Glu Gly Asp Asn Lys Leu
165                              170                              175

Val Thr Thr Phe Lys Gly Ile Lys Ser Val Thr Glu Phe Asn Gly Asp
180                              185                              190

Thr Val Thr Ser Thr Met Thr Lys Gly Asp Val Val Phe Lys Arg Val
195                              200                              205

Ser Lys Arg Ile

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210

<210> SEQ ID NO 6

<211> LENGTH: 492

<212> TYPE: PRT

<213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 6

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Met Asp Pro Tyr Lys Tyr Arg Pro Ala Ser Ser Tyr Asn Ser Pro Phe
 1          5          10          15

Phe Thr Thr Asn Ser Gly Ala Pro Val Trp Asn Asn Asn Ser Ser Met
      20          25          30

Thr Val Gly Pro Arg Gly Leu Ile Leu Leu Glu Asp Tyr His Leu Val
      35          40          45

Glu Lys Leu Ala Asn Phe Asp Arg Glu Arg Ile Pro Glu Arg Val Val
      50          55          60

His Ala Arg Gly Ala Ser Ala Lys Gly Phe Phe Glu Val Thr His Asp
      65          70          75          80

Ile Ser Asn Leu Thr Cys Ala Asp Phe Leu Arg Ala Pro Gly Val Gln
      85          90          95

Thr Pro Val Ile Val Arg Phe Ser Thr Val Ile His Ala Arg Gly Ser
      100         105         110

Pro Glu Thr Leu Arg Asp Pro Arg Gly Phe Ala Val Lys Phe Tyr Thr
      115         120         125

Arg Glu Gly Asn Phe Asp Leu Val Gly Asn Asn Phe Pro Val Phe Phe
      130         135         140

Ile Arg Asp Gly Met Lys Phe Pro Asp Ile Val His Ala Leu Lys Pro
      145         150         155         160

Asn Pro Lys Ser His Ile Gln Glu Asn Trp Arg Ile Leu Asp Phe Phe
      165         170         175

Ser His His Pro Glu Ser Leu Asn Met Phe Thr Phe Leu Phe Asp Asp
      180         185         190

Ile Gly Ile Pro Gln Asp Tyr Arg His Met Asp Gly Ser Gly Val Asn
      195         200         205

Thr Tyr Met Leu Ile Asn Lys Ala Gly Lys Ala His Tyr Val Lys Phe
      210         215         220

His Trp Lys Pro Thr Cys Gly Val Lys Ser Leu Leu Glu Glu Asp Ala
      225         230         235         240

Ile Arg Leu Gly Gly Thr Asn His Ser His Ala Thr Gln Asp Leu Tyr
      245         250         255

Asp Ser Ile Ala Ala Gly Asn Tyr Pro Glu Trp Lys Leu Phe Ile Gln
      260         265         270

Ile Ile Asp Pro Ala Asp Glu Asp Lys Phe Asp Phe Asp Pro Leu Asp
      275         280         285

Val Thr Lys Thr Trp Pro Glu Asp Ile Leu Pro Leu Gln Pro Val Gly
      290         295         300

Arg Met Val Leu Asn Lys Asn Ile Asp Asn Phe Phe Ala Glu Asn Glu
      305         310         315         320

Gln Leu Ala Phe Cys Pro Ala Ile Ile Val Pro Gly Ile His Tyr Ser
      325         330         335

Asp Asp Lys Leu Leu Gln Thr Arg Val Phe Ser Tyr Ala Asp Thr Gln
      340         345         350

Arg His Arg Leu Gly Pro Asn Tyr Leu Gln Leu Pro Val Asn Ala Pro
      355         360         365

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Lys Cys Ala His His Asn Asn His His Glu Gly Phe Met Asn Phe Met
 370 375 380
 His Arg Asp Glu Glu Val Asn Tyr Phe Pro Ser Arg Tyr Asp Gln Val
 385 390 395 400
 Arg His Ala Glu Lys Tyr Pro Thr Pro Pro Ala Val Cys Ser Gly Lys
 405 410 415
 Arg Glu Arg Cys Ile Ile Glu Lys Glu Asn Asn Phe Lys Glu Pro Gly
 420 425 430
 Glu Arg Tyr Arg Thr Phe Thr Pro Glu Arg Gln Glu Arg Phe Ile Gln
 435 440 445
 Arg Trp Ile Asp Ala Leu Ser Asp Pro Arg Ile Thr His Glu Ile Arg
 450 455 460
 Ser Ile Trp Ile Ser Tyr Trp Ser Gln Ala Asp Lys Ser Leu Gly Gln
 465 470 475 480
 Lys Leu Ala Ser Arg Leu Asn Val Arg Pro Ser Ile
 485 490

<210> SEQ ID NO 7
 <211> LENGTH: 753
 <212> TYPE: PRT
 <213> ORGANISM: Escherichia coli

<400> SEQUENCE: 7

Met Ser Gln His Asn Glu Lys Asn Pro His Gln His Gln Ser Pro Leu
 1 5 10 15
 His Asp Ser Ser Glu Ala Lys Pro Gly Met Asp Ser Leu Ala Pro Glu
 20 25 30
 Asp Gly Ser His Arg Pro Ala Ala Glu Pro Thr Pro Pro Gly Ala Gln
 35 40 45
 Pro Thr Ala Pro Gly Ser Leu Lys Ala Pro Asp Thr Arg Asn Glu Lys
 50 55 60
 Leu Asn Ser Leu Glu Asp Val Arg Lys Gly Ser Glu Asn Tyr Ala Leu
 65 70 75 80
 Thr Thr Asn Gln Gly Val Arg Ile Ala Asp Asp Gln Asn Ser Leu Arg
 85 90 95
 Ala Gly Ser Arg Gly Pro Thr Leu Leu Glu Asp Phe Ile Leu Arg Glu
 100 105 110
 Lys Ile Thr His Phe Asp His Glu Arg Ile Pro Glu Arg Ile Val His
 115 120 125
 Ala Arg Gly Ser Ala Ala His Gly Tyr Phe Gln Pro Tyr Lys Ser Leu
 130 135 140
 Ser Asp Ile Thr Lys Ala Asp Phe Leu Ser Asp Pro Asn Lys Ile Thr
 145 150 155 160
 Pro Val Phe Val Arg Phe Ser Thr Val Gln Gly Gly Ala Gly Ser Ala
 165 170 175
 Asp Thr Val Arg Asp Ile Arg Gly Phe Ala Thr Lys Phe Tyr Thr Glu
 180 185 190
 Glu Gly Ile Phe Asp Leu Val Gly Asn Asn Thr Pro Ile Phe Phe Ile
 195 200 205
 Gln Asp Ala His Lys Phe Pro Asp Phe Val His Ala Val Lys Pro Glu
 210 215 220
 Pro His Trp Ala Ile Pro Gln Gly Gln Ser Ala His Asp Thr Phe Trp
 225 230 235 240
 Asp Tyr Val Ser Leu Gln Pro Glu Thr Leu His Asn Val Met Trp Ala
 245 250 255

Met	Ser	Asp	Arg	Gly	Ile	Pro	Arg	Ser	Tyr	Arg	Thr	Met	Glu	Gly	Phe
			260				265						270		
Gly	Ile	His	Thr	Phe	Arg	Leu	Ile	Asn	Ala	Glu	Gly	Lys	Ala	Thr	Phe
			275				280						285		
Val	Arg	Phe	His	Trp	Lys	Pro	Leu	Ala	Gly	Lys	Ala	Ser	Leu	Val	Trp
			290				295						300		
Asp	Glu	Ala	Gln	Lys	Leu	Thr	Gly	Arg	Asp	Pro	Asp	Phe	His	Arg	Arg
			305				310						320		
Glu	Leu	Trp	Glu	Ala	Ile	Glu	Ala	Gly	Asp	Phe	Pro	Glu	Tyr	Glu	Leu
			325				330						335		
Gly	Phe	Gln	Leu	Ile	Pro	Glu	Glu	Asp	Glu	Phe	Lys	Phe	Asp	Phe	Asp
			340				345						350		
Leu	Leu	Asp	Pro	Thr	Lys	Leu	Ile	Pro	Glu	Glu	Leu	Val	Pro	Val	Gln
			355				360						365		
Arg	Val	Gly	Lys	Met	Val	Leu	Asn	Arg	Asn	Pro	Asp	Asn	Phe	Phe	Ala
			370				375						380		
Glu	Asn	Glu	Gln	Ala	Ala	Phe	His	Pro	Gly	His	Ile	Val	Pro	Gly	Leu
			385				390						400		
Asp	Phe	Thr	Asn	Asp	Pro	Leu	Leu	Gln	Gly	Arg	Leu	Phe	Ser	Tyr	Thr
			405				410						415		
Asp	Thr	Gln	Ile	Ser	Arg	Leu	Gly	Gly	Pro	Asn	Phe	His	Glu	Ile	Pro
			420				425						430		
Ile	Asn	Arg	Pro	Thr	Cys	Pro	Tyr	His	Asn	Phe	Gln	Arg	Asp	Gly	Met
			435				440						445		
His	Arg	Met	Gly	Ile	Asp	Thr	Asn	Pro	Ala	Asn	Tyr	Glu	Pro	Asn	Ser
			450				455						460		
Ile	Asn	Asp	Asn	Trp	Pro	Arg	Glu	Thr	Pro	Pro	Gly	Pro	Lys	Arg	Gly
			465				470						480		
Gly	Phe	Glu	Ser	Tyr	Gln	Glu	Arg	Val	Glu	Gly	Asn	Lys	Val	Arg	Glu
			485				490						495		
Arg	Ser	Pro	Ser	Phe	Gly	Glu	Tyr	Tyr	Ser	His	Pro	Arg	Leu	Phe	Trp
			500				505						510		
Leu	Ser	Gln	Thr	Pro	Phe	Glu	Gln	Arg	His	Ile	Val	Asp	Gly	Phe	Ser
			515				520						525		
Phe	Glu	Leu	Ser	Lys	Val	Val	Arg	Pro	Tyr	Ile	Arg	Glu	Arg	Val	Val
			530				535						540		
Asp	Gln	Leu	Ala	His	Ile	Asp	Leu	Thr	Leu	Ala	Gln	Ala	Val	Ala	Lys
			545				550						560		
Asn	Leu	Gly	Ile	Glu	Leu	Thr	Asp	Asp	Gln	Leu	Asn	Ile	Thr	Pro	Pro
			565				570						575		
Pro	Asp	Val	Asn	Gly	Leu	Lys	Lys	Asp	Pro	Ser	Leu	Ser	Leu	Tyr	Ala
			580				585						590		
Ile	Pro	Asp	Gly	Asp	Val	Lys	Gly	Arg	Val	Val	Ala	Ile	Leu	Leu	Asn
			595				600						605		
Asp	Glu	Val	Arg	Ser	Ala	Asp	Leu	Leu	Ala	Ile	Leu	Lys	Ala	Leu	Lys
			610				615						620		
Ala	Lys	Gly	Val	His	Ala	Lys	Leu	Leu	Tyr	Ser	Arg	Met	Gly	Glu	Val
			625				630						640		
Thr	Ala	Asp	Asp	Gly	Thr	Val	Leu	Pro	Ile	Ala	Ala	Thr	Phe	Ala	Gly
			645				650						655		
Ala	Pro	Ser	Leu	Thr	Val	Asp	Ala	Val	Ile	Val	Pro	Cys	Gly	Asn	Ile

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Ala Asp Ile Ala Asp Asn Gly Asp Ala Asn Tyr Tyr Leu Met Glu Ala
675 680 685

Tyr Lys His Leu Lys Pro Ile Ala Leu Ala Gly Asp Ala Arg Lys Phe
690 695 700

Lys Ala Thr Ile Lys Ile Ala Asp Gln Gly Glu Glu Gly Ile Val Glu
705 710 715 720

Ala Asp Ser Ala Asp Gly Ser Phe Met Asp Glu Leu Leu Thr Leu Met
725 730 735

Ala Ala His Arg Val Trp Ser Arg Ile Pro Lys Ile Asp Lys Ile Pro
740 745 750

Ala

<210> SEQ ID NO 8

<211> LENGTH: 492

<212> TYPE: PRT

<213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 8

Met Asp Pro Tyr Arg Val Arg Pro Ser Ser Ala His Asp Ser Pro Phe
1 5 10 15

Phe Thr Thr Asn Ser Gly Ala Pro Val Trp Asn Asn Asn Ser Ser Leu
20 25 30

Thr Val Gly Thr Arg Gly Pro Ile Leu Leu Glu Asp Tyr His Leu Leu
35 40 45

Glu Lys Leu Ala Asn Phe Asp Arg Glu Arg Ile Pro Glu Arg Val Val
50 55 60

His Ala Arg Gly Ala Ser Ala Lys Gly Phe Phe Glu Val Thr His Asp
65 70 75 80

Ile Thr Gln Leu Thr Ser Ala Asp Phe Leu Arg Gly Pro Gly Val Gln
85 90 95

Thr Pro Val Ile Val Arg Phe Ser Thr Val Ile His Glu Arg Gly Ser
100 105 110

Pro Glu Thr Leu Arg Asp Pro Arg Gly Phe Ala Val Lys Phe Tyr Thr
115 120 125

Arg Glu Gly Asn Phe Asp Leu Val Gly Asn Asn Phe Pro Val Phe Phe
130 135 140

Val Arg Asp Gly Met Lys Phe Pro Asp Met Val His Ala Leu Lys Pro
145 150 155 160

Asn Pro Lys Ser His Ile Gln Glu Asn Trp Arg Ile Leu Asp Phe Phe
165 170 175

Ser His His Pro Glu Ser Leu His Met Phe Ser Phe Leu Phe Asp Asp
180 185 190

Leu Gly Ile Pro Gln Asp Tyr Arg His Met Glu Gly Ala Gly Val Asn
195 200 205

Thr Tyr Met Leu Ile Asn Lys Ala Gly Lys Ala His Tyr Val Lys Phe
210 215 220

His Trp Lys Pro Thr Cys Gly Ile Lys Cys Leu Ser Asp Glu Glu Ala
225 230 235 240

Ile Arg Val Gly Gly Ala Asn His Ser His Ala Thr Lys Asp Leu Tyr
245 250 255

Asp Ser Ile Ala Ala Gly Asn Tyr Pro Gln Trp Asn Leu Phe Val Gln
260 265 270

Val Met Asp Pro Ala His Glu Asp Lys Phe Asp Phe Asp Pro Leu Asp
275 280 285

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Val	Thr	Lys	Ile	Trp	Pro	Glu	Asp	Ile	Leu	Pro	Leu	Gln	Pro	Val	Gly
290						295					300				
Arg	Leu	Val	Leu	Asn	Lys	Asn	Ile	Asp	Asn	Phe	Phe	Asn	Glu	Asn	Glu
305				310						315					320
Gln	Ile	Ala	Phe	Cys	Pro	Ala	Leu	Val	Val	Pro	Gly	Ile	His	Tyr	Ser
			325						330					335	
Asp	Asp	Lys	Leu	Leu	Gln	Thr	Arg	Ile	Phe	Ser	Tyr	Ala	Asp	Ser	Gln
		340						345					350		
Arg	His	Arg	Leu	Gly	Pro	Asn	Tyr	Leu	Gln	Leu	Pro	Val	Asn	Ala	Pro
		355					360					365			
Lys	Cys	Ala	His	His	Asn	Asn	His	His	Asp	Gly	Phe	Met	Asn	Phe	Met
	370				375						380				
His	Arg	Asp	Glu	Glu	Val	Asn	Tyr	Phe	Pro	Ser	Arg	Leu	Asp	Pro	Val
385					390					395					400
Arg	His	Ala	Glu	Lys	Tyr	Pro	Thr	Thr	Pro	Ile	Val	Cys	Ser	Gly	Asn
			405						410					415	
Arg	Glu	Lys	Cys	Phe	Ile	Gly	Lys	Glu	Asn	Asn	Phe	Lys	Gln	Pro	Gly
			420				425						430		
Glu	Arg	Tyr	Arg	Ser	Trp	Asp	Ser	Asp	Arg	Gln	Glu	Arg	Phe	Val	Lys
	435					440						445			
Arg	Phe	Val	Glu	Ala	Leu	Ser	Glu	Pro	Arg	Val	Thr	His	Glu	Ile	Arg
	450				455						460				
Ser	Ile	Trp	Ile	Ser	Tyr	Trp	Ser	Gln	Ala	Asp	Lys	Ser	Leu	Gly	Gln
465				470						475					480
Lys	Leu	Ala	Thr	Arg	Leu	Asn	Val	Arg	Pro	Asn	Phe				
		485						490							

<210> SEQ ID NO 9

<211> LENGTH: 492

<212> TYPE: PRT

<213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 9

Met	Asp	Pro	Tyr	Lys	Tyr	Arg	Pro	Ala	Ser	Ser	Tyr	Asn	Ser	Pro	Phe
1				5					10					15	
Phe	Thr	Thr	Asn	Ser	Gly	Ala	Pro	Val	Trp	Asn	Asn	Asn	Ser	Ser	Met
			20				25						30		
Thr	Val	Gly	Pro	Arg	Gly	Pro	Ile	Leu	Leu	Glu	Asp	Tyr	His	Leu	Val
	35					40						45			
Glu	Lys	Leu	Ala	Asn	Phe	Asp	Arg	Glu	Arg	Ile	Pro	Glu	Arg	Val	Val
	50				55					60					
His	Ala	Arg	Gly	Ala	Ser	Ala	Lys	Gly	Phe	Phe	Glu	Val	Thr	His	Asp
65				70					75					80	
Ile	Ser	Asn	Leu	Thr	Cys	Ala	Asp	Phe	Leu	Arg	Ala	Pro	Gly	Val	Gln
			85					90						95	
Thr	Pro	Val	Ile	Val	Arg	Phe	Ser	Thr	Val	Ile	His	Glu	Arg	Gly	Ser
		100					105					110			
Pro	Glu	Thr	Leu	Arg	Asp	Pro	Arg	Gly	Phe	Ala	Val	Lys	Phe	Tyr	Thr
	115					120						125			
Arg	Glu	Gly	Asn	Phe	Asp	Leu	Val	Gly	Asn	Asn	Phe	Pro	Val	Phe	Phe
	130					135					140				
Ile	Arg	Asp	Gly	Met	Lys	Phe	Pro	Asp	Met	Val	His	Ala	Leu	Lys	Pro
145					150				155					160	
Asn	Pro	Lys	Ser	His	Ile	Gln	Glu	Asn	Trp	Arg	Ile	Leu	Asp	Phe	Phe
				165					170					175	

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Ser His His Pro Glu Ser Leu Asn Met Phe Thr Phe Leu Phe Asp Asp
    180                                185                                190

Ile Gly Ile Pro Gln Asp Tyr Arg His Met Asp Gly Ser Gly Val Asn
    195                                200                                205

Thr Tyr Met Leu Ile Asn Lys Ala Gly Lys Ala His Tyr Val Lys Phe
    210                                215                                220

His Trp Lys Pro Thr Cys Gly Val Lys Ser Leu Leu Glu Glu Asp Ala
    225                                230                                235                                240

Ile Arg Val Gly Gly Thr Asn His Ser His Ala Thr Gln Asp Leu Tyr
    245                                250                                255

Asp Ser Ile Ala Ala Gly Asn Tyr Pro Glu Trp Lys Leu Phe Ile Gln
    260                                265                                270

Ile Ile Asp Pro Ala Asp Glu Asp Lys Phe Asp Phe Asp Pro Leu Asp
    275                                280                                285

Val Thr Lys Thr Trp Pro Glu Asp Ile Leu Pro Leu Gln Pro Val Gly
    290                                295                                300

Arg Met Val Leu Asn Lys Asn Ile Asp Asn Phe Phe Ala Glu Asn Glu
    305                                310                                315                                320

Gln Leu Ala Phe Cys Pro Ala Ile Ile Val Pro Gly Ile His Tyr Ser
    325                                330                                335

Asp Asp Lys Leu Leu Gln Thr Arg Val Phe Ser Tyr Ala Asp Thr Gln
    340                                345                                350

Arg His Arg Leu Gly Pro Asn Tyr Leu Gln Leu Pro Val Asn Ala Pro
    355                                360                                365

Lys Cys Ala His His Asn Asn His His Glu Gly Phe Met Asn Phe Met
    370                                375                                380

His Arg Asp Glu Glu Val Asn Tyr Phe Pro Ser Arg Tyr Asp Gln Val
    385                                390                                395                                400

Arg His Ala Glu Lys Tyr Pro Thr Pro Pro Ala Val Cys Ser Gly Lys
    405                                410                                415

Arg Glu Arg Cys Ile Ile Glu Lys Glu Asn Asn Phe Lys Glu Pro Gly
    420                                425                                430

Glu Arg Tyr Arg Thr Phe Thr Pro Glu Arg Gln Glu Arg Phe Ile Gln
    435                                440                                445

Arg Trp Ile Asp Ala Leu Ser Asp Pro Arg Ile Thr His Glu Ile Arg
    450                                455                                460

Ser Ile Trp Ile Ser Tyr Trp Ser Gln Ala Asp Lys Ser Leu Gly Gln
    465                                470                                475                                480

Lys Leu Ala Ser Arg Leu Asn Val Arg Pro Ser Ile
    485                                490

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<210> SEQ ID NO 10

<211> LENGTH: 492

<212> TYPE: PRT

<213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 10

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Met Asp Pro Tyr Lys Tyr Arg Pro Ser Ser Ala Tyr Asn Ala Pro Phe
1      5      10      15

Tyr Thr Thr Asn Gly Gly Ala Pro Val Ser Asn Asn Ile Ser Ser Leu
20     25     30

Thr Ile Gly Glu Arg Gly Pro Val Leu Leu Glu Asp Tyr His Leu Ile
35     40     45

Glu Lys Val Ala Asn Phe Thr Arg Glu Arg Ile Pro Glu Arg Val Val

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50	55	60
His Ala Arg Gly Ile Ser Ala Lys Gly Phe Phe Glu Val Thr His Asp 65 70 75 80		
Ile Ser Asn Leu Thr Cys Ala Asp Phe Leu Arg Ala Pro Gly Val Gln 85 90 95		
Thr Pro Val Ile Val Arg Phe Ser Thr Val Val His Glu Arg Ala Ser 100 105 110		
Pro Glu Thr Met Arg Asp Ile Arg Gly Phe Ala Val Lys Phe Tyr Thr 115 120 125		
Arg Glu Gly Asn Phe Asp Leu Val Gly Asn Asn Thr Pro Val Phe Phe 130 135 140		
Ile Arg Asp Gly Ile Gln Phe Pro Asp Val Val His Ala Leu Lys Pro 145 150 155 160		
Asn Pro Lys Thr Asn Ile Gln Glu Tyr Trp Arg Ile Leu Asp Tyr Met 165 170 175		
Ser His Leu Pro Glu Ser Leu Leu Thr Trp Cys Trp Met Phe Asp Asp 180 185 190		
Val Gly Ile Pro Gln Asp Tyr Arg His Met Glu Gly Phe Gly Val His 195 200 205		
Thr Tyr Thr Leu Ile Ala Lys Ser Gly Lys Val Leu Phe Val Lys Phe 210 215 220		
His Trp Lys Pro Thr Cys Gly Ile Lys Asn Leu Thr Asp Glu Glu Ala 225 230 235 240		
Lys Val Val Gly Gly Ala Asn His Ser His Ala Thr Lys Asp Leu His 245 250 255		
Asp Ala Ile Ala Ser Gly Asn Tyr Pro Glu Trp Lys Leu Phe Ile Gln 260 265 270		
Thr Met Asp Pro Ala Asp Glu Asp Lys Phe Asp Phe Asp Pro Leu Asp 275 280 285		
Val Thr Lys Ile Trp Pro Glu Asp Ile Leu Pro Leu Gln Pro Val Gly 290 295 300		
Arg Leu Val Leu Asn Arg Thr Ile Asp Asn Phe Phe Asn Glu Thr Glu 305 310 315 320		
Gln Leu Ala Phe Asn Pro Gly Leu Val Val Pro Gly Ile Tyr Tyr Ser 325 330 335		
Asp Asp Lys Leu Leu Gln Cys Arg Ile Phe Ala Tyr Gly Asp Thr Gln 340 345 350		
Arg His Arg Leu Gly Pro Asn Tyr Leu Gln Leu Pro Val Asn Ala Pro 355 360 365		
Lys Cys Ala His His Asn Asn His His Glu Gly Phe Met Asn Phe Met 370 375 380		
His Arg Asp Glu Glu Ile Asn Tyr Tyr Pro Ser Lys Phe Asp Pro Val 385 390 395 400		
Arg Cys Ala Glu Lys Val Pro Thr Pro Thr Asn Ser Tyr Thr Gly Ile 405 410 415		
Arg Thr Lys Cys Val Ile Lys Lys Glu Asn Asn Phe Lys Gln Ala Gly 420 425 430		
Asp Arg Tyr Arg Ser Trp Ala Pro Asp Arg Gln Asp Arg Phe Val Lys 435 440 445		
Arg Trp Val Glu Ile Leu Ser Glu Pro Arg Leu Thr His Glu Ile Arg 450 455 460		
Gly Ile Trp Ile Ser Tyr Trp Ser Gln Ala Asp Arg Ser Leu Gly Gln 465 470 475 480		

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Lys Leu Ala Ser Arg Leu Asn Val Arg Pro Ser Ile
485 490

<210> SEQ ID NO 11
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Cannabis

<400> SEQUENCE: 11

Met Asn Cys Ser Ala Phe Ser Phe Trp Phe Val Cys Lys Ile Ile Phe
1 5 10 15

Phe Phe Leu Ser Phe His Ile Gln Ile Ser Ile Ala
20 25

<210> SEQ ID NO 12
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: Cannabis

<400> SEQUENCE: 12

Met Lys Cys Ser Thr Phe Ser Phe Trp Phe Val Cys Lys Ile Ile Phe
1 5 10 15

Phe Phe Phe Ser Phe Asn Ile Gln Thr Ser Ile Ala
20 25

<210> SEQ ID NO 13
<211> LENGTH: 517
<212> TYPE: PRT
<213> ORGANISM: Cannabis

<400> SEQUENCE: 13

Asn Pro Arg Glu Asn Phe Leu Lys Cys Phe Ser Lys His Ile Pro Asn
1 5 10 15

Asn Val Ala Asn Pro Lys Leu Val Tyr Thr Gln His Asp Gln Leu Tyr
20 25 30

Met Ser Ile Leu Asn Ser Thr Ile Gln Asn Leu Arg Phe Ile Ser Asp
35 40 45

Thr Thr Pro Lys Pro Leu Val Ile Val Thr Pro Ser Asn Asn Ser His
50 55 60

Ile Gln Ala Thr Ile Leu Cys Ser Lys Lys Val Gly Leu Gln Ile Arg
65 70 75 80

Thr Arg Ser Gly Gly His Asp Ala Glu Gly Met Ser Tyr Ile Ser Gln
85 90 95

Val Pro Phe Val Val Val Asp Leu Arg Asn Met His Ser Ile Lys Ile
100 105 110

Asp Val His Ser Gln Thr Ala Trp Val Glu Ala Gly Ala Thr Leu Gly
115 120 125

Glu Val Tyr Tyr Trp Ile Asn Glu Lys Asn Glu Asn Leu Ser Phe Pro
130 135 140

Gly Gly Tyr Cys Pro Thr Val Gly Val Gly Gly His Phe Ser Gly Gly
145 150 155 160

Gly Tyr Gly Ala Leu Met Arg Asn Tyr Gly Leu Ala Ala Asp Asn Ile
165 170 175

Ile Asp Ala His Leu Val Asn Val Asp Gly Lys Val Leu Asp Arg Lys
180 185 190

Ser Met Gly Glu Asp Leu Phe Trp Ala Ile Arg Gly Gly Gly Gly Glu
195 200 205

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Asn Phe Gly Ile Ile Ala Ala Trp Lys Ile Lys Leu Val Asp Val Pro
 210 215 220
 Ser Lys Ser Thr Ile Phe Ser Val Lys Lys Asn Met Glu Ile His Gly
 225 230 235 240
 Leu Val Lys Leu Phe Asn Lys Trp Gln Asn Ile Ala Tyr Lys Tyr Asp
 245 250 255
 Lys Asp Leu Val Leu Met Thr His Phe Ile Thr Lys Asn Ile Thr Asp
 260 265 270
 Asn His Gly Lys Asn Lys Thr Thr Val His Gly Tyr Phe Ser Ser Ile
 275 280 285
 Phe His Gly Gly Val Asp Ser Leu Val Asp Leu Met Asn Lys Ser Phe
 290 295 300
 Pro Glu Leu Gly Ile Lys Lys Thr Asp Cys Lys Glu Phe Ser Trp Ile
 305 310 315 320
 Asp Thr Thr Ile Phe Tyr Ser Gly Val Val Asn Phe Asn Thr Ala Asn
 325 330 335
 Phe Lys Lys Glu Ile Leu Leu Asp Arg Ser Ala Gly Lys Lys Thr Ala
 340 345 350
 Phe Ser Ile Lys Leu Asp Tyr Val Lys Lys Pro Ile Pro Glu Thr Ala
 355 360 365
 Met Val Lys Ile Leu Glu Lys Leu Tyr Glu Glu Asp Val Gly Ala Gly
 370 375 380
 Met Tyr Val Leu Tyr Pro Tyr Gly Gly Ile Met Glu Glu Ile Ser Glu
 385 390 395 400
 Ser Ala Ile Pro Phe Pro His Arg Ala Gly Ile Met Tyr Glu Leu Trp
 405 410 415
 Tyr Thr Ala Ser Trp Glu Lys Gln Glu Asp Asn Glu Lys His Ile Asn
 420 425 430
 Trp Val Arg Ser Val Tyr Asn Phe Thr Thr Pro Tyr Val Ser Gln Asn
 435 440 445
 Pro Arg Leu Ala Tyr Leu Asn Tyr Arg Asp Leu Asp Leu Gly Lys Thr
 450 455 460
 Asn His Ala Ser Pro Asn Asn Tyr Thr Gln Ala Arg Ile Trp Gly Glu
 465 470 475 480
 Lys Tyr Phe Gly Lys Asn Phe Asn Arg Leu Val Lys Val Lys Thr Lys
 485 490 495
 Val Asp Pro Asn Asn Phe Phe Arg Asn Glu Gln Ser Ile Pro Pro Leu
 500 505 510
 Pro Pro His His His
 515

<210> SEQ ID NO 14
 <211> LENGTH: 1554
 <212> TYPE: DNA
 <213> ORGANISM: Cannabis sativa

<400> SEQUENCE: 14

atgaatcctc gagaaactt ccttaaagtc ttctcgcaat atattcccaa taatgcaaca	60
aatctaaaac tcgtatacac tcaaaacaac ccattgtata tgtctgtcct aaattcgaca	120
atacacaatc ttagattcac ctctgacaca accccaaaac cacttggtat cgtcactcct	180
tcacatgtct ctcatatcca aggcactatt ctatgtcca agaaagttgg cttgcagatt	240
cgaactcgaa gtggtggtca tgattctgag ggcatgtcct acatatctca agtcccattt	300
gttatagtag acttgagaaa catgcggtca atcaaaatag atgttcatag ccaaactgca	360

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tgggttgaag cgggagctac ccttgagaa gtttattatt gggttaatga gaaaaatgag 420
aatcttagtt tggcggtctg gtattgccct actgtttgcg caggtggaca ctttgggtga 480
ggaggtatg gaccattgat gagaaactat ggccctcgcg ctgataatat cattgatgca 540
cacttagtca acgttcattg aaaagtgcga gatcgaaaat ctatggggga agatctcttt 600
tgggctttac gtggtggtgg agcagaaagc ttcggaatca ttgtagcatg gaaaattaga 660
ctggttgctg toccaaagtc tactatgttt agtggttaaa agatcatgga gatacatgag 720
cttgtcaagt tagttaacaa atggcaaaat attgcttaca agtatgacaa agatttatta 780
ctcatgactc acttcataac taggaacatt acagataatc aagggaagaa taagacagca 840
atacacactt acttctcttc agttttcctt ggtggagtgg atagtctagt cgacttgatg 900
aacaagagtt ttcctgagtt ggggtattaaa aaaacggatt gcagacaatt gagctggatt 960
gatactatca tcttctatag tgggtgtgta aattacgaca ctgataattt taacaaggaa 1020
atthtgcctg atagatccgc tgggcagaaac ggtgctttca agattaagtt agactacgtt 1080
aagaaaccaa ttcagaaatc tgtatttgtc caaattttgg aaaaattata tgaagaagat 1140
ataggagctg ggatgtatgc gttgtaccct tacggtggta taatggatga gatttcagaa 1200
tcagcaatc cattccctca tcgagctgga atcttgatg agttatggta catatgtagt 1260
tgggagaagc aagaagataa cgaagagcat ctaaactgga ttagaaatat ttataacttc 1320
atgactcctt atgtgtccaa aaatccaaga ttggcatatc tcaattatag agaccttgat 1380
ataggaataa atgatcccaa gaatccaaat aattacacac aagcacgtat ttggggtgag 1440
aagtattttg gtaaaaattt tgacaggcta gtaaaagtga aaacctgggt tgatcccaat 1500
aactttttta gaaacgaaca aagcatccca cctctaccac ggcacgtca ttaa 1554

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<210> SEQ ID NO 15

<211> LENGTH: 517

<212> TYPE: PRT

<213> ORGANISM: Cannabis sativa

<400> SEQUENCE: 15

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Met Asn Pro Arg Glu Asn Phe Leu Lys Cys Phe Ser Gln Tyr Ile Pro
1           5           10          15

Asn Asn Ala Thr Asn Leu Lys Leu Val Tyr Thr Gln Asn Asn Pro Leu
20          25          30

Tyr Met Ser Val Leu Asn Ser Thr Ile His Asn Leu Arg Phe Thr Ser
35          40          45

Asp Thr Thr Pro Lys Pro Leu Val Ile Val Thr Pro Ser His Val Ser
50          55          60

His Ile Gln Gly Thr Ile Leu Cys Ser Lys Lys Val Gly Leu Gln Ile
65          70          75          80

Arg Thr Arg Ser Gly Gly His Asp Ser Glu Gly Met Ser Tyr Ile Ser
85          90          95

Gln Val Pro Phe Val Ile Val Asp Leu Arg Asn Met Arg Ser Ile Lys
100         105         110

Ile Asp Val His Ser Gln Thr Ala Trp Val Glu Ala Gly Ala Thr Leu
115         120         125

Gly Glu Val Tyr Tyr Trp Val Asn Glu Lys Asn Glu Asn Leu Ser Leu
130         135         140

Ala Ala Gly Tyr Cys Pro Thr Val Cys Ala Gly Gly His Phe Gly Gly
145         150         155         160

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aacgacatcg tatcatcatc atcatcaaca acaacaacat catcaacaac tacagcaaca 120
tcatcatttc ataatgagaa agttactgtc agtactgata atattattaa tcttgatgat 180
aagcagaaac gacaattatg tcgttgctgt ttagaaaaag aagaagaaga agaaggaagt 240
ggtggttggt gtgagacagt agtaatgatg ctagggtcag tatctcctgc tgctgctact 300
gctgctgcag ctgggggctc atcaagttgt gatgaagaca tggtgggtgg tcatgatcaa 360
ctgttggtgt tgtgtgttgc tgagaaaaaa acgacagaaa tttcatcagt ggtgaacttt 420
aataataata ataataataa taaggaaaat ggtgacgaag tttcaggacc gtacgattat 480
catcatcata aagaagagga agaagaagaa gaagaagatg aagcatctgc atcagtagca 540
gctgttgatg aagggatggt gttgtgcttt gatgacataa tagatagcca cttgctaaat 600
ccaaatgagg ttttgacttt aagagaagat agccataatg aaggtggggc agctgatcag 660
attgacaaga ctacttgtaa taatactact attactacta atgatgatta taacaataac 720
ttgatgatgt tgagctgcaa taataacgga gattatgtta ttagtgatga tcatgatgat 780
cagtactgga tagacgacgt cgttgaggtt gacttttgga gttgggagag ttcgactact 840
actgttatta cccaagaaca agaacaagaa caagatcaag ttcaagaaca gaagaatatg 900
tgggataatg agaagagaa actgttgtct ttgctatggg ataatagtga taacagcagc 960
agttgggagt tacaagataa aagcaataat aataataata ataatgttcc taacaaatgt 1020
caagagatta cctctgataa agaaaatgct atggttgcac ggcttctctc ctga 1074

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<210> SEQ ID NO 17

<211> LENGTH: 357

<212> TYPE: PRT

<213> ORGANISM: Cannabis

<400> SEQUENCE: 17

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Met Lys Lys Asn Lys Ser Thr Ser Asn Asn Lys Asn Asn Ser Asn
1          5          10          15
Asn Ile Ile Lys Asn Asp Ile Val Ser Ser Ser Ser Ser Thr Thr Thr
20          25          30
Thr Ser Ser Thr Thr Thr Ala Thr Ser Ser Phe His Asn Glu Lys Val
35          40          45
Thr Val Ser Thr Asp His Ile Ile Asn Leu Asp Asp Lys Gln Lys Arg
50          55          60
Gln Leu Cys Arg Cys Arg Leu Glu Lys Glu Glu Glu Glu Gly Ser
65          70          75          80
Gly Gly Cys Gly Glu Thr Val Val Met Met Leu Gly Ser Val Ser Pro
85          90          95
Ala Ala Ala Thr Ala Ala Ala Ala Gly Gly Ser Ser Ser Cys Asp Glu
100         105         110
Asp Met Leu Gly Gly His Asp Gln Leu Leu Leu Leu Cys Cys Ser Glu
115        120        125
Lys Lys Thr Thr Glu Ile Ser Ser Val Val Asn Phe Asn Asn Asn Asn
130        135        140
Asn Asn Asn Lys Glu Asn Gly Asp Glu Val Ser Gly Pro Tyr Asp Tyr
145        150        155        160
His His His Lys Glu Glu Glu Glu Glu Glu Glu Asp Glu Ala Ser
165        170        175
Ala Ser Val Ala Ala Val Asp Glu Gly Met Leu Leu Cys Phe Asp Asp
180        185        190
Ile Ile Asp Ser His Leu Leu Asn Pro Asn Glu Val Leu Thr Leu Arg

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195	200	205
Glu Asp Ser His Asn Glu Gly Gly Ala Ala Asp Gln Ile Asp Lys Thr		
210	215	220
Thr Cys Asn Asn Thr Thr Ile Thr Thr Asn Asp Asp Tyr Asn Asn Asn		
225	230	235
Leu Met Met Leu Ser Cys Asn Asn Asn Gly Asp Tyr Val Ile Ser Asp		
	245	250
Asp His Asp Asp Gln Tyr Trp Ile Asp Asp Val Val Gly Val Asp Phe		
	260	265
Trp Ser Trp Glu Ser Ser Thr Thr Thr Val Ile Thr Gln Glu Gln Glu		
	275	280
Gln Glu Gln Asp Gln Val Gln Glu Gln Lys Asn Met Trp Asp Asn Glu		
	290	295
Lys Glu Lys Leu Leu Ser Leu Leu Trp Asp Asn Ser Asp Asn Ser Ser		
305	310	315
Ser Trp Glu Leu Gln Asp Lys Ser Asn Asn Asn Asn Asn Asn Val		
	325	330
Pro Asn Lys Cys Gln Glu Ile Thr Ser Asp Lys Glu Asn Ala Met Val		
	340	345
Ala Trp Leu Leu Ser		
355		
<210> SEQ ID NO 18		
<211> LENGTH: 462		
<212> TYPE: PRT		
<213> ORGANISM: Humulus lupulus		
<400> SEQUENCE: 18		
Met Gly Arg Ala Pro Cys Cys Glu Lys Val Gly Leu Lys Lys Gly Arg		
1	5	10
Trp Thr Ser Glu Glu Asp Glu Ile Leu Thr Lys Tyr Ile Gln Ser Asn		
	20	25
Gly Glu Gly Cys Trp Arg Ser Leu Pro Lys Asn Ala Gly Leu Leu Arg		
	35	40
Cys Gly Lys Ser Cys Arg Leu Arg Trp Ile Asn Tyr Leu Arg Ala Asp		
	50	55
Leu Lys Arg Gly Asn Ile Ser Ser Glu Glu Glu Asp Ile Ile Ile Lys		
	65	70
Leu His Ser Thr Leu Gly Asn Arg Trp Ser Leu Ile Ala Ser His Leu		
	85	90
Pro Gly Arg Thr Asp Asn Glu Ile Lys Asn Tyr Trp Asn Ser His Leu		
	100	105
Ser Arg Lys Ile His Thr Phe Arg Arg Cys Asn Asn Thr Thr Thr His		
	115	120
His His His Leu Pro Asn Leu Val Thr Val Thr Lys Val Asn Leu Pro		
	130	135
Ile Pro Lys Arg Lys Gly Gly Arg Thr Ser Arg Leu Ala Met Lys Lys		
	145	150
Asn Lys Ser Ser Thr Ser Asn Gln Asn Ser Ser Val Ile Lys Asn Asp		
	165	170
Val Gly Ser Ser Ser Ser Thr Thr Thr Thr Ser Val His Gln Arg Thr		
	180	185
Thr Thr Thr Thr Pro Thr Met Asp Asp Gln Gln Lys Arg Gln Leu Ser		
	195	200
		205

Arg 210	Cys	Arg	Leu	Glu	Glu	Lys 215	Glu	Asp	Gln	Asp	Gly 220	Ala	Ser	Thr	Gly
Thr 225	Val	Val	Met	Met	Leu	Gly 230	Gln	Ala	Ala	Ala	Val	Gly	Ser	Ser	Cys 240
Asp	Glu	Asp	Met	Leu 245	Gly	His	Asp	Gln	Leu 250	Ser	Phe	Leu	Cys	Cys 255	Ser
Glu	Glu	Lys	Thr 260	Thr	Glu	Asn	Ser	Met	Thr	Asn	Leu	Lys	Glu	Asn	Gly
Asp	His	Glu 275	Val	Ser	Gly	Pro	Tyr 280	Asp	Tyr	Asp	His	Arg 285	Tyr	Glu	Lys
Glu 290	Thr	Ser	Val	Asp	Glu	Gly 295	Met	Leu	Leu	Cys	Phe 300	Asn	Asp	Ile	Ile
Asp 305	Ser	Asn	Leu	Leu	Asn 310	Pro	Asn	Glu	Val	Leu 315	Thr	Leu	Ser	Glu	Glu 320
Ser	Leu	Asn	Leu	Gly 325	Gly	Ala	Leu	Met	Asp 330	Thr	Thr	Thr	Ser	Thr 335	Thr
Thr	Asn	Asn	Asn 340	Asn	Tyr	Ser	Leu	Ser	Tyr 345	Asn	Asn	Asn	Gly 350	Asp	Cys
Val	Ile	Ser 355	Asp	Asp	His	Asp	Gln 360	Tyr	Trp	Leu	Asp 365	Asp	Val	Val	Gly
Val 370	Asp	Phe	Trp	Ser	Trp	Glu 375	Ser	Ser	Thr	Thr	Val 380	Thr	Gln	Glu	Gln
Glu 385	Gln	Glu	Gln	Glu	Gln 390	Glu	Gln	Glu	Gln	Glu 395	Gln	Glu	Gln	Glu	Gln 400
Glu	Gln	Glu	His 405	His	His	Gln	Gln	Asp	Gln 410	Lys	Lys	Asn	Thr	Trp 415	Asp
Asn	Glu	Lys 420	Glu	Lys	Met	Leu	Ala	Leu	Leu 425	Trp	Asp	Ser	Asp	Asn	Ser
Asn	Trp	Glu 435	Leu	Gln	Asp	Asn	Asn	Asn	Tyr 440	His	Lys	Cys 445	Gln	Glu	Ile
Thr 450	Ser	Asp	Lys	Glu	Asn	Ala 455	Met	Val	Ala	Trp	Leu 460	Leu	Ser		

<400> SEQUENCE: 19

Met	Gly	Arg	Ala	Pro	Cys	Cys	Glu	Lys	Val	Gly	Ile	Lys	Arg	Gly	Arg
1				5					10					15	
Trp	Thr	Ala	Glu	Glu	Asp	Gln	Ile	Leu	Ser	Asn	Tyr	Ile	Gln	Ser	Asn
			20					25					30		
Gly	Glu	Gly	Ser	Trp	Arg	Ser	Leu	Pro	Lys	Asn	Ala	Gly	Leu	Lys	Arg
		35					40					45			
Cys	Gly	Lys	Ser	Cys	Arg	Leu	Arg	Trp	Ile	Asn	Tyr	Leu	Arg	Ser	Asp
	50					55					60				
Leu	Lys	Arg	Gly	Asn	Ile	Thr	Pro	Glu	Glu	Glu	Glu	Leu	Val	Val	Lys
65					70					75					80
Leu	His	Ser	Thr	Leu	Gly	Asn	Arg	Trp	Ser	Leu	Ile	Ala	Gly	His	Leu
				85					90					95	
Pro	Gly	Arg	Thr	Asp	Asn	Glu	Ile	Lys	Asn	Tyr	Trp	Asn	Ser	His	Leu
			100					105					110		
Ser	Arg	Lys	Leu	His	Asn	Phe	Ile	Arg	Lys	Pro	Ser	Ile	Ser	Gln	Asp
		115					120					125			

Val	Ser	Ala	Val	Ile	Met	Thr	Asn	Ala	Ser	Ser	Ala	Pro	Pro	Pro	Pro
130						135						140			
Gln	Ala	Lys	Arg	Arg	Leu	Gly	Arg	Thr	Ser	Arg	Ser	Ala	Met	Lys	Pro
145						150						155			
Lys	Ile	His	Arg	Thr	Lys	Thr	Arg	Lys	Thr	Lys	Lys	Thr	Ser	Ala	Pro
165						170						175			
Pro	Glu	Pro	Asn	Ala	Asp	Val	Ala	Gly	Ala	Asp	Lys	Glu	Ala	Leu	Met
180						185						190			
Val	Glu	Ser	Ser	Gly	Ala	Glu	Ala	Glu	Leu	Gly	Arg	Pro	Cys	Asp	Tyr
195						200						205			
Tyr	Gly	Asp	Asp	Cys	Asn	Lys	Asn	Leu	Met	Ser	Ile	Asn	Gly	Asp	Asn
210						215						220			
Gly	Val	Leu	Thr	Phe	Asp	Asp	Asp	Ile	Ile	Asp	Leu	Leu	Leu	Asp	Glu
225						230						235			
Ser	Asp	Pro	Gly	His	Leu	Tyr	Thr	Asn	Thr	Thr	Cys	Gly	Gly	Asp	Gly
245						250						255			
Glu	Leu	His	Asn	Ile	Arg	Asp	Ser	Glu	Gly	Ala	Arg	Gly	Phe	Ser	Asp
260						265						270			
Thr	Trp	Asn	Gln	Gly	Asn	Leu	Asp	Cys	Leu	Leu	Gln	Ser	Cys	Pro	Ser
275						280						285			
Val	Glu	Ser	Phe	Leu	Asn	Tyr	Asp	His	Gln	Val	Asn	Asp	Ala	Ser	Thr
290						295						300			
Asp	Glu	Phe	Ile	Asp	Trp	Asp	Cys	Val	Trp	Gln	Glu	Gly	Ser	Asp	Asn
305						310						315			
Asn	Leu	Trp	His	Glu	Lys	Glu	Asn	Pro	Asp	Ser	Met	Val	Ser	Trp	Leu
325						330						335			
Leu	Asp	Gly	Asp	Asp	Glu	Ala	Thr	Ile	Gly	Asn	Ser	Asn	Cys	Glu	Asn
340						345						350			
Phe	Gly	Glu	Pro	Leu	Asp	His	Asp	Asp	Glu	Ser	Ala	Leu	Val	Ala	Trp
355						360						365			
Leu	Leu	Ser													
370															

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<210> SEQ ID NO 20
<211> LENGTH: 243
<212> TYPE: PRT
<213> ORGANISM: Arabidopsis thaliana
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<400> SEQUENCE: 20

Met	Asn	Ile	Ser	Arg	Thr	Glu	Phe	Ala	Asn	Cys	Lys	Thr	Leu	Ile	Asn
1				5					10					15	
His	Lys	Glu	Glu	Val	Glu	Glu	Val	Glu	Lys	Lys	Met	Glu	Ile	Glu	Ile
			20					25					30		
Arg	Arg	Gly	Pro	Trp	Thr	Val	Glu	Glu	Asp	Met	Lys	Leu	Val	Ser	Tyr
		35					40					45			
Ile	Ser	Leu	His	Gly	Glu	Gly	Arg	Trp	Asn	Ser	Leu	Ser	Arg	Ser	Ala
	50					55					60				
Gly	Leu	Asn	Arg	Thr	Gly	Lys	Ser	Cys	Arg	Leu	Arg	Trp	Leu	Asn	Tyr
65					70					75				80	
Leu	Arg	Pro	Asp	Ile	Arg	Arg	Gly	Asp	Ile	Ser	Leu	Gln	Glu	Gln	Phe
				85					90					95	
Ile	Ile	Leu	Glu	Leu	His	Ser	Arg	Trp	Gly	Asn	Arg	Trp	Ser	Lys	Ile
		100						105					110		
Ala	Gln	His	Leu	Pro	Gly	Arg	Thr	Asp	Asn	Glu	Ile	Lys	Asn	Tyr	Trp

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115	120	125
Arg Thr Arg Val Gln Lys His Ala Lys Leu Leu Lys Cys Asp Val Asn		
130	135	140
Ser Lys Gln Phe Lys Asp Thr Ile Lys His Leu Trp Met Pro Arg Leu		
145	150	155
Ile Glu Arg Ile Ala Ala Thr Gln Ser Val Gln Phe Thr Ser Asn His		
165	170	175
Tyr Ser Pro Glu Asn Ser Ser Val Ala Thr Ala Thr Ser Ser Thr Ser		
180	185	190
Ser Ser Glu Ala Val Arg Ser Ser Phe Tyr Gly Gly Asp Gln Val Glu		
195	200	205
Phe Gly Thr Leu Asp His Met Thr Asn Gly Gly Tyr Trp Phe Asn Gly		
210	215	220
Gly Asp Thr Phe Glu Thr Leu Cys Ser Phe Asp Glu Leu Asn Lys Trp		
225	230	235
240		
Leu Ile Gln		

<210> SEQ ID NO 21
 <211> LENGTH: 127
 <212> TYPE: PRT
 <213> ORGANISM: Homo Sapiens

<400> SEQUENCE: 21

Met Ser Phe Ser Gly Lys Tyr Gln Leu Gln Ser Gln Glu Asn Phe Glu	
1	15
Ala Phe Met Lys Ala Ile Gly Leu Pro Glu Glu Leu Ile Gln Lys Gly	
20	30
Lys Asp Ile Lys Gly Val Ser Glu Ile Val Gln Asn Gly Lys His Phe	
35	45
Lys Phe Thr Ile Thr Ala Gly Ser Lys Val Ile Gln Asn Glu Phe Thr	
50	60
Val Gly Glu Glu Cys Glu Leu Glu Thr Met Thr Gly Glu Lys Val Lys	
65	80
Thr Val Val Gln Leu Glu Gly Asp Asn Lys Leu Val Thr Thr Phe Lys	
85	95
Asn Ile Lys Ser Val Thr Glu Leu Asn Gly Asp Ile Ile Thr Asn Thr	
100	110
Met Thr Leu Gly Asp Ile Val Phe Lys Arg Ile Ser Lys Arg Ile	
115	125

<210> SEQ ID NO 22
 <211> LENGTH: 1515
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 22

atgaacttgt tttctgcttt gtctttggat actttgggtt tgttggtat tattttgggt	60
ttgttgtaca gatacgttac tagaactcat ggtttgttta agaagcaagg tattccaggt	120
ccaaagccat tgccattttt gggtactgtt ttgaactact aactgggtat ttggaagttt	180
gatatggaat gttacgaaaa gtacggtaag acttgggggtt tgtttgatgg tcaaactcca	240
ttgttggtta ttactgatcc agaaactatt aagaacgttt tggtaagga ttgtttgtct	300
gtttttacta acagaagaga atttgggtcca gttggtatta tgtctaaggc tatttctatt	360
tctaaggatg aagaatggaa gagatacaga gctttgttgt ctccaacttt tactttcgtt	420

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agattgaagg aaatgtttcc agttattgaa caatacgggtg atattttggt taagtacttg 480
agacaagaag ctgaaaaggg tatgccagtt gctatgaagg atgttttggg tgcttactct 540
atggatgtta ttacttttac ttcttttggg gttaacgttg attctttgaa caaccagaa 600
gatccatttg ttgaagaagc taagaagttt ttgagagttg atttttttga tccattgttg 660
ttttctgttg tttgtttcc attgttgact ccagtttacg aaatgttgaa catttgatatg 720
tttccaaagc attctattga attttttaag aagtttgttg atagaatgca agaactctaga 780
ttggattcta accaaaagca tagagttgat tttttgcaat tgatgatgaa ctctcataac 840
aactctaagg ataaggattc tcataaggct ttttctaaca tggaaattac tgttcaatct 900
attattttta tttctgtggt ttacgaaact acttcttcta ctttgtcttt tactttgtac 960
tgtttggtgta ctcatccaga tattcaaaag aagttgcaag ctgaaattga taaggctttg 1020
ccaaacaagg ctactccaac ttgtgatact gttatggaaa tggaaatactt ggatattggtt 1080
ttgaacgaaa ctttgagatt gtaccaatt gttactagat tggaaagagt ttgtaagaag 1140
gatgttgaat tgaacggtgt ttacattcca aagggttcta tggttatgat tccatcttac 1200
gctttgcacg atgatccaca acattggcca gatccagaag aatttcaacc agaaagattt 1260
tctaaggaaa acaagggttc tattgatcca tacgtttact tgccatttgg tattggtcca 1320
agaaactgta ttggtatgag atttgctttg atgaacatga agttggtgtg tactaagggt 1380
ttgcaaaact tttcttttca accatgtcaa gaaactcaaa ttccattgaa gttgtctaga 1440
caaggtatgt tgcaaccaga aaagccaatt gttttgaagg ttgttccaag agatgctgtt 1500
attactggtg cttaa 1515

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<210> SEQ ID NO 23

<211> LENGTH: 504

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 23

```

Met Asn Leu Phe Ser Ala Leu Ser Leu Asp Thr Leu Val Leu Leu Ala
1           5           10          15

Ile Ile Leu Val Leu Leu Tyr Arg Tyr Gly Thr Arg Thr His Gly Leu
20          25          30

Phe Lys Lys Gln Gly Ile Pro Gly Pro Lys Pro Leu Pro Phe Leu Gly
35          40          45

Thr Val Leu Asn Tyr Tyr Thr Gly Ile Trp Lys Phe Asp Met Glu Cys
50          55          60

Tyr Glu Lys Tyr Gly Lys Thr Trp Gly Leu Phe Asp Gly Gln Thr Pro
65          70          75          80

Leu Leu Val Ile Thr Asp Pro Glu Thr Ile Lys Asn Val Leu Val Lys
85          90          95

Asp Cys Leu Ser Val Phe Thr Asn Arg Arg Glu Phe Gly Pro Val Gly
100         105         110

Ile Met Ser Lys Ala Ile Ser Ile Ser Lys Asp Glu Glu Trp Lys Arg
115         120         125

Tyr Arg Ala Leu Leu Ser Pro Thr Phe Thr Ser Gly Arg Leu Lys Glu
130         135         140

Met Phe Pro Val Ile Glu Gln Tyr Gly Asp Ile Leu Val Lys Tyr Leu
145         150         155         160

Arg Gln Glu Ala Glu Lys Gly Met Pro Val Ala Met Lys Asp Val Leu
165         170         175

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Gly Ala Tyr Ser Met Asp Val Ile Thr Ser Thr Ser Phe Gly Val Asn
 180 185 190
 Val Asp Ser Leu Asn Asn Pro Glu Asp Pro Phe Val Glu Glu Ala Lys
 195 200 205
 Lys Phe Leu Arg Val Asp Phe Phe Asp Pro Leu Leu Phe Ser Val Val
 210 215 220
 Leu Phe Pro Leu Leu Thr Pro Val Tyr Glu Met Leu Asn Ile Cys Met
 225 230 235 240
 Phe Pro Asn Asp Ser Ile Glu Phe Phe Lys Lys Phe Val Asp Arg Met
 245 250 255
 Gln Glu Ser Arg Leu Asp Ser Asn Gln Lys His Arg Val Asp Phe Leu
 260 265 270
 Gln Leu Met Met Asn Ser His Asn Asn Ser Lys Asp Lys Asp Ser His
 275 280 285
 Lys Ala Phe Ser Asn Met Glu Ile Thr Val Gln Ser Ile Ile Phe Ile
 290 295 300
 Ser Ala Gly Tyr Glu Thr Thr Ser Ser Thr Leu Ser Phe Thr Leu Tyr
 305 310 315 320
 Cys Leu Ala Thr His Pro Asp Ile Gln Lys Lys Leu Gln Ala Glu Ile
 325 330 335
 Asp Lys Ala Leu Pro Asn Lys Ala Thr Pro Thr Cys Asp Thr Val Met
 340 345 350
 Glu Met Glu Tyr Leu Asp Met Val Leu Asn Glu Thr Leu Arg Leu Tyr
 355 360 365
 Pro Ile Val Thr Arg Leu Glu Arg Val Cys Lys Lys Asp Val Glu Leu
 370 375 380
 Asn Gly Val Tyr Ile Pro Lys Gly Ser Met Val Met Ile Pro Ser Tyr
 385 390 395 400
 Ala Leu His His Asp Pro Gln His Trp Pro Asp Pro Glu Glu Phe Gln
 405 410 415
 Pro Glu Arg Phe Ser Lys Glu Asn Lys Gly Ser Ile Asp Pro Tyr Val
 420 425 430
 Tyr Leu Pro Phe Gly Ile Gly Pro Arg Asn Cys Ile Gly Met Arg Phe
 435 440 445
 Ala Leu Met Asn Met Lys Leu Ala Val Thr Lys Val Leu Gln Asn Phe
 450 455 460
 Ser Phe Gln Pro Cys Gln Glu Thr Gln Ile Pro Leu Lys Leu Ser Arg
 465 470 475 480
 Gln Gly Ile Leu Gln Pro Glu Lys Pro Ile Val Leu Lys Val Val Pro
 485 490 495
 Arg Asp Ala Val Ile Thr Gly Ala
 500

<210> SEQ ID NO 24

<211> LENGTH: 2037

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 24

```

atgggtgatt ctcataaga tacttctgct actgttccag aagctgttgc tgaagaagtt    60
tctttgtttt ctactactga tattgttttg ttttctttga ttgttggtgt ttgacttac    120
tggtttatatt ttaagaagaa gaaggaagaa attccagaat tttctaagat tcaaactact    180
gtccaccag ttaaggaatc ttcttttggg gaaaagatga agaagactgg tagaaacatt    240

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atgtttttt acggttctca aactggtact gctgaagaat ttgctaacag attgtctaag 300
gatgctcata gatacggat gagaggatg tctgctgac cagaagaata cgatttggct 360
gatttgtctt ctttgccaga aattgataag tctttggtg tttttgtat ggctacttac 420
ggtgaagggtg atccaactga taacgctcaa gatttttacg attggttgca agaaactgat 480
gttgatttga ctggtgttaa gtttgcgtt tttggttgg gtaacaagac ttacgaacat 540
tttaacgcta tgggtaagta cgttgatcaa agattggaac aattgggtgc tcaaagaatt 600
tttgaattgg gtttgggtga tgatgatgg aacttgaag aagattttat tacttggaga 660
gaacaatttt ggccagctgt ttgtgaattt tttggtgtg aagctactgg tgaagaatct 720
tctattagac aatacgaatt ggtgttcat gaagatagg atactgctaa ggtttacact 780
ggtgaaatgg gtagattgaa gtcttacgaa aaccaaagc caccatttga tgctaagaac 840
ccatttttgg ctgctgttac tactaacaga aagttgaacc aaggtagtga aagacatttg 900
atgcatttgg aattggatat ttctgattct aagattagat acgaatctgg tgatcatgtt 960
gctgtttacc cagctaacga ttctactttg gtttaacaaa ttggtgaaat tttgggtgct 1020
gatttggatg ttattatgtc tttgaacaac ttggatgaag aatctaacaa gaagcatcca 1080
tttccatgtc caactactta cagaactgct ttgacttact acttgatat tactaaccca 1140
ccaagaacta acgttttgta cgaattggct caatacgtt ctgaaccatc tgaacaagaa 1200
catttgcata agatggcttc ttcttctggt gaaggtaagg aattgtactt gtcttgggtt 1260
gttgaagcta gaagacatat tttggctatt ttgcaagatt acccatcttt gagaccacca 1320
attgatcatt tgtgtgaatt gttgccaaga ttgcaagcta gatactactc tattgttctt 1380
tcttctaagg ttcaccaaaa ctctgttcat atttgtgctg ttgctgttga atacgaagct 1440
aagtctggtg gagttaacaa ggggtgttgc acttcttggg tgagaactaa ggaaccagct 1500
ggtgaaaacg gtagaagagc tttggtcca atgtttgtta gaaagtctca atttagattg 1560
ccatttaagc caactactcc agttattatg gttggtccag gtactggtgt tgctccattt 1620
atgggtttta ttcaagaaag agcttgggtg agagaacaag gtaagggaagt tggtgaaact 1680
ttgtgttact acggtttagt aagatctgat gaagattact tgtacagaga agaattggct 1740
agatttcata aggatgtgtc tttgactcaa ttgaacgttg ctttttctag agaacaagct 1800
cataagggtt acgttcaaca tttgttgaag agagataagg aacatttgtg gaagttgatt 1860
catgaagggt gtgctcatat ttacgtttgt ggtgatgcta gaaacatggc taaggatgtt 1920
caaaacactt ttacgatat tgtgtctgaa tttggtccaa tggaacatac tcaagctgtt 1980
gattacgtta agaagttgat gactaagggt agataactct tggatgtttg gtcttaa 2037

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<210> SEQ ID NO 25

<211> LENGTH: 678

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 25

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Met Gly Asp Ser His Glu Asp Thr Ser Ala Thr Val Pro Glu Ala Val
1           5           10          15

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Ala Glu Glu Val Ser Leu Phe Ser Thr Thr Asp Ile Val Leu Phe Ser
20          25          30

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Leu Ile Val Gly Val Leu Thr Tyr Trp Phe Ile Phe Lys Lys Lys Lys
35          40          45

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Glu Glu Ile Pro Glu Phe Ser Lys Ile Gln Thr Thr Ala Pro Pro Val
50          55          60

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Lys Glu Ser Ser Phe Val Glu Lys Met Lys Lys Thr Gly Arg Asn Ile
 65 70 75 80
 Ile Val Phe Tyr Gly Ser Gln Thr Gly Thr Ala Glu Glu Phe Ala Asn
 85 90 95
 Arg Leu Ser Lys Asp Ala His Arg Tyr Gly Met Arg Gly Met Ser Ala
 100 105 110
 Asp Pro Glu Glu Tyr Asp Leu Ala Asp Leu Ser Ser Leu Pro Glu Ile
 115 120 125
 Asp Lys Ser Leu Val Val Phe Cys Met Ala Thr Tyr Gly Glu Gly Asp
 130 135 140
 Pro Thr Asp Asn Ala Gln Asp Phe Tyr Asp Trp Leu Gln Glu Thr Asp
 145 150 155 160
 Val Asp Leu Thr Gly Val Lys Phe Ala Val Phe Gly Leu Gly Asn Lys
 165 170 175
 Thr Tyr Glu His Phe Asn Ala Met Gly Lys Tyr Val Asp Gln Arg Leu
 180 185 190
 Glu Gln Leu Gly Ala Gln Arg Ile Phe Glu Leu Gly Leu Gly Asp Asp
 195 200 205
 Asp Gly Asn Leu Glu Glu Asp Phe Ile Thr Trp Arg Glu Gln Phe Trp
 210 215 220
 Pro Ala Val Cys Glu Phe Phe Gly Val Glu Ala Thr Gly Glu Glu Ser
 225 230 235 240
 Ser Ile Arg Gln Tyr Glu Leu Val Val His Glu Asp Met Asp Thr Ala
 245 250 255
 Lys Val Tyr Thr Gly Glu Met Gly Arg Leu Lys Ser Tyr Glu Asn Gln
 260 265 270
 Lys Pro Pro Phe Asp Ala Lys Asn Pro Phe Leu Ala Ala Val Thr Thr
 275 280 285
 Asn Arg Lys Leu Asn Gln Gly Thr Glu Arg His Leu Met His Leu Glu
 290 295 300
 Leu Asp Ile Ser Asp Ser Lys Ile Arg Tyr Glu Ser Gly Asp His Val
 305 310 315 320
 Ala Val Tyr Pro Ala Asn Asp Ser Thr Leu Val Asn Gln Ile Gly Glu
 325 330 335
 Ile Leu Gly Ala Asp Leu Asp Val Ile Met Ser Leu Asn Asn Leu Asp
 340 345 350
 Glu Glu Ser Asn Lys Lys His Pro Phe Pro Cys Pro Thr Thr Tyr Arg
 355 360 365
 Thr Ala Leu Thr Tyr Tyr Leu Asp Ile Thr Asn Pro Pro Arg Thr Asn
 370 375 380
 Val Leu Tyr Glu Leu Ala Gln Tyr Ala Ser Glu Pro Ser Glu Gln Glu
 385 390 395 400
 His Leu His Lys Met Ala Ser Ser Ser Gly Glu Gly Lys Glu Leu Tyr
 405 410 415
 Leu Ser Trp Val Val Glu Ala Arg Arg His Ile Leu Ala Ile Leu Gln
 420 425 430
 Asp Tyr Pro Ser Leu Arg Pro Pro Ile Asp His Leu Cys Glu Leu Leu
 435 440 445
 Pro Arg Leu Gln Ala Arg Tyr Tyr Ser Ile Ala Ser Ser Ser Lys Val
 450 455 460
 His Pro Asn Ser Val His Ile Cys Ala Val Ala Val Glu Tyr Glu Ala
 465 470 475 480

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Lys Ser Gly Arg Val Asn Lys Gly Val Ala Thr Ser Trp Leu Arg Thr
 485 490 495
 Lys Glu Pro Ala Gly Glu Asn Gly Arg Arg Ala Leu Val Pro Met Phe
 500 505 510
 Val Arg Lys Ser Gln Phe Arg Leu Pro Phe Lys Pro Thr Thr Pro Val
 515 520 525
 Ile Met Val Gly Pro Gly Thr Gly Val Ala Pro Phe Met Gly Phe Ile
 530 535 540
 Gln Glu Arg Ala Trp Leu Arg Glu Gln Gly Lys Glu Val Gly Glu Thr
 545 550 555 560
 Leu Leu Tyr Tyr Gly Cys Arg Arg Ser Asp Glu Asp Tyr Leu Tyr Arg
 565 570 575
 Glu Glu Leu Ala Arg Phe His Lys Asp Gly Ala Leu Thr Gln Leu Asn
 580 585 590
 Val Ala Phe Ser Arg Glu Gln Ala His Lys Val Tyr Val Gln His Leu
 595 600 605
 Leu Lys Arg Asp Lys Glu His Leu Trp Lys Leu Ile His Glu Gly Gly
 610 615 620
 Ala His Ile Tyr Val Cys Gly Asp Ala Arg Asn Met Ala Lys Asp Val
 625 630 635 640
 Gln Asn Thr Phe Tyr Asp Ile Val Ala Glu Phe Gly Pro Met Glu His
 645 650 655
 Thr Gln Ala Val Asp Tyr Val Lys Lys Leu Met Thr Lys Gly Arg Tyr
 660 665 670
 Ser Leu Asp Val Trp Ser
 675

<210> SEQ ID NO 26
 <211> LENGTH: 1509
 <212> TYPE: DNA
 <213> ORGANISM: Human

<400> SEQUENCE: 26

atggccttga ttcctgattt ggctatggaa actagattgt tgttggtgtt ttcattgggt	60
ttgttggtatt tgtatggaac tcattcacat ggattgttta aaaaattggg aattcctgga	120
cctactcctt tgcctttttt gggaaatatt ttgtcatatc ataaaggatt ttgcattgtt	180
gatatggaat gccataaaaa atatggaaaa gtttggggat tttatgatgg acaacaacct	240
gtttttggcta ttactgatcc tgatagattt aaaactgttt tggttaaaga atgctattca	300
gtttttacta atagaagacc ttttggacct gttggattta tgaaatcagc tatttcaatt	360
gctgaagatg aagaatggaa aagattgaga tcattgttgt cacctacttt tacttcagga	420
aaattgaaag aaatggttcc tattattgct caatatggag atgttttggg tagaaatttg	480
agaagagaag ctgaaactgg aaaacctgtt actttgaaag atgtttttgg agcttattca	540
atggatgtta ttacttcaac ttcatttgga gttaaatattg attcattgaa taatcctcaa	600
gatacctttt ttgaaaatac taaaaaattg ttgagatttg attttttgga tccttttttt	660
ttgtcaatta ctgtttttcc ttttttgatt cctatttttg aagttttgaa tatttgcgtt	720
tttctagag aagttactaa ttttttgaga aaatcagtta aaagaatgaa agaatacaaga	780
ttggaagata ctcaaaaaca tagagttgat tttttgcaat tgatgattga ttcacaaaat	840
tcaaaagaaa ctgaatcaca taaagctttg tcagattttg aattgggtgc tcaatcaatt	900
attttttatt ttgctggatg cgaaactact tcatcagttt tgtcatttat tatgtatgaa	960

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ttggctactc atcctgatgt tcaacaaaaa ttgcaagaag aaattgatgc tgttttgct 1020
aataaagctc ctcctactta tgatactggt ttgcaaatgg aatatttggga tatgggtggt 1080
aatgaaactt tgagattggt tcctattgct atgagattgg aaagagtttg caaaaaagat 1140
gttgaaatta atggaatggt tattcctaaa ggagttgttg ttatgattcc ttcatatgct 1200
ttgcatagag atcctaaata ttggactgaa cctgaaaaat ttttgctga aagattttca 1260
aaaaaaaata aagataatat tgatccttat atttatactc cttttggatc aggacctaga 1320
aattgcattg gaatgagatt tgctttgatg aatatgaaat tggctttgat tagagttttg 1380
caaaattttt catttaaacc ttgcaagaa actcaaattc ctttgaaatt gtcattggga 1440
ggattgttgc aacctgaaaa acctgttgtt ttgaaagttg aatcaagaga tggaactgtt 1500
tcaggagct 1509

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<210> SEQ ID NO 27
<211> LENGTH: 503
<212> TYPE: PRT
<213> ORGANISM: Human

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<400> SEQUENCE: 27

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Met Ala Leu Ile Pro Asp Leu Ala Met Glu Thr Arg Leu Leu Leu Ala
 1             5             10             15
Val Ser Leu Val Leu Leu Tyr Leu Tyr Gly Thr His Ser His Gly Leu
      20             25             30
Phe Lys Lys Leu Gly Ile Pro Gly Pro Thr Pro Leu Pro Phe Leu Gly
      35             40             45
Asn Ile Leu Ser Tyr His Lys Gly Phe Cys Met Phe Asp Met Glu Cys
      50             55             60
His Lys Lys Tyr Gly Lys Val Trp Gly Phe Tyr Asp Gly Gln Gln Pro
      65             70             75             80
Val Leu Ala Ile Thr Asp Pro Asp Met Ile Lys Thr Val Leu Val Lys
      85             90             95
Glu Cys Tyr Ser Val Phe Thr Asn Arg Arg Pro Phe Gly Pro Val Gly
      100            105            110
Phe Met Lys Ser Ala Ile Ser Ile Ala Glu Asp Glu Glu Trp Lys Arg
      115            120            125
Leu Arg Ser Leu Leu Ser Pro Thr Phe Thr Ser Gly Lys Leu Lys Glu
      130            135            140
Met Val Pro Ile Ile Ala Gln Tyr Gly Asp Val Leu Val Arg Asn Leu
      145            150            155            160
Arg Arg Glu Ala Glu Thr Gly Lys Pro Val Thr Leu Lys Asp Val Phe
      165            170            175
Gly Ala Tyr Ser Met Asp Val Ile Thr Ser Thr Ser Phe Gly Val Asn
      180            185            190
Ile Asp Ser Leu Asn Asn Pro Gln Asp Pro Phe Val Glu Asn Thr Lys
      195            200            205
Lys Leu Leu Arg Phe Asp Phe Leu Asp Pro Phe Phe Leu Ser Ile Thr
      210            215            220
Val Phe Pro Phe Leu Ile Pro Ile Leu Glu Val Leu Asn Ile Cys Val
      225            230            235            240
Phe Pro Arg Glu Val Thr Asn Phe Leu Arg Lys Ser Val Lys Arg Met
      245            250            255
Lys Glu Ser Arg Leu Glu Asp Thr Gln Lys His Arg Val Asp Phe Leu
      260            265            270

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Gln Leu Met Ile Asp Ser Gln Asn Ser Lys Glu Thr Glu Ser His Lys
 275 280 285
 Ala Leu Ser Asp Leu Glu Leu Val Ala Gln Ser Ile Ile Phe Ile Phe
 290 295 300
 Ala Gly Cys Glu Thr Thr Ser Ser Val Leu Ser Phe Ile Met Tyr Glu
 305 310 315 320
 Leu Ala Thr His Pro Asp Val Gln Gln Lys Leu Gln Glu Glu Ile Asp
 325 330 335
 Ala Val Leu Pro Asn Lys Ala Pro Pro Thr Tyr Asp Thr Val Leu Gln
 340 345 350
 Met Glu Tyr Leu Asp Met Val Val Asn Glu Thr Leu Arg Leu Phe Pro
 355 360 365
 Ile Ala Met Arg Leu Glu Arg Val Cys Lys Lys Asp Val Glu Ile Asn
 370 375 380
 Gly Met Phe Ile Pro Lys Gly Val Val Val Met Ile Pro Ser Tyr Ala
 385 390 395 400
 Leu His Arg Asp Pro Lys Tyr Trp Thr Glu Pro Glu Lys Phe Leu Pro
 405 410 415
 Glu Arg Phe Ser Lys Lys Asn Lys Asp Asn Ile Asp Pro Tyr Ile Tyr
 420 425 430
 Thr Pro Phe Gly Ser Gly Pro Arg Asn Cys Ile Gly Met Arg Phe Ala
 435 440 445
 Leu Met Asn Met Lys Leu Ala Leu Ile Arg Val Leu Gln Asn Phe Ser
 450 455 460
 Phe Lys Pro Cys Lys Glu Thr Gln Ile Pro Leu Lys Leu Ser Leu Gly
 465 470 475 480
 Gly Leu Leu Gln Pro Glu Lys Pro Val Val Leu Lys Val Glu Ser Arg
 485 490 495
 Asp Gly Thr Val Ser Gly Ala
 500

<210> SEQ ID NO 28
 <211> LENGTH: 2040
 <212> TYPE: DNA
 <213> ORGANISM: Human

<400> SEQUENCE: 28

atgattaata tgggagattc acatgttgat acttcatcaa ctgtttcaga agctgttgct	60
gaagaagttt cattgttttc aatgactgat atgattttgt tttcattgat tgttggattg	120
ttgacttatt ggtttttgtt tagaaaaaaa aaagaagaag ttcctgaatt tactaaaatt	180
caaaactttga cttcatcagt tagagaatca tcatttgttg aaaaaatgaa aaaaactgga	240
agaaatatta ttgtttttta tggatcacaa actggaactg ctgaagaatt tgctaataga	300
ttgtcaaaag atgctcatag atatggaatg agaggaatgt cagctgatcc tgaagaatat	360
gatttggctg atttgtcatc attgcctgaa attgataatg ctttggttgt tttttgcatg	420
gctacttatg gagaaggaga tctactgat aatgctcaag atttttatga ttggttgcaa	480
gaaactgatg ttgatttgtc aggagttaaa tttgctgttt ttggattggg aaataaaaact	540
tatgaacatt ttaatgctat gggaaaaatat gttgataaaa gattggaaca attgggagct	600
caaagaattt ttgaattggg attgggagat gatgatggaa atttgaaga agattttatt	660
acttggagag aacaattttg gttggctgtt tgccaacatt ttggagtga agctactgga	720
gaagaatcat caattagaca atatgaattg gttgttcata ctgatattga tgctgctaaa	780

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gtttatatgg gagaaatggg aagattgaaa tcatatgaaa atcaaaaacc tccttttgat 840
gctaaaaato cttttttggc tgctgttact actaatagaa aattgaatca aggaactgaa 900
agacatttga tgcatttggg attggatatt tcagattcaa aaattagata tgaatcagga 960
gatcatgttg ctgtttatcc tgctaattgat tcagctttgg ttaatcaatt gggaaaaatt 1020
ttgggagctg atttggatgt tgttatgtca ttgaataatt tggatgaaga atcaataaaa 1080
aaacatcctt ttccttgcgc tacttcatat agaactgctt tgacttatta tttggatatt 1140
actaatcctc ctagaactaa tgttttgat gaattggctc aatatgcttc agaacttca 1200
gaacaagaat tgttgagaaa aatggcttca tcatcaggag aaggaaaaga attgtatttg 1260
tcatgggttg ttgaagctag aagacatatt ttggctattt tgcaagattg ccttcattg 1320
agacctccta ttgatcattt gtgcgaattg ttgcctagat tgcaagctag atattattca 1380
attgcttcat catcaaaagt tcacctaatt tcagttcata tttgcgctgt tgttgttgaa 1440
tatgaaacta aagctggaag aattaataaa ggagttgcta ctaattgggt gagagctaaa 1500
gaacctgttg gagaaaatgg aggaagagct ttggctccta tgtttgtag aaaatcacia 1560
tttagattgc cttttaaacg tactactcct gttattatgg ttggacctgg aactggagtt 1620
gtccttttta ttggatttat tcaagaaaga gcttggttga gacaacaagg aaaagaagtt 1680
ggagaaactt tgttgtatta tggatgcaga agatcagatg aagattatth gtatagagaa 1740
gaattggctc aatttcatag agatggagct ttgactcaat tgaatgttg tttttcaaga 1800
gaacaatcac ataaagttha tgttcaacat ttgttgaac aagatagaga acatttgttg 1860
aaattgattg aaggaggagc tcatatttat gtttgcggag atgctagaaa tatggctaga 1920
gatgttcaaa atacttttta tgatattggt gctgaattgg gagctatgga acatgctcaa 1980
gctgttgatt atattaaaaa attgatgact aaaggaagat attcattgga tgtttggtca 2040

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<210> SEQ ID NO 29
<211> LENGTH: 680
<212> TYPE: PRT
<213> ORGANISM: Human

<400> SEQUENCE: 29

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Met Ile Asn Met Gly Asp Ser His Val Asp Thr Ser Ser Thr Val Ser
1         5             10             15

Glu Ala Val Ala Glu Glu Val Ser Leu Phe Ser Met Thr Asp Met Ile
20        25             30

Leu Phe Ser Leu Ile Val Gly Leu Leu Thr Tyr Trp Phe Leu Phe Arg
35        40             45

Lys Lys Lys Glu Glu Val Pro Glu Phe Thr Lys Ile Gln Thr Leu Thr
50        55             60

Ser Ser Val Arg Glu Ser Ser Phe Val Glu Lys Met Lys Lys Thr Gly
65        70             75             80

Arg Asn Ile Ile Val Phe Tyr Gly Ser Gln Thr Gly Thr Ala Glu Glu
85        90             95

Phe Ala Asn Arg Leu Ser Lys Asp Ala His Arg Tyr Gly Met Arg Gly
100       105            110

Met Ser Ala Asp Pro Glu Glu Tyr Asp Leu Ala Asp Leu Ser Ser Leu
115       120            125

Pro Glu Ile Asp Asn Ala Leu Val Val Phe Cys Met Ala Thr Tyr Gly
130       135            140

Glu Gly Asp Pro Thr Asp Asn Ala Gln Asp Phe Tyr Asp Trp Leu Gln
145       150            155            160

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Glu Thr Asp Val Asp Leu Ser Gly Val Lys Phe Ala Val Phe Gly Leu	165	170	175
Gly Asn Lys Thr Tyr Glu His Phe Asn Ala Met Gly Lys Tyr Val Asp	180	185	190
Lys Arg Leu Glu Gln Leu Gly Ala Gln Arg Ile Phe Glu Leu Gly Leu	195	200	205
Gly Asp Asp Asp Gly Asn Leu Glu Glu Asp Phe Ile Thr Trp Arg Glu	210	215	220
Gln Phe Trp Leu Ala Val Cys Glu His Phe Gly Val Glu Ala Thr Gly	225	230	235
Glu Glu Ser Ser Ile Arg Gln Tyr Glu Leu Val Val His Thr Asp Ile	245	250	255
Asp Ala Ala Lys Val Tyr Met Gly Glu Met Gly Arg Leu Lys Ser Tyr	260	265	270
Glu Asn Gln Lys Pro Pro Phe Asp Ala Lys Asn Pro Phe Leu Ala Ala	275	280	285
Val Thr Thr Asn Arg Lys Leu Asn Gln Gly Thr Glu Arg His Leu Met	290	295	300
His Leu Glu Leu Asp Ile Ser Asp Ser Lys Ile Arg Tyr Glu Ser Gly	305	310	315
Asp His Val Ala Val Tyr Pro Ala Asn Asp Ser Ala Leu Val Asn Gln	325	330	335
Leu Gly Lys Ile Leu Gly Ala Asp Leu Asp Val Val Met Ser Leu Asn	340	345	350
Asn Leu Asp Glu Glu Ser Asn Lys Lys His Pro Phe Pro Cys Pro Thr	355	360	365
Ser Tyr Arg Thr Ala Leu Thr Tyr Tyr Leu Asp Ile Thr Asn Pro Pro	370	375	380
Arg Thr Asn Val Leu Tyr Glu Leu Ala Gln Tyr Ala Ser Glu Pro Ser	385	390	395
Glu Gln Glu Leu Leu Arg Lys Met Ala Ser Ser Ser Gly Glu Gly Lys	405	410	415
Glu Leu Tyr Leu Ser Trp Val Val Glu Ala Arg Arg His Ile Leu Ala	420	425	430
Ile Leu Gln Asp Cys Pro Ser Leu Arg Pro Pro Ile Asp His Leu Cys	435	440	445
Glu Leu Leu Pro Arg Leu Gln Ala Arg Tyr Tyr Ser Ile Ala Ser Ser	450	455	460
Ser Lys Val His Pro Asn Ser Val His Ile Cys Ala Val Val Val Glu	465	470	475
Tyr Glu Thr Lys Ala Gly Arg Ile Asn Lys Gly Val Ala Thr Asn Trp	485	490	495
Leu Arg Ala Lys Glu Pro Val Gly Glu Asn Gly Gly Arg Ala Leu Val	500	505	510
Pro Met Phe Val Arg Lys Ser Gln Phe Arg Leu Pro Phe Lys Ala Thr	515	520	525
Thr Pro Val Ile Met Val Gly Pro Gly Thr Gly Val Ala Pro Phe Ile	530	535	540
Gly Phe Ile Gln Glu Arg Ala Trp Leu Arg Gln Gln Gly Lys Glu Val	545	550	555
Gly Glu Thr Leu Leu Tyr Tyr Gly Cys Arg Arg Ser Asp Glu Asp Tyr	565	570	575

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Leu Tyr Arg Glu Glu Leu Ala Gln Phe His Arg Asp Gly Ala Leu Thr
 580 585 590

 Gln Leu Asn Val Ala Phe Ser Arg Glu Gln Ser His Lys Val Tyr Val
 595 600 605

 Gln His Leu Leu Lys Gln Asp Arg Glu His Leu Trp Lys Leu Ile Glu
 610 615 620

 Gly Gly Ala His Ile Tyr Val Cys Gly Asp Ala Arg Asn Met Ala Arg
 625 630 635 640

 Asp Val Gln Asn Thr Phe Tyr Asp Ile Val Ala Glu Leu Gly Ala Met
 645 650 655

 Glu His Ala Gln Ala Val Asp Tyr Ile Lys Lys Leu Met Thr Lys Gly
 660 665 670

 Arg Tyr Ser Leu Asp Val Trp Ser
 675 680

<210> SEQ ID NO 30
 <211> LENGTH: 1554
 <212> TYPE: DNA
 <213> ORGANISM: Cannabis sativa

<400> SEQUENCE: 30

atgaatcctc gagaaaactt ccttaaatgc ttctcgcaat atattcccaa taatgcaaca	60
aatctaaaac tcgtatacac tcaaaacaac ccattgtata tgtctgtcct aaattcgaca	120
atacacaatc ttagattcac ctctgacaca accccaaaac cacttggtat cgtcactcct	180
tcacatgtct ctcatatcca aggcaactatt ctatgctcca agaaagtgg cttgcagatt	240
cgaactcgaa gtgggtggta tgattctgag ggcattgctt acatatctca agtcccattt	300
ggtatagtag acttgagaaa catgcgttca atcaaaatag atgttcata ccaaactgca	360
tgggttgaag ccggagctac ccttgagaaa gtttattatt gggttaatga gaaaaatgag	420
aatcttagtt tggcggtggt gtattgccct actgtttggt caggtggaca ctttgggtga	480
ggaggtatg gaccattgat gagaaactat ggctcgcgg ctgataatat cattgatgca	540
cacttagtca acgttcattg aaaagtgtca gatcgaaaat ctatggggga agatctcttt	600
tgggctttac gtgggtgggt agcagaaaagc ttcggaatca ttgtagcatg gaaaattaga	660
ctggttgctg tcccaaagtc tactatgttt agtggttaaa agatcatgga gatacatgag	720
cttgtcaagt tagttaacaa atggcaaaat attgcttaca agtatgaca agatttatta	780
ctcatgactc acttcataac taggaacatt acagataatc aagggaagaa taagacagca	840
atacacactt acttctcttc agttttcctt ggtggagtgg atagtctagt cgacttgatg	900
aacaagagtt ttcctgagtt ggggtattaaa aaacggatt gcagacaatt gagctggatt	960
gatacatca tcttctatag tgggtgtgta aattacgaca ctgataattt taacaaggaa	1020
atgttgcttg atagatccgc tgggcagaaac ggtgctttca agattaagtt agactacgtt	1080
aagaaaccaa ttccagaatc tgtatttgtc caaattttgg aaaaattata tgaagaagat	1140
ataggagctg ggatgtatgc gttgtaccct tacggtggta taatggatga gatttcagaa	1200
tcagcaatc cattocctca tcgagctgga atcttgatg agttatggta catatgtagt	1260
tgggagaagc aagaagataa cgaagagcat ctaaactgga ttagaaatat ttataacttc	1320
atgactcctt atgtgtccaa aaattcaaga ttggcatatc tcaattatag agaccttgat	1380
ataggaataa atgatcccaa gaatccaaat aattacacac aagcacgtat ttgggtgag	1440
aagtattttg gtaaaaattt tgacaggcta gtaaaagtga aaacctgggt tgatcccaat	1500

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aactttttta gaaacgaaca aagcatccca cctcaaccac ggcacgtca ttaa 1554

<210> SEQ ID NO 31

<211> LENGTH: 517

<212> TYPE: PRT

<213> ORGANISM: Cannabis sativa

<400> SEQUENCE: 31

Met Asn Pro Arg Glu Asn Phe Leu Lys Cys Phe Ser Gln Tyr Ile Pro
1 5 10 15

Asn Asn Ala Thr Asn Leu Lys Leu Val Tyr Thr Gln Asn Asn Pro Leu
20 25 30

Tyr Met Ser Val Leu Asn Ser Thr Ile His Asn Leu Arg Phe Thr Ser
35 40 45

Asp Thr Thr Pro Lys Pro Leu Val Ile Val Thr Pro Ser His Val Ser
50 55 60

His Ile Gln Gly Thr Ile Leu Cys Ser Lys Lys Val Gly Leu Gln Ile
65 70 75 80

Arg Thr Arg Ser Gly Gly His Asp Ser Glu Gly Met Ser Tyr Ile Ser
85 90 95

Gln Val Pro Phe Val Ile Val Asp Leu Arg Asn Met Arg Ser Ile Lys
100 105 110

Ile Asp Val His Ser Gln Thr Ala Trp Val Glu Ala Gly Ala Thr Leu
115 120 125

Gly Glu Val Tyr Tyr Trp Val Asn Glu Lys Asn Glu Asn Leu Ser Leu
130 135 140

Ala Ala Gly Tyr Cys Pro Thr Val Cys Ala Gly Gly His Phe Gly Gly
145 150 155 160

Gly Gly Tyr Gly Pro Leu Met Arg Asn Tyr Gly Leu Ala Ala Asp Asn
165 170 175

Ile Ile Asp Ala His Leu Val Asn Val His Gly Lys Val Leu Asp Arg
180 185 190

Lys Ser Met Gly Glu Asp Leu Phe Trp Ala Leu Arg Gly Gly Gly Ala
195 200 205

Glu Ser Phe Gly Ile Ile Val Ala Trp Lys Ile Arg Leu Val Ala Val
210 215 220

Pro Lys Ser Thr Met Phe Ser Val Lys Lys Ile Met Glu Ile His Glu
225 230 235 240

Leu Val Lys Leu Val Asn Lys Trp Gln Asn Ile Ala Tyr Lys Tyr Asp
245 250 255

Lys Asp Leu Leu Leu Met Thr His Phe Ile Thr Arg Asn Ile Thr Asp
260 265 270

Asn Gln Gly Lys Asn Lys Thr Ala Ile His Thr Tyr Phe Ser Ser Val
275 280 285

Phe Leu Gly Gly Val Asp Ser Leu Val Asp Leu Met Asn Lys Ser Phe
290 295 300

Pro Glu Leu Gly Ile Lys Lys Thr Asp Cys Arg Gln Leu Ser Trp Ile
305 310 315 320

Asp Thr Ile Ile Phe Tyr Ser Gly Val Val Asn Tyr Asp Thr Asp Asn
325 330 335

Phe Asn Lys Glu Ile Leu Leu Asp Arg Ser Ala Gly Gln Asn Gly Ala
340 345 350

Phe Lys Ile Lys Leu Asp Tyr Val Lys Lys Pro Ile Pro Glu Ser Val
355 360 365

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Phe Val Gln Ile Leu Glu Lys Leu Tyr Glu Glu Asp Ile Gly Ala Gly
 370 375 380
 Met Tyr Ala Leu Tyr Pro Tyr Gly Gly Ile Met Asp Glu Ile Ser Glu
 385 390 395 400
 Ser Ala Ile Pro Phe Pro His Arg Ala Gly Ile Leu Tyr Glu Leu Trp
 405 410 415
 Tyr Ile Cys Ser Trp Glu Lys Gln Glu Asp Asn Glu Lys His Leu Asn
 420 425 430
 Trp Ile Arg Asn Ile Tyr Asn Phe Met Thr Pro Tyr Val Ser Lys Asn
 435 440 445
 Ser Arg Leu Ala Tyr Leu Asn Tyr Arg Asp Leu Asp Ile Gly Ile Asn
 450 455 460
 Asp Pro Lys Asn Pro Asn Asn Tyr Thr Gln Ala Arg Ile Trp Gly Glu
 465 470 475 480
 Lys Tyr Phe Gly Lys Asn Phe Asp Arg Leu Val Lys Val Lys Thr Leu
 485 490 495
 Val Asp Pro Asn Asn Phe Phe Arg Asn Glu Gln Ser Ile Pro Pro Gln
 500 505 510
 Pro Arg His Arg His
 515

<210> SEQ ID NO 32
 <211> LENGTH: 1374
 <212> TYPE: DNA
 <213> ORGANISM: Stevia rebaudiana

<400> SEQUENCE: 32

atggaaaata aaactgaac tactgttaga agaagaagaa gaattatattt gtttcctggt	60
ccttttcaag gacatattaa tcctattttg caattggcta atgttttgta ttcaaaagga	120
ttttcaatta ctatttttca tactaatttt aataaaccta aaacttcaaa ttatcctcat	180
tttactttta gatttatattt ggataatgat cctcaagatg aaagaatttc aaatttgcct	240
actcatggac ctttggctgg aatgagaatt cctattatta atgaacatgg agctgatgaa	300
ttgagaagag aattggaatt gttgatgttg gcttcagaag aagatgaaga agtttcatgc	360
ttgattactg atgctttgtg gtattttgct caatcagttg ctgattcatt gaatttgaga	420
agattggttt tgaatgacttc atcattgttt aattttcatg ctcatgtttc attgcctcaa	480
tttgatgaat tgggatattt ggatcctgat gataaaacta gattggaaga acaagcttca	540
ggatttccta tgttgaaagt taaagatatt aaatcagctt attcaaattg gcaaattttg	600
aaagaaattt tgggaaaaat gattaacaa actagagctt catcaggagt tatttggaat	660
tcatttaaag aattggaaga atcagaattg gaaactgtta ttagagaaat tcctgctcct	720
tcatttttga ttcctttgcc taaacatttg actgcttcat catcatcatt gttggatcat	780
gatagaactg tttttcaatg gttggatcaa caacctcctt catcagtttt gtatgtttca	840
tttgatcaa cttcagaagt tgatgaaaaa gatttttttg aaattgctag aggattggtt	900
gattcaaaac aatcattttt gtgggttggt agacctggat ttgttaaagg atcaacttgg	960
gttgaaacct tgctgatgg atttttggga gaaagaggaa gaattgttaa atgggttcct	1020
caacaagaag ttttggctca tggagctatt ggagcttttt ggactcattc aggatggaat	1080
tcaacttttg aatcagtttg cgaaggagtt cctatgattt tttcagattt tggattggat	1140
caacctttga atgctagata tatgtcagat gttttgaaag ttggagttaa tttggaaaat	1200
ggatgggaaa gaggagaaat tgctaagtct attagaagag ttatggttga tgaagaagga	1260

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gaatatatta gacaaaatgc tagagttttg aaacaaaaag ctgatgtttc attgatgaaa 1320

ggaggatcat catatgaatc attggaatca ttggtttcat atatttcacg attg 1374

<210> SEQ ID NO 33

<211> LENGTH: 458

<212> TYPE: PRT

<213> ORGANISM: Stevia rebaudiana

<400> SEQUENCE: 33

Met Glu Asn Lys Thr Glu Thr Thr Val Arg Arg Arg Arg Arg Ile Ile
1 5 10 15

Leu Phe Pro Val Pro Phe Gln Gly His Ile Asn Pro Ile Leu Gln Leu
20 25 30

Ala Asn Val Leu Tyr Ser Lys Gly Phe Ser Ile Thr Ile Phe His Thr
35 40 45

Asn Phe Asn Lys Pro Lys Thr Ser Asn Tyr Pro His Phe Thr Phe Arg
50 55 60

Phe Ile Leu Asp Asn Asp Pro Gln Asp Glu Arg Ile Ser Asn Leu Pro
65 70 75 80

Thr His Gly Pro Leu Ala Gly Met Arg Ile Pro Ile Ile Asn Glu His
85 90 95

Gly Ala Asp Glu Leu Arg Arg Glu Leu Glu Leu Leu Met Leu Ala Ser
100 105 110

Glu Glu Asp Glu Glu Val Ser Cys Leu Ile Thr Asp Ala Leu Trp Tyr
115 120 125

Phe Ala Gln Ser Val Ala Asp Ser Leu Asn Leu Arg Arg Leu Val Leu
130 135 140

Met Thr Ser Ser Leu Phe Asn Phe His Ala His Val Ser Leu Pro Gln
145 150 155 160

Phe Asp Glu Leu Gly Tyr Leu Asp Pro Asp Asp Lys Thr Arg Leu Glu
165 170 175

Glu Gln Ala Ser Gly Phe Pro Met Leu Lys Val Lys Asp Ile Lys Ser
180 185 190

Ala Tyr Ser Asn Trp Gln Ile Leu Lys Glu Ile Leu Gly Lys Met Ile
195 200 205

Lys Gln Thr Arg Ala Ser Ser Gly Val Ile Trp Asn Ser Phe Lys Glu
210 215 220

Leu Glu Glu Ser Glu Leu Glu Thr Val Ile Arg Glu Ile Pro Ala Pro
225 230 235 240

Ser Phe Leu Ile Pro Leu Pro Lys His Leu Thr Ala Ser Ser Ser Ser
245 250 255

Leu Leu Asp His Asp Arg Thr Val Phe Gln Trp Leu Asp Gln Gln Pro
260 265 270

Pro Ser Ser Val Leu Tyr Val Ser Phe Gly Ser Thr Ser Glu Val Asp
275 280 285

Glu Lys Asp Phe Leu Glu Ile Ala Arg Gly Leu Val Asp Ser Lys Gln
290 295 300

Ser Phe Leu Trp Val Val Arg Pro Gly Phe Val Lys Gly Ser Thr Trp
305 310 315 320

Val Glu Pro Leu Pro Asp Gly Phe Leu Gly Glu Arg Gly Arg Ile Val
325 330 335

Lys Trp Val Pro Gln Gln Glu Val Leu Ala His Gly Ala Ile Gly Ala
340 345 350

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Phe	Trp	Thr	His	Ser	Gly	Trp	Asn	Ser	Thr	Leu	Glu	Ser	Val	Cys	Glu
	355						360					365			
Gly	Val	Pro	Met	Ile	Phe	Ser	Asp	Phe	Gly	Leu	Asp	Gln	Pro	Leu	Asn
	370					375					380				
Ala	Arg	Tyr	Met	Ser	Asp	Val	Leu	Lys	Val	Gly	Val	Tyr	Leu	Glu	Asn
	385				390					395					400
Gly	Trp	Glu	Arg	Gly	Glu	Ile	Ala	Asn	Ala	Ile	Arg	Arg	Val	Met	Val
				405					410					415	
Asp	Glu	Glu	Gly	Glu	Tyr	Ile	Arg	Gln	Asn	Ala	Arg	Val	Leu	Lys	Gln
			420					425					430		
Lys	Ala	Asp	Val	Ser	Leu	Met	Lys	Gly	Gly	Ser	Ser	Tyr	Glu	Ser	Leu
		435					440					445			
Glu	Ser	Leu	Val	Ser	Tyr	Ile	Ser	Ser	Leu						
	450					455									

<210> SEQ ID NO 34
 <211> LENGTH: 485
 <212> TYPE: PRT
 <213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 34

Met	Gly	Ser	Ile	Gly	Ala	Glu	Leu	Thr	Lys	Pro	His	Ala	Val	Cys	Ile
1			5					10					15		
Pro	Tyr	Pro	Ala	Gln	Gly	His	Ile	Asn	Pro	Met	Leu	Lys	Leu	Ala	Lys
		20					25					30			
Ile	Leu	His	His	Lys	Gly	Phe	His	Ile	Thr	Phe	Val	Asn	Thr	Glu	Phe
	35					40					45				
Asn	His	Arg	Arg	Leu	Leu	Lys	Ser	Arg	Gly	Pro	Asp	Ser	Leu	Lys	Gly
	50				55					60					
Leu	Ser	Ser	Phe	Arg	Phe	Glu	Thr	Ile	Pro	Asp	Gly	Leu	Pro	Pro	Cys
65			70					75						80	
Glu	Ala	Asp	Ala	Thr	Gln	Asp	Ile	Pro	Ser	Leu	Cys	Glu	Ser	Thr	Thr
		85					90						95		
Asn	Thr	Cys	Leu	Ala	Pro	Phe	Arg	Asp	Leu	Leu	Ala	Lys	Leu	Asn	Asp
	100					105						110			
Thr	Asn	Thr	Ser	Asn	Val	Pro	Pro	Val	Ser	Cys	Ile	Val	Ser	Asp	Gly
	115				120						125				
Val	Met	Ser	Phe	Thr	Leu	Ala	Ala	Ala	Gln	Glu	Leu	Gly	Val	Pro	Glu
	130				135					140					
Val	Leu	Phe	Trp	Thr	Thr	Ser	Ala	Cys	Gly	Phe	Leu	Gly	Tyr	Met	His
145			150					155						160	
Tyr	Cys	Lys	Val	Ile	Glu	Lys	Gly	Tyr	Ala	Pro	Leu	Lys	Asp	Ala	Ser
		165					170						175		
Asp	Leu	Thr	Asn	Gly	Tyr	Leu	Glu	Thr	Thr	Leu	Asp	Phe	Ile	Pro	Gly
	180					185					190				
Met	Lys	Asp	Val	Arg	Leu	Arg	Asp	Leu	Pro	Ser	Phe	Leu	Arg	Thr	Thr
	195					200					205				
Asn	Pro	Asp	Glu	Phe	Met	Ile	Lys	Phe	Val	Leu	Gln	Glu	Thr	Glu	Arg
	210				215						220				
Ala	Arg	Lys	Ala	Ser	Ala	Ile	Ile	Leu	Asn	Thr	Phe	Glu	Thr	Leu	Glu
	225				230					235				240	
Ala	Glu	Val	Leu	Glu	Ser	Leu	Arg	Asn	Leu	Leu	Pro	Pro	Val	Tyr	Pro
		245					250						255		
Ile	Gly	Pro	Leu	His	Phe	Leu	Val	Lys	His	Val	Asp	Asp	Glu	Asn	Leu
		260					265						270		

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Lys Gly Leu Arg Ser Ser Leu Trp Lys Glu Glu Pro Glu Cys Ile Gln
 275 280 285
 Trp Leu Asp Thr Lys Glu Pro Asn Ser Val Val Tyr Val Asn Phe Gly
 290 295 300
 Ser Ile Thr Val Met Thr Pro Asn Gln Leu Ile Glu Phe Ala Trp Gly
 305 310 315 320
 Leu Ala Asn Ser Gln Gln Thr Phe Leu Trp Ile Ile Arg Pro Asp Ile
 325 330 335
 Val Ser Gly Asp Ala Ser Ile Leu Pro Pro Glu Phe Val Glu Glu Thr
 340 345 350
 Lys Asn Arg Gly Met Leu Ala Ser Trp Cys Ser Gln Glu Glu Val Leu
 355 360 365
 Ser His Pro Ala Ile Val Gly Phe Leu Thr His Ser Gly Trp Asn Ser
 370 375 380
 Thr Leu Glu Ser Ile Ser Ser Gly Val Pro Met Ile Cys Trp Pro Phe
 385 390 395 400
 Phe Ala Glu Gln Gln Thr Asn Cys Trp Phe Ser Val Thr Lys Trp Asp
 405 410 415
 Val Gly Met Glu Ile Asp Ser Asp Val Lys Arg Asp Glu Val Glu Ser
 420 425 430
 Leu Val Arg Glu Leu Met Val Gly Gly Lys Gly Lys Lys Met Lys Lys
 435 440 445
 Lys Ala Met Glu Trp Lys Glu Leu Ala Glu Ala Ser Ala Lys Glu His
 450 455 460
 Ser Gly Ser Ser Tyr Val Asn Ile Glu Lys Leu Val Asn Asp Ile Leu
 465 470 475 480
 Leu Ser Ser Lys His
 485

<210> SEQ ID NO 35

<211> LENGTH: 1458

<212> TYPE: DNA

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 35

atgggttcca ttggtgctga attaacaag ccacatgcag tttgcatacc atatcccgcc	60
caaggccata ttaaccccat gttaaagcta gccaaaatcc ttcatacaca aggctttcac	120
atcacttttg tcaatactga atttaaccac cgacgtctcc ttaaatctcg tggeccctgat	180
tctctcaagg gtctttcttc ttccggtttt gagaccattc ctgatggact tccgccatgt	240
gaggcagatg ccacacaaga tataccttct ttgtgtgaat ctacaaccaa tacttgcttg	300
gtctctttta gggatcttct tgcgaaactc aatgatacta acacatctaa cgtgccaccc	360
gtttcgtgca tcgtctcgga tgggtgcatg agcttcacct tagccgctgc acaagaattg	420
ggagtccttg aagttctgtt ttggaccact agtgcttgtg gtttcttagg ttacatgcat	480
tactgcaagg ttattgaaaa aggatatgct ccacttaaag atgcgagtga cttgacaaat	540
ggatacctag agacaacatt ggattttata ccaggcatga aagacgtacg ttttaaggat	600
cttccaagtt tcttgagaac taaaaatcca gatgaattca tgatcaaatt tgctctccaa	660
gaaacagaga gagcaagaaa ggcttctgca attatcctca acacatttga aacactagag	720
gctgaagttc ttgaatcgct ccgaaatctt ctctctccag tctaccccat agggcccttg	780
cattttctag tgaacatgt tgatgatgag aatttgaagg gacttagatc cagcctttgg	840

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aaagaggaac cagagtgtat acaatggctt gataccaaag aaccaaattc tgttgtttat    900
gttaactttg gaagcattac tgttatgact cctaatacgc ttattgagtt tgcttgggga    960
cttgcaaaca gccagcaaac attcttatgg atcataagac ctgatattgt ttcaggtgat   1020
gcacgcgattc ttccaccgca attcgtggaa gaaacgaaga acagagggtat gcttgctagt   1080
tgggtgttcac aagaagaagt acttagtcac cctgcaatag taggattctt gactcacagt   1140
ggatggaatt cgacactcga aagtataagc agtgggggtgc ctatgatttg ctggccattt   1200
ttcgcgtgaac agcaaacaaa ttgttggttt tccgtcacta aatgggatgt tggaatggag   1260
attgacagtg atgtgaagag agatgaagtg gaaagccttg taagggaatt gatggttggg   1320
ggaaaaggca aaaagatgaa gaaaaaggca atggaatgga aggaattggc tgaagcatct   1380
gctaaagaac attcagggtc atcttatgtg aacattgaaa agttggtcaa tgatattctt   1440
ctttcatcca aacattaa                                     1458

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<210> SEQ ID NO 36

<211> LENGTH: 485

<212> TYPE: PRT

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 36

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Met Gly Ser Ile Gly Ala Glu Phe Thr Lys Pro His Ala Val Cys Ile
1           5           10          15

Pro Tyr Pro Ala Gln Gly His Ile Asn Pro Met Leu Lys Leu Ala Lys
20          25          30

Ile Leu His His Lys Gly Phe His Ile Thr Phe Val Asn Thr Glu Phe
35          40          45

Asn His Arg Arg Leu Leu Lys Ser Arg Gly Pro Asp Ser Leu Lys Gly
50          55          60

Leu Ser Ser Phe Arg Phe Glu Thr Ile Pro Asp Gly Leu Pro Pro Cys
65          70          75          80

Asp Ala Asp Ala Thr Gln Asp Ile Pro Ser Leu Cys Glu Ser Thr Thr
85          90          95

Asn Thr Cys Leu Gly Pro Phe Arg Asp Leu Leu Ala Lys Leu Asn Asp
100         105         110

Thr Asn Thr Ser Asn Val Pro Pro Val Ser Cys Ile Ile Ser Asp Gly
115         120         125

Val Met Ser Phe Thr Leu Ala Ala Ala Gln Glu Leu Gly Val Pro Glu
130         135         140

Val Leu Phe Trp Thr Thr Ser Ala Cys Gly Phe Leu Gly Tyr Met His
145         150         155         160

Tyr Tyr Lys Val Ile Glu Lys Gly Tyr Ala Pro Leu Lys Asp Ala Ser
165         170         175

Asp Leu Thr Asn Gly Tyr Leu Glu Thr Thr Leu Asp Phe Ile Pro Cys
180         185         190

Met Lys Asp Val Arg Leu Arg Asp Leu Pro Ser Phe Leu Arg Thr Thr
195         200         205

Asn Pro Asp Glu Phe Met Ile Lys Phe Val Leu Gln Glu Thr Glu Arg
210         215         220

Ala Arg Lys Ala Ser Ala Ile Ile Leu Asn Thr Tyr Glu Thr Leu Glu
225         230         235         240

Ala Glu Val Leu Glu Ser Leu Arg Asn Leu Leu Pro Pro Val Tyr Pro
245         250         255

Ile Gly Pro Leu His Phe Leu Val Lys His Val Asp Asp Glu Asn Leu

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260	265	270
Lys Gly Leu Arg Ser Ser Leu Trp Lys Glu Glu Pro Glu Cys Ile Gln 275 280 285		
Trp Leu Asp Thr Lys Glu Pro Asn Ser Val Val Tyr Val Asn Phe Gly 290 295 300		
Ser Ile Thr Val Met Thr Pro Asn Gln Leu Ile Glu Phe Ala Trp Gly 305 310 315 320		
Leu Ala Asn Ser Gln Gln Ser Phe Leu Trp Ile Ile Arg Pro Asp Ile 325 330 335		
Val Ser Gly Asp Ala Ser Ile Leu Pro Pro Glu Phe Val Glu Glu Thr 340 345 350		
Lys Lys Arg Gly Met Leu Ala Ser Trp Cys Ser Gln Glu Glu Val Leu 355 360 365		
Ser His Pro Ala Ile Gly Gly Phe Leu Thr His Ser Gly Trp Asn Ser 370 375 380		
Thr Leu Glu Ser Ile Ser Ser Gly Val Pro Met Ile Cys Trp Pro Phe 385 390 395 400		
Phe Ala Glu Gln Gln Thr Asn Cys Trp Phe Ser Val Thr Lys Trp Asp 405 410 415		
Val Gly Met Glu Ile Asp Cys Asp Val Lys Arg Asp Glu Val Glu Ser 420 425 430		
Leu Val Arg Glu Leu Met Val Gly Gly Lys Gly Lys Lys Met Lys Lys 435 440 445		
Lys Ala Met Glu Trp Lys Glu Leu Ala Glu Ala Ser Ala Lys Glu His 450 455 460		
Ser Gly Ser Ser Tyr Val Asn Ile Glu Lys Val Val Asn Asp Ile Leu 465 470 475 480		
Leu Ser Ser Lys His 485		

<210> SEQ ID NO 37

<211> LENGTH: 1458

<212> TYPE: DNA

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 37

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atgggttcca ttggtgctga atttacaag ccacatgcag tttgcatacc atatcccgcc      60
caaggccata ttaaccccat gttaaagcta gccaaaatcc ttcatacaca aggttttcac      120
atcacttttg tcaatactga atttaaccac agacgtctgc ttaaattctg tggccctgat      180
tctctcaagg gtctttcttc ttcccgtttt gagacaattc ctgatggact tccgccatgt      240
gatgcagatg ccacacaaga tataccttct ttgtgtgaat ctacaaccaa tacttgcttg      300
ggtcctttta gggatcttct tgcgaaactc aatgatacta acacatctaa cgtgccaccc      360
gtttcgtgca tcattctcaga tgggtgtcatg agcttcacct tagccgctgc acaagaattg      420
ggagtcacct aagttctgtt ttggaccact agtgcttggtg gttctctagg ttacatgcat      480
tattacaagg ttattgaaaa aggatacgtc ccacttaaag atgcgagtga cttgacaaat      540
ggatacctag agacaacatt ggattttata ccatgcatga aagacgtacg ttttaagggat      600
cttccaagtt tcttgagaac tacaaatcca gatgaattca tgatcaaatt tgtctcccaa      660
gaaacagaga gagcaagaaa ggcttctgca attatcttca acacatatga aacactagag      720
gctgaagttc ttgaatcgct ccgaaatctt ctctctccag tctaccccat tgggcccttg      780
cattttctag tgaacatgtg tgatgatgag aatttgaagg gacttagatc cagcctttgg      840

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aaagaggaac cagagtgtat acaatggctt gataccaaag aaccaaattc tgtgttttat    900
gttaactttg gaagcattac tgttatgact cctaatacaac ttattgaatt tgcttgggga    960
cttgcaaaca gccacaatc attcttatgg atcataagac ctgatattgt ttcaggtgat    1020
gcatcgattc ttcccccga attcgtggaa gaaacgaaga agagaggatg gcttgctagt    1080
tggtgttcac aagaagaagt acttagtcac cctgcaatag gaggattctt gactcacagt    1140
ggatggaatt cgacactcga aagtataagc agtgggggtc ctatgatttg ctggccattt    1200
ttcgctgaac agcaaacaaa ttgttggttt tccgtcacta aatgggatgt tggaatggag    1260
attgactgtg atgtgaagag ggatgaagtg gaaagccttg taagggaatt gatggttggg    1320
ggaaaaggca aaaagatgaa gaaaaaggca atggaatgga aggaattggc tgaagcatct    1380
gctaagaac attcagggtc atcttatgtg aacattgaga aggtggtcaa tgatattctt    1440
ctttcgtcca aacattaa                                     1458

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<210> SEQ ID NO 38

<211> LENGTH: 496

<212> TYPE: PRT

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 38

```

Met Ala Thr Gln Val His Lys Leu His Phe Ile Leu Phe Pro Leu Met
 1              5              10              15

Ala Pro Gly His Met Ile Pro Met Ile Asp Ile Ala Lys Leu Leu Ala
      20              25              30

Asn Arg Gly Val Ile Thr Thr Ile Ile Thr Thr Pro Val Asn Ala Asn
      35              40              45

Arg Phe Ser Ser Thr Ile Thr Arg Ala Ile Lys Ser Gly Leu Arg Ile
      50              55              60

Gln Ile Leu Thr Leu Lys Phe Pro Ser Val Glu Val Gly Leu Pro Glu
      65              70              75              80

Gly Cys Glu Asn Ile Asp Met Leu Pro Ser Leu Asp Leu Ala Ser Lys
      85              90              95

Phe Phe Ala Ala Ile Ser Met Leu Lys Gln Gln Val Glu Asn Leu Leu
      100             105             110

Glu Gly Ile Asn Pro Ser Pro Ser Cys Val Ile Ser Asp Met Gly Phe
      115             120             125

Pro Trp Thr Thr Gln Ile Ala Gln Asn Phe Asn Ile Pro Arg Ile Val
      130             135             140

Phe His Gly Thr Cys Cys Phe Ser Leu Leu Cys Ser Tyr Lys Ile Leu
      145             150             155             160

Ser Ser Asn Ile Leu Glu Asn Ile Thr Ser Asp Ser Glu Tyr Phe Val
      165             170             175

Val Pro Asp Leu Pro Asp Arg Val Glu Leu Thr Lys Ala Gln Val Ser
      180             185             190

Gly Ser Thr Lys Asn Thr Thr Ser Val Ser Ser Ser Val Leu Lys Glu
      195             200             205

Val Thr Glu Gln Ile Arg Leu Ala Glu Glu Ser Ser Tyr Gly Val Ile
      210             215             220

Val Asn Ser Phe Glu Glu Leu Glu Gln Val Tyr Glu Lys Glu Tyr Arg
      225             230             235             240

Lys Ala Arg Gly Lys Lys Val Trp Cys Val Gly Pro Val Ser Leu Cys
      245             250             255

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gaagatttgg ttacaagggg taataaaact gcaattgata atcaagattg cttgaaatgg      840
ttagataatt ttgaacaga atctgtggtt tatgcaagtc ttggaagttt atctcgtttg      900
acattattgc aaatggtgga acttgggtctt ggtttagaag agtcaaatag gccttttgta     960
tgggtattag gaggaggtga taaattaaat gatttagaga aatggattct tgagaatgga    1020
tttgagcaaa gaattaaaga aagaggagtt ttgattagag gatgggctcc tcaagtgcct     1080
atactttcac accctgcaat tggtaggagta ttgactcatt gcggatggaa ttctacattg    1140
gaaggtatct cagcaggatt accaatggta acatggccac tatttgcgtga gcaattttgc     1200
aatgagaagt tagtagtcca agtgctaaaa attggagtgga gcctagggtg gaaggtgcct     1260
gtcaaatggg gagatgagga aaatgttgga gttttggtta aaaaggatga tgtaagaaa     1320
gcattagaca aactaatgga tgaaggagaa gaaggacaag taagaagaac aaaagcaaaa     1380
gagttaggag aattggctaa aaaggcatct ggagaagggtg gttcttctta tgtaactta     1440
acatctctga ttgaagacat cattgagcaa caaaatcaca aggaaaaata g              1491

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<210> SEQ ID NO 40

<211> LENGTH: 479

<212> TYPE: PRT

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 40

```

Met Lys Thr Ala Glu Leu Val Phe Ile Pro Ala Pro Gly Met Gly His
1           5           10           15
Leu Val Pro Thr Val Glu Val Ala Lys Gln Leu Val Asp Arg His Glu
20          25          30
Gln Leu Ser Ile Thr Val Leu Ile Met Thr Ile Pro Leu Glu Thr Asn
35          40          45
Ile Pro Ser Tyr Thr Lys Ser Leu Ser Ser Asp Tyr Ser Ser Arg Ile
50          55          60
Thr Leu Leu Pro Leu Ser Gln Pro Glu Thr Ser Val Thr Met Ser Ser
65          70          75          80
Phe Asn Ala Ile Asn Phe Phe Glu Tyr Ile Ser Ser Tyr Lys Gly Arg
85          90          95
Val Lys Asp Ala Val Ser Glu Thr Ser Phe Ser Ser Ser Asn Ser Val
100         105         110
Lys Leu Ala Gly Phe Val Ile Asp Met Phe Cys Thr Ala Met Ile Asp
115         120         125
Val Ala Asn Glu Phe Gly Ile Pro Ser Tyr Val Phe Tyr Thr Ser Ser
130         135         140
Ala Ala Met Leu Gly Leu Gln Leu His Phe Gln Ser Leu Ser Ile Glu
145         150         155         160
Cys Ser Pro Lys Val His Asn Tyr Val Glu Pro Glu Ser Glu Val Leu
165         170         175
Ile Ser Thr Tyr Met Asn Pro Val Pro Val Lys Cys Leu Pro Gly Ile
180         185         190
Ile Leu Val Asn Asp Glu Ser Ser Thr Met Phe Val Asn His Ala Arg
195         200         205
Arg Phe Arg Glu Thr Lys Gly Ile Met Val Asn Thr Phe Thr Glu Leu
210         215         220
Glu Ser His Ala Leu Lys Ala Leu Ser Asp Asp Glu Lys Ile Pro Pro
225         230         235         240
Ile Tyr Pro Val Gly Pro Ile Leu Asn Leu Glu Asn Gly Asn Glu Asp
245         250         255

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His Asn Gln Glu Tyr Asp Ala Ile Met Lys Trp Leu Asp Glu Lys Pro
 260 265 270
 Asn Ser Ser Val Val Phe Leu Cys Phe Gly Ser Lys Gly Ser Phe Glu
 275 280 285
 Glu Asp Gln Val Lys Glu Ile Ala Asn Ala Leu Glu Ser Ser Gly Tyr
 290 295 300
 His Phe Leu Trp Ser Leu Arg Arg Pro Pro Pro Lys Asp Lys Leu Gln
 305 310 315 320
 Phe Pro Ser Glu Phe Glu Asn Pro Glu Glu Val Leu Pro Glu Gly Phe
 325 330 335
 Phe Gln Arg Thr Lys Gly Arg Gly Lys Val Ile Gly Trp Ala Pro Gln
 340 345 350
 Leu Ala Ile Leu Ser His Pro Ser Val Gly Gly Phe Val Ser His Cys
 355 360 365
 Gly Trp Asn Ser Thr Leu Glu Ser Val Arg Ser Gly Val Pro Ile Ala
 370 375 380
 Thr Trp Pro Leu Tyr Ala Glu Gln Gln Ser Asn Ala Phe Gln Leu Val
 385 390 395 400
 Lys Asp Leu Gly Met Ala Val Glu Ile Lys Met Asp Tyr Arg Glu Asp
 405 410 415
 Phe Asn Thr Arg Asn Pro Pro Leu Val Lys Ala Glu Glu Ile Glu Asp
 420 425 430
 Gly Ile Arg Lys Leu Met Asp Ser Glu Asn Lys Ile Arg Ala Lys Val
 435 440 445
 Thr Glu Met Lys Asp Lys Ser Arg Ala Ala Leu Leu Glu Gly Gly Ser
 450 455 460
 Ser Tyr Val Ala Leu Gly His Phe Val Glu Thr Val Met Lys Asn
 465 470 475

<210> SEQ ID NO 41
 <211> LENGTH: 1440
 <212> TYPE: DNA
 <213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 41

atgaagacag cagagttagt attcattcct gctcctggga tgggtcacct tgtaccaact	60
gtggagggtgg caaagcaact agtcgacaga cacgagcagc tttcgatcac agttctaact	120
atgacaattc ctttggaaac aaatattcca tcatatacta aatcactgtc ctgagactac	180
agttctcgta taacgctgct tccactctct caacctgaga cctctgttac tatgagcagt	240
tttaatgcc acaatttttt tgagtacatc tccagctaca agggctcgtgt caaagatgct	300
gttagtgaaa cctcctttag ttctgtcaa tctgtgaaac ttgcaggatt tgtaatagac	360
atgtttctgca ctgcgatgat tgatgtagcg aacgagtttg gaatcccaag ttatgtgttc	420
tacactttcta gtgcagctat gcttggacta caactgcatt ttcaaagtct tagcattgaa	480
tgacagtcga aagttcataa ctacgttgaa cctgaatcag aagttctgat ctcaacttac	540
atgaatccgg ttccagtc aa atgtttgcc ggaattatac tagtaaatga tgaaagtagc	600
accatgtttg tcaatcatgc acgaagattc agggagacga aaggaattat ggtgaacacg	660
ttcactgagc ttgaatcaca cgctttgaaa gccctttccg atgatgaaaa aatcccacca	720
atctacccag ttggacatt acttaacctt gaaaatggga atgaagatca caatcaagaa	780
tatgatgcga ttatgaagtg gcttgacgag aagcctaatt catcagtggt gttcttatgc	840

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tttgaagca aggggtcttt cgaagaagat caggtgaagg aaatagcaaa tgctctagag    900
agcagtggct accacttctt gtggtcgcta aggcgaccgc caccaaaaga caagctacaa    960
ttccaagcg aattogagaa tccagaggaa gtcttaccag agggattctt tcaaaggact   1020
aaaggaagag gaaaggtgat aggatgggca cccagttgg ctattttgtc tcatccttca   1080
gtaggaggat tcgtgtcgca ttgtgggtgg aattcaactc tggagagcgt tcgaagtgga   1140
gtgccgatag caacatggcc attgtatgca gagcaacaga gcaatgcatt tcaactggtg   1200
aaggatttgg gtatggcagt agagattaag atggattaca gggaagattt taatacgaga   1260
aatccaccac tggttaaagc tgaggagata gaagatggaa ttaggaagct gatggattca   1320
gagaataaaa tcagggtctaa ggtgacggag atgaaggaca aaagtagagc agcactgctg   1380
gagggcggat catcatatgt agctcttggg cattttgttg agactgtcat gaaaaactag   1440

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<210> SEQ ID NO 42

<211> LENGTH: 478

<212> TYPE: PRT

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 42

```

Met Lys Thr Thr Glu Leu Val Phe Ile Pro Ala Pro Gly Met Gly His
 1             5             10            15
Leu Val Pro Thr Val Glu Val Ala Lys Gln Leu Val Asp Arg Asp Glu
      20             25             30
Gln Leu Ser Ile Thr Val Leu Ile Met Thr Leu Pro Leu Glu Thr Asn
      35             40             45
Ile Pro Ser Tyr Thr Lys Ser Leu Ser Ser Asp Tyr Ser Ser Arg Ile
      50             55             60
Thr Leu Leu Gln Leu Ser Gln Pro Glu Thr Ser Val Ser Met Ser Ser
      65             70             75             80
Phe Asn Ala Ile Asn Phe Phe Glu Tyr Ile Ser Ser Tyr Lys Asp Arg
      85             90             95
Val Lys Asp Ala Val Asn Glu Thr Phe Ser Ser Ser Ser Ser Val Lys
      100            105            110
Leu Lys Gly Phe Val Ile Asp Met Phe Cys Thr Ala Met Ile Asp Val
      115            120            125
Ala Asn Glu Phe Gly Ile Pro Ser Tyr Val Phe Tyr Thr Ser Asn Ala
      130            135            140
Ala Met Leu Gly Leu Gln Leu His Phe Gln Ser Leu Ser Ile Glu Tyr
      145            150            155            160
Ser Pro Lys Val His Asn Tyr Leu Asp Pro Glu Ser Glu Val Ala Ile
      165            170            175
Ser Thr Tyr Ile Asn Pro Ile Pro Val Lys Cys Leu Pro Gly Ile Ile
      180            185            190
Leu Asp Asn Asp Lys Ser Gly Thr Met Phe Val Asn His Ala Arg Arg
      195            200            205
Phe Arg Glu Thr Lys Gly Ile Met Val Asn Thr Phe Ala Glu Leu Glu
      210            215            220
Ser His Ala Leu Lys Ala Leu Ser Asp Asp Glu Lys Ile Pro Pro Ile
      225            230            235            240
Tyr Pro Val Gly Pro Ile Leu Asn Leu Gly Asp Gly Asn Glu Asp His
      245            250            255
Asn Gln Glu Tyr Asp Met Ile Met Lys Trp Leu Asp Glu Gln Pro His
      260            265            270

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Ser	Ser	Val	Val	Phe	Leu	Cys	Phe	Gly	Ser	Lys	Gly	Ser	Phe	Glu	Glu
		275						280						285	
Asp	Gln	Val	Lys	Glu	Ile	Ala	Asn	Ala	Leu	Glu	Arg	Ser	Gly	Asn	Arg
	290					295					300				
Phe	Leu	Trp	Ser	Leu	Arg	Arg	Pro	Pro	Pro	Lys	Asp	Thr	Leu	Gln	Phe
305					310					315				320	
Pro	Ser	Glu	Phe	Glu	Asn	Pro	Glu	Glu	Val	Leu	Pro	Val	Gly	Phe	Phe
			325						330					335	
Gln	Arg	Thr	Lys	Gly	Arg	Gly	Lys	Val	Ile	Gly	Trp	Ala	Pro	Gln	Leu
			340					345					350		
Ala	Ile	Leu	Ser	His	Pro	Ala	Val	Gly	Gly	Phe	Val	Ser	His	Cys	Gly
	355						360					365			
Trp	Asn	Ser	Thr	Leu	Glu	Ser	Val	Arg	Ser	Gly	Val	Pro	Ile	Ala	Thr
	370					375					380				
Trp	Pro	Leu	Tyr	Ala	Glu	Gln	Gln	Ser	Asn	Ala	Phe	Gln	Leu	Val	Lys
385					390					395					400
Asp	Leu	Gly	Met	Ala	Val	Glu	Ile	Lys	Met	Asp	Tyr	Arg	Glu	Asp	Phe
			405						410					415	
Asn	Lys	Thr	Asn	Pro	Pro	Leu	Val	Lys	Ala	Glu	Glu	Ile	Glu	Asp	Gly
			420					425					430		
Ile	Arg	Lys	Leu	Met	Asp	Ser	Glu	Asn	Lys	Ile	Arg	Ala	Lys	Val	Met
	435						440					445			
Glu	Met	Lys	Asp	Lys	Ser	Arg	Ala	Ala	Leu	Leu	Glu	Gly	Gly	Ser	Ser
	450					455					460				
Tyr	Val	Ala	Leu	Gly	His	Phe	Val	Glu	Thr	Val	Met	Lys	Asn		
465					470					475					

<210> SEQ ID NO 43

<211> LENGTH: 1437

<212> TYPE: DNA

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 43

atgaagacaa cagagttagt attcattcct gctcctggca tgggtcacct tgtaccact	60
gtggagggtgg caaagcaact agtcgacaga gacgaacagc tttcaatcac agttctcatc	120
atgacgcttc ctttggaac aaatattcca tcatatacta aatcactgtc ctcagactac	180
agttctcgta taacgctgct tcaactttct caacctgaga cctctgttag tatgagcagt	240
tttaatgcc acaatttttt tgagtacatc tccagctaca aggatcgtgt caaagatgct	300
gttaatgaaa ccttagtttc gtcaagttct gtgaaactca aaggatttgt aatagacatg	360
ttctgcactg cgatgattga tgtggcgaac gagtttggaa tcccaagtta tgtctctac	420
acttctaag cagctatgct tggactccaa ctccattttc aaagtcttag tattgaatac	480
agtccgaaag ttcataatta cctagaccct gaatcagaag tagcgatctc aacttacatt	540
aatccgattc cagtcaaatg tttgccggg attatactag acaatgataa aagtggcacc	600
atgttcgtca atcatgcacg aagattcagg gagacgaaag gaattatggt gaacacattc	660
gctgagcttg aatcacacgc tttgaaagcc ctttccgatg atgagaaaat cccaccaatc	720
taccagttg ggcctatact taaccttga gatgggaatg aagatcaca tcaagaatat	780
gatatgatta tgaagtggct cgacgagcag cctcattcat cagtgggtgt cctatgcttt	840
ggaagcaagg gatctttcga agaagatcaa gtgaaggaaa tagcaaatgc tctagagaga	900
agtggtaacc ggttctgtg gtcgctaaga cgaccgccac caaagacac gctacaattc	960

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ccaagcgaat tgcagaatcc agaggaagtc ttgccggtgg gattctttca aaggactaaa 1020
ggaagaggaa aggtgatagg atgggcaccc cagttggcta tttgtctca tctgcagta 1080
ggaggattcg tgtcgattg tgggtggaat tcaactttgg agagtgttcg tagtgagta 1140
ccgatagcaa catggccatt gtatgcagag caacagagca atgcatttca actggtgaag 1200
gatttgggga tggcagtgga gattaagatg gattacaggg aagattttaa taagacaaat 1260
ccaccactgg ttaaagctga ggagatagaa gatggaatta ggaagctgat ggattcagag 1320
aataaaatca gggctaaggt gatggagatg aaggacaaaa gtagagcagc gttattagaa 1380
ggcggatcat catatgtagc tctcgggcat tttgttgaga ctgtcatgaa aaactaa 1437

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<210> SEQ ID NO 44

<211> LENGTH: 482

<212> TYPE: PRT

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 44

```

Met Lys Glu Thr Lys Lys Ile Glu Leu Val Phe Ile Pro Ser Pro Gly
1           5           10           15
Ile Gly His Leu Val Ser Thr Val Glu Met Ala Lys Leu Leu Ile Ala
20          25          30
Arg Glu Glu Gln Leu Ser Ile Thr Val Leu Ile Ile Gln Trp Pro Asn
35          40          45
Asp Lys Lys Leu Asp Ser Tyr Ile Gln Ser Val Ala Asn Phe Ser Ser
50          55          60
Arg Leu Lys Phe Ile Arg Leu Pro Gln Asp Asp Ser Ile Met Gln Leu
65          70          75          80
Leu Lys Ser Asn Ile Phe Thr Thr Phe Ile Ala Ser His Lys Pro Ala
85          90          95
Val Arg Asp Ala Val Ala Asp Ile Leu Lys Ser Glu Ser Asn Asn Thr
100         105         110
Leu Ala Gly Ile Val Ile Asp Leu Phe Cys Thr Ser Met Ile Asp Val
115         120         125
Ala Asn Glu Phe Glu Leu Pro Thr Tyr Val Phe Tyr Thr Ser Gly Ala
130         135         140
Ala Thr Leu Gly Leu His Tyr His Ile Gln Asn Leu Arg Asp Glu Phe
145         150         155         160
Asn Lys Asp Ile Thr Lys Tyr Lys Asp Glu Pro Glu Glu Lys Leu Ser
165         170         175
Ile Ala Thr Tyr Leu Asn Pro Phe Pro Ala Lys Cys Leu Pro Ser Val
180         185         190
Ala Leu Asp Lys Glu Gly Gly Ser Thr Met Phe Leu Asp Leu Ala Lys
195         200         205
Arg Phe Arg Glu Thr Lys Gly Ile Met Ile Asn Thr Phe Leu Glu Leu
210         215         220
Glu Ser Tyr Ala Leu Asn Ser Leu Ser Arg Asp Lys Asn Leu Pro Pro
225         230         235         240
Ile Tyr Pro Val Gly Pro Val Leu Asn Leu Asn Asn Val Glu Gly Asp
245         250         255
Asn Leu Gly Ser Ser Asp Gln Asn Thr Met Lys Trp Leu Asp Asp Gln
260         265         270
Pro Ala Ser Ser Val Val Phe Leu Cys Phe Gly Ser Gly Gly Ser Phe
275         280         285
Glu Lys His Gln Val Lys Glu Ile Ala Tyr Ala Leu Glu Ser Ser Gly

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290	295	300
Cys Arg Phe Leu Trp Ser Leu Arg Arg Pro Pro Thr Glu Asp Ala Arg		
305	310	315 320
Phe Pro Ser Asn Tyr Glu Asn Leu Glu Glu Ile Leu Pro Glu Gly Phe		
	325	330 335
Leu Glu Arg Thr Lys Gly Ile Gly Lys Val Ile Gly Trp Ala Pro Gln		
	340	345 350
Leu Ala Ile Leu Ser His Lys Ser Thr Gly Gly Phe Val Ser His Cys		
	355	360 365
Gly Trp Asn Ser Thr Leu Glu Ser Thr Tyr Phe Gly Val Pro Ile Ala		
	370	375 380
Thr Trp Pro Met Tyr Ala Glu Gln Gln Ala Asn Ala Phe Gln Leu Val		
	385	390 395 400
Lys Asp Leu Arg Met Gly Val Glu Ile Lys Met Asp Tyr Arg Lys Asp		
	405	410 415
Met Lys Val Met Gly Lys Glu Val Ile Val Lys Ala Glu Glu Ile Glu		
	420	425 430
Lys Ala Ile Arg Glu Ile Met Asp Ser Glu Ser Glu Ile Arg Val Lys		
	435	440 445
Val Lys Glu Met Lys Glu Lys Ser Arg Ala Ala Gln Met Glu Gly Gly		
	450	455 460
Ser Ser Tyr Thr Ser Ile Gly Gly Phe Ile Gln Ile Ile Met Glu Asn		
	465	470 475 480
Ser Gln		

<210> SEQ ID NO 45

<211> LENGTH: 1449

<212> TYPE: DNA

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 45

```

atgaaagaaa ccaagaaaat agagtttagtc ttcattcctt caccaggaat tggccattta    60
gtatccacag ttgaaatggc aaagcttctt atagctagag aagagcagct atctatcaca    120
gtctcatca tccaatggcc taacgacaag aagctcgatt cttatatcca atcagtcgcc    180
aatttcagct cgcgtttgaa attcattcga ctccctcagg atgattccat tatgcagcta    240
ctcaaaagca acattttcac caggtttatt gccagtcata agcctgcagt tagagatgct    300
gttgctgata ttctcaagtc agaatacaat aatacgttag caggtattgt tatcgacttg    360
ttctgcacct caatgataga cgtggccaat gagttcgagc taccaaccta tgttttctac    420
acgtctgggtg cagcaaccct tgggtcttcat tatcatatac agaattctcag ggatgaattt    480
aacaagata ttaccaagta caaagacgaa cctgaagaaa aactctctat agcaacatat    540
ctcaatccat ttccagcaaa atgtttgccc tctgtagcct tagacaaaga aggtggttca    600
acaatgtttc ttgatctcgc aaaaagggtt cgagaaacca aaggtattat gataaacaca    660
tttctagagc tcgaatccta tgcattaaac tcgctctcac gagacaagaa tcttccacct    720
atataccctg tcggaccagt attgaacctt aacaatgttg aaggtgacaa cttaggttca    780
tctgaccaga atactatgaa atgggttagat gatcagcccg cttcatctgt agtggtcctt    840
tgttttggta gtggtggaag ctttgaaaaa catcaagtta aggaaatagc ctatgctctg    900
gagagcagtg ggtgtcggtt tttgtggtcg ttaaggcgac caccaaccga agatgcaaga    960
tttccaagca actatgaaaa tcttgaagaa attttgccag aaggattctt ggaaagaaca   1020

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aaagggattg gaaaagtgat aggatgggca cctcagttgg cgattttgtc acataaatcg 1080
acgggggggat ttgtgtcgca ctgtggatgg aattcgactt tggaagtac atattttgga 1140
gtgccaatag caacctggcc aatgtacgag gagcaacaag cgaatgcatt tcaattgggt 1200
aaggatttga gaatgggagt tgagattaag atggattata ggaaggatat gaaagtgatg 1260
ggcaagaag ttatagttaa agctgaggag attgagaaag caataagaga aattatggat 1320
tccgagatg aaattcgggt gaagtgaaa gagatgaagg agaagagcag agcagcaca 1380
atggaagggt gctcttctta cacttctatt ggaggtttca tccaaattat catggagaat 1440
tctcaataa 1449

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<210> SEQ ID NO 46

<211> LENGTH: 470

<212> TYPE: PRT

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 46

```

Met Val Gln Pro His Val Leu Leu Val Thr Phe Pro Ala Gln Gly His
1          5          10         15
Ile Asn Pro Cys Leu Gln Phe Ala Lys Arg Leu Ile Arg Met Gly Ile
20        25        30
Glu Val Thr Phe Ala Thr Ser Val Phe Ala His Arg Arg Met Ala Lys
35        40        45
Thr Thr Thr Ser Thr Leu Ser Lys Gly Leu Asn Phe Ala Ala Phe Ser
50        55        60
Asp Gly Tyr Asp Asp Gly Phe Lys Ala Asp Glu His Asp Ser Gln His
65        70        75        80
Tyr Met Ser Glu Ile Lys Ser Arg Gly Ser Lys Thr Leu Lys Asp Ile
85        90        95
Ile Leu Lys Ser Ser Asp Glu Gly Arg Pro Val Thr Ser Leu Val Tyr
100       105       110
Ser Leu Leu Leu Pro Trp Ala Ala Lys Val Ala Arg Glu Phe His Ile
115       120       125
Pro Cys Ala Leu Leu Trp Ile Gln Pro Ala Thr Val Leu Asp Ile Tyr
130       135       140
Tyr Tyr Tyr Phe Asn Gly Tyr Glu Asp Ala Ile Lys Gly Ser Thr Asn
145       150       155       160
Asp Pro Asn Trp Cys Ile Gln Leu Pro Arg Leu Pro Leu Leu Lys Ser
165       170       175
Gln Asp Leu Pro Ser Phe Leu Leu Ser Ser Ser Asn Glu Glu Lys Tyr
180       185       190
Ser Phe Ala Leu Pro Thr Phe Lys Glu Gln Leu Asp Thr Leu Asp Val
195       200       205
Glu Glu Asn Pro Lys Val Leu Val Asn Thr Phe Asp Ala Leu Glu Pro
210       215       220
Lys Glu Leu Lys Ala Ile Glu Lys Tyr Asn Leu Ile Gly Ile Gly Pro
225       230       235       240
Leu Ile Pro Ser Thr Phe Leu Asp Gly Lys Asp Pro Leu Asp Ser Ser
245       250       255
Phe Gly Gly Asp Leu Phe Gln Lys Ser Asn Asp Tyr Ile Glu Trp Leu
260       265       270
Asn Ser Lys Ala Asn Ser Ser Val Val Tyr Ile Ser Phe Gly Ser Leu
275       280       285
Leu Asn Leu Ser Lys Asn Gln Lys Glu Glu Ile Ala Lys Gly Leu Ile

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290	295	300
Glu Ile Lys Lys Pro Phe Leu Trp Val Ile Arg Asp Gln Glu Asn Gly		
305	310	315 320
Lys Gly Asp Glu Lys Glu Glu Lys Leu Ser Cys Met Met Glu Leu Glu		
	325	330 335
Lys Gln Gly Lys Ile Val Pro Trp Cys Ser Gln Leu Glu Val Leu Thr		
	340	345 350
His Pro Ser Ile Gly Cys Phe Val Ser His Cys Gly Trp Asn Ser Thr		
	355	360 365
Leu Glu Ser Leu Ser Ser Gly Val Ser Val Val Ala Phe Pro His Trp		
	370	375 380
Thr Asp Gln Gly Thr Asn Ala Lys Leu Ile Glu Asp Val Trp Lys Thr		
385	390	395 400
Gly Val Arg Leu Lys Lys Asn Glu Asp Gly Val Val Glu Ser Glu Glu		
	405	410 415
Ile Lys Arg Cys Ile Glu Met Val Met Asp Gly Gly Glu Lys Gly Glu		
	420	425 430
Glu Met Arg Arg Asn Ala Gln Lys Trp Lys Glu Leu Ala Arg Glu Ala		
	435	440 445
Val Lys Glu Gly Gly Ser Ser Glu Met Asn Leu Lys Ala Phe Val Gln		
	450	455 460
Glu Val Gly Lys Gly Cys		
465	470	

<210> SEQ ID NO 47

<211> LENGTH: 1413

<212> TYPE: DNA

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 47

```

atggtgcaac cccatgtcct cttggtgact ttccagcac aaggccatat taatccatgt      60
ctccaatttg ccaagaggct aattagaatg ggcattgagg taacttttgc cagagcggt      120
ttcgcccatc gtcgtatggc aaaaactacg acttccactc tatccaaggg cttaaathtt      180
ggcgcatctc ctgatgggta cgaagatggg ttcaaggccg atgagcatga ttctcaacat      240
tacatgtcgg agataaaaag tcgcggttct aaaaccctaa aagatatcat ttgaagagc      300
tcagacgagg gacgtcctgt gacatccctc gtctattctc tttgtcttcc atgggctgca      360
aaggtagcgc gtgaatttca cataccgtgc gcgttactat ggattcaacc agcaactgtg      420
ctagacatat attattatta cttcaatggc tatgaggatg ccataaaagg tagcaccaat      480
gatccaaatt ggtgtattca attgcctagg cttccactac taaaagcca agatcttctc      540
tcttttttac tttcttctag taatgaagaa aaatatagct ttgtcttacc aacattttaa      600
gagcaacttg acacattaga tgttgaagaa aatcctaaag tacttgtgaa cacatttgat      660
gcattagagc caaaggaact caaagctatt gaaaagtaca atttaattgg gattggacca      720
ttgattcctt caacattttt ggacggaaaa gaccctttgg attcttctt ttgtgtgtgat      780
ctttttcaaa agtctaata ctatattgaa tgggtgaact caaaggctaa ctcatctgtg      840
gtttatatct catttgggag tctcttgaat ttgtcaaaaa atcaaaagga ggagattgca      900
aaagggttga tagagattaa aaagccattc ttgtgggtaa taagagatca agaaaatggt      960
aagggagatg aaaaagaaga gaaattaagt tgtatgatgg agttggaaaa gcaagggaaa     1020
atagtaccat ggtgttcaca acttgaagtc ttaacacatc catctatagg atgtttcgtg     1080

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tcacattgtg gatggaattc gactctggaa agtttatcgt caggcgtgtc agtagtggca 1140
tttctcatt ggacggatca agggacaaat gctaaactaa ttgaagatgt ttggaagaca 1200
ggtgtaaggt tgaaaaagaa tgaagatggt gtggttgaga gtgaagagat aaaaagggtgc 1260
atagaaatgg taatggatgg tggagagaaa ggagaagaaa tgagaagaaa tgctcaaaaa 1320
tggaagaat tggcaaggga agctgtaaaa gaaggcggat cttcggaat gaattcaaaa 1380
gcttttgttc aagaagttgg caaaggttgc tga 1413

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<210> SEQ ID NO 48
<211> LENGTH: 545
<212> TYPE: PRT
<213> ORGANISM: Cannabis

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<400> SEQUENCE: 48

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```

Met Asn Cys Ser Ala Phe Ser Phe Trp Phe Val Cys Lys Ile Ile Phe
1      5      10      15
Phe Phe Leu Ser Phe His Ile Gln Ile Ser Ile Ala Asn Pro Arg Glu
20      25      30
Asn Phe Leu Lys Cys Phe Ser Lys His Ile Pro Asn Asn Val Ala Asn
35      40      45
Pro Lys Leu Val Tyr Thr Gln His Asp Gln Leu Tyr Met Ser Ile Leu
50      55      60
Asn Ser Thr Ile Gln Asn Leu Arg Phe Ile Ser Asp Thr Thr Pro Lys
65      70      75      80
Pro Leu Val Ile Val Thr Pro Ser Asn Asn Ser His Ile Gln Ala Thr
85      90      95
Ile Leu Cys Ser Lys Lys Val Gly Leu Gln Ile Arg Thr Arg Ser Gly
100     105     110
Gly His Asp Ala Glu Gly Met Ser Tyr Ile Ser Gln Val Pro Phe Val
115     120     125
Val Val Asp Leu Arg Asn Met His Ser Ile Lys Ile Asp Val His Ser
130     135     140
Gln Thr Ala Trp Val Glu Ala Gly Ala Thr Leu Gly Glu Val Tyr Tyr
145     150     155     160
Trp Ile Asn Glu Lys Asn Glu Asn Leu Ser Phe Pro Gly Gly Tyr Cys
165     170     175
Pro Thr Val Gly Val Gly Gly His Phe Ser Gly Gly Gly Tyr Gly Ala
180     185     190
Leu Met Arg Asn Tyr Gly Leu Ala Ala Asp Asn Ile Ile Asp Ala His
195     200     205
Leu Val Asn Val Asp Gly Lys Val Leu Asp Arg Lys Ser Met Gly Glu
210     215     220
Asp Leu Phe Trp Ala Ile Arg Gly Gly Gly Gly Glu Asn Phe Gly Ile
225     230     235     240
Ile Ala Ala Trp Lys Ile Lys Leu Val Asp Val Pro Ser Lys Ser Thr
245     250     255
Ile Phe Ser Val Lys Lys Asn Met Glu Ile His Gly Leu Val Lys Leu
260     265     270
Phe Asn Lys Trp Gln Asn Ile Ala Tyr Lys Tyr Asp Lys Asp Leu Val
275     280     285
Leu Met Thr His Phe Ile Thr Lys Asn Ile Thr Asp Asn His Gly Lys
290     295     300
Asn Lys Thr Thr Val His Gly Tyr Phe Ser Ser Ile Phe His Gly Gly
305     310     315     320

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Val Asp Ser Leu Val Asp Leu Met Asn Lys Ser Phe Pro Glu Leu Gly
325 330 335

Ile Lys Lys Thr Asp Cys Lys Glu Phe Ser Trp Ile Asp Thr Thr Ile
340 345 350

Phe Tyr Ser Gly Val Val Asn Phe Asn Thr Ala Asn Phe Lys Lys Glu
355 360 365

Ile Leu Leu Asp Arg Ser Ala Gly Lys Lys Thr Ala Phe Ser Ile Lys
370 375 380

Leu Asp Tyr Val Lys Lys Pro Ile Pro Glu Thr Ala Met Val Lys Ile
385 390 395 400

Leu Glu Lys Leu Tyr Glu Glu Asp Val Gly Ala Gly Met Tyr Val Leu
405 410 415

Tyr Pro Tyr Gly Gly Ile Met Glu Glu Ile Ser Glu Ser Ala Ile Pro
420 425 430

Phe Pro His Arg Ala Gly Ile Met Tyr Glu Leu Trp Tyr Thr Ala Ser
435 440 445

Trp Glu Lys Gln Glu Asp Asn Glu Lys His Ile Asn Trp Val Arg Ser
450 455 460

Val Tyr Asn Phe Thr Thr Pro Tyr Val Ser Gln Asn Pro Arg Leu Ala
465 470 475 480

Tyr Leu Asn Tyr Arg Asp Leu Asp Leu Gly Lys Thr Asn His Ala Ser
485 490 495

Pro Asn Asn Tyr Thr Gln Ala Arg Ile Trp Gly Glu Lys Tyr Phe Gly
500 505 510

Lys Asn Phe Asn Arg Leu Val Lys Val Lys Thr Lys Val Asp Pro Asn
515 520 525

Asn Phe Phe Arg Asn Glu Gln Ser Ile Pro Pro Leu Pro Pro His His
530 535 540

His
545

<210> SEQ ID NO 49

<211> LENGTH: 1437

<212> TYPE: DNA

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 49

atgaaaacaa cagaacttgt cttcatacc gcccccgta tgggtcacct tgtaccaca	60
gtcgaagtcg ccaacaact agttgataga gacgaacagt tgtctattac cgtcttgata	120
atgacgttac cctgggagac taatatccca agttacacca agagtttgtc ctctgactat	180
tcatcccgta tcacgttggtt acaactaagt caacctgaga cgagtgtctc aatgagtagt	240
tttaacgcca taaacttctt cgaatacatt agttcctata aggatcgtgt taaagatgcc	300
gtaaacgaga cattctcttc ttcattctcc gtcaaaacta aaggatttgt aatcgacatg	360
ttttgcacgg caatgataga cgtggccaac gagttcggta ttccatctta tgtattctac	420
acgtccaacg ctgccatgct aggcctacaa cttcaacttc aatccttgtc catcgaatat	480
tcacctaagg ttcataatta tttagaccct gaactctgagg tagctatatc aacgtacatt	540
aacccaatac cagtaaaatg cttaccgggt ataattcttg acaatgataa gagtggcact	600
atgttcgtaa accatgccag gagattccgt gaaacaaagg gtataatggg aaatactttt	660
gcagaattag aaagtcacgc cctaaaggca cttagtgacg atgagaaaat tcctccaatc	720
tatcccgctg gaccattctt aaacttgggt gatggtaatg aggatcataa ccaagagtac	780

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gacatgataa tgaaatggct ggatgaacaa ccacacagtt cagtggtttt cctgtgcttc      840
ggttccaaag gttcatttga agaagaccag gttaaagaga tagcaaatgc tttagagaga      900
tcaggcaata ggttcctgtg gagtttaaga cgtccccctc ccaaggatac tcttcaattc      960
ccttcogaat ttgaaaaccc cgaggaagtg ctacctgtag gattttttca aagaacaaaa    1020
ggcagaggaa aagtcacgag atgggcacca cagcttgcaa ttctatctca ccctgccgtc    1080
ggtggattcg tttcccactg cggtcggaat agtactttgg aatcagttag atcaggtgta    1140
cccatagcaa catggcctct ttatgcagag cagcagtcca atgcatttca attggtcaag    1200
gatctaggta tggccgtcga aattaaatg gattaccgtg aggactttaa caagactaat    1260
cctccattgg taaaggcaga ggaatagaa gacggcatta ggaagttgat ggactccgag    1320
aataagatta gggcaaaggt gatggaaatg aaagataagt ccagagctgc attactggaa    1380
ggaggatcct cctatgttgc actgggtcac ttcgtggaga ccgtaatgaa gaactaa      1437

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<210> SEQ ID NO 50

<211> LENGTH: 478

<212> TYPE: PRT

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 50

```

Met Lys Thr Thr Glu Leu Val Phe Ile Pro Ala Pro Gly Met Gly His
 1             5             10             15
Leu Val Pro Thr Val Glu Val Ala Lys Gln Leu Val Asp Arg Asp Glu
      20             25             30
Gln Leu Ser Ile Thr Val Leu Ile Met Thr Leu Pro Leu Glu Thr Asn
      35             40             45
Ile Pro Ser Tyr Thr Lys Ser Leu Ser Ser Asp Tyr Ser Ser Arg Ile
      50             55             60
Thr Leu Leu Gln Leu Ser Gln Pro Glu Thr Ser Val Ser Met Ser Ser
      65             70             75             80
Phe Asn Ala Ile Asn Phe Phe Glu Tyr Ile Ser Ser Tyr Lys Asp Arg
      85             90             95
Val Lys Asp Ala Val Asn Glu Thr Phe Ser Ser Ser Ser Ser Val Lys
      100            105            110
Leu Lys Gly Phe Val Ile Asp Met Phe Cys Thr Ala Met Ile Asp Val
      115            120            125
Ala Asn Glu Phe Gly Ile Pro Ser Tyr Val Phe Tyr Thr Ser Asn Ala
      130            135            140
Ala Met Leu Gly Leu Gln Leu His Phe Gln Ser Leu Ser Ile Glu Tyr
      145            150            155            160
Ser Pro Lys Val His Asn Tyr Leu Asp Pro Glu Ser Glu Val Ala Ile
      165            170            175
Ser Thr Tyr Ile Asn Pro Ile Pro Val Lys Cys Leu Pro Gly Ile Ile
      180            185            190
Leu Asp Asn Asp Lys Ser Gly Thr Met Phe Val Asn His Ala Arg Arg
      195            200            205
Phe Arg Glu Thr Lys Gly Ile Met Val Asn Thr Phe Ala Glu Leu Glu
      210            215            220
Ser His Ala Leu Lys Ala Leu Ser Asp Asp Glu Lys Ile Pro Pro Ile
      225            230            235            240
Tyr Pro Val Gly Pro Ile Leu Asn Leu Gly Asp Gly Asn Glu Asp His
      245            250            255

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Asn Gln Glu Tyr Asp Met Ile Met Lys Trp Leu Asp Glu Gln Pro His
 260 265 270
 Ser Ser Val Val Phe Leu Cys Phe Gly Ser Lys Gly Ser Phe Glu Glu
 275 280 285
 Asp Gln Val Lys Glu Ile Ala Asn Ala Leu Glu Arg Ser Gly Asn Arg
 290 295 300
 Phe Leu Trp Ser Leu Arg Arg Pro Pro Pro Lys Asp Thr Leu Gln Phe
 305 310 315 320
 Pro Ser Glu Phe Glu Asn Pro Glu Glu Val Leu Pro Val Gly Phe Phe
 325 330 335
 Gln Arg Thr Lys Gly Arg Gly Lys Val Ile Gly Trp Ala Pro Gln Leu
 340 345 350
 Ala Ile Leu Ser His Pro Ala Val Gly Gly Phe Val Ser His Cys Gly
 355 360 365
 Trp Asn Ser Thr Leu Glu Ser Val Arg Ser Gly Val Pro Ile Ala Thr
 370 375 380
 Trp Pro Leu Tyr Ala Glu Gln Gln Ser Asn Ala Phe Gln Leu Val Lys
 385 390 395 400
 Asp Leu Gly Met Ala Val Glu Ile Lys Met Asp Tyr Arg Glu Asp Phe
 405 410 415
 Asn Lys Thr Asn Pro Pro Leu Val Lys Ala Glu Glu Ile Glu Asp Gly
 420 425 430
 Ile Arg Lys Leu Met Asp Ser Glu Asn Lys Ile Arg Ala Lys Val Met
 435 440 445
 Glu Met Lys Asp Lys Ser Arg Ala Ala Leu Leu Glu Gly Gly Ser Ser
 450 455 460
 Tyr Val Ala Leu Gly His Phe Val Glu Thr Val Met Lys Asn
 465 470 475

<210> SEQ ID NO 51
 <211> LENGTH: 1413
 <212> TYPE: DNA
 <213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 51

atggttcaac cacacgtctt actggttact ttccagcac aaggccatat caacccttgc	60
ctacaattcg ccaaaagact aataaggatg ggcacgaag taacttttgc cagagtgtgta	120
ttcgacata ggcgtatggc taaaactacg acatcaactt tgtccaaagg actaaacttc	180
gccgccttca gtgatggcta tgacgatgga ttcaaagcgg acgaacatga cagtcaacac	240
tacatgagtg aaataaagtc cgtggatct aaaacactta aggatattat acttaaatcc	300
tccgatgagg gaagaccctg tacctcttta gtttattcac tgttactgcc ctgggctgca	360
aaagtcgcca gagagtttca tattccttgc gctttattgt ggatccaacc agctacggta	420
ttagacatct actattacta cttcaatgga tacgaggatg caataaaggg atcaacaaac	480
gaccccaact ggtgtattca actgcctaga cttcctctat taaaaagtca ggacttacct	540
agttttttac tgtcatccag taacgaagaa aaatattcat tcgctttacc caccttcaaa	600
gagcagcttg acactttgga tgttgaagag aaccccaagg ttttggtaa tacttttgac	660
gctttggagc caaaagagct aaaggctatt gaaaaatata accttatcgg cataggacct	720
ttaatcccct ctactttctt agatggcaaa gacctctag attcaagttt cggaggtgat	780
ttgtttcaaa agagtaacga ttatatcgag tggctaaata gtaaagccaa ctccagtgtg	840
gtctacatctt ctttoggag tcttctgaat ttatcaaaaa accaaaagga agagatcgca	900

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aaaggactga tagagataaa aaaacctttc ttatgggtga tcagagacca ggaaaacggt    960
aaaggcgatg agaaggagga aaaactgtcc tgtatgatgg agctagagaa acaaggaaaa    1020
atcggtccct ggtgttcaca gttagaagtg ttaacccatc catccatagg ttgcttcgta    1080
tcacattgtg gttggaatag tacacttgaa agtctttcat caggcgtctc tgcgtcgca    1140
ttccccact ggacggacca gggcacaaac gccaaactga tcgaagatgt atggaagacg    1200
ggcgtcaggc taaaaaaaa tgaggatggc gtggtagaga gtgaagagat aaagcgttgc    1260
atagaaatgg tcattgatgg cggtgaaaag ggagaggaaa tgaggcgtaa cgcacaaaag    1320
tggaaggaac tagcccgta agcagtgaag gaaggaggtt ctagtgagat gaatttaaaa    1380
gctttcgtgc aggaagtgg aaaaggctgc tga                                  1413

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<210> SEQ ID NO 52

<211> LENGTH: 470

<212> TYPE: PRT

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 52

```

Met Val Gln Pro His Val Leu Leu Val Thr Phe Pro Ala Gln Gly His
 1              5              10              15

Ile Asn Pro Cys Leu Gln Phe Ala Lys Arg Leu Ile Arg Met Gly Ile
      20              25              30

Glu Val Thr Phe Ala Thr Ser Val Phe Ala His Arg Arg Met Ala Lys
      35              40              45

Thr Thr Thr Ser Thr Leu Ser Lys Gly Leu Asn Phe Ala Ala Phe Ser
      50              55              60

Asp Gly Tyr Asp Asp Gly Phe Lys Ala Asp Glu His Asp Ser Gln His
      65              70              75              80

Tyr Met Ser Glu Ile Lys Ser Arg Gly Ser Lys Thr Leu Lys Asp Ile
      85              90              95

Ile Leu Lys Ser Ser Asp Glu Gly Arg Pro Val Thr Ser Leu Val Tyr
      100             105             110

Ser Leu Leu Leu Pro Trp Ala Ala Lys Val Ala Arg Glu Phe His Ile
      115             120             125

Pro Cys Ala Leu Leu Trp Ile Gln Pro Ala Thr Val Leu Asp Ile Tyr
      130             135             140

Tyr Tyr Tyr Phe Asn Gly Tyr Glu Asp Ala Ile Lys Gly Ser Thr Asn
      145             150             155             160

Asp Pro Asn Trp Cys Ile Gln Leu Pro Arg Leu Pro Leu Leu Lys Ser
      165             170             175

Gln Asp Leu Pro Ser Phe Leu Leu Ser Ser Ser Asn Glu Glu Lys Tyr
      180             185             190

Ser Phe Ala Leu Pro Thr Phe Lys Glu Gln Leu Asp Thr Leu Asp Val
      195             200             205

Glu Glu Asn Pro Lys Val Leu Val Asn Thr Phe Asp Ala Leu Glu Pro
      210             215             220

Lys Glu Leu Lys Ala Ile Glu Lys Tyr Asn Leu Ile Gly Ile Gly Pro
      225             230             235             240

Leu Ile Pro Ser Thr Phe Leu Asp Gly Lys Asp Pro Leu Asp Ser Ser
      245             250             255

Phe Gly Gly Asp Leu Phe Gln Lys Ser Asn Asp Tyr Ile Glu Trp Leu
      260             265             270

Asn Ser Lys Ala Asn Ser Ser Val Val Tyr Ile Ser Phe Gly Ser Leu

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275	280	285
Leu Asn Leu Ser Lys Asn Gln Lys Glu Glu Ile Ala Lys Gly Leu Ile 290 295 300		
Glu Ile Lys Lys Pro Phe Leu Trp Val Ile Arg Asp Gln Glu Asn Gly 305 310 315 320		
Lys Gly Asp Glu Lys Glu Glu Lys Leu Ser Cys Met Met Glu Leu Glu 325 330 335		
Lys Gln Gly Lys Ile Val Pro Trp Cys Ser Gln Leu Glu Val Leu Thr 340 345 350		
His Pro Ser Ile Gly Cys Phe Val Ser His Cys Gly Trp Asn Ser Thr 355 360 365		
Leu Glu Ser Leu Ser Ser Gly Val Ser Val Val Ala Phe Pro His Trp 370 375 380		
Thr Asp Gln Gly Thr Asn Ala Lys Leu Ile Glu Asp Val Trp Lys Thr 385 390 395 400		
Gly Val Arg Leu Lys Lys Asn Glu Asp Gly Val Val Glu Ser Glu Glu 405 410 415		
Ile Lys Arg Cys Ile Glu Met Val Met Asp Gly Gly Glu Lys Gly Glu 420 425 430		
Glu Met Arg Arg Asn Ala Gln Lys Trp Lys Glu Leu Ala Arg Glu Ala 435 440 445		
Val Lys Glu Gly Gly Ser Ser Glu Met Asn Leu Lys Ala Phe Val Gln 450 455 460		
Glu Val Gly Lys Gly Cys 465 470		

<210> SEQ ID NO 53

<211> LENGTH: 1449

<212> TYPE: DNA

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 53

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atgaaagaga ctaaaaaaat tgagttagtt tttatcccca gtcttggtat aggacactta    60
gtctcaactg tggagatggc caaactgttg atagcccggtg aagagcaact ttctattact    120
gtcttgatta tacaatggcc taatgataaa aagctagaca gttatatcca gtccgtcgca    180
aactttagtt ctgactgaa gttttacgt ctgccccaaag atgactcaat catgcaactt    240
ttgaaatcaa acattttcac gacattcatc gcctctcaca agccagctgt aagagacgcc    300
gttgctgaca tactaaagag tgaagtaat aacacattgg caggcattgt aatcgatctt    360
ttctgcacat ccatgatcga tgtagccaat gagtttgagc tgcctactta tgtgttttac    420
actagtggcg cagccacggt gggctctgcac taccatattc aaaatctgcg tgatgagttt    480
aataaagaca ttaccaaata taaggatgag ccagaagaaa aattaagtat agccacgtac    540
cttaaccat tcctgtctaa gtgtctaccc tccgtggcat tggataagga aggaggatca    600
acgatgttcc tagacttagc taagagggtc aggagacca aaggcataat gattaacact    660
tttcttgagc tggaatcata cgctctaaac tcattgtcta gagataaaaa cttgccccct    720
atataccctg taggccctgt tttgaacttg aacaacgttg agggtgataa cttgggctct    780
agtgatcaaa ataccatgaa atggctggac gaccagccag cttcttccgt tgtgttccta    840
tgttttggtc caggaggaag ttctgaaaaa caccaagtca aagaaatage ttatgcotta    900
gaatcttccg gatgcagggt cttgtggagt ttgcgtagac cccccacgga agatgctagg    960
ttcccttcta attacgaaaa cttagaggaa attttaccag agggatttct ggaaagaacg   1020

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aaaggcattg gtaaggtcat tggatgggcc ccacagttag caatcttgtc tcacaagtcc 1080
acaggaggat tcgtgtctca ttgcggatgg aactctaccc ttgaaagtac ctatttcggc 1140
gttcctattg ctacttggcc aatgtatgct gaacaacagg ccaacgcttt tcaacttgtt 1200
aaagatttga ggatgggtgt tgagatcaaa atggattata ggaaggatat gaaggaatg 1260
ggcaaggagg ttatcgtaa ggcagaagaa attgaaaagg ccataaggga aatcatggac 1320
tcagaatcag aaatcagggt caaggtcaaa gagatgaagg agaaaagtcg tgcagcccaa 1380
atggaaggag gatcatcata tacctctatc ggcggttca ttcaaataat catggagaac 1440
tcacagtaa 1449

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<210> SEQ ID NO 54

<211> LENGTH: 482

<212> TYPE: PRT

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 54

```

Met Lys Glu Thr Lys Lys Ile Glu Leu Val Phe Ile Pro Ser Pro Gly
 1             5             10             15
Ile Gly His Leu Val Ser Thr Val Glu Met Ala Lys Leu Leu Ile Ala
      20             25             30
Arg Glu Glu Gln Leu Ser Ile Thr Val Leu Ile Ile Gln Trp Pro Asn
      35             40             45
Asp Lys Lys Leu Asp Ser Tyr Ile Gln Ser Val Ala Asn Phe Ser Ser
      50             55             60
Arg Leu Lys Phe Ile Arg Leu Pro Gln Asp Asp Ser Ile Met Gln Leu
      65             70             75             80
Leu Lys Ser Asn Ile Phe Thr Thr Phe Ile Ala Ser His Lys Pro Ala
      85             90             95
Val Arg Asp Ala Val Ala Asp Ile Leu Lys Ser Glu Ser Asn Asn Thr
      100            105            110
Leu Ala Gly Ile Val Ile Asp Leu Phe Cys Thr Ser Met Ile Asp Val
      115            120            125
Ala Asn Glu Phe Glu Leu Pro Thr Tyr Val Phe Tyr Thr Ser Gly Ala
      130            135            140
Ala Thr Leu Gly Leu His Tyr His Ile Gln Asn Leu Arg Asp Glu Phe
      145            150            155            160
Asn Lys Asp Ile Thr Lys Tyr Lys Asp Glu Pro Glu Glu Lys Leu Ser
      165            170            175
Ile Ala Thr Tyr Leu Asn Pro Phe Pro Ala Lys Cys Leu Pro Ser Val
      180            185            190
Ala Leu Asp Lys Glu Gly Gly Ser Thr Met Phe Leu Asp Leu Ala Lys
      195            200            205
Arg Phe Arg Glu Thr Lys Gly Ile Met Ile Asn Thr Phe Leu Glu Leu
      210            215            220
Glu Ser Tyr Ala Leu Asn Ser Leu Ser Arg Asp Lys Asn Leu Pro Pro
      225            230            235            240
Ile Tyr Pro Val Gly Pro Val Leu Asn Leu Asn Asn Val Glu Gly Asp
      245            250            255
Asn Leu Gly Ser Ser Asp Gln Asn Thr Met Lys Trp Leu Asp Asp Gln
      260            265            270
Pro Ala Ser Ser Val Val Phe Leu Cys Phe Gly Ser Gly Gly Ser Phe
      275            280            285

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Glu Lys His Gln Val Lys Glu Ile Ala Tyr Ala Leu Glu Ser Ser Gly
 290 295 300
 Cys Arg Phe Leu Trp Ser Leu Arg Arg Pro Pro Thr Glu Asp Ala Arg
 305 310 315 320
 Phe Pro Ser Asn Tyr Glu Asn Leu Glu Glu Ile Leu Pro Glu Gly Phe
 325 330 335
 Leu Glu Arg Thr Lys Gly Ile Gly Lys Val Ile Gly Trp Ala Pro Gln
 340 345 350
 Leu Ala Ile Leu Ser His Lys Ser Thr Gly Gly Phe Val Ser His Cys
 355 360 365
 Gly Trp Asn Ser Thr Leu Glu Ser Thr Tyr Phe Gly Val Pro Ile Ala
 370 375 380
 Thr Trp Pro Met Tyr Ala Glu Gln Gln Ala Asn Ala Phe Gln Leu Val
 385 390 395 400
 Lys Asp Leu Arg Met Gly Val Glu Ile Lys Met Asp Tyr Arg Lys Asp
 405 410 415
 Met Lys Val Met Gly Lys Glu Val Ile Val Lys Ala Glu Glu Ile Glu
 420 425 430
 Lys Ala Ile Arg Glu Ile Met Asp Ser Glu Ser Glu Ile Arg Val Lys
 435 440 445
 Val Lys Glu Met Lys Glu Lys Ser Arg Ala Ala Gln Met Glu Gly Gly
 450 455 460
 Ser Ser Tyr Thr Ser Ile Gly Gly Phe Ile Gln Ile Ile Met Glu Asn
 465 470 475 480
 Ser Gln

<210> SEQ ID NO 55
 <211> LENGTH: 1491
 <212> TYPE: DNA
 <213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 55

atggctactc aggtgcataa attgcatttc attctgttcc cactgatggc tcccgggtcac	60
atgatcccta tgatagacat cgcaaaacta ttggctaacc gtggcgtgat aactaccata	120
ataactacgc cggtaacgc caatcgtttt tctctacga tcaactagggc cattaatca	180
ggcctaagaa tccagatttt aaccttaaaa tteccatcag ttgaggtagg cctgcctgaa	240
ggatgtgaaa acatcgacat gttgccatct ttggacttag cctctaaatt ctttgctgct	300
atttctatgc ttaacaaca agtggagaac ttgctagagg gtattaaccc tagtccctca	360
tgcgttattt ctgacatggg ctteccatgg acgacacaga tcgctcaaaa tttcaatatt	420
cctcgtatcg tatttcatgg cacgtgttgc ttttctcttc tttgttctta caaaatcctg	480
tcatccaata tcttagagaa cattactagt gactcagagt attttgcgt gccagatctg	540
ccagaccgtg tcgagctaac taaggcccaa gtctctggat ctacaaagaa tactacatca	600
gtaagtagtt cagtactgaa ggaggttaca gagcagatca ggcttcgaga ggaatcatcc	660
tacggtgtga tagttaattc cttcgaagaa ctggaacagg tgtatgaaaa agagtacaga	720
aaagccaggg gcaaaaaggc ctggtgcgtg ggtcctgtct ctttgtgcaa caaggagatt	780
gaagatcttg ttactagagg aaacaaaacc gctatagaca atcaggattg tcttaagtgg	840
ttagacaact tcgagactga atccgtcgtc tatgcaagtt taggctcact aagtaggctt	900
acgttactgc aaatggttga gctgggattg ggactggagg agagtaatag gccatttgta	960
tgggttctgg gaggaggaga caaactaaat gatcttgaga aatggatatt ggagaatggc	1020

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tttgaacagc gtataaagga gagaggtgtc ctgatacgtg gctgggcacc tcaagtattg 1080
attttaagtc accccgcaat tggaggagtt ttaacgcatt gtggatggaa ctctacatta 1140
gagggcattt cagccggact acccatggtc acctggccac tatttgccga acagttctgt 1200
aacgaaaaat tagtagtgca ggttcttaaa atcggtgtct cacttgagtg gaaggtccct 1260
gttaagtggg gtgacgaaga gaacgtaggt gtcttagtga aaaaggatga cgttaaaaaa 1320
gcactggata agctaattga tgagggtgag gagggccagg ttaggaggac caaagccaaa 1380
gagcttggtg agttagctaa aaaagccttt ggagaggcg gatcatccta cgtgaaccta 1440
acgtccctaa ttgaagatat aatcgagcag cagaaccata aggagaagta g 1491

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<210> SEQ ID NO 56

<211> LENGTH: 496

<212> TYPE: PRT

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 56

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Met Ala Thr Gln Val His Lys Leu His Phe Ile Leu Phe Pro Leu Met
1           5           10          15

Ala Pro Gly His Met Ile Pro Met Ile Asp Ile Ala Lys Leu Leu Ala
20          25          30

Asn Arg Gly Val Ile Thr Thr Ile Ile Thr Thr Pro Val Asn Ala Asn
35          40          45

Arg Phe Ser Ser Thr Ile Thr Arg Ala Ile Lys Ser Gly Leu Arg Ile
50          55          60

Gln Ile Leu Thr Leu Lys Phe Pro Ser Val Glu Val Gly Leu Pro Glu
65          70          75          80

Gly Cys Glu Asn Ile Asp Met Leu Pro Ser Leu Asp Leu Ala Ser Lys
85          90          95

Phe Phe Ala Ala Ile Ser Met Leu Lys Gln Gln Val Glu Asn Leu Leu
100         105         110

Glu Gly Ile Asn Pro Ser Pro Ser Cys Val Ile Ser Asp Met Gly Phe
115         120         125

Pro Trp Thr Thr Gln Ile Ala Gln Asn Phe Asn Ile Pro Arg Ile Val
130         135         140

Phe His Gly Thr Cys Cys Phe Ser Leu Leu Cys Ser Tyr Lys Ile Leu
145         150         155         160

Ser Ser Asn Ile Leu Glu Asn Ile Thr Ser Asp Ser Glu Tyr Phe Val
165         170         175

Val Pro Asp Leu Pro Asp Arg Val Glu Leu Thr Lys Ala Gln Val Ser
180         185         190

Gly Ser Thr Lys Asn Thr Thr Ser Val Ser Ser Ser Val Leu Lys Glu
195         200         205

Val Thr Glu Gln Ile Arg Leu Ala Glu Glu Ser Ser Tyr Gly Val Ile
210         215         220

Val Asn Ser Phe Glu Glu Leu Glu Gln Val Tyr Glu Lys Glu Tyr Arg
225         230         235         240

Lys Ala Arg Gly Lys Lys Val Trp Cys Val Gly Pro Val Ser Leu Cys
245         250         255

Asn Lys Glu Ile Glu Asp Leu Val Thr Arg Gly Asn Lys Thr Ala Ile
260         265         270

Asp Asn Gln Asp Cys Leu Lys Trp Leu Asp Asn Phe Glu Thr Glu Ser
275         280         285

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Val	Val	Tyr	Ala	Ser	Leu	Gly	Ser	Leu	Ser	Arg	Leu	Thr	Leu	Leu	Gln
	290					295					300				
Met	Val	Glu	Leu	Gly	Leu	Gly	Leu	Glu	Glu	Ser	Asn	Arg	Pro	Phe	Val
305					310					315					320
Trp	Val	Leu	Gly	Gly	Gly	Asp	Lys	Leu	Asn	Asp	Leu	Glu	Lys	Trp	Ile
			325					330						335	
Leu	Glu	Asn	Gly	Phe	Glu	Gln	Arg	Ile	Lys	Glu	Arg	Gly	Val	Leu	Ile
		340						345					350		
Arg	Gly	Trp	Ala	Pro	Gln	Val	Leu	Ile	Leu	Ser	His	Pro	Ala	Ile	Gly
	355					360						365			
Gly	Val	Leu	Thr	His	Cys	Gly	Trp	Asn	Ser	Thr	Leu	Glu	Gly	Ile	Ser
	370					375					380				
Ala	Gly	Leu	Pro	Met	Val	Thr	Trp	Pro	Leu	Phe	Ala	Glu	Gln	Phe	Cys
385					390					395					400
Asn	Glu	Lys	Leu	Val	Val	Gln	Val	Leu	Lys	Ile	Gly	Val	Ser	Leu	Gly
			405					410						415	
Val	Lys	Val	Pro	Val	Lys	Trp	Gly	Asp	Glu	Glu	Asn	Val	Gly	Val	Leu
			420				425						430		
Val	Lys	Lys	Asp	Asp	Val	Lys	Lys	Ala	Leu	Asp	Lys	Leu	Met	Asp	Glu
	435					440						445			
Gly	Glu	Glu	Gly	Gln	Val	Arg	Arg	Thr	Lys	Ala	Lys	Glu	Leu	Gly	Glu
	450					455					460				
Leu	Ala	Lys	Lys	Ala	Phe	Gly	Glu	Gly	Gly	Ser	Ser	Tyr	Val	Asn	Leu
465					470					475					480
Thr	Ser	Leu	Ile	Glu	Asp	Ile	Ile	Glu	Gln	Gln	Asn	His	Lys	Glu	Lys
			485					490						495	

<210> SEQ ID NO 57

<211> LENGTH: 1458

<212> TYPE: DNA

<213> ORGANISM: Nicotiana tabacum

<400> SEQUENCE: 57

atgggctcta tcggtgcaga actaaccaag ccacacgccg tatgcattcc ctatcccgcc	60
cagggaacaca taaatcctat gctgaagtta gctaagatac tgcatacaca gggcttccat	120
ataaccttcg taaatacga atttaatcac aggcgtctgc tgaagtccag aggtcctgac	180
tccctgaaaag gtctttcaag ttccagggtc gagacgatac ctgacggact gcccccatgc	240
gaagctgacg ctacacagga cattccttca ctgtgtgaat ccacgactaa tacatgtcta	300
gctcctttta gagacctact tgctaagcta aatgatacga atacttctaa cgtccctccc	360
gtaagttgta ttgtcagtga cggagtgatg tcatttacct ttgcagctgc acaggaactg	420
ggtgtcccag aggttttatt ttggactaca tctgcttggt gattcttagg ttacatgcac	480
tattgcaaag tcattgaaaa aggatatgct ccattaaaag acgcatcaga cctgacgaat	540
ggctatcttg agacaacctt ggacttcac cccggcatga aggacgtcag gctgagagac	600
ttaccttctt ttcttaggac caccaatcca gacgaattta tgattaagtt tgtactacag	660
gaaactgagc gtgctcgtaa ggccagtgcc ataatactta ataccttga aaccttagag	720
gcagaggtat tagaatcatt aaggaacctt ctaccccccg tctatccaat cgcccccttg	780
catttccttg tcaaacacgt agacgatgag aacctaaaag gtctacgttc ctactttgg	840
aaggaggaac ctgaatgtat tcaatggta gacaccaaag aacctaac tcgtcgtgtac	900
gtgaatttcg gatccattac tgtgatgact cccaatcaat taatagagtt cgcttgggga	960

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ctggcaaaact ctcaacagac cttcctttgg atcataaggc ctgacatcgt aagtggatgat 1020
gcttccatat tacctccga gtttgttgag gagactaaga acagaggcat gcttgccctcc 1080
tggtgctctc aggaggaggt actatcccat cccgcaatag tgggattttt gacgcactct 1140
ggttggaact caactttaga atcaatttct agtggcgctc ccatgatctg ttggccttcc 1200
tttgctgagc agcaaacgaa ctgctggttt tcagtgaaga agtgggacgt tggaatggaa 1260
attgattcag atgtgaagag agatgaagta gagagtttag taagagagtt aatgggtggg 1320
ggtaaaggca agaagatgaa gaagaaggca atggagtggg aggaactggc cgaggcttca 1380
gcaaagaac actctggctc ctcttacgtc aatatcgaga agttgggttaa cgatatatta 1440
ctatctagta agcactaa 1458

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<210> SEQ ID NO 58
<211> LENGTH: 485
<212> TYPE: PRT
<213> ORGANISM: Nicotiana tabacum

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<400> SEQUENCE: 58

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Met Gly Ser Ile Gly Ala Glu Leu Thr Lys Pro His Ala Val Cys Ile
1      5      10      15
Pro Tyr Pro Ala Gln Gly His Ile Asn Pro Met Leu Lys Leu Ala Lys
20     25     30
Ile Leu His His Lys Gly Phe His Ile Thr Phe Val Asn Thr Glu Phe
35     40     45
Asn His Arg Arg Leu Leu Lys Ser Arg Gly Pro Asp Ser Leu Lys Gly
50     55     60
Leu Ser Ser Phe Arg Phe Glu Thr Ile Pro Asp Gly Leu Pro Pro Cys
65     70     75     80
Glu Ala Asp Ala Thr Gln Asp Ile Pro Ser Leu Cys Glu Ser Thr Thr
85     90     95
Asn Thr Cys Leu Ala Pro Phe Arg Asp Leu Leu Ala Lys Leu Asn Asp
100    105    110
Thr Asn Thr Ser Asn Val Pro Pro Val Ser Cys Ile Val Ser Asp Gly
115    120    125
Val Met Ser Phe Thr Leu Ala Ala Ala Gln Glu Leu Gly Val Pro Glu
130    135    140
Val Leu Phe Trp Thr Thr Ser Ala Cys Gly Phe Leu Gly Tyr Met His
145    150    155    160
Tyr Cys Lys Val Ile Glu Lys Gly Tyr Ala Pro Leu Lys Asp Ala Ser
165    170    175
Asp Leu Thr Asn Gly Tyr Leu Glu Thr Thr Leu Asp Phe Ile Pro Gly
180    185    190
Met Lys Asp Val Arg Leu Arg Asp Leu Pro Ser Phe Leu Arg Thr Thr
195    200    205
Asn Pro Asp Glu Phe Met Ile Lys Phe Val Leu Gln Glu Thr Glu Arg
210    215    220
Ala Arg Lys Ala Ser Ala Ile Ile Leu Asn Thr Phe Glu Thr Leu Glu
225    230    235    240
Ala Glu Val Leu Glu Ser Leu Arg Asn Leu Leu Pro Pro Val Tyr Pro
245    250    255
Ile Gly Pro Leu His Phe Leu Val Lys His Val Asp Asp Glu Asn Leu
260    265    270
Lys Gly Leu Arg Ser Ser Leu Trp Lys Glu Glu Pro Glu Cys Ile Gln
275    280    285

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Trp Leu Asp Thr Lys Glu Pro Asn Ser Val Val Tyr Val Asn Phe Gly
 290 295 300
 Ser Ile Thr Val Met Thr Pro Asn Gln Leu Ile Glu Phe Ala Trp Gly
 305 310 315 320
 Leu Ala Asn Ser Gln Gln Thr Phe Leu Trp Ile Ile Arg Pro Asp Ile
 325 330 335
 Val Ser Gly Asp Ala Ser Ile Leu Pro Pro Glu Phe Val Glu Glu Thr
 340 345 350
 Lys Asn Arg Gly Met Leu Ala Ser Trp Cys Ser Gln Glu Glu Val Leu
 355 360 365
 Ser His Pro Ala Ile Val Gly Phe Leu Thr His Ser Gly Trp Asn Ser
 370 375 380
 Thr Leu Glu Ser Ile Ser Ser Gly Val Pro Met Ile Cys Trp Pro Phe
 385 390 395 400
 Phe Ala Glu Gln Gln Thr Asn Cys Trp Phe Ser Val Thr Lys Trp Asp
 405 410 415
 Val Gly Met Glu Ile Asp Ser Asp Val Lys Arg Asp Glu Val Glu Ser
 420 425 430
 Leu Val Arg Glu Leu Met Val Gly Gly Lys Gly Lys Lys Met Lys Lys
 435 440 445
 Lys Ala Met Glu Trp Lys Glu Leu Ala Glu Ala Ser Ala Lys Glu His
 450 455 460
 Ser Gly Ser Ser Tyr Val Asn Ile Glu Lys Leu Val Asn Asp Ile Leu
 465 470 475 480
 Leu Ser Ser Lys His
 485

<210> SEQ ID NO 59
 <211> LENGTH: 1377
 <212> TYPE: DNA
 <213> ORGANISM: Stevia rebaudiana

<400> SEQUENCE: 59

atggagaaca aaaccgagac aaccgtagg cgtagacgta ggataatatt gtttcccggtg	60
ccctttcaag gccatataaa cccaatcctg cagctagcca acgtattgta ctcaaagggc	120
ttcagataaa cgatcttcca caccaacttt aataagccaa aaacgtctaa ttatccacac	180
ttcacattta gatttatact tgataacgac ccacaggatg aaagaatatt aaacttgccc	240
acgcacggcc cactagccgg aatgagaata ccaataatca atgagcatgg cgccgacgag	300
ttgcgtagag agctggaatt gttgatgcta gccagtggag aagacgaaga ggtgtcctgc	360
ttaataacgg atgcactttg gtattttgct caatctgtgg ccgactccct taacctgagg	420
cgtcttgtcc ttatgacctc cagtctattc aactttcatg cccatgtctc attgccccaa	480
tttgatgagc ttggctatatt ggatcctgat gacaaaacta ggctggagga acaggcttcc	540
ggttttccca tgctaaaggt taaggacatc aaatccgcct actcaaactg gcagatcctt	600
aaggaaatcc ttggcaaat gatcaaacag acgagggcat ccagtggcgt catctggaac	660
tcctttaagg aacttgaaga atcagaactt gaaacagtaa tcagagaaat acctgcccc	720
agtttcttga tccctctacc taagcacctt acggcttcta gttcttcttt gttggaccac	780
gatcgtactg tctttcaatg gttagatcag caacccccct catcagtgtc atatgtgtca	840
ttcggtagta catcagaagt ggacgaaaag gatttccttg agatagcccg tggattggtg	900
gactctaaac agtccttttt atgggttgtg agacctggat ttgtaaaggg atccacgtgg	960

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gtcgaaccct tgcccgatgg ttctctgggt gaaagaggaa ggatagtgaa gtgggtccct 1020
cagcaagagg tactggccca tgggtctata ggtgctttct ggaccactc cggttggaat 1080
agtacactag aatccgtttg cgagggtgtc cctatgattt tttctgattt tggtttagat 1140
caaccctga atgctaggta catgtcagac gtccttaaag tcggcgctcta cctagaaaat 1200
ggctgggaga ggggtgagat agcaaacgct atcagacgtg ttatggtaga cgaagaggga 1260
gagtacataa ggcaaaacgc cagggtcctg aaacaaaaag ccgatgtgtc cttgatgaag 1320
ggcggtctct catacgaaag tctagaaagt cttgtttctt atatttcctc actataa 1377

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<210> SEQ ID NO 60
<211> LENGTH: 458
<212> TYPE: PRT
<213> ORGANISM: Stevia rebaudiana

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<400> SEQUENCE: 60

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Met Glu Asn Lys Thr Glu Thr Thr Val Arg Arg Arg Arg Ile Ile
1      5      10      15
Leu Phe Pro Val Pro Phe Gln Gly His Ile Asn Pro Ile Leu Gln Leu
20     25     30
Ala Asn Val Leu Tyr Ser Lys Gly Phe Ser Ile Thr Ile Phe His Thr
35     40     45
Asn Phe Asn Lys Pro Lys Thr Ser Asn Tyr Pro His Phe Thr Phe Arg
50     55     60
Phe Ile Leu Asp Asn Asp Pro Gln Asp Glu Arg Ile Ser Asn Leu Pro
65     70     75     80
Thr His Gly Pro Leu Ala Gly Met Arg Ile Pro Ile Ile Asn Glu His
85     90     95
Gly Ala Asp Glu Leu Arg Arg Glu Leu Glu Leu Leu Met Leu Ala Ser
100    105    110
Glu Glu Asp Glu Glu Val Ser Cys Leu Ile Thr Asp Ala Leu Trp Tyr
115    120    125
Phe Ala Gln Ser Val Ala Asp Ser Leu Asn Leu Arg Arg Leu Val Leu
130    135    140
Met Thr Ser Ser Leu Phe Asn Phe His Ala His Val Ser Leu Pro Gln
145    150    155    160
Phe Asp Glu Leu Gly Tyr Leu Asp Pro Asp Asp Lys Thr Arg Leu Glu
165    170    175
Glu Gln Ala Ser Gly Phe Pro Met Leu Lys Val Lys Asp Ile Lys Ser
180    185    190
Ala Tyr Ser Asn Trp Gln Ile Leu Lys Glu Ile Leu Gly Lys Met Ile
195    200    205
Lys Gln Thr Arg Ala Ser Ser Gly Val Ile Trp Asn Ser Phe Lys Glu
210    215    220
Leu Glu Glu Ser Glu Leu Glu Thr Val Ile Arg Glu Ile Pro Ala Pro
225    230    235    240
Ser Phe Leu Ile Pro Leu Pro Lys His Leu Thr Ala Ser Ser Ser Ser
245    250    255
Leu Leu Asp His Asp Arg Thr Val Phe Gln Trp Leu Asp Gln Gln Pro
260    265    270
Pro Ser Ser Val Leu Tyr Val Ser Phe Gly Ser Thr Ser Glu Val Asp
275    280    285
Glu Lys Asp Phe Leu Glu Ile Ala Arg Gly Leu Val Asp Ser Lys Gln
290    295    300

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Ser Phe Leu Trp Val Val Arg Pro Gly Phe Val Lys Gly Ser Thr Trp
 305 310 315 320

Val Glu Pro Leu Pro Asp Gly Phe Leu Gly Glu Arg Gly Arg Ile Val
 325 330 335

Lys Trp Val Pro Gln Gln Glu Val Leu Ala His Gly Ala Ile Gly Ala
 340 345 350

Phe Trp Thr His Ser Gly Trp Asn Ser Thr Leu Glu Ser Val Cys Glu
 355 360 365

Gly Val Pro Met Ile Phe Ser Asp Phe Gly Leu Asp Gln Pro Leu Asn
 370 375 380

Ala Arg Tyr Met Ser Asp Val Leu Lys Val Gly Val Tyr Leu Glu Asn
 385 390 395 400

Gly Trp Glu Arg Gly Glu Ile Ala Asn Ala Ile Arg Arg Val Met Val
 405 410 415

Asp Glu Glu Gly Glu Tyr Ile Arg Gln Asn Ala Arg Val Leu Lys Gln
 420 425 430

Lys Ala Asp Val Ser Leu Met Lys Gly Gly Ser Ser Tyr Glu Ser Leu
 435 440 445

Glu Ser Leu Val Ser Tyr Ile Ser Ser Leu
 450 455

<210> SEQ ID NO 61

<211> LENGTH: 1434

<212> TYPE: DNA

<213> ORGANISM: Lycium barbarum

<400> SEQUENCE: 61

```

atgggtcaat tgcatttttt tttgtttcca atgatggctc aaggtcatat gattccaact    60
ttggatatgg ctaagttgat tgccttctaga ggtgttaagg ctactattat tactactcca    120
ttgaacgaat ctgttttttc taaggctatt caaagaaaca agcaattggg tattgaaatt    180
gaaattgaaa ttagattgat taagtttcca gctttggaaa acgatttgcc agaagattgt    240
gaaagattgg atttgattcc aactgaagct catttgccaa acttttttaa ggctgctgct    300
atgatgcaag aaccattgga acaattgatt caagaatgta gaccagattg tttggtttct    360
gatatgtttt tgccatggac tactgatact gctgctaagt ttaacattcc aagaattggt    420
tttcatggta ctaactactt tgccttgtgt gttggtgatt ctatgagaag aaacaagcca    480
tttaagaacg tttcttctga ttctgaaact tttgttggtc caaacttgcc acatgaaatt    540
aagttgacta gaactcaagt ttctccattt gaacaatctg atgaagaatc tgttatgtct    600
agagttttga aggaagttag agaatctgat ttgaagtctt acggtgttat ttttaactct    660
ttttacgaat tggaaccaga ttacgttgaa cattacacta aggttatggg tagaaagtct    720
tgggctattg gtccattgtc tttgtgtaac agagatgttg aagataaggc tgaaagaggt    780
aagaagtctt ctattgataa gcatgaatgt ttggaatggg tggattctaa gaagccatct    840
tctattgttt acgtttgttt tggttctgtt gctaacttta ctgttactca aatgagagaa    900
ttggcttttg gtttggaagc ttctggtttg gattttattt gggctgttag agctgataac    960
gaagattggg tgccagaagg ttttgaagaa agaactaagg aaaagggttt gattattaga   1020
ggttgggctc cacaagtttt gattttggat catgaatctg ttggtgcttt tgttactcat   1080
tgtggttgga actctacttt ggaaggtatt tctgctggtg ttccaatggt tacttggccca   1140
gtttttgctg aacaattttt taacgaaaag ttggttactc aagttatgag aactggtgct   1200

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gggtgttggt ctgttcaatg gaagagatct gcttctgaag gtgttgaaaa ggaagctatt 1260
gctaaggcta ttaagagagt tatggtttct gaagaagctg aaggttttag aaacagagct 1320
agagcttaca aggaaatggc tagacaagct attgaagaag gtggttcttc ttacactggt 1380
ttgactactt tgttggaaga tatttcttct tacgaatctt tgtcttctga ttaa      1434

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<210> SEQ ID NO 62

<211> LENGTH: 477

<212> TYPE: PRT

<213> ORGANISM: *Lycium barbarum*

<400> SEQUENCE: 62

```

Met Gly Gln Leu His Phe Phe Leu Phe Pro Met Met Ala Gln Gly His
 1              5              10             15
Met Ile Pro Thr Leu Asp Met Ala Lys Leu Ile Ala Ser Arg Gly Val
              20              25              30
Lys Ala Thr Ile Ile Thr Thr Pro Leu Asn Glu Ser Val Phe Ser Lys
              35              40              45
Ala Ile Gln Arg Asn Lys Gln Leu Gly Ile Glu Ile Glu Ile Glu Ile
 50              55              60
Arg Leu Ile Lys Phe Pro Ala Leu Glu Asn Asp Leu Pro Glu Asp Cys
65              70              75              80
Glu Arg Leu Asp Leu Ile Pro Thr Glu Ala His Leu Pro Asn Phe Phe
              85              90              95
Lys Ala Ala Ala Met Met Gln Glu Pro Leu Glu Gln Leu Ile Gln Glu
              100             105             110
Cys Arg Pro Asp Cys Leu Val Ser Asp Met Phe Leu Pro Trp Thr Thr
              115             120             125
Asp Thr Ala Ala Lys Phe Asn Ile Pro Arg Ile Val Phe His Gly Thr
              130             135             140
Asn Tyr Phe Ala Leu Cys Val Gly Asp Ser Met Arg Arg Asn Lys Pro
145              150             155             160
Phe Lys Asn Val Ser Ser Asp Ser Glu Thr Phe Val Val Pro Asn Leu
              165             170             175
Pro His Glu Ile Lys Leu Thr Arg Thr Gln Val Ser Pro Phe Glu Gln
              180             185             190
Ser Asp Glu Glu Ser Val Met Ser Arg Val Leu Lys Glu Val Arg Glu
              195             200             205
Ser Asp Leu Lys Ser Tyr Gly Val Ile Phe Asn Ser Phe Tyr Glu Leu
              210             215             220
Glu Pro Asp Tyr Val Glu His Tyr Thr Lys Val Met Gly Arg Lys Ser
225              230             235             240
Trp Ala Ile Gly Pro Leu Ser Leu Cys Asn Arg Asp Val Glu Asp Lys
              245             250             255
Ala Glu Arg Gly Lys Lys Ser Ser Ile Asp Lys His Glu Cys Leu Glu
              260             265             270
Trp Leu Asp Ser Lys Lys Pro Ser Ser Ile Val Tyr Val Cys Phe Gly
              275             280             285
Ser Val Ala Asn Phe Thr Val Thr Gln Met Arg Glu Leu Ala Leu Gly
              290             295             300
Leu Glu Ala Ser Gly Leu Asp Phe Ile Trp Ala Val Arg Ala Asp Asn
305              310             315             320
Glu Asp Trp Leu Pro Glu Gly Phe Glu Glu Arg Thr Lys Glu Lys Gly
              325             330             335

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Leu Ile Ile Arg Gly Trp Ala Pro Gln Val Leu Ile Leu Asp His Glu
 340 345 350
 Ser Val Gly Ala Phe Val Thr His Cys Gly Trp Asn Ser Thr Leu Glu
 355 360 365
 Gly Ile Ser Ala Gly Val Pro Met Val Thr Trp Pro Val Phe Ala Glu
 370 375 380
 Gln Phe Phe Asn Glu Lys Leu Val Thr Gln Val Met Arg Thr Gly Ala
 385 390 395 400
 Gly Val Gly Ser Val Gln Trp Lys Arg Ser Ala Ser Glu Gly Val Glu
 405 410 415
 Lys Glu Ala Ile Ala Lys Ala Ile Lys Arg Val Met Val Ser Glu Glu
 420 425 430
 Ala Glu Gly Phe Arg Asn Arg Ala Arg Ala Tyr Lys Glu Met Ala Arg
 435 440 445
 Gln Ala Ile Glu Glu Gly Gly Ser Ser Tyr Thr Gly Leu Thr Thr Leu
 450 455 460
 Leu Glu Asp Ile Ser Ser Tyr Glu Ser Leu Ser Ser Asp
 465 470 475

<210> SEQ ID NO 63

<211> LENGTH: 1377

<212> TYPE: DNA

<213> ORGANISM: Stevia rebaudiana

<400> SEQUENCE: 63

atggaaaata aaacggaac caccgtccgc cgtcgtcgcc gtatcattct gttcccggtc	60
cgttcacagg gccacatcaa ccgattctg caactggcga acgtgctgta ttcgaaagg	120
ttcagcatca ccatcttcca tacgaacttc aacaagccga agaccagcaa ttaccgcac	180
tttacgttcc gttttattct ggataacgac ccgcaggatg aacgcatctc taatctgccg	240
acccacggcc cgctggcggg tatgctgatt ccgattatca acgaacacgg cgcagatgaa	300
ctgctgcgcg aactggaact gctgatgctg gccagcgaag aagatgaaga agtttcttgc	360
ctgatcaccg acgcaactgt gtattttgcc cagtctgttg cagatagtct gaacctgcgt	420
cgctctggtc tgatgaccag cagcctgttc aattttcatg ccacggttag tctgcgcgag	480
ttcgatgaac tgggttatct ggaccggat gacaaaacc gcctggaaga acaggcgagc	540
ggctttccga tgctgaaagt caaggatatt aagtcagcgt actcgaactg gcagattctg	600
aaagaaatcc tgggtaaaat gattaagcaa accaaagcaa gttccggcgt catctggaat	660
agtttcaaag aactggaaga atccgaactg gaaacgggta ttcgtgaaat cccggctccg	720
agttttctga ttccgctgcc gaagcatctg accgcgagca gcagcagcct gctggatcac	780
gaccgcacgg tgtttcagtg gctggatcag caaccgccga gttccgtgct gtatgttagc	840
ttcggtagta cctcggaagt ggatgaaaag gactttctgg aaatcgctcg tggcctgggt	900
gatagcaaac aatctttcct gtgggtgggt cgcccggtt ttgtgaaggg ctctacgtgg	960
gttgaaccgc tgccggacgg ctctcctgggt gaacgtggcc gcattgtcaa atgggtgccg	1020
cagcaagaag tgctggcgca tggcgcgatt ggcgcgtttt ggaccactc cggttggaac	1080
tcaacgctgg aatcggtttg tgaaggtgtc ccgatgattt tctcagattt tggcctggac	1140
cagccgctga atgcacgtta tatgtcggat gttctgaaag tcggtgtgta cctggaaaac	1200
ggttggaac gcggcgaaat tgcgaatgcc atccgtcgcg ttatggtcga tgaagaaggc	1260
gaatacatto gtcagaatgc tcgcgtcctg aaacaaaagg cggacgtgag cctgatgaaa	1320

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ggcgggttcac cgtatgaaag tctggaatcc ctggtttcat acatcagctc tctgtaa 1377

<210> SEQ ID NO 64

<211> LENGTH: 458

<212> TYPE: PRT

<213> ORGANISM: Stevia rebaudiana

<400> SEQUENCE: 64

```

Met Glu Asn Lys Thr Glu Thr Thr Val Arg Arg Arg Arg Arg Ile Ile
 1          5          10          15

Leu Phe Pro Val Pro Phe Gln Gly His Ile Asn Pro Ile Leu Gln Leu
 20          25          30

Ala Asn Val Leu Tyr Ser Lys Gly Phe Ser Ile Thr Ile Phe His Thr
 35          40          45

Asn Phe Asn Lys Pro Lys Thr Ser Asn Tyr Pro His Phe Thr Phe Arg
 50          55          60

Phe Ile Leu Asp Asn Asp Pro Gln Asp Glu Arg Ile Ser Asn Leu Pro
 65          70          75          80

Thr His Gly Pro Leu Ala Gly Met Arg Ile Pro Ile Ile Asn Glu His
 85          90          95

Gly Ala Asp Glu Leu Arg Arg Glu Leu Glu Leu Leu Met Leu Ala Ser
100          105          110

Glu Glu Asp Glu Glu Val Ser Cys Leu Ile Thr Asp Ala Leu Trp Tyr
115          120          125

Phe Ala Gln Ser Val Ala Asp Ser Leu Asn Leu Arg Arg Leu Val Leu
130          135          140

Met Thr Ser Ser Leu Phe Asn Phe His Ala His Val Ser Leu Pro Gln
145          150          155          160

Phe Asp Glu Leu Gly Tyr Leu Asp Pro Asp Asp Lys Thr Arg Leu Glu
165          170          175

Glu Gln Ala Ser Gly Phe Pro Met Leu Lys Val Lys Asp Ile Lys Ser
180          185          190

Ala Tyr Ser Asn Trp Gln Ile Leu Lys Glu Ile Leu Gly Lys Met Ile
195          200          205

Lys Gln Thr Lys Ala Ser Ser Gly Val Ile Trp Asn Ser Phe Lys Glu
210          215          220

Leu Glu Glu Ser Glu Leu Glu Thr Val Ile Arg Glu Ile Pro Ala Pro
225          230          235          240

Ser Phe Leu Ile Pro Leu Pro Lys His Leu Thr Ala Ser Ser Ser Ser
245          250          255

Leu Leu Asp His Asp Arg Thr Val Phe Gln Trp Leu Asp Gln Gln Pro
260          265          270

Pro Ser Ser Val Leu Tyr Val Ser Phe Gly Ser Thr Ser Glu Val Asp
275          280          285

Glu Lys Asp Phe Leu Glu Ile Ala Arg Gly Leu Val Asp Ser Lys Gln
290          295          300

Ser Phe Leu Trp Val Val Arg Pro Gly Phe Val Lys Gly Ser Thr Trp
305          310          315          320

Val Glu Pro Leu Pro Asp Gly Phe Leu Gly Glu Arg Gly Arg Ile Val
325          330          335

Lys Trp Val Pro Gln Gln Glu Val Leu Ala His Gly Ala Ile Gly Ala
340          345          350

Phe Trp Thr His Ser Gly Trp Asn Ser Thr Leu Glu Ser Val Cys Glu
355          360          365

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-continued

Gly Val Pro Met Ile Phe Ser Asp Phe Gly Leu Asp Gln Pro Leu Asn
 370 375 380

Ala Arg Tyr Met Ser Asp Val Leu Lys Val Gly Val Tyr Leu Glu Asn
 385 390 395 400

Gly Trp Glu Arg Gly Glu Ile Ala Asn Ala Ile Arg Arg Val Met Val
 405 410 415

Asp Glu Glu Gly Glu Tyr Ile Arg Gln Asn Ala Arg Val Leu Lys Gln
 420 425 430

Lys Ala Asp Val Ser Leu Met Lys Gly Gly Ser Ser Tyr Glu Ser Leu
 435 440 445

Glu Ser Leu Val Ser Tyr Ile Ser Ser Leu
 450 455

<210> SEQ ID NO 65
 <211> LENGTH: 88
 <212> TYPE: PRT
 <213> ORGANISM: *S. cerevisiae*

<400> SEQUENCE: 65

Met Arg Gln Val Trp Phe Ser Trp Ile Val Gly Leu Phe Leu Cys Phe
 1 5 10 15

Phe Asn Val Ser Ser Ala Ala Pro Val Asn Thr Thr Thr Glu Asp Glu
 20 25 30

Thr Ala Gln Ile Pro Ala Glu Ala Val Ile Gly Tyr Ser Asp Leu Glu
 35 40 45

Gly Asp Phe Asp Val Ala Val Leu Pro Phe Ser Asn Ser Thr Asn Asn
 50 55 60

Gly Leu Leu Phe Ile Asn Thr Thr Ile Ala Ser Ile Ala Ala Lys Glu
 65 70 75 80

Glu Gly Val Ser Leu Glu Lys Arg
 85

<210> SEQ ID NO 66
 <211> LENGTH: 22
 <212> TYPE: PRT
 <213> ORGANISM: *S. cerevisiae*

<400> SEQUENCE: 66

Met Arg Gln Val Trp Phe Ser Trp Ile Val Gly Leu Phe Leu Cys Phe
 1 5 10 15

Phe Asn Val Ser Ser Ala
 20

<210> SEQ ID NO 67
 <211> LENGTH: 215
 <212> TYPE: PRT
 <213> ORGANISM: *Mus musculus*

<400> SEQUENCE: 67

Met Arg Gln Val Trp Phe Ser Trp Ile Val Gly Leu Phe Leu Cys Phe
 1 5 10 15

Phe Asn Val Ser Ser Ala Ala Pro Val Asn Thr Thr Thr Glu Asp Glu
 20 25 30

Thr Ala Gln Ile Pro Ala Glu Ala Val Ile Gly Tyr Ser Asp Leu Glu
 35 40 45

Gly Asp Phe Asp Val Ala Val Leu Pro Phe Ser Asn Ser Thr Asn Asn
 50 55 60

Gly Leu Leu Phe Ile Asn Thr Thr Ile Ala Ser Ile Ala Ala Lys Glu

-continued

65	70	75	80
Glu Gly Val Ser Leu Glu Lys Arg Met Asn Phe Ser Gly Lys Tyr Gln	85	90	95
Leu Gln Ser Gln Glu Asn Phe Glu Pro Phe Met Lys Ala Ile Gly Leu	100	105	110
Pro Glu Asp Leu Ile Gln Lys Gly Lys Asp Ile Lys Gly Val Ser Glu	115	120	125
Ile Val His Glu Gly Lys Lys Ile Lys Leu Thr Ile Thr Tyr Gly Pro	130	135	140
Lys Val Val Arg Asn Glu Phe Thr Leu Gly Glu Glu Cys Glu Leu Glu	145	150	155
Thr Met Thr Gly Glu Lys Val Lys Ala Val Val Lys Leu Glu Gly Asp	165	170	175
Asn Lys Met Val Thr Thr Phe Lys Gly Ile Lys Ser Val Thr Glu Leu	180	185	190
Asn Gly Asp Thr Ile Thr Asn Thr Met Thr Leu Gly Asp Ile Val Tyr	195	200	205
Lys Arg Val Ser Lys Arg Ile	210	215	

<210> SEQ ID NO 68

<211> LENGTH: 215

<212> TYPE: PRT

<213> ORGANISM: Bos taurus

<400> SEQUENCE: 68

Met Arg Gln Val Trp Phe Ser Trp Ile Val Gly Leu Phe Leu Cys Phe	1	5	10	15
Phe Asn Val Ser Ser Ala Ala Pro Val Asn Thr Thr Thr Glu Asp Glu	20	25	30	
Thr Ala Gln Ile Pro Ala Glu Ala Val Ile Gly Tyr Ser Asp Leu Glu	35	40	45	
Gly Asp Phe Asp Val Ala Val Leu Pro Phe Ser Asn Ser Thr Asn Asn	50	55	60	
Gly Leu Leu Phe Ile Asn Thr Thr Ile Ala Ser Ile Ala Ala Lys Glu	65	70	75	80
Glu Gly Val Ser Leu Glu Lys Arg Met Asn Phe Ser Gly Lys Tyr Gln	85	90	95	
Val Gln Thr Gln Glu Asn Tyr Glu Ala Phe Met Lys Ala Val Gly Met	100	105	110	
Pro Asp Asp Ile Ile Gln Lys Gly Lys Asp Ile Lys Gly Val Ser Glu	115	120	125	
Ile Val Gln Asn Gly Lys His Phe Lys Phe Ile Ile Thr Ala Gly Ser	130	135	140	
Lys Val Ile Gln Asn Glu Phe Thr Leu Gly Glu Glu Cys Glu Met Glu	145	150	155	160
Phe Met Thr Gly Glu Lys Ile Lys Ala Val Val Gln Gln Glu Gly Asp	165	170	175	
Asn Lys Leu Val Thr Thr Phe Lys Gly Ile Lys Ser Val Thr Glu Phe	180	185	190	
Asn Gly Asp Thr Val Thr Ser Thr Met Thr Lys Gly Asp Val Val Phe	195	200	205	
Lys Arg Val Ser Lys Arg Ile	210	215		

-continued

<210> SEQ ID NO 69
<211> LENGTH: 22
<212> TYPE: PRT
<213> ORGANISM: E. Coli

<400> SEQUENCE: 69

Met Glu Phe Phe Lys Lys Thr Ala Leu Ala Ala Leu Val Met Gly Phe
1 5 10 15
Ser Gly Ala Ala Leu Ala
20

<210> SEQ ID NO 70
<211> LENGTH: 36
<212> TYPE: PRT
<213> ORGANISM: E. Coli

<400> SEQUENCE: 70

Met Lys Leu Ser Arg Arg Ser Phe Met Lys Ala Asn Ala Val Ala Ala
1 5 10 15
Ala Ala Ala Ala Ala Gly Leu Ser Val Pro Gly Val Ala Arg Ala Val
20 25 30
Val Gly Gln Gln
35

<210> SEQ ID NO 71
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 71

Met Ser Ser Ser Phe Leu Ser Ser Thr Ala Phe Phe Leu Leu Leu Cys
1 5 10 15
Leu Gly Phe Cys His Val Ser Ser Ser
20 25

<210> SEQ ID NO 72
<211> LENGTH: 23
<212> TYPE: PRT
<213> ORGANISM: barley (Hordeum vulgare)

<400> SEQUENCE: 72

Met Gly Lys Lys Ser His Ile Cys Cys Phe Ser Leu Leu Leu Leu Leu
1 5 10 15
Phe Ala Gly Leu Ala Ser Gly
20

<210> SEQ ID NO 73
<211> LENGTH: 28
<212> TYPE: PRT
<213> ORGANISM: rice

<400> SEQUENCE: 73

Met Lys Asn Thr Ser Ser Leu Cys Leu Leu Leu Leu Val Val Leu Cys
1 5 10 15
Ser Leu Thr Cys Asn Ser Gly Gln Ala Ala Gln Val
20 25

<210> SEQ ID NO 74
<211> LENGTH: 132
<212> TYPE: PRT
<213> ORGANISM: Gallus gallus

-continued

<400> SEQUENCE: 74

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Met Val Glu Ala Phe Cys Ala Thr Trp Lys Leu Ala Asp Ser His Asn
1           5           10           15
Phe Asp Glu Tyr Met Lys Ala Leu Gly Val Gly Phe Ala Met Arg Gln
20           25           30
Val Gly Asn Val Thr Lys Pro Thr Val Ile Ile Ser Ser Glu Gly Asp
35           40           45
Lys Val Val Ile Arg Thr Gln Ser Thr Phe Lys Asn Thr Glu Ile Ser
50           55           60
Phe Lys Leu Gly Glu Glu Phe Asp Glu Thr Thr Pro Asp Asp Arg Asn
65           70           75           80
Cys Lys Ser Val Val Thr Leu Asp Gly Asp Lys Leu Val His Val Gln
85           90           95
Lys Trp Asp Gly Lys Glu Thr Asn Phe Val Arg Glu Ile Lys Asp Gly
100          105          110
Arg Met Val Met Thr Leu Thr Phe Gly Asp Val Val Ala Val Arg His
115          120          125
Tyr Glu Lys Ala
130

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<210> SEQ ID NO 75

<211> LENGTH: 127

<212> TYPE: PRT

<213> ORGANISM: Sus scrofa

<400> SEQUENCE: 75

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Met Asn Phe Ser Gly Lys Tyr Gln Val Gln Ser Gln Glu Asn Phe Glu
1           5           10           15
Ala Phe Met Lys Ala Val Gly Leu Pro Asp Glu Leu Ile Gln Lys Gly
20           25           30
Lys Asp Ile Lys Gly Thr Ser Glu Ile Val Gln Asn Gly Lys His Phe
35           40           45
Lys Leu Thr Ile Thr Thr Gly Ser Lys Val Val Gln Asn Glu Phe Thr
50           55           60
Leu Gly Glu Glu Cys Glu Met Glu Thr Leu Thr Gly Glu Lys Val Lys
65           70           75           80
Thr Val Val Gln Leu Glu Gly Asp Asn Lys Leu Val Thr Thr Phe Lys
85           90           95
Gly Ile Lys Ser Val Thr Glu Leu Asn Gly Asp Ile Ile Thr Ser Thr
100          105          110
Met Thr Leu Gly Asp Ile Val Phe Lys Arg Ile Ser Lys Arg Ile
115          120          125

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<210> SEQ ID NO 76

<211> LENGTH: 155

<212> TYPE: PRT

<213> ORGANISM: Equus caballus

<400> SEQUENCE: 76

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Met Asn Phe Ser Gly Lys Tyr Gln Leu Gln Ser Gln Glu Asn Phe Glu
1           5           10           15
Pro Phe Met Lys Ala Val Gly Met Ser Asp Asp Leu Ile Gln Lys Gly
20           25           30
Lys Asp Leu Lys Glu Thr Ser Glu Ile Val His Asn Gly Asn His Phe
35           40           45
Lys Ile Thr Ile Thr Thr Gly Pro Lys Val Val His His Glu Phe Thr

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-continued

50	55	60	
Leu Gly Glu Glu Phe Glu Leu Glu Ser Phe Thr Gly Glu Lys Val Lys			
65	70	75	80
Ala Ile Thr His Met Glu Gly Asp Lys Leu Val Ser Thr Ile Lys Gly			
	85	90	95
Ile Lys Ser Val Thr Glu Leu Lys Gly Asp Ile Ile Thr Asn Gln Leu			
	100	105	110
Ser Val Trp His Thr Pro Val Val Cys Ala Thr Ser Ala Leu Val Pro			
	115	120	125
Ala Cys Arg Pro Gly Pro Val Leu Asp Leu Gly Thr Met Thr Leu Gly			
	130	135	140
Asp Ile Val Phe Lys Arg Val Ser Lys Arg Ile			
145	150	155	

<210> SEQ ID NO 77

<211> LENGTH: 135

<212> TYPE: PRT

<213> ORGANISM: *Rhodamnia argentea*

<400> SEQUENCE: 77

Met Thr Glu Phe Ala Gly Val Tyr Glu Phe Val Lys Asp Asp Gly Lys			
1	5	10	15
Phe Glu Glu Ile Leu Lys Ala Met Asp Val Asn Phe Leu Ile Arg Lys			
	20	25	30
Val Ala Gly Lys Met Asn Pro Asn Thr Ala Ile Asp Val Leu Asp Asp			
	35	40	45
Gly Lys Met Ala Phe Lys Thr Ile Thr Pro Leu Lys Thr Val Glu Ile			
	50	55	60
His Phe Glu Leu Gly Lys Glu Tyr Glu Asn Lys Arg Leu Asp Gly Thr			
65	70	75	80
Val Val Lys Gly Ile Val Thr Arg Asp Gly Asn Lys Leu Ile Gln Glu			
	85	90	95
Gln Met Ser Glu Pro Lys Phe Lys Val Ile Arg Glu Leu Asp Gly Pro			
	100	105	110
Glu Lys Leu Ile Val Thr Trp Met Cys Lys Asp Ile Val Cys Val Arg			
	115	120	125
Glu Tyr Lys Arg Ile Gln Thr			
130	135		

<210> SEQ ID NO 78

<211> LENGTH: 127

<212> TYPE: PRT

<213> ORGANISM: *Mus musculus*

<400> SEQUENCE: 78

Met Asn Phe Ser Gly Lys Tyr Gln Leu Gln Ser Gln Glu Asn Phe Glu			
1	5	10	15
Pro Phe Met Lys Ala Ile Gly Leu Pro Glu Asp Leu Ile Gln Lys Gly			
	20	25	30
Lys Asp Ile Lys Gly Val Ser Glu Ile Val His Glu Gly Lys Lys Ile			
	35	40	45
Lys Leu Thr Ile Thr Tyr Gly Pro Lys Val Val Arg Asn Glu Phe Thr			
	50	55	60
Leu Gly Glu Glu Cys Glu Leu Glu Thr Met Thr Gly Glu Lys Val Lys			
65	70	75	80
Ala Val Val Lys Leu Glu Gly Asp Asn Lys Met Val Thr Thr Phe Lys			

85								90				95			
Gly	Ile	Lys	Ser	Val	Thr	Glu	Leu	Asn	Gly	Asp	Thr	Ile	Thr	Asn	Thr
100								105				110			
Met	Thr	Leu	Gly	Asp	Ile	Val	Tyr	Lys	Arg	Val	Ser	Lys	Arg	Ile	
115								120				125			

<400> SEQUENCE: 79

[illegible]

<400> SEQUENCE: 80

[illegible]

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<210> SEQ ID NO 81
 <211> LENGTH: 126
 <212> TYPE: PRT
 <213> ORGANISM: *Sus scrofa*

<400> SEQUENCE: 81

```

Met Asn Phe Ser Gly Lys Tyr Gln Val Gln Ser Gln Glu Asn Phe Glu
 1             5             10             15

Ala Phe Met Lys Ala Val Gly Gln Pro Asp Glu Leu Ile Gln Lys Gly
      20             25             30

Lys Asp Ile Lys Gly Thr Ser Glu Ile Val Gln Asn Gly Lys His Phe
      35             40             45

Lys Val Thr Ile Thr Thr Gly Ser Lys Val Val Gln Ala Glu Phe Thr
      50             55             60

Leu Gly Glu Glu Cys Glu Ala Glu Thr Asp Thr Gly Glu Lys Val Lys
65             70             75             80

Val Val Val Gln Leu Glu Gly Asp Asn Lys Ala Val Val Thr Ala Lys
      85             90             95

Gly Ile Lys Ala Val Phe Glu Leu Asn Gly Asp Ile Ile Thr Ala Thr
      100            105            110

Ala Thr Leu Gly Asp Ile Val Phe Lys Arg Ile Ser Lys Arg
      115            120            125

```

<210> SEQ ID NO 82
 <211> LENGTH: 126
 <212> TYPE: PRT
 <213> ORGANISM: *Sus scrofa*

<400> SEQUENCE: 82

```

Met Asn Phe Ser Gly Lys Tyr Gln Val Gln Ser Gln Glu Asn Phe Glu
 1             5             10             15

Ala Ala Ala Lys Ala Ala Gly Val Pro Asp Glu Glu Ile Gln Lys Gly
      20             25             30

Lys Asp Ile Lys Gly Thr Leu Glu Ile Val Gln Asn Gly Lys His Phe
      35             40             45

Lys Leu Thr Leu Thr Thr Gly Ser Lys Val Val Gln Asn Glu Phe Thr
      50             55             60

Leu Gly Glu Glu Cys Glu Met Glu Thr Leu Gly Gly Glu Lys Ala Lys
65             70             75             80

Thr Val Val Gln Leu Glu Gly Asp Asn Lys Ala Val Trp Thr Phe Lys
      85             90             95

Gly Ile Lys Ala Val Ile Glu Leu Asn Gly Asp Ile Ile Thr Leu Thr
      100            105            110

Val Thr Leu Gly Asp Ile Val Phe Lys Arg Ile Ser Lys Arg
      115            120            125

```

<210> SEQ ID NO 83
 <211> LENGTH: 155
 <212> TYPE: PRT
 <213> ORGANISM: *Equus caballus*

<400> SEQUENCE: 83

```

Met Asn Phe Ser Gly Lys Tyr Gln Leu Gln Ser Gln Glu Asn Ala Glu
 1             5             10             15

Pro Phe Phe Lys Ala Val Gly Met Ser Asp Asp Leu Ile Gln Leu Ala
      20             25             30

Lys Asp Leu Lys Val Thr Gly Glu Ile Val His Asn Gly Asn His Phe
      35             40             45

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-continued

Lys Ile Thr His Thr Ser Gly Ala Ala Val Val His His Glu Phe Thr
 50 55 60
 Leu Gly Glu Glu Phe Glu Leu Glu Ser Leu Thr Gly Glu Lys Val Lys
 65 70 75 80
 Ala Ile Thr His Met Glu Gly Asp Lys Leu Val Ser Thr Ile Lys Gly
 85 90 95
 Ile Lys Ser Val Thr Glu Leu Lys Gly Asp Ile Ile Thr Asn Gln Leu
 100 105 110
 Ser Val Trp His Thr Pro Val Val Cys Ala Thr Ser Ala Leu Val Pro
 115 120 125
 Ala Cys Arg Pro Gly Pro Val Leu Asp Leu Gly Thr Met Thr Leu Gly
 130 135 140
 Asp Ile Val Met Lys Phe Val Ser Lys Arg Ile
 145 150 155

<210> SEQ ID NO 84
 <211> LENGTH: 155
 <212> TYPE: PRT
 <213> ORGANISM: Equus caballus

<400> SEQUENCE: 84

Met Asn Phe Ser Gly Lys Tyr Gln Leu Gln Ser Ser Glu Asn Val Glu
 1 5 10 15
 Pro Phe Asn Lys Ala Val Gly Met Ser Asp Asp Leu Ile Gln Lys Ala
 20 25 30
 Lys Asp Leu Lys Phe Thr Gly Glu Ile Val His Asn Gly Asn His Phe
 35 40 45
 Lys Ile Thr Ile Thr Thr Gly Pro Lys Val Val His His Glu Phe Thr
 50 55 60
 Leu Gly Glu Glu Phe Glu Leu Glu His Phe Thr Gly Glu Lys Val Lys
 65 70 75 80
 Ala Ile Thr His Met Glu Gly Asp Lys Leu Val Met Thr Val Lys Gly
 85 90 95
 Ile Lys Ser Val Met Val Trp Lys Gly Asp Ile Ile Thr Asn Gln Leu
 100 105 110
 Ser Val Trp His Thr Pro Val Val Cys Ala Thr Ser Ala Leu Val Pro
 115 120 125
 Ala Cys Arg Pro Gly Pro Val Leu Asp Leu Gly Thr Gly Val Leu Gly
 130 135 140
 Asp Ile Val Met Lys His Val Ser Lys Arg Ile
 145 150 155

<210> SEQ ID NO 85
 <211> LENGTH: 135
 <212> TYPE: PRT
 <213> ORGANISM: Rhodamnia argentea

<400> SEQUENCE: 85

Met Thr Glu Phe Ala Gly Val Tyr Glu Phe Val Lys Asp Asp Gly Lys
 1 5 10 15
 Phe Glu Glu Ile Leu Lys Ala Met Asp Val Asn Phe Leu Ile Arg Lys
 20 25 30
 Val Gly Met Lys Met Asn Ala Asn Val Ala Ile Asp Val Leu Asp Asp
 35 40 45
 Gly Lys Met Ala Ile Lys Val Ile Thr Pro Leu Lys Thr Val Glu Ile
 50 55 60

-continued

His Phe Glu Leu Gly Lys Glu Tyr Glu Asn Lys Arg Phe Asp Gly Thr
 65 70 75 80
 Val Val Lys Gly Ile Val Thr Arg Asp Gly Asn Lys Leu Ile Ile Glu
 85 90 95
 Gln Met Ser Glu Pro Lys Phe Lys Val Ile Ala Glu Leu Asp Gly Pro
 100 105 110
 Glu Lys Leu Ile Phe Thr Glu Met Cys Lys Asp Ile Val Cys Val Ala
 115 120 125
 Glu Tyr Lys Arg Ile Gln Thr
 130 135

<210> SEQ ID NO 86
 <211> LENGTH: 135
 <212> TYPE: PRT
 <213> ORGANISM: Rhodamnia argentea

<400> SEQUENCE: 86

Met Thr Glu Phe Ala Gly Val Tyr Glu Phe Val Lys Asp Asp Gly Lys
 1 5 10 15
 Phe Glu Glu Ile Leu Lys Ala Met Asp Val Asn Phe Leu Ile Arg Lys
 20 25 30
 Val Ala Gly Lys Met Asn Pro Asn Ile Ala Ile Asp Val Leu Asp Asp
 35 40 45
 Gly Lys Met Ala Ile Lys Met Ile Thr Pro Leu Lys Thr Val Glu Val
 50 55 60
 His Phe Glu Leu Gly Lys Glu Tyr Glu Ala Lys Ala Leu Asp Gly Thr
 65 70 75 80
 Val Val Lys Ala Ile Val Thr Arg Asp Gly Asn Lys Leu Ile Ile Glu
 85 90 95
 Asn Met Ser Glu Pro Lys Phe Thr Val Ile Arg Glu Leu Asp Gly Pro
 100 105 110
 Glu Lys Leu Ile Val Thr Trp Met Cys Lys Asp Ile Val Cys Val Ala
 115 120 125
 Glu Tyr Lys Arg Ile Gln Thr
 130 135

What is claimed is:

1. A method of solubilizing a cannabinoid comprising the steps of:

modifying a Fatty-Acid Binding Protein (FABP)-carrier protein having affinity towards a cannabinoid to produce a recombinant FABP-carrier protein having enhanced affinity towards said cannabinoid;

introducing said recombinant FABP-carrier protein to said cannabinoid, wherein said recombinant FABP-carrier protein binds said cannabinoid forming a water-soluble protein-cannabinoid complex; and wherein the recombinant FABP-carrier protein comprises a recombinant FABP-carrier protein having the amino acid sequence selected from the group consisting of: SEQ ID NOs. 79-86, and wherein said cannabinoid is cannabidiol (CBD), or D⁹-tetrahydrocannabinol (THC).

2. The method of claim 1, wherein said step of genetically modifying the FABP-carrier protein comprises at least one of the following:

replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket with side chains orientated toward the binding cavity;

45 replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket having a hydrophilic side chain with amino acid residues having a hydrophobic side chain; and
 50 replacing one or more small hydrophobic amino acid residues of the FABP-carrier protein cannabinoid binding pocket with larger hydrophobic amino acid residues.

3. The method of claim 1, wherein the recombinant FABP-carrier protein is expressed in a protein production system selected from the group consisting of:

55 a bacterial cell culture;
 a yeast cell culture;
 a plant cell culture;
 a fungi cell culture;
 an algae cell culture;
 a bioreactor production system; and
 a plant.

4. The method of claim 1, wherein the recombinant FABP-carrier protein is coupled with a secretion signal.

5. The method of claim 4, wherein said secretion signal is selected from the group consisting of: SEQ ID NO. 3, SEQ ID NO. 65, and SEQ ID NOs. 69-73.

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6. The method of claim 1, wherein said recombinant FABP-carrier protein is further genetically modified to decrease aggregation propensity and potential antigenicity.

7. The method of claim 6, wherein said recombinant FABP-carrier protein comprises a recombinant FABP-carrier protein having one or both of the following mutations: V58E, and V42R.

8. A method of solubilizing a cannabinoid comprising the steps of:

expressing in a protein production system a heterologous nucleotide sequence, operably linked to a promoter, encoding generating a Fatty-Acid Binding Protein (FABP)-carrier protein having affinity towards a cannabinoid; and

introducing said FABP-carrier protein to said cannabinoid, wherein said FABP-carrier protein binds said cannabinoid to form a water-soluble protein-cannabinoid complex, wherein the FABP-carrier protein comprises a FABP-carrier protein having the amino acid sequence selected from the group of consisting of: SEQ

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ID NOs. 79-86, or a sequence having at least 95% or more sequence identity with any of the SEQ ID NOs. 79-86, and wherein said cannabinoid is cannabidiol (CBD), or Δ^9 -tetrahydrocannabinol (THC).

9. The method of claim 8, wherein the recombinant FABP-carrier protein is expressed in a protein production system selected from the group consisting of:

a bacterial cell culture;
a yeast cell culture;
a plant cell culture;
a fungi cell culture;
an algae cell culture;
a bioreactor production system; and
a plant.

10. The method of claim 8, wherein said recombinant FABP-carrier protein comprises a recombinant FABP-carrier protein having one or both of the following mutations: V58E, and V42R.

* * * * *